Course Notes

Abstract Algebra III

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Chapter 1

Lecture 1: Modules

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1.1 Basic Definitions and Examples

Definition 1.1.1. *Let R be a ring with 1, and let M be an abelian group. We say that M is a left R-module if there is a map*

$$R \times M \to M, \quad (r, m) \mapsto rm,$$

such that

$$\begin{aligned} r(m+n) &= rm + rn, \\ (r+s)m &= rm + sm, \\ (rs)m &= r(sm), \\ 1m &= m \end{aligned}$$

for all $r, s \in R$ and $m, n \in M$.

*A **right R-module** is defined similarly, using a map $(m, r) \mapsto mr$ and the axiom $m(rs) = (mr)s$. If R is commutative, left R-modules and right R-modules coincide. In general, a right R-module may be viewed as a left R^{op}-module.*

For each $r \in R$, the map $\varphi_r \colon M \to M$ given by $\varphi_r(m) = rm$ is a homomorphism of abelian groups. Let

$$\mathrm{End}(M) = \{\, \varphi \colon M \to M \mid \varphi \text{ is a homomorphism of abelian groups} \,\}.$$

Then $\mathrm{End}(M)$ is a ring under composition.

Theorem 1.1.2. *Let M be an abelian group. Then M is an R-module if and only if there is a ring homomorphism*

$$\rho \colon R \to \mathrm{End}(M).$$

Proof. (⇒) Suppose M is an R-module. Define $\rho\colon R \to \mathrm{End}(M)$ by

$$\rho(r)(m) = rm \qquad (r \in R,\ m \in M).$$

For each $r \in R$, the map $\rho(r)\colon M \to M$ is a homomorphism of abelian groups since

$$\rho(r)(m_1 + m_2) = r(m_1 + m_2) = rm_1 + rm_2.$$

Moreover, ρ is a ring homomorphism:

- $\rho(r+s)(m) = (r+s)m = rm + sm = \rho(r)(m) + \rho(s)(m)$, so $\rho(r+s) = \rho(r) + \rho(s)$;
- $\rho(rs)(m) = (rs)m = r(sm) = \rho(r)(\rho(s)(m))$, so $\rho(rs) = \rho(r) \circ \rho(s)$;
- $\rho(1)(m) = 1m = m$, so $\rho(1) = \mathrm{id}_M$.

(⇐) Suppose $\rho\colon R \to \mathrm{End}(M)$ is a ring homomorphism. Define scalar multiplication by

$$(r, m) \mapsto \rho(r)(m).$$

Then

$$\begin{aligned} r(m_1 + m_2) &= \rho(r)(m_1 + m_2) = \rho(r)(m_1) + \rho(r)(m_2), \\ (r+s)m &= \rho(r+s)(m) = \rho(r)(m) + \rho(s)(m), \\ (rs)m &= \rho(rs)(m) = \rho(r)(\rho(s)(m)) = r(sm), \\ 1m &= \rho(1)(m) = \mathrm{id}_M(m) = m. \end{aligned}$$

Hence M is an R-module. □

Example 1.1.3. *(1) A vector space over a field is a module over that field.*

(2) An abelian group is a $\mathbb{Z}$-module.

(3) Any ideal of a ring R is naturally an R-module.

(4) Let M be an abelian group. Then M is an $\mathrm{End}(M)$*-module via*

$$\mathrm{End}(M) \times M \to M, \quad (\varphi, m) \mapsto \varphi(m).$$

(5) Let V be a vector space over a field K, and let $A\colon V \to V$ be a linear transformation. Then V is a $K[x]$-module via

$$K[x] \times V \to V, \quad (f(x), v) \mapsto f(A)v.$$

(6) If $|M| = 1$, then M is called the zero module.

(7) Let R be a ring with 1 and let $A \in M_n(R)$. Then the solution set of $AX = 0$, where $X \in R^n$, is a right R-module.

Definition 1.1.4 (R-submodule)**.** *Let R be a ring with 1, and let M be a left R-module. A subset $N \subseteq M$ is called an* R-submodule *of M if*

1. $0 \in N$*;*

2. $n_1, n_2 \in N$ *implies* $n_1 + n_2 \in N$*;*

3. $n \in N$ *implies* $-n \in N$*;*

4. $r \in R$ *and* $n \in N$ *imply* $rn \in N$*.*

Equivalently, N is a subgroup of the abelian group $(M, +)$ that is closed under the R-action.

Example 1.1.5. *(1) If V is a vector space over a field K, then every subspace of V is a K-submodule of V.*

(2) If G is an abelian group, then every subgroup of G is a $\mathbb{Z}$-submodule of G.

(3) The left regular module ${}_RR$ has precisely the left ideals of R as its submodules.

(4) Let G be an abelian group, viewed as an $\mathrm{End}(G)$*-module. Then the* $\mathrm{End}(G)$*-submodules of G are exactly the fully invariant subgroups of G.*

Remark 1.1.6. *A natural question is the following: are submodules of finitely generated modules again finitely generated? In general the answer is no. This property holds for all submodules if and only if the ring is Noetherian.*

Example 1.1.7 (Submodule of a finitely generated module that is not finitely generated)**.** *Let k be a field, and consider the polynomial ring*

$$R = k[x_1, x_2, x_3, \dots]$$

in infinitely many variables.

- *The ring R is a cyclic left R-module, generated by* 1*.*
- *Let $I = \langle x_1, x_2, x_3, \dots \rangle$. Then I is a submodule of the left R-module R.*
- *The submodule I is not finitely generated. Indeed, suppose*

$$I = \langle f_1, \dots, f_n \rangle$$

  for some polynomials $f_1, \dots, f_n \in R$. Let N be the largest index of any variable appearing in any of the f_i. Then every element of $\langle f_1, \dots, f_n \rangle$ lies in the subring $k[x_1, \dots, x_N]$, but $x_{N+1} \in I$ does not. This is a contradiction.

Thus I is a submodule of the finitely generated R-module R which is not finitely generated.

Definition 1.1.8. *Let M be an R-module, and let $(M_i)_{i \in I}$ be a family of submodules of M. Then*

$$\bigcap_{i \in I} M_i$$

is an R-submodule of M. Moreover,

$$\sum_{i\in I} M_i = \left\{ \sum_{i\in I}^{\text{finite}} m_i \,\middle|\, m_i \in M_i \right\}$$

is also an R-submodule of M.

Definition 1.1.9. *Let M be an R-module, and let $S \subseteq M$. The submodule generated by S is*

$$\langle S\rangle = \left\{ \sum_{j=1}^{n} a_j s_j \,\middle|\, n \geq 0,\ a_j \in R,\ s_j \in S \right\}.$$

Equivalently,

$$\langle S\rangle = \bigcap_{\substack{N\leq M\\ S\subseteq N}} N.$$

If $M = \langle S\rangle$, then we say that M is generated by S. In particular, if $|S| = 1$, then M is called cyclic*; if $|S| < \infty$, then M is called* finitely generated.

Definition 1.1.10. *Let R be a ring, and let M be a left R-module. We say that M is* Noetherian *if every ascending chain of submodules*

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$

stabilizes; that is, there exists $n \geq 1$ such that

$$M_n = M_{n+1} = M_{n+2} = \cdots .$$

Equivalently, M is Noetherian if every submodule of M is finitely generated.

1.2 Quotient Modules and Homomorphism Theorems

Definition 1.2.1. *Let R be a ring, let M be a left R-module, and let N be a submodule of M. The* quotient module *(or* factor module*) of M by N is the quotient abelian group*

$$M/N = \{\, x + N \mid x \in M \,\},$$

with scalar multiplication defined by

$$r(x+N) = rx + N \qquad (r \in R,\ x \in M).$$

This is well defined, and with this operation M/N becomes a left R-module.

Example 1.2.2. *(1) If I is a left ideal of a ring R, then R/I is a quotient R-module of the left regular module ${}_RR$.*

(2) For $n \in \mathbb{Z}$, the subgroup $n\mathbb{Z}$ is a submodule of the $\mathbb{Z}$-module $\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$ is a cyclic quotient module.

Definition 1.2.3. *Let R be a ring, and let M and N be left R-modules. A group homomorphism*

$$f\colon M \to N$$

*is called an R-*module homomorphism *(or R-*linear map*) if*

$$f(rx) = rf(x)$$

for all $r \in R$ and $x \in M$.

Definition 1.2.4. *Let $f\colon M \to N$ be an R-module homomorphism. The* kernel *of f is the set*

$$\ker f = \{\, x \in M \mid f(x) = 0 \,\}.$$

The kernel is a submodule of M. The image *of f is the set*

$$\operatorname{Im} f = \{\, f(x) \mid x \in M \,\}.$$

The image is a submodule of N.

If $\ker f = 0$, then f is injective. If $\operatorname{Im} f = N$, then f is surjective. A bijective R-module homomorphism is called an isomorphism.

Example 1.2.5. *Let R be a commutative ring, and let M and N be R-modules. Then $\operatorname{Hom}_R(M, N)$ is naturally an R-module.*

Theorem 1.2.6. *If $f\colon M \to N$ is an R-module homomorphism, then there is a canonical R-module isomorphism*

$$M/\ker f \to \operatorname{Im} f.$$

Theorem 1.2.7. *Let N be an R-submodule of M, and let $\pi_N\colon M \to M/N$ be the quotient map. Then π_N induces a natural bijection*

$$\{\, R\text{-submodules of } M \text{ containing } N \,\} \to \{\, R\text{-submodules of } M/N \,\}.$$

Theorem 1.2.8. *If $N \subseteq H$ are two R-submodules of M, then*

$$M/H \cong (M/N)/(H/N).$$

Theorem 1.2.9. *If N and H are R-submodules of M, then*

$$\frac{H+N}{N} \cong \frac{H}{H \cap N}.$$

Proposition 1.2.10. *Let M be an R-module, let N be a submodule of M, let M' be an R-module, and let $\varphi\colon M \to M'$ be a morphism such that $N \subseteq \ker\varphi$. Then there exists a unique morphism*

$$\varphi_N\colon M/N \to M'$$

such that the following diagram commutes. Moreover,

$$\ker\varphi_N = (\ker\varphi)/N.$$

$$\begin{tikzcd} M \arrow[r, "\varphi"] \arrow[d, "\pi_N"'] & M' \\ M/N \arrow[ur, dashed, "\varphi_N"'] & \end{tikzcd}$$

Remark 1.2.11. *The pair $(M/N, \pi_N)$ is unique in the following sense: if (S, θ) is a pair such that*

- *S is an R-module,*
- *$\theta : M \to S$ is a surjective morphism satisfying the property of $(M/N, \pi_N)$ described in the above proposition,*

then there exists a unique isomorphism $\psi : M/N \to S$ such that the following diagram commutes.

$$\begin{tikzcd} M \arrow[r, two heads, "\theta"] \arrow[d, two heads, "\pi_N"'] & S \\ M/N \arrow[ur, dashed, "\exists!\,\psi"'] & \end{tikzcd}$$

Proof. Since (S, θ) satisfies the same universal property as $(M/N, \pi_N)$, we may apply that property to the canonical projection

$$\pi_N : M \to M/N.$$

Because $N \subseteq \ker(\pi_N)$, there exists a unique morphism

$$u : S \to M/N$$

such that

$$u \circ \theta = \pi_N.$$

On the other hand, since $(M/N, \pi_N)$ has the universal property of the quotient by N, and since $N \subseteq \ker(\theta)$, there exists a unique morphism

$$\psi : M/N \to S$$

such that

$$\psi \circ \pi_N = \theta.$$

We now show that ψ and u are inverse to each other.

First, compute
$$(u \circ \psi) \circ \pi_N = u \circ (\psi \circ \pi_N) = u \circ \theta = \pi_N.$$
But the identity morphism $\mathrm{id}_{M/N} : M/N \to M/N$ also satisfies
$$\mathrm{id}_{M/N} \circ \pi_N = \pi_N.$$
By the uniqueness part of the universal property of $(M/N, \pi_N)$, it follows that
$$u \circ \psi = \mathrm{id}_{M/N}.$$

Similarly,
$$(\psi \circ u) \circ \theta = \psi \circ (u \circ \theta) = \psi \circ \pi_N = \theta.$$
But the identity morphism $\mathrm{id}_S : S \to S$ also satisfies
$$\mathrm{id}_S \circ \theta = \theta.$$
By the uniqueness part of the universal property of (S, θ), we obtain
$$\psi \circ u = \mathrm{id}_S.$$

Therefore ψ is an isomorphism, with inverse u.

Finally, the uniqueness of ψ follows directly from the universal property of $(M/N, \pi_N)$: indeed, ψ is the unique morphism
$$\psi : M/N \to S$$
such that
$$\psi \circ \pi_N = \theta.$$
Hence there exists a unique isomorphism
$$\psi : M/N \to S$$
for which the diagram commutes. □

Proposition 1.2.12. *There is a unique isomorphism $\bar{\varphi} : \operatorname{coim} \varphi \to \operatorname{Im} \varphi$ such that the following diagram commutes:*

$$\begin{tikzcd}
M \arrow[r, "\varphi"] \arrow[d, "\pi_{\ker(\varphi)}"'] & M' \\
\operatorname{coim}(\varphi) \arrow[r, "\overline{\varphi}"] & \operatorname{Im}(\varphi) \arrow[u, hook]
\end{tikzcd}$$

Recall:

$$\varphi : M \to M', \quad R\text{-morphism}$$
$$\operatorname{coker}(\varphi) = M'/\operatorname{Im}(\varphi)$$
$$\operatorname{coim}(\varphi) = M/\ker\varphi$$

Definition 1.2.13. *Let M be an R-module and I an ideal of R. Denote by IM the submodule of M generated by all elements am where $a \in I$, $m \in M$.*

$$IM = \left\{ \sum_{\text{finite}} a_i m_i \mid a_i \in I,\ m_i \in M \right\}$$

Proposition 1.2.14. *Let M be an R-module. Then $R \to \operatorname{End}(M)$ is a ring homomorphism.*

1. *$R \to \operatorname{End}(M)$ factors through the natural surjection $R \to R/I$ if and only if $IM = 0$:*
$$\begin{tikzcd} R \arrow[rr, "\rho"] \arrow[dr, "\pi_I"'] & & \operatorname{End}(M) \\ & R/I \arrow[ur, dashed, "\exists\,\rho_I"'] & \end{tikzcd}$$
Where ρ factors through π_I means $\exists \rho_I : R/I \to \operatorname{End}(M)$ such that the diagram above commutes.
2. *The module M/IM is naturally endowed with a structure of R/I-module.*
3. *Let $\varphi : M \to M'$ be an R-homomorphism. Then $\varphi(IM) \subseteq IM'$, so φ induces a morphism of R/I-modules*
$$M/IM \to M'/IM'.$$
4. *If $M = M_1 \oplus \cdots \oplus M_n$, then $M/IM = M_1/IM_1 \oplus \cdots \oplus M_n/IM_n$.*
5. *$R^n/IR^n \cong (R/IR)^n$.*

Proof. **(1)** Suppose first that ρ factors through π_I. Then there exists a ring homomorphism

$$\rho_I : R/I \to \operatorname{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

If $a \in I$, then $\pi_I(a) = 0$, so

$$\rho(a) = \rho_I(0) = 0.$$

Therefore $am = 0$ for every $m \in M$. Since IM is generated by elements of the form am

with $a \in I$ and $m \in M$, it follows that

$$IM = 0.$$

Conversely, assume that $IM = 0$. Then for every $a \in I$ and every $m \in M$ we have

$$\rho(a)(m) = am = 0.$$

Thus $\rho(a) = 0$, so $I \subseteq \ker \rho$. Hence ρ factors uniquely through the quotient ring R/I: there exists a unique ring homomorphism

$$\rho_I : R/I \to \mathrm{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

(2) Define an action of R/I on M/IM by

$$(r + I) \cdot (m + IM) := rm + IM.$$

We must show that this is well-defined.

If $r + I = r' + I$, then $r - r' \in I$, so

$$(r - r')m \in IM,$$

hence

$$rm + IM = r'm + IM.$$

If $m + IM = m' + IM$, then $m - m' \in IM$. Since IM is a submodule of M, we have

$$r(m - m') \in IM,$$

so

$$rm + IM = rm' + IM.$$

Thus the action is well-defined. The module axioms follow immediately from the module axioms on M. Therefore M/IM is naturally an R/I-module.

(3) Let $\varphi : M \to M'$ be an R-module homomorphism. Take any element of IM, say

$$x = \sum_{j=1}^{t} a_j m_j \qquad (a_j \in I,\ m_j \in M).$$

Then

$$\varphi(x) = \sum_{j=1}^{t} \varphi(a_j m_j) = \sum_{j=1}^{t} a_j \varphi(m_j).$$

Since each $a_j \in I$ and each $\varphi(m_j) \in M'$, it follows that

$$a_j\varphi(m_j) \in IM'.$$

Hence $\varphi(x) \in IM'$, and therefore

$$\varphi(IM) \subseteq IM'.$$

Thus we may define

$$\overline{\varphi} : M/IM \to M'/IM', \qquad \overline{\varphi}(m + IM) = \varphi(m) + IM'.$$

This is well-defined: if $m + IM = m' + IM$, then $m - m' \in IM$, so

$$\varphi(m) - \varphi(m') = \varphi(m - m') \in IM',$$

hence

$$\varphi(m) + IM' = \varphi(m') + IM'.$$

Finally, $\overline{\varphi}$ is R/I-linear, since

$$\overline{\varphi}\big((r+I)\cdot(m+IM)\big) = \overline{\varphi}(rm+IM) = \varphi(rm)+IM' = r\varphi(m)+IM' = (r+I)\cdot\overline{\varphi}(m+IM).$$

(4) Assume that

$$M = M_1 \oplus \cdots \oplus M_n.$$

We first show that

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

If $x \in IM$, then x is a finite sum of elements of the form am with $a \in I$ and $m \in M$. Writing

$$m = m_1 + \cdots + m_n \qquad (m_i \in M_i),$$

we get

$$am = am_1 + \cdots + am_n,$$

and each $am_i \in IM_i$. Hence $x \in IM_1 + \cdots + IM_n$, so

$$IM \subseteq IM_1 + \cdots + IM_n.$$

The reverse inclusion is immediate because each $M_i \subseteq M$, hence each $IM_i \subseteq IM$. Therefore

$$IM = IM_1 + \cdots + IM_n.$$

To see that the sum is direct, suppose

$$x_1 + \cdots + x_n = 0 \qquad \text{with } x_i \in IM_i \subseteq M_i.$$

Since

$$M = M_1 \oplus \cdots \oplus M_n$$

is a direct sum, we must have $x_i = 0$ for all i. Thus

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

Now define

$$\Psi : M_1/IM_1 \oplus \cdots \oplus M_n/IM_n \to M/IM$$

by

$$\Psi(m_1 + IM_1, \dots, m_n + IM_n) = (m_1 + \cdots + m_n) + IM.$$

This map is well-defined and R/I-linear. It is surjective because every element of M can be written as $m_1 + \cdots + m_n$. It is injective because

$$(m_1 + \cdots + m_n) + IM = IM$$

means that

$$m_1 + \cdots + m_n \in IM = IM_1 \oplus \cdots \oplus IM_n,$$

so each $m_i \in IM_i$. Hence

$$m_i + IM_i = IM_i$$

for all i, and the kernel is trivial. Therefore Ψ is an isomorphism:

$$M/IM \cong M_1/IM_1 \oplus \cdots \oplus M_n/IM_n.$$

(5) Since

$$R^n = R \oplus \cdots \oplus R$$

(n copies), part (4) yields

$$R^n/IR^n \cong R/IR \oplus \cdots \oplus R/IR.$$

Therefore

$$R^n/IR^n \cong (R/IR)^n.$$

This completes the proof. $\square$

1.3 Direct Sums and Free Modules

Definition 1.3.1. *Let $(M_i)_{i\in I}$ be a family of R-modules. The* direct product *of $(M_i)_{i\in I}$ is the module*

$$\prod_{i\in I} M_i = \left\{ \varphi\colon I \to \bigcup_{i\in I} M_i \;\middle|\; \varphi(i) \in M_i \text{ for all } i \in I \right\},$$

with componentwise addition and scalar multiplication:

$$R \times \prod_{i\in I} M_i \to \prod_{i\in I} M_i, \quad (r, \varphi) \mapsto (r\varphi(i))_{i\in I}$$

The direct sum *of* $(M_i)_{i\in I}$ *is the submodule*

$$\bigoplus_{i\in I} M_i = \left\{ \varphi \in \prod_{i\in I} M_i \;\middle|\; \#\{\, i \in I \mid \varphi(i) \neq 0 \,\} < \infty \right\}.$$

These two modules agree when $|I| < \infty$*. In the category of* R*-modules, the direct product is the categorical product and the direct sum is the categorical coproduct.*

Theorem 1.3.2. *Suppose that* $M_1, \ldots, M_n$ *are* R*-submodules of* M *such that* $M = M_1 + \cdots + M_n$*. Let* $I = [n]$*. Then the following conditions are equivalent:*

1. *The map* $\varphi\colon \bigoplus_{i\in I} M_i \to M$ *given by* $\varphi((m_i)_{i\in I}) = \sum_{i\in I} m_i$ *is an isomorphism.*
2. *Any element of* M *can be uniquely written as a sum of elements of* M_i*.*
3. *The zero element can be uniquely written as a sum of elements of the* M_i*.*
4. *For each* $i \in I$*, one has*
$$M_i \cap \sum_{j\neq i} M_j = 0.$$

Proof. We prove

$$(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (1).$$

(1) $\Rightarrow$ (2)**.** If φ is an isomorphism, then for every $m \in M$ there exists a unique

$$(m_1, \ldots, m_n) \in \bigoplus_{i=1}^{n} M_i$$

such that

$$m = \varphi(m_1, \ldots, m_n) = m_1 + \cdots + m_n.$$

Hence every element of M can be written uniquely as a sum of elements of the M_i.

(2) $\Rightarrow$ (3)**.** Apply (2) to the element $0 \in M$.

(3) $\Rightarrow$ (4)**.** Fix $i \in I$, and let

$$x \in M_i \cap \sum_{j\neq i} M_j.$$

Then $x \in M_i$, and there exist $m_j \in M_j$ for $j \neq i$ such that

$$x = \sum_{j\neq i} m_j.$$

Therefore

$$0 = x - \sum_{j \neq i} m_j$$

is a decomposition of 0 as a sum of elements from the submodules $M_1, \ldots, M_n$. By the uniqueness in (3), this decomposition must be the trivial one. In particular,

$$x = 0.$$

Hence

$$M_i \cap \sum_{j \neq i} M_j = 0.$$

(4) ⇒ (1). Since

$$M = M_1 + \cdots + M_n,$$

the map φ is surjective. It remains to show that φ is injective.

Suppose that

$$\varphi(m_1, \ldots, m_n) = 0,$$

that is,

$$m_1 + \cdots + m_n = 0 \qquad \text{with } m_i \in M_i.$$

For each $i \in I$, we have

$$m_i = -\sum_{j \neq i} m_j \in \sum_{j \neq i} M_j.$$

Since also $m_i \in M_i$, it follows that

$$m_i \in M_i \cap \sum_{j \neq i} M_j = 0.$$

Thus $m_i = 0$ for all i, so

$$\ker \varphi = 0.$$

Therefore φ is injective, and hence an isomorphism.

This proves that the four conditions are equivalent. □

Definition 1.3.3 (free module)**.** *Let M be an R-module.*

1. $x_1, \ldots, x_n \in M$ *are linearly independent if*

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0 \to a_1 = a_2 = \cdots = a_n = 0.$$

2. *M is called free if it is generated by a linearly independent subset. This subset is called a basis of M.*

3. *If M is an R-module with a basis $x_1, \ldots, x_n$, then $M = \langle x_1 \rangle \oplus \cdots \oplus \langle x_n \rangle$.*

Example 1.3.4. $R \oplus \cdots \oplus R = R^n$.

Theorem 1.3.5. *Let R be a commutative ring with 1. Let M be a finitely generated free R-module. Then any basis of M has the same cardinality. The number of elements in a basis is called the rank of the free module.*

Proof. Let $B = \{x_i\}_{i\in I}$ be a basis of M. We first show that any basis is finite. Since M is finitely generated, choose generators $y_1, \dots, y_n$ of M. Each y_k is a finite R–linear combination of elements of B, so there exists a finite subset $I_0 \subset I$ such that

$$y_1, \dots, y_n \in \sum_{i\in I_0} Rx_i.$$

Hence

$$M = \sum_{k=1}^{n} Ry_k \subseteq \sum_{i\in I_0} Rx_i \subseteq M,$$

so $M = \sum_{i\in I_0} Rx_i$. If $I \setminus I_0 \neq \varnothing$, pick $j \in I \setminus I_0$. Then

$$x_j \in M = \sum_{i\in I_0} Rx_i,$$

contradicting that B is linearly independent. Thus $I = I_0$ is finite.

Now let $x_1, \dots, x_r$ and $y_1, \dots, y_s$ be two bases of M. Let $J \subset R$ be a maximal ideal and set $F := R/J$, a field. Then M/JM is naturally an F-vector space. Write $\overline{m}$ for the class of $m \in M$ in M/JM, and $\overline{a}$ for the class of $a \in R$ in F. We claim that $\overline{x_1}, \dots, \overline{x_r}$ form an F–basis of M/JM.

Spanning. For any $m \in M$, write $m = \sum_{i=1}^{r} a_i x_i$. Then

$$\overline{m} = \sum_{i=1}^{r} \overline{a_i}\,\overline{x_i},$$

so $\overline{x_1}, \dots, \overline{x_r}$ span M/JM.

Linear independence. Suppose

$$\sum_{i=1}^{r} \overline{a_i}\,\overline{x_i} = 0 \quad \text{in } M/JM.$$

Then $\sum_{i=1}^{r} a_i x_i \in JM$, so there exist $a'_1, \dots, a'_r \in J$ such that

$$\sum_{i=1}^{r} a_i x_i = \sum_{i=1}^{r} a'_i x_i.$$

Hence

$$\sum_{i=1}^{r} (a_i - a'_i)\, x_i = 0.$$

By linear independence of $x_1, \dots, x_r$, we get $a_i - a'_i = 0$ for all i, so $a_i \in J$, thus $\overline{a_i} = 0$ for all i. Therefore $\overline{x_1}, \dots, \overline{x_r}$ are F-linearly independent, and form an F–basis of M/JM.

By the same argument, $\overline{y_1}, \ldots, \overline{y_s}$ also form an F–basis of M/JM. Hence

$$r = \dim_F(M/JM) = s.$$

Therefore any two bases of M have the same cardinality. $\square$

Chapter 2

Lecture 2: Tensor Product

2.1 Annihilator, Torsion and Projectivity of Free Modules

Definition 2.1.1. *Let M be an R-module, and let $m \in M$. The cyclic submodule generated by m is*

$$Rm = \langle m \rangle = \{\, rm \mid r \in R \,\}.$$

The annihilator *of m in R is*

$$\operatorname{Ann}_R(m) = \{\, r \in R \mid rm = 0 \,\}.$$

It is a left ideal of R.

Proposition 2.1.2. *For every $m \in M$, there is an isomorphism of left R-modules*

$$R/\operatorname{Ann}_R(m) \cong Rm.$$

In particular, every cyclic module is isomorphic to a quotient of R.

Definition 2.1.3. *An element $m \in M$ is called*

- ***torsion-free*** *if* $\operatorname{Ann}_R(m) = 0$;
- ***torsion*** *if* $\operatorname{Ann}_R(m) \neq 0$.

A module is called torsion-free *if every nonzero element is torsion-free.*

Definition 2.1.4. *The* torsion submodule *of M is*

$$\operatorname{Tor}(M) = \{\, m \in M \mid \operatorname{Ann}_R(m) \neq 0 \,\}.$$

Lemma 2.1.5. *If R is an integral domain, then the torsion part of M is a submodule of M.*

Proof. We verify the submodule criteria.

Non-empty. Since $r \cdot 0_M = 0_M$ for every $r \in R$ and $R \neq 0$, we have $\mathrm{Ann}_R(0_M) = R \neq 0$, so $0_M \in \mathrm{Tor}(M)$.

Closed under addition. Let $m, n \in \mathrm{Tor}(M)$. Then there exist $\lambda, \mu \in R \setminus \{0\}$ such that $\lambda m = 0$ and $\mu n = 0$. Hence

$$(\lambda\mu)(m + n) = \mu(\lambda m) + \lambda(\mu n) = 0.$$

Since R is an integral domain and $\lambda, \mu \neq 0$, we have $\lambda\mu \neq 0$, so $\mathrm{Ann}_R(m + n) \neq 0$ and $m + n \in \mathrm{Tor}(M)$.

Closed under scalar multiplication. Let $m \in \mathrm{Tor}(M)$ and $r \in R$. Choose $\lambda \in R \setminus \{0\}$ with $\lambda m = 0$. Then

$$\lambda(rm) = r(\lambda m) = 0,$$

so $\mathrm{Ann}_R(rm) \neq 0$ and $rm \in \mathrm{Tor}(M)$.

Therefore $\mathrm{Tor}(M)$ is a submodule of M. □

Example 2.1.6. *(1) The element $0 \in M$ is torsion.*

(2) Let R be a commutative ring with 1. Then R is an integral domain if and only if the left regular module ${}_RR$ is torsion-free. In this case $\mathrm{Tor}({}_RR) = 0$ and $\mathrm{Tor}(R^n) = 0$ for all $n \geq 1$.

(3) If I is a nonzero ideal of R, then $\mathrm{Tor}(R/I) = R/I$ when R/I is regarded as an R-module. If we regard R/I as an R/I-module, then it is torsion-free if and only if I is a prime ideal.

(4) More generally, let I be any nonzero ideal of R, and let M be an R-module. Then $\mathrm{Tor}(M/IM) = M/IM$ when M/IM is regarded as an R-module.

(5) Even when I is prime, M/IM need not be torsion-free as an R/I-module. For $\overline{m} \in M/IM$, one has $\mathrm{Ann}_{R/I}(\overline{m}) \neq 0$ if and only if there exists $r \in R \setminus I$ with $rm \in IM$. Thus primeness of I does not rule out nonzero operators in R/I annihilating nonzero classes in M/IM.

For instance, let $R = k[x, y]$, let $I = (y)$, and let $M = R/(x) \cong k[y]$. Then $IM = (y)$ and $M/IM \cong k$, while $x \cdot M = 0$ but $x \notin I$; hence $x + I \in R/I$ is nonzero and annihilates all of M/IM. If instead I is maximal, then R/I is a field, nonzero scalars act invertibly, and M/IM is always torsion-free over R/I. Item (3) is the case $M = R$.

Remark 2.1.7. *Let R be a commutative ring with 1, let M be an R-module, and fix $r \in R$. Then the map*

$$f \colon M \to M, \qquad f(m) = rm,$$

is an R-module homomorphism. If moreover M is torsion-free and $r \neq 0$, then f is a monomorphism.

Lemma 2.1.8. *If R is an integral domain and M is an R-module, then*

1. $\mathrm{Tor}(\mathrm{Tor}(M)) = \mathrm{Tor}(M)$.

2. $\mathrm{Tor}(M/\mathrm{Tor}(M)) = 0$.

Proof.

1. Since $\mathrm{Tor}(M)$ is a submodule of M, its torsion submodule is

$$\mathrm{Tor}(\mathrm{Tor}(M)) = \{\, m \in \mathrm{Tor}(M) \mid \mathrm{Ann}_R(m) \neq 0 \,\}.$$

Every $m \in \mathrm{Tor}(M)$ already satisfies $\mathrm{Ann}_R(m) \neq 0$, so $\mathrm{Tor}(\mathrm{Tor}(M)) \subseteq \mathrm{Tor}(M)$. Conversely, $\mathrm{Tor}(M) \subseteq \mathrm{Tor}(\mathrm{Tor}(M))$ because the annihilator of m in R does not depend on the ambient module: for any $m \in \mathrm{Tor}(M)$, its annihilator computed in the submodule $\mathrm{Tor}(M)$ is the same as $\mathrm{Ann}_R(m) \neq 0$, so $m \in \mathrm{Tor}(\mathrm{Tor}(M))$. Hence $\mathrm{Tor}(\mathrm{Tor}(M)) = \mathrm{Tor}(M)$.

2. Let $\pi\colon M \to M/\mathrm{Tor}(M)$ be the canonical projection. Write $\overline{m} = \pi(m) = m + \mathrm{Tor}(M)$. Suppose $\overline{m} \in \mathrm{Tor}(M/\mathrm{Tor}(M))$, i.e. there exists $\lambda \in R \setminus \{0\}$ such that

$$\lambda\,\overline{m} = \overline{0} \quad \text{in } M/\mathrm{Tor}(M).$$

This means $\lambda m \in \mathrm{Tor}(M)$, so there exists $\mu \in R \setminus \{0\}$ with $\mu(\lambda m) = 0$, i.e.

$$(\mu\lambda)\, m = 0.$$

Since R is an integral domain and $\mu, \lambda \neq 0$, we have $\mu\lambda \neq 0$, hence $m \in \mathrm{Tor}(M)$, so $\overline{m} = \overline{0}$. Therefore $\mathrm{Tor}(M/\mathrm{Tor}(M)) = 0$.

□

Definition 2.1.9. *Let J be a subscript set, denote $R^J = \prod_{j\in J} R.R^{(J)} = \bigoplus_{j\in J} R$.*

Let M be an R-module. Let $x_j \in M$ where $j \in J$ be a family of elements of M. Then the submodule generated by $(x_j)_{j\in J}$ is denoted by $\sum_{j\in J} Rx_j$.

Corollary 2.1.10. *We have a morphism of R-modules $\phi : R^{(J)} \to M$ such that $\phi(r_j)_{j\in J} = \sum_{j\in J} r_j x_j$ for all $(r_j)_{j\in J} \in R^{(J)}$.*

- *ϕ is injective if and only if $\{x_j\}_{j\in J}$ are linearly independent.*
- *ϕ is surjective if and only if $\{x_j\}_{j\in J}$ generate M.*
- *ϕ is an isomorphism if and only if M is free with basis $\{x_j\}_{j\in J}$.*

Lemma 2.1.11. *Let L be a free R-module with basis $\{x_j\}_{j\in J}$. For any R-module M, the map*

$$\Phi : \mathrm{Hom}_R(L, M) \to M^J, \qquad f \mapsto (f(x_j))_{j\in J}$$

is a bijection.

Proof. We prove that Φ is injective and surjective.

Injectivity. Let $f, g \in \mathrm{Hom}_R(L, M)$ and suppose that

$$\Phi(f) = \Phi(g).$$

Then for every $j \in J$ we have

$$f(x_j) = g(x_j).$$

Since $\{x_j\}_{j \in J}$ is a basis of L, every element $x \in L$ can be written uniquely as a finite sum

$$x = \sum_{j \in J} a_j x_j$$

with $a_j \in R$ and all but finitely many a_j equal to 0. Hence

$$f(x) = f\left(\sum_{j \in J} a_j x_j\right) = \sum_{j \in J} a_j f(x_j) = \sum_{j \in J} a_j g(x_j) = g\left(\sum_{j \in J} a_j x_j\right) = g(x).$$

Thus $f = g$, so Φ is injective.

Surjectivity. Let $(m_j)_{j \in J} \in M^J$ be given. We define a map

$$f : L \to M$$

by declaring that for

$$x = \sum_{j \in J} a_j x_j \qquad (a_j \in R, \text{ all but finitely many } a_j = 0),$$

we set

$$f(x) := \sum_{j \in J} a_j m_j.$$

This is well defined because each element of L has a unique expression as a finite linear combination of the basis elements.

Now let

$$x = \sum_{j \in J} a_j x_j, \qquad y = \sum_{j \in J} b_j x_j, \qquad r \in R.$$

Then

$$f(x + y) = f\left(\sum_{j \in J} (a_j + b_j) x_j\right) = \sum_{j \in J} (a_j + b_j) m_j = \sum_{j \in J} a_j m_j + \sum_{j \in J} b_j m_j = f(x) + f(y),$$

and

$$f(rx) = f\left(\sum_{j \in J} (r a_j) x_j\right) = \sum_{j \in J} (r a_j) m_j = r \sum_{j \in J} a_j m_j = r f(x).$$

So f is an R-module homomorphism. Moreover, for every $j \in J$,

$$f(x_j) = m_j.$$

Therefore

$$\Phi(f) = (m_j)_{j\in J}.$$

Thus Φ is surjective.

Since Φ is both injective and surjective, it is a bijection. ☐

Proposition 2.1.12. *Let L be a free R-module.*

1. *Let $\pi : X \to Y$ be an epimorphism of R-modules. For any R-module homomorphism $\varphi : L \to Y$ there exists a morphism $\tilde{\varphi} : L \to X$ such that the following diagram commutes:*

$$\begin{tikzcd} & L \arrow[dl, "\tilde{\varphi}"'] \arrow[d, "\varphi"] & \\ X \arrow[r, "\pi"'] & Y \arrow[r] & 0 \end{tikzcd}$$

2. *In particular, for any R-module M, any epimorphism $\pi : M \to L$ has a section, that is, there exists a morphism $\sigma : L \to M$ such that $\pi \circ \sigma = \mathrm{id}_L$. In that case σ is injective, $\mathrm{Im}(\sigma) \cong L$, and*

$$M = \ker \pi \oplus \mathrm{Im}(\sigma).$$

$$\begin{tikzcd} 0 \arrow[r] & \ker \pi \arrow[r] & M \arrow[r, two heads, shift left, "\pi"] & L \arrow[l, dashed, shift left, "\sigma"] \arrow[r] & 0 \end{tikzcd}$$

Proof. 1. Since L is free, let $\{x_j\}_{j\in J}$ be a basis of L.

For each $j \in J$, since $\pi : X \to Y$ is surjective, there exists an element

$$\tilde{x}_j \in X$$

such that

$$\pi(\tilde{x}_j) = \varphi(x_j).$$

Because L is free with basis $\{x_j\}_{j\in J}$, there exists a unique R-module homomorphism

$$\tilde{\varphi} : L \to X$$

satisfying

$$\tilde{\varphi}(x_j) = \tilde{x}_j \qquad \text{for all } j \in J.$$

We now check that $\pi \circ \tilde{\varphi} = \varphi$. For each basis element x_j,

$$(\pi \circ \tilde{\varphi})(x_j) = \pi(\tilde{\varphi}(x_j)) = \pi(\tilde{x}_j) = \varphi(x_j).$$

Hence $\pi\circ\tilde{\varphi}$ and φ agree on a basis of L, so they are equal as R-module homomorphisms. Therefore the diagram commutes.

2. Apply part (1) with
$$X = M, \qquad Y = L, \qquad \varphi = \mathrm{id}_L\,.$$
Since $\pi : M \to L$ is an epimorphism, there exists an R-module homomorphism
$$\sigma : L \to M$$
such that
$$\pi \circ \sigma = \mathrm{id}_L\,.$$
Thus σ is a section of π.

We first show that σ is injective. If $\sigma(\ell) = 0$ for some $\ell \in L$, then
$$\ell = (\pi \circ \sigma)(\ell) = \pi(0) = 0.$$
Hence $\ker \sigma = 0$, so σ is injective.

Next, since σ is injective, it induces an isomorphism
$$L \xrightarrow{\sim} \mathrm{Im}(\sigma), \qquad \ell \mapsto \sigma(\ell).$$
Thus $\mathrm{Im}(\sigma) \cong L$.

It remains to prove that
$$M = \ker \pi \oplus \mathrm{Im}(\sigma).$$
First, let $m \in M$. Then
$$m = \big(m - \sigma(\pi(m))\big) + \sigma(\pi(m)).$$
Now
$$\pi\big(m - \sigma(\pi(m))\big) = \pi(m) - (\pi \circ \sigma)(\pi(m)) = \pi(m) - \pi(m) = 0,$$
so
$$m - \sigma(\pi(m)) \in \ker \pi,$$
while clearly
$$\sigma(\pi(m)) \in \mathrm{Im}(\sigma).$$
Hence
$$M = \ker \pi + \mathrm{Im}(\sigma).$$
Finally, suppose
$$z \in \ker \pi \cap \mathrm{Im}(\sigma).$$
Then $z = \sigma(\ell)$ for some $\ell \in L$, and also $\pi(z) = 0$. Therefore
$$0 = \pi(z) = \pi(\sigma(\ell)) = (\pi \circ \sigma)(\ell) = \ell.$$

So $z = \sigma(\ell) = 0$. Thus

$$\ker \pi \cap \mathrm{Im}(\sigma) = 0.$$

Therefore

$$M = \ker \pi \oplus \mathrm{Im}(\sigma).$$

$\square$

Exercise 2.1.13. *Let L be a free R-module. Let P be a direct summand of L. Show that whenever*

$$\pi : X \to Y$$

is an epimorphism and

$$\varphi : P \to Y$$

is a morphism, there exists a morphism

$$\tilde{\varphi} : P \to X$$

such that

$$\varphi = \pi \circ \tilde{\varphi}.$$

Proof. Since P is a direct summand of L, there exist R-module homomorphisms

$$i : P \to L \qquad \text{and} \qquad p : L \to P$$

such that

$$p \circ i = \mathrm{id}_P .$$

(Here i is the inclusion and p is the projection.)

Consider the composite

$$\psi := \varphi \circ p : L \to Y.$$

Since L is free and $\pi : X \to Y$ is an epimorphism, by the lifting property of free modules there exists an R-module homomorphism

$$\widetilde{\psi} : L \to X$$

such that

$$\pi \circ \widetilde{\psi} = \psi.$$

Now define

$$\widetilde{\varphi} := \widetilde{\psi} \circ i : P \to X.$$

Then

$$\pi \circ \widetilde{\varphi} = \pi \circ \widetilde{\psi} \circ i = \psi \circ i = \varphi \circ p \circ i = \varphi \circ \mathrm{id}_P = \varphi.$$

Therefore there exists a morphism $\widetilde{\varphi} : P \to X$ such that

$$\varphi = \pi \circ \widetilde{\varphi}.$$

$\square$

2.2 Tensor product

Let R be a commutative ring, and let M and N be R-modules. Roughly speaking, the tensor product $M \otimes_R N$ linearizes bilinear maps out of $M \times N$.

Definition 2.2.1. *Let R be a commutative ring, and let M, N, and K be R-modules. A map*

$$f\colon M \times N \to K$$

is called bilinear *if for all $r \in R$, $m, m_1, m_2 \in M$, and $n, n_1, n_2 \in N$, one has*

- $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$;
- $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$;
- $f(rm, n) = rf(m, n)$;
- $f(m, rn) = rf(m, n)$.

Example 2.2.2. *If $f\colon M \times N \to K$ is bilinear, then*

$$f(x, 0) = 0 = f(0, y)$$

for all $x \in M$ and $y \in N$. Moreover,

$$f(-x, y) = -f(x, y), \qquad f(x, -y) = -f(x, y)$$

for all $x \in M$ and $y \in N$.

Definition 2.2.3. *The* tensor product *of M and N is a pair $(M \otimes_R N, t_{M,N})$, where $M \otimes_R N$ is an R-module and*

$$t_{M,N}\colon M \times N \to M \otimes_R N$$

is a bilinear map, usually written $t_{M,N}(m, n) = m \otimes n$, such that the following universal property holds:

whenever X is an R-module and $f\colon M \times N \to X$ is bilinear, there exists a unique R-module homomorphism

$$\tilde{f}\colon M \otimes_R N \to X$$

such that the following diagram commutes:

$$\begin{tikzcd} M \times N \arrow[r, "f"] \arrow[d, "t_{M,N}"'] & X \\ M \otimes_R N \arrow[ur, dashed, "\tilde{f}"'] & \end{tikzcd}$$

Theorem 2.2.4. *The tensor product of M and N exists and is unique up to isomorphism.*

Proof. We first prove uniqueness.

Suppose (T_1, t_1) and (T_2, t_2) are two tensor products of M and N. By the universal property, there exist unique R-module homomorphisms

$$\varphi\colon T_1 \to T_2, \qquad \psi\colon T_2 \to T_1$$

such that

$$t_2 = \varphi \circ t_1, \qquad t_1 = \psi \circ t_2.$$

Hence

$$\varphi \circ \psi \circ t_2 = \varphi \circ t_1 = t_2, \qquad \psi \circ \varphi \circ t_1 = \psi \circ t_2 = t_1.$$

By uniqueness in the universal property, it follows that

$$\varphi \circ \psi = \mathrm{id}_{T_2}, \qquad \psi \circ \varphi = \mathrm{id}_{T_1}.$$

Therefore φ and ψ are inverse isomorphisms, so $T_1 \cong T_2$.

Now we prove existence.

Let F be the free R-module with basis $M \times N$, say

$$F = \bigoplus_{(m,n)\in M\times N} Re_{m,n}.$$

Let B be the submodule of F generated by the following elements:

- $e_{m_1+m_2,n} - e_{m_1,n} - e_{m_2,n}$;
- $e_{m,n_1+n_2} - e_{m,n_1} - e_{m,n_2}$;
- $e_{rm,n} - re_{m,n}$;
- $e_{m,rn} - re_{m,n}$,

where $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, and $r \in R$.

Define

$$M \otimes_R N = F/B, \qquad t_{M,N}(m,n) = e_{m,n} + B.$$

Then $t_{M,N}\colon M \times N \to M \otimes_R N$ is a well-defined bilinear map.

It remains to verify the universal property.

Let X be an R-module, and let $f\colon M \times N \to X$ be bilinear. Since F is free on the set $M \times N$, there exists a unique R-module homomorphism

$$\hat{f}\colon F \to X$$

such that $\hat{f}(e_{m,n}) = f(m,n)$ for all $(m,n) \in M \times N$.

We claim that $B \subseteq \ker \hat{f}$. Indeed, since f is bilinear,

$$\begin{aligned}
\hat{f}(e_{m_1+m_2,n} - e_{m_1,n} - e_{m_2,n}) &= f(m_1+m_2,n) - f(m_1,n) - f(m_2,n) = 0, \\
\hat{f}(e_{m,n_1+n_2} - e_{m,n_1} - e_{m,n_2}) &= f(m,n_1+n_2) - f(m,n_1) - f(m,n_2) = 0, \\
\hat{f}(e_{rm,n} - re_{m,n}) &= f(rm,n) - rf(m,n) = 0, \\
\hat{f}(e_{m,rn} - re_{m,n}) &= f(m,rn) - rf(m,n) = 0.
\end{aligned}$$

Since B is generated by these elements and $\ker \hat{f}$ is a submodule, we obtain $B \subseteq \ker \hat{f}$.

By the universal property of quotient modules, $\hat{f}$ induces a unique R-module homomorphism

$$\tilde{f}\colon M \otimes_R N = F/B \to X$$

such that $\tilde{f} \circ \pi = \hat{f}$, where $\pi\colon F \to F/B$ is the canonical projection. In particular,

$$\tilde{f}(m \otimes n) = \tilde{f}(e_{m,n} + B) = \hat{f}(e_{m,n}) = f(m,n),$$

so $\tilde{f} \circ t_{M,N} = f$.

For uniqueness, suppose $g\colon M \otimes_R N \to X$ is another R-module homomorphism such that $g \circ t_{M,N} = f$. Then

$$g(m \otimes n) = f(m,n) = \tilde{f}(m \otimes n)$$

for all $m \in M$ and $n \in N$. Since the elementary tensors generate $M \otimes_R N$ as an R-module, it follows that $g = \tilde{f}$.

Therefore $(M \otimes_R N, t_{M,N})$ is a tensor product of M and N. □

Definition 2.2.5. *The set of elements of the form $m \otimes n$ for $m \in M$ and $n \in N$ is called the set of elementary tensors. This set generates $M \otimes_R N$ as an R-module.*

Example 2.2.6. *Fix $m \in M$. The map $N \to M \otimes_R N, n \mapsto m \otimes n$ is a morphism of R-modules. Its image $m \otimes_R N = \{m \otimes n \mid n \in N\}$ is a submodule of $M \otimes_R N$ which is isomorphic to a quotient $N/\operatorname{Ann}_R(m)N$.*

Proposition 2.2.7. *We have the following isomorphisms:*

1. *$(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$ where $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$.*
2. *$M_1 \otimes_R M_2 \cong M_2 \otimes_R M_1$ where $m_1 \otimes m_2 \mapsto m_2 \otimes m_1$.*
3. *$M_1 \otimes_R (M_2 \oplus M_3) \cong (M_1 \otimes_R M_2) \oplus (M_1 \otimes_R M_3)$ where $m_1 \otimes (m_2 \oplus m_3) \mapsto (m_1 \otimes m_2) + (m_1 \otimes m_3)$. And more generally,*

$$M_1 \otimes_R (\bigoplus_{i=1}^n M_i) \cong \bigoplus_{i=1}^n (M_1 \otimes_R M_i)$$

where $m_1 \otimes (m_2 \oplus \cdots \oplus m_n) \mapsto (m_1 \otimes m_2) + \cdots + (m_1 \otimes m_n)$.

4. $R \otimes_R M \cong M$ where $r \otimes m \mapsto rm$. And more generally, $R^{(I)} \otimes_R M \cong M^{(I)}$ where $(r_i)_{i\in I} \otimes m \mapsto (r_i m)_{i\in I}$.

Proof. We prove (1):$(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$.

Step 1. We construct $\varphi : (M_1 \otimes_R M_2) \otimes_R M_3 \to M_1 \otimes_R (M_2 \otimes_R M_3)$.

Fix $m_3 \in M_3$. For each m_3, define $f_{m_3} : M_1 \times M_2 \to M_1 \otimes_R (M_2 \otimes_R M_3)$ by $f_{m_3}(m_1, m_2) = m_1 \otimes (m_2 \otimes m_3)$. This is bilinear in (m_1, m_2), so by the universal property of $M_1 \otimes_R M_2$, there exists a unique R-module homomorphism $\tilde{f}_{m_3} : M_1 \otimes_R M_2 \to M_1 \otimes_R (M_2 \otimes_R M_3)$ such that $\tilde{f}_{m_3}(m_1 \otimes m_2) = m_1 \otimes (m_2 \otimes m_3)$:

$$\begin{tikzcd} M_1 \times M_2 \arrow[r, "f_{m_3}"] \arrow[d, "t_{M_1,M_2}"'] & M_1 \otimes_R (M_2 \otimes_R M_3) \\ M_1 \otimes_R M_2 \arrow[ur, dashed, "\tilde{f}_{m_3}"'] & \end{tikzcd}$$

Now define $g : (M_1 \otimes_R M_2) \times M_3 \to M_1 \otimes_R (M_2 \otimes_R M_3)$ by $g(\alpha, m_3) = \tilde{f}_{m_3}(\alpha)$. We verify that g is bilinear. Linearity in the first argument follows because each $\tilde{f}_{m_3}$ is an R-module homomorphism. For the second argument, on elementary tensors:

$$\begin{aligned} g(m_1 \otimes m_2, m_3 + m_3') &= m_1 \otimes (m_2 \otimes (m_3 + m_3')) \\ &= m_1 \otimes (m_2 \otimes m_3 + m_2 \otimes m_3') \\ &= m_1 \otimes (m_2 \otimes m_3) + m_1 \otimes (m_2 \otimes m_3') \\ &= g(m_1 \otimes m_2, m_3) + g(m_1 \otimes m_2, m_3'), \\ g(m_1 \otimes m_2, rm_3) &= m_1 \otimes (m_2 \otimes rm_3) \\ &= m_1 \otimes (r(m_2 \otimes m_3)) \\ &= r(m_1 \otimes (m_2 \otimes m_3)) \\ &= r\, g(m_1 \otimes m_2, m_3). \end{aligned}$$

Since elementary tensors generate $M_1 \otimes_R M_2$, bilinearity holds on all of $(M_1 \otimes_R M_2) \times M_3$.

By the universal property of $(M_1 \otimes_R M_2) \otimes_R M_3$, there exists a unique R-module homomorphism

$$\varphi : (M_1 \otimes_R M_2) \otimes_R M_3 \to M_1 \otimes_R (M_2 \otimes_R M_3)$$

such that $\varphi((m_1 \otimes m_2) \otimes m_3) = m_1 \otimes (m_2 \otimes m_3)$:

$$\begin{tikzcd} (M_1 \otimes_R M_2) \times M_3 \arrow[r, "g"] \arrow[d, "t"'] & M_1 \otimes_R (M_2 \otimes_R M_3) \\ (M_1 \otimes_R M_2) \otimes_R M_3 \arrow[ur, dashed, "\varphi"'] & \end{tikzcd}$$

Step 2. By an entirely symmetric argument (fixing m_1 first, then extending), we obtain an R-module homomorphism

$$\psi : M_1 \otimes_R (M_2 \otimes_R M_3) \to (M_1 \otimes_R M_2) \otimes_R M_3$$

such that $\psi(m_1 \otimes (m_2 \otimes m_3)) = (m_1 \otimes m_2) \otimes m_3$.

Step 3. We show φ and ψ are inverses. On elementary tensors:

$$(\psi \circ \varphi)((m_1 \otimes m_2) \otimes m_3) = \psi(m_1 \otimes (m_2 \otimes m_3)) = (m_1 \otimes m_2) \otimes m_3,$$

$$(\varphi \circ \psi)(m_1 \otimes (m_2 \otimes m_3)) = \varphi((m_1 \otimes m_2) \otimes m_3) = m_1 \otimes (m_2 \otimes m_3).$$

Since elementary tensors generate both modules, $\psi \circ \varphi = \mathrm{id}$ and $\varphi \circ \psi = \mathrm{id}$. The overall situation is summarized by the following commutative diagram:

$$\begin{tikzcd}
& M_1 \times M_2 \times M_3 \arrow[dl] \arrow[dr] & \\
(M_1 \otimes_R M_2) \otimes_R M_3 \arrow[rr, shift left, "\varphi"] & & M_1 \otimes_R (M_2 \otimes_R M_3) \arrow[ll, shift left, "\psi"]
\end{tikzcd}$$

where the diagonal arrows send (m_1, m_2, m_3) to $(m_1 \otimes m_2) \otimes m_3$ and $m_1 \otimes (m_2 \otimes m_3)$ respectively. Hence φ is an isomorphism with $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$. $\square$

Theorem 2.2.8. *Let M, N, P be R-modules. We have a canonical isomorphism:*

$$\phi_{M,N,P} : \mathrm{Hom}_R(M \otimes_R N, P) \to \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$$

such that $\phi_{M,N,P}(f)(m)(n) = f(m \otimes n)$. Furthermore, if R is commutative, then $\phi_{M,N,P}$ is an R-module isomorphism. If R is not commutative, then $\phi_{M,N,P}$ is only a $\mathbb{Z}$-module isomorphism.

Proof. Let $f \in \mathrm{Hom}_R(M \otimes_R N, P)$. For each $m \in M$, define $\phi_{M,N,P}(f)(m) : N \to P$ by

$$\phi_{M,N,P}(f)(m)(n) = f(m \otimes n).$$

We verify that $\phi_{M,N,P}(f)(m) \in \mathrm{Hom}_R(N, P)$: for $n_1, n_2 \in N$ and $r \in R$,

$$\begin{aligned}
\phi_{M,N,P}(f)(m)(n_1 + n_2) &= f(m \otimes (n_1 + n_2)) = f(m \otimes n_1 + m \otimes n_2) \\
&= f(m \otimes n_1) + f(m \otimes n_2), \\
\phi_{M,N,P}(f)(m)(rn) &= f(m \otimes rn) = f(r(m \otimes n)) \\
&= r f(m \otimes n) = r\, \phi_{M,N,P}(f)(m)(n).
\end{aligned}$$

Next, we verify that $\phi_{M,N,P}(f) \in \mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$: for $m_1, m_2 \in M$, $r \in R$, and any $n \in N$,

$$\begin{aligned}
\phi_{M,N,P}(f)(m_1 + m_2)(n) &= f((m_1 + m_2) \otimes n) \\
&= f(m_1 \otimes n) + f(m_2 \otimes n) \\
\phi_{M,N,P}(f)(rm)(n) &= f(rm \otimes n) \\
&= f(r(m \otimes n)) \\
&= r f(m \otimes n) \\
&= (r\, \phi_{M,N,P}(f)(m))(n).
\end{aligned}$$

So

$$\phi_{M,N,P}(f)(m_1 + m_2) = \phi_{M,N,P}(f)(m_1) + \phi_{M,N,P}(f)(m_2)$$

and $\phi_{M,N,P}(f)(rm) = r\,\phi_{M,N,P}(f)(m)$, confirming R-linearity.

Let $g \in \operatorname{Hom}_R(M, \operatorname{Hom}_R(N,P))$. Define $\hat{g} : M \times N \to P$ by $\hat{g}(m,n) = g(m)(n)$. We check that $\hat{g}$ is R-bilinear:

$$\begin{aligned}
\hat{g}(m_1 + m_2, n) &= g(m_1 + m_2)(n) = g(m_1)(n) + g(m_2)(n) = \hat{g}(m_1, n) + \hat{g}(m_2, n),\\
\hat{g}(m, n_1 + n_2) &= g(m)(n_1 + n_2) = g(m)(n_1) + g(m)(n_2) = \hat{g}(m, n_1) + \hat{g}(m, n_2),\\
\hat{g}(rm, n) &= g(rm)(n) = (rg(m))(n) = rg(m)(n) = r\,\hat{g}(m,n),\\
\hat{g}(m, rn) &= g(m)(rn) = rg(m)(n) = r\,\hat{g}(m,n).
\end{aligned}$$

By the universal property of $M \otimes_R N$, there exists a unique R-module homomorphism $\psi_{M,N,P}(g) : M \otimes_R N \to P$ such that $\psi_{M,N,P}(g)(m \otimes n) = g(m)(n)$:

$$\begin{tikzcd}
M \times N \arrow[r, "\hat{g}"] \arrow[d, "t_{M,N}"'] & P \\
M \otimes_R N \arrow[ur, dashed, "\psi_{M,N,P}(g)"'] &
\end{tikzcd}$$

For any $f \in \operatorname{Hom}_R(M \otimes_R N, P)$:

$$\psi_{M,N,P}(\phi_{M,N,P}(f))(m \otimes n) = \phi_{M,N,P}(f)(m)(n) = f(m \otimes n).$$

Since elementary tensors generate $M \otimes_R N$, we have $\psi_{M,N,P}(\phi_{M,N,P}(f)) = f$.

For any $g \in \operatorname{Hom}_R(M, \operatorname{Hom}_R(N,P))$, for all $m \in M$ and $n \in N$:

$$\phi_{M,N,P}(\psi_{M,N,P}(g))(m)(n) = \psi_{M,N,P}(g)(m \otimes n) = g(m)(n).$$

So $\phi_{M,N,P}(\psi_{M,N,P}(g))(m) = g(m)$ for all m, hence $\phi_{M,N,P}(\psi_{M,N,P}(g)) = g$.

Therefore $\phi_{M,N,P}$ is a bijection of sets (in fact, of abelian groups).

It remains to show that $\phi_{M,N,P}$ respects the appropriate module structure.

Both $\operatorname{Hom}_R(M \otimes_R N, P)$ and $\operatorname{Hom}_R(M, \operatorname{Hom}_R(N,P))$ are abelian groups under pointwise addition. We verify $\phi_{M,N,P}$ is a group homomorphism: for $f_1, f_2 \in \operatorname{Hom}_R(M \otimes_R N, P)$,

$$\begin{aligned}
\phi_{M,N,P}(f_1 + f_2)(m)(n) &= (f_1 + f_2)(m \otimes n)\\
&= f_1(m \otimes n) + f_2(m \otimes n)\\
&= \phi_{M,N,P}(f_1)(m)(n) + \phi_{M,N,P}(f_2)(m)(n)
\end{aligned}$$

so $\phi_{M,N,P}(f_1 + f_2) = \phi_{M,N,P}(f_1) + \phi_{M,N,P}(f_2)$. Hence $\phi_{M,N,P}$ is a $\mathbb{Z}$-module isomorphism.

If R is commutative, $\operatorname{Hom}_R(M \otimes_R N, P)$ carries an R-module structure via $(r \cdot f)(x) =$

$rf(x)$, and similarly for $\mathrm{Hom}_R(M, \mathrm{Hom}_R(N, P))$. Then

$$\begin{aligned}\phi_{M,N,P}(rf)(m)(n) &= (rf)(m \otimes n)\\ &= rf(m \otimes n)\\ &= r\,\phi_{M,N,P}(f)(m)(n)\\ &= (r\,\phi_{M,N,P}(f))(m)(n)\end{aligned}$$

so $\phi_{M,N,P}(rf) = r\,\phi_{M,N,P}(f)$, and $\phi_{M,N,P}$ is an R-module isomorphism.

If R is not commutative, $\mathrm{Hom}_R(M \otimes_R N, P)$ does not naturally carry an R-module structure (scalar multiplication by r need not commute with R-linearity), so $\phi_{M,N,P}$ is only a $\mathbb{Z}$-module isomorphism. □

Definition 2.2.9. *Let $f : M \to M'$ and $g : N \to N'$ be R-module homomorphisms. The tensor product of f and g is the R-module homomorphism:*

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

defined by $(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$.

Definition 2.2.10. *Let $M_i, i \in I$ be a family of R-modules. Use the following notation:*

$$L^n(M_1, \ldots, M_n; N) = \{f : M_1 \times \cdots \times M_n \to N \mid f \text{ is } n\text{-linear}\}$$

In particular,

$$L^2(M_1, M_2; N) \cong L(M_1 \otimes M_2, N) \cong \mathrm{Hom}_R(M_1 \otimes M_2, N)$$

or equivalently,

$$L^2(M_1 \otimes M_2, N) \cong L(M_1, L(M_2, N))$$

Proposition 2.2.11. *Let M, N be two R-modules where M is finitely generated and free. Denote*

$$M^\vee = \mathrm{Hom}_R(M, R)$$

Then we have a canonical isomorphism:

$$M^\vee \otimes N \cong \mathrm{Hom}_R(M, N)$$

Proof. Since M is a finitely generated free R-module, we have $M \cong R^m$ for some $m \geq 1$. We construct an explicit isomorphism

$$\Phi : M^\vee \otimes_R N \to \mathrm{Hom}_R(M, N).$$

Define a bilinear map $\beta : M^\vee \times N \to \mathrm{Hom}_R(M, N)$ by

$$\beta(f, n)(m) = f(m)\, n, \qquad f \in M^\vee,\ n \in N,\ m \in M.$$

We check R-bilinearity. For $f_1, f_2 \in M^\vee$, $n, n_1, n_2 \in N$, $m \in M$, and $r \in R$:

$$\begin{aligned}
\beta(f_1 + f_2, n)(m) &= (f_1 + f_2)(m)\, n = f_1(m)\, n + f_2(m)\, n \\
&= \beta(f_1, n)(m) + \beta(f_2, n)(m), \\
\beta(f, n_1 + n_2)(m) &= f(m)(n_1 + n_2) = f(m)\, n_1 + f(m)\, n_2 \\
&= \beta(f, n_1)(m) + \beta(f, n_2)(m), \\
\beta(rf, n)(m) &= (rf)(m)\, n = r\, f(m)\, n = r\, \beta(f, n)(m), \\
\beta(f, rn)(m) &= f(m)\, (rn) = r\, f(m)\, n = r\, \beta(f, n)(m).
\end{aligned}$$

By the universal property of the tensor product, there exists a unique R-module homomorphism

$$\Phi : M^\vee \otimes_R N \to \mathrm{Hom}_R(M, N), \qquad \Phi(f \otimes n)(m) = f(m)\, n.$$

Now let $\{e_1, \ldots, e_m\}$ be a basis of $M \cong R^m$, and let $\{e_1^*, \ldots, e_m^*\}$ be the dual basis; that is,

$$e_i^* \in M^\vee, \qquad e_i^*(e_j) = \delta_{ij}.$$

Define

$$\Psi : \mathrm{Hom}_R(M, N) \to M^\vee \otimes_R N, \qquad \Psi(g) = \sum_{i=1}^{m} e_i^* \otimes g(e_i).$$

We verify Ψ is an R-module homomorphism: for $g_1, g_2 \in \mathrm{Hom}_R(M, N)$ and $r \in R$,

$$\begin{aligned}
\Psi(g_1 + g_2) &= \sum_{i=1}^{m} e_i^* \otimes (g_1 + g_2)(e_i) = \sum_{i=1}^{m} e_i^* \otimes (g_1(e_i) + g_2(e_i)) \\
&= \sum_{i=1}^{m} e_i^* \otimes g_1(e_i) + \sum_{i=1}^{m} e_i^* \otimes g_2(e_i) = \Psi(g_1) + \Psi(g_2), \\
\Psi(rg) &= \sum_{i=1}^{m} e_i^* \otimes (rg)(e_i) = \sum_{i=1}^{m} e_i^* \otimes r\, g(e_i) \\
&= r \sum_{i=1}^{m} e_i^* \otimes g(e_i) = r\, \Psi(g).
\end{aligned}$$

We show Φ and Ψ are mutual inverses. For any $g \in \mathrm{Hom}_R(M, N)$ and $m = \sum_j r_j e_j \in M$:

$$\begin{aligned}
\Phi(\Psi(g))(m) &= \Phi\Big(\sum_{i=1}^{m} e_i^* \otimes g(e_i)\Big)\Big(\sum_j r_j e_j\Big) \\
&= \sum_{i=1}^{m} e_i^*\Big(\sum_j r_j e_j\Big) g(e_i) = \sum_{i=1}^{m} r_i\, g(e_i) \\
&= g\Big(\sum_{i=1}^{m} r_i e_i\Big) = g(m).
\end{aligned}$$

So $\Phi \circ \Psi = \mathrm{id}$.

For any elementary tensor $f \otimes n \in M^\vee \otimes_R N$:

$$\begin{aligned}\Psi(\Phi(f \otimes n)) &= \sum_{i=1}^m e_i^* \otimes \Phi(f \otimes n)(e_i) = \sum_{i=1}^m e_i^* \otimes f(e_i)\, n \\ &= \sum_{i=1}^m f(e_i)\,(e_i^* \otimes n) = \Big(\sum_{i=1}^m f(e_i)\, e_i^*\Big) \otimes n = f \otimes n,\end{aligned}$$

where the last equality uses $f = \sum_{i=1}^m f(e_i)\, e_i^*$(expansion of f in the dual basis). Since elementary tensors generate $M^\vee \otimes_R N$ and $\Psi \circ \Phi$ is R-linear, we have $\Psi \circ \Phi = \mathrm{id}$.

Therefore Φ is an R-module isomorphism, giving $M^\vee \otimes_R N \cong \mathrm{Hom}_R(M, N)$. □

Lemma 2.2.12. *Let P be a finitely generated projective R-module. Then there exists an R-module Q and an integer $t \geq 0$ such that*

$$P \oplus Q \cong R^t.$$

Proof. Since P is finitely generated, there exist elements

$$p_1, \ldots, p_t \in P$$

such that

$$P = Rp_1 + \cdots + Rp_t.$$

Therefore we have a surjective R-module homomorphism

$$\pi : R^t \longrightarrow P$$

defined by

$$\pi(a_1, \ldots, a_t) = a_1 p_1 + \cdots + a_t p_t.$$

Equivalently, if $e_1, \ldots, e_t$ denotes the standard basis of R^t, then

$$\pi(e_i) = p_i \qquad (1 \leq i \leq t).$$

Since $p_1, \ldots, p_t$ generate P, the map π is surjective.

Now we use that P is projective. Since $\pi : R^t \to P$ is an epimorphism and P is projective, the map π splits. That is, there exists an R-module homomorphism

$$s : P \longrightarrow R^t$$

such that

$$\pi \circ s = \mathrm{id}_P\,.$$

In particular, s is injective, because if $s(p) = 0$, then

$$p = (\pi \circ s)(p) = \pi(0) = 0.$$

We claim that

$$R^t = \mathrm{Im}(s) \oplus \ker \pi.$$

First, let $x \in R^t$. Then

$$x = s(\pi(x)) + \big(x - s(\pi(x))\big).$$

Clearly,

$$s(\pi(x)) \in \mathrm{Im}(s).$$

Moreover,

$$\pi\big(x - s(\pi(x))\big) = \pi(x) - (\pi \circ s)(\pi(x)) = \pi(x) - \pi(x) = 0.$$

Hence

$$x - s(\pi(x)) \in \ker \pi.$$

Therefore

$$R^t = \mathrm{Im}(s) + \ker \pi.$$

It remains to show that the sum is direct. Suppose

$$y \in \mathrm{Im}(s) \cap \ker \pi.$$

Then $y = s(p)$ for some $p \in P$, and also $\pi(y) = 0$. Hence

$$0 = \pi(y) = \pi(s(p)) = (\pi \circ s)(p) = p.$$

Therefore

$$y = s(p) = s(0) = 0.$$

Thus

$$\mathrm{Im}(s) \cap \ker \pi = 0.$$

Hence

$$R^t = \mathrm{Im}(s) \oplus \ker \pi.$$

Since $s : P \to \mathrm{Im}(s)$ is an isomorphism, we obtain

$$R^t \cong P \oplus \ker \pi.$$

Taking

$$Q := \ker \pi,$$

we get

$$P \oplus Q \cong R^t.$$

□

Remark 2.2.13. *This does not mean that P itself is free. It only says that P is a direct summand of a finite free module. Thus a finitely generated projective module is "part of" a* *finite free module:*

$$R^t \cong P \oplus Q.$$

Corollary 2.2.14. *Let R be a commutative ring with 1, let P be a finitely generated projective R-module, and let N be any R-module. Denote*

$$P^\vee := \mathrm{Hom}_R(P, R).$$

Then the canonical R-module homomorphism

$$\tau_{P,N} : P^\vee \otimes_R N \longrightarrow \mathrm{Hom}_R(P, N)$$

defined by

$$\tau_{P,N}(\varphi \otimes n)(p) = \varphi(p)n \qquad (\varphi \in P^\vee,\ n \in N,\ p \in P)$$

is an isomorphism.

Proof. Since P is finitely generated and projective, there exist elements

$$p_1, \ldots, p_t \in P$$

and homomorphisms

$$\varphi_1, \ldots, \varphi_t \in P^\vee$$

such that

$$p = \sum_{i=1}^{t} \varphi_i(p) p_i \qquad \text{for every } p \in P.$$

Indeed, since P is finitely generated projective, there exists an R-module Q and an integer $t \geq 0$ such that

$$P \oplus Q \cong R^t.$$

Let

$$\iota : P \longrightarrow R^t \qquad \text{and} \qquad \rho : R^t \longrightarrow P$$

be the corresponding inclusion and projection, so that

$$\rho \circ \iota = \mathrm{id}_P .$$

Let $e_1, \ldots, e_t$ be the standard basis of R^t, and let $e_1^\vee, \ldots, e_t^\vee$ be the dual basis of $(R^t)^\vee$. Define

$$p_i := \rho(e_i) \in P, \qquad \varphi_i := e_i^\vee \circ \iota \in P^\vee.$$

Then, for every $p \in P$,

$$p = \rho(\iota(p)) = \rho\left(\sum_{i=1}^{t} e_i^\vee(\iota(p)) e_i\right) = \sum_{i=1}^{t} \varphi_i(p) p_i.$$

Now define a map

$$\sigma_{P,N} : \mathrm{Hom}_R(P, N) \longrightarrow P^\vee \otimes_R N$$

by

$$\sigma_{P,N}(f) = \sum_{i=1}^{t} \varphi_i \otimes f(p_i).$$

We show that $\sigma_{P,N}$ is the inverse of $\tau_{P,N}$.

First, let $f \in \mathrm{Hom}_R(P, N)$. For every $p \in P$, we have

$$\big(\tau_{P,N} \circ \sigma_{P,N}\big)(f)(p) = \tau_{P,N} \left(\sum_{i=1}^{t} \varphi_i \otimes f(p_i) \right)(p) = \sum_{i=1}^{t} \varphi_i(p) f(p_i).$$

Since f is R-linear,

$$\sum_{i=1}^{t} \varphi_i(p) f(p_i) = f\left(\sum_{i=1}^{t} \varphi_i(p) p_i \right) = f(p).$$

Hence

$$\tau_{P,N} \circ \sigma_{P,N} = \mathrm{id}_{\mathrm{Hom}_R(P,N)}\,.$$

Conversely, let $\psi \otimes n \in P^\vee \otimes_R N$ be an elementary tensor. Then

$$\big(\sigma_{P,N} \circ \tau_{P,N}\big)(\psi \otimes n) = \sigma_{P,N}\big(\tau_{P,N}(\psi \otimes n)\big) = \sum_{i=1}^{t} \varphi_i \otimes \tau_{P,N}(\psi \otimes n)(p_i).$$

By definition of $\tau_{P,N}$,

$$\tau_{P,N}(\psi \otimes n)(p_i) = \psi(p_i) n.$$

Therefore

$$\big(\sigma_{P,N} \circ \tau_{P,N}\big)(\psi \otimes n) = \sum_{i=1}^{t} \varphi_i \otimes \psi(p_i) n = \left(\sum_{i=1}^{t} \psi(p_i) \varphi_i \right) \otimes n.$$

But for every $p \in P$,

$$\left(\sum_{i=1}^{t} \psi(p_i) \varphi_i \right)(p) = \sum_{i=1}^{t} \psi(p_i) \varphi_i(p) = \sum_{i=1}^{t} \varphi_i(p) \psi(p_i) = \psi\left(\sum_{i=1}^{t} \varphi_i(p) p_i \right) = \psi(p).$$

Hence

$$\sum_{i=1}^{t} \psi(p_i) \varphi_i = \psi.$$

Thus

$$\big(\sigma_{P,N} \circ \tau_{P,N}\big)(\psi \otimes n) = \psi \otimes n.$$

Since elementary tensors generate $P^\vee \otimes_R N$, it follows that

$$\sigma_{P,N} \circ \tau_{P,N} = \mathrm{id}_{P^\vee \otimes_R N}\,.$$

Therefore $\tau_{P,N}$ is an isomorphism, with inverse $\sigma_{P,N}$. □

Proposition 2.2.15. *Assume that R is a subring of S, and M is an R-module. Then the bilinear map:*

$$S \times (S \otimes_R M) \to S \otimes_R M$$

defined by

$$(s, s' \otimes m) \mapsto ss' \otimes m$$

defines a structure of S-module on the R-module $S \otimes_R M$.

More generally, let $f : R \to T$ be a ring homomorphism. And view T as an R-module via $R \times T \to T, (\lambda, \mu) \mapsto f(\lambda)\mu$. Then the bilinear map:

$$T \times (T \otimes_R M) \to T \otimes_R M$$

defined by

$$(t, t' \otimes m) \mapsto tt' \otimes m$$

defines a structure of T-module on the R-module $T \otimes_R M$.

Proof. It suffices to prove the more general statement in (2), since (1) is the special case where $T = S$ and $f : R \hookrightarrow S$ is the inclusion map.

For each fixed $t \in T$, define a map

$$\beta_t : T \times M \longrightarrow T \otimes_R M, \qquad (t', m) \longmapsto tt' \otimes m.$$

This map is R-bilinear. Hence, by the universal property of the tensor product, there exists a unique R-linear map

$$\mu_t : T \otimes_R M \longrightarrow T \otimes_R M$$

such that

$$\mu_t(t' \otimes m) = tt' \otimes m \qquad \text{for all } t' \in T,\ m \in M.$$

Now define

$$t \cdot x := \mu_t(x), \qquad t \in T,\ x \in T \otimes_R M.$$

We claim that this gives a left T-module structure on $T \otimes_R M$.

Since the elementary tensors $t' \otimes m$ generate $T \otimes_R M$ as an abelian group, it is enough to verify the module axioms on elementary tensors.

Let $t_1, t_2, t' \in T$ and $m \in M$. Then

$$(t_1 + t_2) \cdot (t' \otimes m) = (t_1 + t_2)t' \otimes m = t_1 t' \otimes m + t_2 t' \otimes m = t_1 \cdot (t' \otimes m) + t_2 \cdot (t' \otimes m).$$

Also,

$$(t_1 t_2) \cdot (t' \otimes m) = t_1 t_2 t' \otimes m = t_1 \cdot (t_2 t' \otimes m) = t_1 \cdot \big(t_2 \cdot (t' \otimes m)\big).$$

Moreover,

$$1_T \cdot (t' \otimes m) = 1_T t' \otimes m = t' \otimes m.$$

Finally, for any $x, y \in T \otimes_R M$,

$$t \cdot (x + y) = \mu_t(x + y) = \mu_t(x) + \mu_t(y) = t \cdot x + t \cdot y,$$

because μ_t is R-linear, hence additive.

Therefore the operation

$$T \times (T \otimes_R M) \longrightarrow T \otimes_R M, \qquad (t, t' \otimes m) \longmapsto tt' \otimes m$$

defines a left T-module structure on $T \otimes_R M$.

This proves (2), and hence also (1). □

Proposition 2.2.16. *Let I be an ideal of R. Let M be an R-module. Then the morphism of R-modules:*

$$M \to (R/I) \otimes_R M$$

defined by

$$m \mapsto 1_{R/I} \otimes m$$

induces an isomorphism of R-modules:

$$M/(IM) \cong (R/I) \otimes_R M$$

as R/I-modules.

Proof. Define

$$\varphi : M \longrightarrow (R/I) \otimes_R M, \qquad m \longmapsto 1_{R/I} \otimes m.$$

This is an R-module homomorphism. For any $a \in I$ and $m \in M$, we have

$$\varphi(am) = 1_{R/I} \otimes am = (1_{R/I} \cdot a) \otimes m = (a + I) \otimes m = 0,$$

since $a + I = 0$ in R/I. Hence $\varphi(IM) = 0$, so $IM \subseteq \ker(\varphi)$.

Therefore, by the universal property of the quotient, φ induces an R-module homomorphism

$$\overline{\varphi} : M/IM \longrightarrow (R/I) \otimes_R M, \qquad m + IM \longmapsto 1_{R/I} \otimes m.$$

Now define

$$\beta : (R/I) \times M \longrightarrow M/IM, \qquad (r + I, m) \longmapsto rm + IM.$$

We first check that β is well defined. If $r + I = r' + I$, then $r - r' \in I$, so

$$(r - r')m \in IM,$$

hence

$$rm + IM = r'm + IM.$$

Also, β is R-balanced, since for any $a \in R$,

$$\beta((r+I)a, m) = \beta(ra+I, m) = ram + IM = \beta(r+I, am).$$

Thus, by the universal property of the tensor product, there exists a unique homomorphism

$$\psi : (R/I) \otimes_R M \longrightarrow M/IM$$

such that

$$\psi((r+I) \otimes m) = rm + IM \qquad \text{for all } r \in R,\ m \in M.$$

We now show that $\overline{\varphi}$ and ψ are inverse to each other.

For any $m \in M$,

$$(\psi \circ \overline{\varphi})(m + IM) = \psi(1_{R/I} \otimes m) = 1m + IM = m + IM.$$

So

$$\psi \circ \overline{\varphi} = \mathrm{id}_{M/IM}.$$

For any elementary tensor $(r+I) \otimes m \in (R/I) \otimes_R M$,

$$(\overline{\varphi} \circ \psi)\big((r+I) \otimes m\big) = \overline{\varphi}(rm + IM) = 1_{R/I} \otimes rm = (r+I) \otimes m.$$

Since elementary tensors generate $(R/I) \otimes_R M$, it follows that

$$\overline{\varphi} \circ \psi = \mathrm{id}_{(R/I)\otimes_R M}.$$

Hence $\overline{\varphi}$ is an isomorphism:

$$M/IM \cong (R/I) \otimes_R M.$$

Finally, both modules carry natural R/I-module structures, and the maps $\overline{\varphi}$ and ψ are R/I-linear. Therefore this is an isomorphism of R/I-modules. $\square$

2.3 Balanced product

Definition 2.3.1. *Let R be a ring (not necessarily commutative). Let M_R be a right R-module and ${}_RN$ be a left R-module. The balanced product of M and N is defined as a pair (P, f) where P is an abelian group and $f : M \times N \to P$ is a bilinear map satisfies:*

- $f(mr, n) = f(m, rn)$ *for all* $r \in R$, $m \in M$, $n \in N$.
- $f(m + m', n) = f(m, n) + f(m', n)$ *for all* $m, m' \in M$, $n \in N$.
- $f(m, n + n') = f(m, n) + f(m, n')$ *for all* $m \in M$, $n, n' \in N$.

Definition 2.3.2. *A tensor product of M_R and $_RN$ is a balanced product $(M \otimes_R N, t_{M,N})$ such that if (P, f) is any balanced product of M_R and $_RN$, then there exists a unique abelian group homomorphism $\tilde{f} : M \otimes_R N \to P$ such that the following diagram commutes:*

$$\begin{tikzcd}
M \times N \arrow[r, "t_{M,N}"] \arrow[d, "f"'] & M \otimes_R N \arrow[dl, dashed, "\tilde{f}"'] \\
P &
\end{tikzcd}$$

Chapter 3

Lecture 3: Categories, Bimodules

3.1 Categories

Definition 3.1.1. *A category $\mathcal{C}$ consists of:*

- *A class of objects* $\mathrm{ob}(\mathcal{C})$*.*
- *A set of morphisms* $\mathrm{Hom}_{\mathcal{C}}(A, B)$ *for each pair of objects* $A, B \in \mathcal{C}$*.*
- *A composition map* $\mathrm{Hom}_{\mathcal{C}}(B, C) \times \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{C}}(A, C)$*,* $(\phi, \psi) \mapsto \phi \circ \psi$ *for each triple of objects* $A, B, C \in \mathcal{C}$*.*

It is assumed that objects and morphisms satisfy the following conditions:

- If $(A, B) \neq (C, D)$, then
$$\mathrm{Hom}(A, B) \cap \mathrm{Hom}(C, D) = \varnothing.$$
- If
$$f \in \mathrm{Hom}(A, B), \qquad g \in \mathrm{Hom}(B, C), \qquad h \in \mathrm{Hom}(C, D),$$
then
$$(h \circ g) \circ f = h \circ (g \circ f).$$
- (Unit) For every object A we have an element
$$1_A \in \mathrm{Hom}(A, A)$$
such that
$$f \circ 1_A = f \qquad \text{for any } f \in \mathrm{Hom}(A, B),$$
and
$$1_A \circ g = g \qquad \text{for any } g \in \mathrm{Hom}(B, A).$$

Example 3.1.2. *The following are examples of categories.*

1. *The category of sets: the objects are sets, and the morphisms are maps. It is*

denoted by **Set***.*

2. *The category of groups: the objects are groups, and the morphisms are group homomorphisms. It is denoted by* **Grp***.*

3. *The category of abelian groups: the objects are abelian groups, and the morphisms are group homomorphisms. It is denoted by* **Ab***.*

4. *The category of R-modules: the objects are R-modules, and the morphisms are R-module homomorphisms. It is denoted by R-*$\mathbf{Mod}$*(left R-modules, for right R-modules, we use $\mathbf{Mod} - R$).*

5. *The category of finitely generated R-modules: the objects are finitely generated R-modules, and the morphisms are R-module homomorphisms. It is denoted by R-*$\mathbf{mod}$*.*

6. *The category of topological spaces: the objects are topological spaces, and the morphisms are continuous maps. It is denoted by* **Top***.*

Definition 3.1.3 (Subcategory)**.** *A category $\mathcal{D}$ is called a* subcategory *of a category $\mathcal{C}$ if*

1. *the objects of $\mathcal{D}$ form a subclass of the objects of $\mathcal{C}$;*

2. *for any objects A, B of $\mathcal{D}$, we have*
   - $\mathrm{Hom}_{\mathcal{D}}(A, B) \subseteq \mathrm{Hom}_{\mathcal{C}}(A, B)$;
   - *the identity morphisms and the composition of morphisms in $\mathcal{D}$ are the same as those in $\mathcal{C}$.*

The subcategory $\mathcal{D}$ of $\mathcal{C}$ is called full *if*

$$\mathrm{Hom}_{\mathcal{D}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(A, B)$$

for any objects A, B of $\mathcal{D}$.

Example 3.1.4. *The category* **Ab** *is a full subcategory of* **Grp***.*

The category R-$\mathbf{mod}$ *is a full subcategory of R-*$\mathbf{Mod}$*.*

The category **Grp** *is not a full subcategory of* **Set***.*

Definition 3.1.5. *Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A* covariant functor

$$F : \mathcal{C} \to \mathcal{D}$$

consists of the following data:

- *for every object A of $\mathcal{C}$, an object $F(A)$ of $\mathcal{D}$;*

- *for each pair of objects A, B of $\mathcal{C}$, a map*

$$F : \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(F(A), F(B)), \qquad f \mapsto F(f).$$

These data are required to satisfy the following conditions:

1. *If $g \circ f$ is defined in $\mathcal{C}$, then*

$$F(g) \circ F(f) = F(g \circ f).$$

2. *For every object A of $\mathcal{C}$,*

$$F(1_A) = 1_{F(A)}.$$

Proposition 3.1.6. *Functors preserve diagram:*

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "g"'] & B \arrow[d, "h"] & {} \arrow[r, "F"] & {} & F(A) \arrow[r, "F(f)"] \arrow[dr, "F(g)"'] & F(B) \arrow[d, "F(h)"] \\ & C & & & & F(AC \end{tikzcd}$$

Let M and N be R-modules. Then

$$\mathrm{Hom}_R(M, N) = \{\, f : M \to N \mid f \text{ is an } R\text{-module homomorphism} \,\}.$$

This is an abelian group.

Moreover,

$$\mathrm{End}_R(M) = \mathrm{Hom}_R(M, M),$$

and $\mathrm{End}_R(M)$ is a ring.

Example 3.1.7. *Let M be an R-module. Then*

$$\mathrm{Hom}_R(M, -) : R\text{-Mod} \to \mathrm{Ab}$$

is a functor.

More precisely, it acts as follows:

$$N \longmapsto \mathrm{Hom}_R(M, N), \qquad f : N \to N' \longmapsto \mathrm{Hom}_R(M, f).$$

Given an R-module homomorphism

$$f : N \to N',$$

the induced map

$$\mathrm{Hom}_R(M, f) : \mathrm{Hom}_R(M, N) \to \mathrm{Hom}_R(M, N')$$

is defined by

$$\mathrm{Hom}_R(M, f)(h) = f \circ h \qquad (h \in \mathrm{Hom}_R(M, N)).$$

This corresponds to the commutative diagram

$$\begin{tikzcd} & M \arrow[dl, "h"'] \arrow[dr, "f\circ h"] & \\ N \arrow[rr, "f"'] & & N' \end{tikzcd}$$

Hence

$$\mathrm{Hom}_R(M, -)$$

is a covariant functor.

Definition 3.1.8. *Let $\mathcal{C}$ be a category. The* opposite category *(or* dual category*) $\mathcal{C}^{\mathrm{op}}$ is defined as follows:*

- $\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{C})$*;*
- *for any objects A, B,*

$$\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(B, A).$$

The composition in $\mathcal{C}^{\mathrm{op}}$ is defined by reversing the order of composition in $\mathcal{C}$: if

$$f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B), \qquad g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(B, C),$$

then

$$g \circ f \text{ in } \mathcal{C}^{\mathrm{op}}$$

is defined to be

$$f \circ g \text{ in } \mathcal{C}.$$

We call $\mathcal{C}^{\mathrm{op}}$ the dual category of $\mathcal{C}$.

This is illustrated by the following diagrams:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[dr, "g\circ f"'] & B \arrow[d, "g"] \\ & C \end{tikzcd} \qquad \text{in } \mathcal{C}^{\mathrm{op}},$$

while the corresponding picture in $\mathcal{C}$ is

$$\begin{tikzcd} A & B \arrow[l, "f"'] \\ & C \arrow[ul, "f\circ g"] \arrow[u, "g"'] \end{tikzcd} \qquad \text{in } \mathcal{C}.$$

Definition 3.1.9. *A* contravariant functor *from a category $\mathcal{C}$ to a category $\mathcal{D}$ is a covariant functor*

$$\mathcal{C}^{\mathrm{op}} \to \mathcal{D}.$$

Example 3.1.10. *Let M be an R-module. Then*

$$\mathrm{Hom}_R(-, M) : R\text{-Mod} \to \mathrm{Ab}$$

is a contravariant functor.

More precisely, it is defined on objects by

$$N \longmapsto \mathrm{Hom}_R(N, M),$$

and on morphisms by sending an R-module homomorphism

$$f : N \to N'$$

to the induced group homomorphism

$$\mathrm{Hom}_R(f, M) : \mathrm{Hom}_R(N', M) \to \mathrm{Hom}_R(N, M),$$

given by

$$\mathrm{Hom}_R(f, M)(h) = h \circ f \qquad (h \in \mathrm{Hom}_R(N', M)).$$

This corresponds to the commutative diagram

$$\begin{tikzcd}
N \arrow[d, "f"'] \arrow[dr, "h\circ f"] & \\
N' \arrow[r, "h"'] & M
\end{tikzcd}$$

3.2 Exact Sequences

Definition 3.2.1. *Let $\{M_i\}_{i\in\mathbb{Z}}$ be a family of R-modules, and let*

$$\varphi_i : M_i \to M_{i+1}$$

be R-module homomorphisms for each $i \in \mathbb{Z}$. Then

$$\cdots \longrightarrow M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} M_{i+1} \longrightarrow \cdots$$

is called a sequence *of R-modules.*

Definition 3.2.2. *The sequence*

$$\cdots \longrightarrow M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} M_{i+1} \longrightarrow \cdots$$

is said to be exact at M_i *if*

$$\mathrm{Im}(\varphi_{i-1}) = \ker(\varphi_i).$$

It is called an exact sequence *if it is exact at every term.*

A short exact sequence is a sequence of the form

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 0.$$

It is exact if and only if

$$\ker(\iota) = 0, \qquad \mathrm{Im}(\pi) = M'', \qquad \mathrm{Im}(\iota) = \ker(\pi).$$

Equivalently, ι is injective, π is surjective, and

$$\mathrm{Im}(\iota) = \ker(\pi).$$

Hence, for every short exact sequence

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 0,$$

we have

$$M/M' \cong M''.$$

Definition 3.2.3. *Let R, S be rings, and let*

$$F : R\text{-Mod} \to S\text{-Mod}$$

be a covariant functor.

We say that F is left exact *if for any exact sequence*

$$0 \to N \xrightarrow{\varphi} M \xrightarrow{\psi} P,$$

the sequence

$$0 \to F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P)$$

is exact.

We say that F is right exact *if for any exact sequence*

$$N \xrightarrow{\varphi} M \xrightarrow{\psi} P \to 0,$$

the sequence

$$F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P) \to 0$$

is exact.

Definition 3.2.4. *Let*

$$G : R\text{-Mod} \to S\text{-Mod}$$

be a contravariant functor. We say that G is left exact *(respectively,* right exact*) if it is left exact (respectively, right exact) when regarded as a covariant functor*

$$G : (R\text{-Mod})^{\mathrm{op}} \to S\text{-Mod}.$$

A functor is called exact *if it is both left exact and right exact.*

Let M be an R-module. Then:

$$\mathrm{Hom}_R(M, -) : R\text{-Mod} \to \mathrm{Ab} \quad \text{is left exact,}$$

$$\mathrm{Hom}_R(-, M) : R\text{-Mod} \to \mathrm{Ab} \quad \text{is left exact,}$$

and

$$M \otimes_R - : R\text{-Mod} \to \mathrm{Ab} \quad \text{is right exact.}$$

Definition 3.2.5. *A functor $F : \mathcal{C} \to \mathcal{D}$ is called* exact *if it is both left exact and right exact.*

Definition 3.2.6. *Let M be an R-module.*

1. *M is called* projective *if the functor*

$$\mathrm{Hom}_R(M, -)$$

   is exact.

2. *M is called* injective *if the functor*

$$\mathrm{Hom}_R(-, M)$$

   is exact.

3. *M is called* flat *if the functor*

$$- \otimes_R M$$

   is exact.

If R is noncommutative, then in order for

$$M \otimes_R -$$

to be well-defined, M must be a right R-module.

3.3 Bimodules

Definition 3.3.1. *Let R and S be rings. An abelian group M is called a* left-R-right-S-bimodule *if M is both a left R-module and a right S-module such that the two scalar multiplications satisfy*

$$r(xs) = (rx)s \qquad \text{for all } r \in R,\ s \in S,\ x \in M.$$

Notation.

$$_RM \qquad \text{left } R\text{-module,}$$

$$M_S \qquad \text{right } S\text{-module,}$$

$$_RM_S \qquad \text{left } R\text{-right } S\text{-bimodule.}$$

Let R, S be rings, and let $_RU_S$ be a bimodule.

For each left R-module $_RM$, there are two naturally induced S-module structures:

$$\mathrm{Hom}_R({}_RU_S, {}_RM), \qquad \text{a left } S\text{-module,}$$

$$\mathrm{Hom}_R({}_RM, {}_RU_S), \qquad \text{a right } S\text{-module.}$$

More precisely, for each $s \in S$:

if

$$f \in \mathrm{Hom}_R({}_RU_S, {}_RM),$$

define

$$s \cdot f : U \to M, \qquad u \mapsto f(us).$$

Similarly, if

$$f \in \mathrm{Hom}_R({}_RM, {}_RU_S),$$

define

$$f \cdot s : M \to U, \qquad m \mapsto f(m)s.$$

Thus we obtain functors

$$\mathrm{Hom}_R({}_RU_S, -) : R\text{-}\mathbf{Mod} \longrightarrow S\text{-}\mathbf{Mod},$$

given on objects by

$$_RM \longmapsto \mathrm{Hom}_R({}_RU_S, {}_RM),$$

which is covariant, and

$$\mathrm{Hom}_R(-, {}_RU_S) : R\text{-}\mathbf{Mod} \longrightarrow \mathbf{Mod}\text{-}S,$$

given on objects by

$$_RM \longmapsto \mathrm{Hom}_R({}_RM, {}_RU_S),$$

which is contravariant.

On morphisms, for

$$f : M \to M',$$

the induced map

$$\operatorname{Hom}_R({}_RU_S, {}_RM) \longrightarrow \operatorname{Hom}_R({}_RU_S, {}_RM')$$

is given by

$$h \longmapsto f \circ h.$$

the induced map

$$\operatorname{Hom}_R({}_RM', {}_RU_S) \longrightarrow \operatorname{Hom}_R({}_RM, {}_RU_S)$$

is given by

$$h \longmapsto h \circ f.$$

Theorem 3.3.2. *The functor*

$$\operatorname{Hom}_R({}_RU_S, -) : R\text{-}\mathbf{Mod} \longrightarrow S\text{-}\mathbf{Mod}$$

is left exact.

Proof. Let

$$0 \longrightarrow N' \xrightarrow{f} N \xrightarrow{g} N''$$

be an exact sequence in R-**Mod**. Consider the induced sequence

$$0 \longrightarrow \operatorname{Hom}_R({}_RU_S, N') \xrightarrow{\operatorname{Hom}_R(U,f)} \operatorname{Hom}_R({}_RU_S, N) \xrightarrow{\operatorname{Hom}_R(U,g)} \operatorname{Hom}_R({}_RU_S, N'').$$

We show that this sequence is exact.

First, $\operatorname{Hom}_R(U, f)$ is injective. Let

$$\varphi \in \operatorname{Hom}_R(U, N')$$

and suppose that

$$\operatorname{Hom}_R(U, f)(\varphi) = f \circ \varphi = 0.$$

Then for every $u \in U$,

$$f(\varphi(u)) = 0.$$

Since f is injective, it follows that $\varphi(u) = 0$ for all $u \in U$. Hence $\varphi = 0$, so $\operatorname{Hom}_R(U, f)$ is injective.

Next, we show that

$$\operatorname{Im}\bigl(\operatorname{Hom}_R(U, f)\bigr) \subseteq \ker\bigl(\operatorname{Hom}_R(U, g)\bigr).$$

Indeed, if

$$\varphi \in \operatorname{Hom}_R(U, N'),$$

then

$$\operatorname{Hom}_R(U,g)\bigl(\operatorname{Hom}_R(U,f)(\varphi)\bigr) = g \circ f \circ \varphi = 0,$$

because $g \circ f = 0$. Therefore

$$\operatorname{Im}\bigl(\operatorname{Hom}_R(U,f)\bigr) \subseteq \ker\bigl(\operatorname{Hom}_R(U,g)\bigr).$$

Finally, let

$$\psi \in \ker\bigl(\operatorname{Hom}_R(U,g)\bigr).$$

Then $\psi \in \operatorname{Hom}_R(U,N)$ and

$$g \circ \psi = 0.$$

Thus for every $u \in U$,

$$g(\psi(u)) = 0,$$

so $\psi(u) \in \ker g = \operatorname{Im} f$ by exactness. Hence for each $u \in U$ there exists some $y \in N'$ such that

$$f(y) = \psi(u).$$

Since f is injective, such y is unique. Define

$$\varphi : U \to N', \qquad u \mapsto y,$$

where y is the unique element satisfying $f(y) = \psi(u)$.

Then

$$f \circ \varphi = \psi.$$

It remains to check that φ is R-linear. For $u, v \in U$,

$$f(\varphi(u+v)) = \psi(u+v) = \psi(u) + \psi(v) = f(\varphi(u)) + f(\varphi(v)) = f(\varphi(u) + \varphi(v)).$$

Since f is injective,

$$\varphi(u+v) = \varphi(u) + \varphi(v).$$

Similarly, for $r \in R$ and $u \in U$,

$$f(\varphi(ru)) = \psi(ru) = r\psi(u) = rf(\varphi(u)) = f(r\varphi(u)),$$

so again by injectivity of f,

$$\varphi(ru) = r\varphi(u).$$

Thus $\varphi \in \operatorname{Hom}_R(U,N')$, and

$$\operatorname{Hom}_R(U,f)(\varphi) = f \circ \varphi = \psi.$$

Therefore

$$\ker\bigl(\operatorname{Hom}_R(U,g)\bigr) \subseteq \operatorname{Im}\bigl(\operatorname{Hom}_R(U,f)\bigr).$$

Combining the two inclusions, we get

$$\operatorname{Im}\bigl(\operatorname{Hom}_R(U, f)\bigr) = \ker\bigl(\operatorname{Hom}_R(U, g)\bigr).$$

Hence

$$0 \longrightarrow \operatorname{Hom}_R({}_RU_S, N') \xrightarrow{\operatorname{Hom}_R(U,f)} \operatorname{Hom}_R({}_RU_S, N) \xrightarrow{\operatorname{Hom}_R(U,g)} \operatorname{Hom}_R({}_RU_S, N'')$$

is exact, and so the functor $\operatorname{Hom}_R({}_RU_S, -)$ is left exact. □

Theorem 3.3.3. *Let ${}_RU_S$ be an R-S-bimodule. Then the contravariant functor*

$$\operatorname{Hom}_R(-, U) : R\text{-Mod} \longrightarrow \text{Mod-}S$$

is left exact.

Proof. Since $\operatorname{Hom}_R(-, U)$ is contravariant, to say that it is left exact means the following: for every short exact sequence of left R-modules

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0,$$

the induced sequence of right S-modules

$$0 \longrightarrow \operatorname{Hom}_R(C, U) \xrightarrow{\operatorname{Hom}_R(g,U)} \operatorname{Hom}_R(B, U) \xrightarrow{\operatorname{Hom}_R(f,U)} \operatorname{Hom}_R(A, U)$$

is exact.

We verify this.

First, $\operatorname{Hom}_R(g, U)$ is injective. Indeed, let $\varphi \in \operatorname{Hom}_R(C, U)$ and suppose

$$\operatorname{Hom}_R(g, U)(\varphi) = \varphi \circ g = 0.$$

Since g is surjective, it follows that $\varphi = 0$. Hence $\operatorname{Hom}_R(g, U)$ is injective.

Next, we show

$$\operatorname{Im}\bigl(\operatorname{Hom}_R(g, U)\bigr) \subseteq \ker\bigl(\operatorname{Hom}_R(f, U)\bigr).$$

Let $\varphi \in \operatorname{Hom}_R(C, U)$. Then

$$\operatorname{Hom}_R(f, U)\bigl(\operatorname{Hom}_R(g, U)(\varphi)\bigr) = (\varphi \circ g) \circ f = \varphi \circ (g \circ f) = 0,$$

because $g \circ f = 0$. Therefore

$$\operatorname{Im}\bigl(\operatorname{Hom}_R(g, U)\bigr) \subseteq \ker\bigl(\operatorname{Hom}_R(f, U)\bigr).$$

Finally, we prove the reverse inclusion. Let $\psi \in \operatorname{Hom}_R(B, U)$ satisfy

$$\operatorname{Hom}_R(f, U)(\psi) = \psi \circ f = 0.$$

Then ψ vanishes on $\mathrm{Im}(f) = \ker(g)$. Hence ψ factors through the quotient

$$B/\ker(g).$$

Since g is surjective, there exists a unique R-module homomorphism $\overline{\psi} : C \to U$ such that

$$\psi = \overline{\psi} \circ g.$$

Thus

$$\psi \in \mathrm{Im}\big(\mathrm{Hom}_R(g, U)\big),$$

so

$$\ker\big(\mathrm{Hom}_R(f, U)\big) \subseteq \mathrm{Im}\big(\mathrm{Hom}_R(g, U)\big).$$

Combining the two inclusions, we obtain

$$\mathrm{Im}\big(\mathrm{Hom}_R(g, U)\big) = \ker\big(\mathrm{Hom}_R(f, U)\big).$$

Hence the sequence

$$0 \longrightarrow \mathrm{Hom}_R(C, U) \xrightarrow{\mathrm{Hom}_R(g,U)} \mathrm{Hom}_R(B, U) \xrightarrow{\mathrm{Hom}_R(f,U)} \mathrm{Hom}_R(A, U)$$

is exact. This proves that $\mathrm{Hom}_R(-, U)$ is left exact. $\square$

Let ${}_SM_R$ be a left-S-right-R-bimodule, ${}_RN$ be a left R-module, consider the tensor product $M \otimes_R N$.

For each $s \in S$, the mapping

$$M \times N \longrightarrow M \otimes_R N, \qquad (m, n) \longmapsto (sm) \otimes n$$

is balanced.

So there is a unique abelian group homomorphism

$$u(s) : M \otimes_R N \longrightarrow M \otimes_R N$$

such that

$$u(s)(m \otimes n) = (sm) \otimes n.$$

Only need to check:

$$u : S \longrightarrow \mathrm{End}_{\mathbb{Z}}(M \otimes_R N)$$

is a ring homomorphism.

If M_R, ${}_RN_T$, then $M \otimes_R N$ is a right T-module:

$$(m \otimes n)t = m \otimes (nt).$$

Lemma 3.3.4. *Let ${}_S M_R$ and ${}_R N_T$ be bimodules. Then $M \otimes_R N$ carries a natural structure of a left S-right T-bimodule, defined on elementary tensors by*

$$s(m \otimes n) = (sm) \otimes n, \qquad (m \otimes n)t = m \otimes (nt),$$

for all $s \in S$, $t \in T$, $m \in M$, and $n \in N$.

Proof. Fix $s \in S$. Define

$$\beta_s : M \times N \to M \otimes_R N, \qquad \beta_s(m, n) = sm \otimes n.$$

We claim that β_s is R-balanced. Indeed, it is additive in each variable, and for $r \in R$,

$$\beta_s(mr, n) = s(mr) \otimes n = (sm)r \otimes n = sm \otimes rn = \beta_s(m, rn).$$

Hence, by the universal property of the tensor product, there exists a unique group homomorphism

$$\lambda_s : M \otimes_R N \to M \otimes_R N$$

such that

$$\lambda_s(m \otimes n) = sm \otimes n.$$

We write $\lambda_s(x) = sx$. Hence $\lambda : S \to \mathrm{End}_{\mathbb{Z}}(M \otimes_R N)$ is a ring homomorphism. Which means that $M \otimes_R N$ is a left S-module.

Similarly, for fixed $t \in T$, define

$$\gamma_t : M \times N \to M \otimes_R N, \qquad \gamma_t(m, n) = m \otimes nt.$$

Again γ_t is R-balanced, since for $r \in R$,

$$\gamma_t(mr, n) = mr \otimes nt = m \otimes r(nt) = m \otimes (rn)t = \gamma_t(m, rn).$$

Thus there exists a unique group homomorphism

$$\rho_t : M \otimes_R N \to M \otimes_R N$$

such that

$$\rho_t(m \otimes n) = m \otimes nt.$$

We write $\rho_t(x) = xt$. Hence $\rho : T \to \mathrm{End}_{\mathbb{Z}}(M \otimes_R N)$ is a ring homomorphism. Which means that $M \otimes_R N$ is a right T-module.

It remains to verify the bimodule axioms. Since elementary tensors generate $M \otimes_R N$, it suffices to check them on $m \otimes n$:

$$(s_1 + s_2)(m \otimes n) = ((s_1 + s_2)m) \otimes n = (s_1 m) \otimes n + (s_2 m) \otimes n = s_1(m \otimes n) + s_2(m \otimes n),$$

$$(s_1 s_2)(m \otimes n) = ((s_1 s_2)m) \otimes n = s_1((s_2 m) \otimes n) = s_1\big(s_2(m \otimes n)\big),$$

$$1_S(m \otimes n) = (1_S m) \otimes n = m \otimes n,$$

and similarly

$$(m \otimes n)(t_1 + t_2) = m \otimes n(t_1 + t_2) = (m \otimes nt_1) + (m \otimes nt_2),$$

$$((m \otimes n)t_1)t_2 = (m \otimes nt_1)t_2 = m \otimes (nt_1)t_2 = m \otimes n(t_1 t_2) = (m \otimes n)(t_1 t_2),$$

$$(m \otimes n)1_T = m \otimes n.$$

Finally,

$$\big(s(m \otimes n)\big)t = ((sm) \otimes n)t = (sm) \otimes nt = s(m \otimes nt) = s\big((m \otimes n)t\big).$$

Therefore $M \otimes_R N$ is a left S-right T-bimodule. □

Lemma 3.3.5. *Let M_R, M'_R be right R-modules, and let ${}_RN$, ${}_RN'$ be left R-modules. Suppose*

$$f_1, f_2, f \in \operatorname{Hom}_R(M, M'), \qquad g_1, g_2, g \in \operatorname{Hom}_R(N, N').$$

Then for each pair (f, g) there exists a unique group homomorphism

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n) \qquad (m \in M,\ n \in N).$$

Moreover,

1. $(f_1 + f_2) \otimes g = (f_1 \otimes g) + (f_2 \otimes g)$;
2. $f \otimes (g_1 + g_2) = (f \otimes g_1) + (f \otimes g_2)$;
3. $f \otimes 0 = 0$ *and* $0 \otimes g = 0$;
4. $1_M \otimes 1_N = 1_{M \otimes_R N}$.

Proof. Define

$$\beta : M \times N \to M' \otimes_R N', \qquad \beta(m, n) = f(m) \otimes g(n).$$

We check that β is R-balanced. It is additive in each variable because f and g are module homomorphisms. Also, for $r \in R$,

$$\beta(mr, n) = f(mr) \otimes g(n) = f(m)r \otimes g(n) = f(m) \otimes rg(n) = f(m) \otimes g(rn) = \beta(m, rn).$$

Hence, by the universal property of $M \otimes_R N$, there exists a unique group homomorphism

$$f \otimes g : M \otimes_R N \to M' \otimes_R N'$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n).$$

Now each asserted identity is verified on elementary tensors, and therefore on all of $M \otimes_R N$.

For (1), for all $m \in M$ and $n \in N$,

$$\big(((f_1 + f_2) \otimes g)(m \otimes n)\big) = (f_1 + f_2)(m) \otimes g(n) = f_1(m) \otimes g(n) + f_2(m) \otimes g(n),$$

while

$$\big((f_1 \otimes g) + (f_2 \otimes g)\big)(m \otimes n) = (f_1 \otimes g)(m \otimes n) + (f_2 \otimes g)(m \otimes n) = f_1(m) \otimes g(n) + f_2(m) \otimes g(n).$$

Hence

$$(f_1 + f_2) \otimes g = (f_1 \otimes g) + (f_2 \otimes g).$$

The proof of (2) is analogous:

$$(f \otimes (g_1 + g_2))(m \otimes n) = f(m) \otimes (g_1 + g_2)(n) = f(m) \otimes g_1(n) + f(m) \otimes g_2(n),$$

which equals

$$((f \otimes g_1) + (f \otimes g_2))(m \otimes n).$$

For (3),

$$(f \otimes 0)(m \otimes n) = f(m) \otimes 0 = 0, \qquad (0 \otimes g)(m \otimes n) = 0 \otimes g(n) = 0,$$

so both maps are the zero homomorphism.

For (4),

$$(1_M \otimes 1_N)(m \otimes n) = 1_M(m) \otimes 1_N(n) = m \otimes n,$$

hence $1_M \otimes 1_N$ is the identity on $M \otimes_R N$. □

Lemma 3.3.6. *Let M_R, M'_R, M''_R be right R-modules, and let ${}_RN$, ${}_RN'$, ${}_RN''$ be left R-modules. Let*

$$f : M \to M', \qquad f' : M' \to M'', \qquad g : N \to N', \qquad g' : N' \to N''$$

be R-module homomorphisms. Then

$$(f' \otimes g') \circ (f \otimes g) = (f' \circ f) \otimes (g' \circ g).$$

Proof. Again it suffices to check the equality on elementary tensors. For $m \in M$ and $n \in N$,

$$\big((f' \otimes g') \circ (f \otimes g)\big)(m \otimes n) = (f' \otimes g')\big(f(m) \otimes g(n)\big) = f'(f(m)) \otimes g'(g(n)),$$

while

$$\big((f'\circ f)\otimes(g'\circ g)\big)(m\otimes n)=(f'\circ f)(m)\otimes(g'\circ g)(n)=f'(f(m))\otimes g'(g(n)).$$

Thus both homomorphisms agree on the generators $m\otimes n$ of $M\otimes_R N$, hence they are equal. $\square$

Definition 3.3.7. *Let ${}_SU_R$ be a bimodule. Define*

$$U\otimes_R - : R\text{-Mod}\longrightarrow S\text{-Mod}$$

by

$$M\longmapsto U\otimes_R M,\qquad f\longmapsto \mathrm{id}_U\otimes f.$$

Similarly, Let ${}_RU_S$ be a bimodule.

$$-\otimes_R U : \text{Mod-}R\longrightarrow \text{Mod-}S$$

by

$$N\longmapsto N\otimes_R U,\qquad g\longmapsto g\otimes \mathrm{id}_U\,.$$

Proposition 3.3.8. *Let ${}_SU_R$ be a bimodule. Then*

$$U\otimes_R - : R\text{-Mod}\longrightarrow S\text{-Mod}$$

is a covariant functor.

Similarly, if ${}_RU_S$ is a bimodule, then

$$-\otimes_R U : \text{Mod-}R\longrightarrow \text{Mod-}S$$

is a covariant functor.

Proof. We prove the first statement. For each left R-module M, the tensor product

$$U\otimes_R M$$

is naturally a left S-module via

$$s(u\otimes m)=(su)\otimes m\qquad (s\in S,\ u\in U,\ m\in M).$$

For each R-module homomorphism $f:M\to N$, define

$$U\otimes_R f : U\otimes_R M\to U\otimes_R N$$

by

$$(U\otimes_R f)(u\otimes m)=u\otimes f(m).$$

Equivalently,

$$U \otimes_R f = \mathrm{id}_U \otimes f.$$

This is well-defined and is an S-module homomorphism.

It remains to verify the functorial properties. For the identity map $\mathrm{id}_M : M \to M$, we have

$$U \otimes_R \mathrm{id}_M = \mathrm{id}_U \otimes \mathrm{id}_M = \mathrm{id}_{U \otimes_R M}.$$

If

$$M \xrightarrow{f} N \xrightarrow{g} P$$

are R-module homomorphisms, then for every $u \otimes m \in U \otimes_R M$,

$$(U \otimes_R (g \circ f))(u \otimes m) = u \otimes (g \circ f)(m) = g(u \otimes f(m)) = ((U \otimes_R g) \circ (U \otimes_R f))(u \otimes m).$$

Hence

$$U \otimes_R (g \circ f) = (U \otimes_R g) \circ (U \otimes_R f).$$

Therefore $U \otimes_R -$ is a covariant functor from R-Mod to S-Mod.

The proof of the second statement is entirely similar. If ${}_R U_S$ is a bimodule and N is a right R-module, then

$$N \otimes_R U$$

is naturally a right S-module via

$$(n \otimes u)s = n \otimes (us).$$

For a homomorphism $g : N \to N'$ in Mod-R, define

$$g \otimes_R U = g \otimes \mathrm{id}_U : N \otimes_R U \to N' \otimes_R U$$

by

$$(g \otimes_R U)(n \otimes u) = g(n) \otimes u.$$

One checks immediately that identities and compositions are preserved. Thus

$$- \otimes_R U : \text{Mod-}R \to \text{Mod-}S$$

is also a covariant functor. □

Theorem 3.3.9. *Let ${}_S U_R$ be a bimodule. Then the functor*

$$U \otimes_R - \;:\; R\text{-}Mod \longrightarrow S\text{-}Mod$$

is right exact. In other words, if

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \longrightarrow 0$$

is an exact sequence of left R-modules, then

$$U \otimes_R N' \xrightarrow{\mathrm{id}_U \otimes f} U \otimes_R N \xrightarrow{\mathrm{id}_U \otimes g} U \otimes_R N'' \longrightarrow 0$$

is exact.

Proof. Since

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \longrightarrow 0$$

is exact, we have

$$\ker g = \operatorname{Im} f \qquad \text{and} \qquad N'' \cong N / \operatorname{Im} f.$$

Let

$$\pi : N \longrightarrow N / \operatorname{Im} f$$

be the canonical projection. Then there exists an isomorphism

$$\alpha : N / \operatorname{Im} f \longrightarrow N''$$

such that

$$g = \alpha \circ \pi.$$

Now consider the map

$$\Phi : U \otimes_R N \longrightarrow U \otimes_R (N / \operatorname{Im} f), \qquad u \otimes n \longmapsto u \otimes \pi(n).$$

By the universal property of tensor products, Φ is a well-defined S-module homomorphism.

We claim that

$$\ker \Phi = \operatorname{Im}(\mathrm{id}_U \otimes f).$$

First, since $\pi \circ f = 0$, functoriality gives

$$\Phi \circ (\mathrm{id}_U \otimes f) = \mathrm{id}_U \otimes (\pi \circ f) = 0.$$

Hence

$$\operatorname{Im}(\mathrm{id}_U \otimes f) \subseteq \ker \Phi.$$

To prove the reverse inclusion, define

$$\overline{\Phi} : (U \otimes_R N) / \operatorname{Im}(\mathrm{id}_U \otimes f) \longrightarrow U \otimes_R (N / \operatorname{Im} f)$$

by

$$\overline{\Phi}\big((u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f)\big) = u \otimes \pi(n).$$

This is well defined because Φ vanishes on $\operatorname{Im}(\mathrm{id}_U \otimes f)$.

Next define

$$\Psi : U \times (N / \operatorname{Im} f) \longrightarrow (U \otimes_R N) / \operatorname{Im}(\mathrm{id}_U \otimes f)$$

by

$$\Psi(u, \overline{n}) = (u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f),$$

where $\overline{n} = n + \operatorname{Im} f$.

We check that Ψ is well defined. If $\overline{n} = \overline{n'}$, then

$$n - n' \in \operatorname{Im} f,$$

so there exists $x \in N'$ such that

$$n - n' = f(x).$$

Hence

$$(u \otimes n) - (u \otimes n') = u \otimes (n - n') = u \otimes f(x) = (\mathrm{id}_U \otimes f)(u \otimes x) \in \operatorname{Im}(\mathrm{id}_U \otimes f).$$

Thus

$$(u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f) = (u \otimes n') + \operatorname{Im}(\mathrm{id}_U \otimes f).$$

So Ψ is well defined. It is clearly R-balanced, hence induces an S-module homomorphism

$$\overline{\Psi} : U \otimes_R (N/\operatorname{Im} f) \longrightarrow (U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f)$$

such that

$$\overline{\Psi}(u \otimes \overline{n}) = (u \otimes n) + \operatorname{Im}(\mathrm{id}_U \otimes f).$$

It is immediate that $\overline{\Phi}$ and $\overline{\Psi}$ are inverse to each other. Therefore

$$(U \otimes_R N)/\operatorname{Im}(\mathrm{id}_U \otimes f) \cong U \otimes_R (N/\operatorname{Im} f).$$

It follows that

$$\ker \Phi = \operatorname{Im}(\mathrm{id}_U \otimes f).$$

Finally, since $\alpha : N/\operatorname{Im} f \to N''$ is an isomorphism, the induced map

$$\mathrm{id}_U \otimes \alpha : U \otimes_R (N/\operatorname{Im} f) \longrightarrow U \otimes_R N''$$

is an isomorphism. Moreover,

$$\mathrm{id}_U \otimes g = (\mathrm{id}_U \otimes \alpha) \circ \Phi.$$

Hence

$$\ker(\mathrm{id}_U \otimes g) = \ker \Phi = \operatorname{Im}(\mathrm{id}_U \otimes f),$$

and $\mathrm{id}_U \otimes g$ is surjective because Φ is surjective and $\mathrm{id}_U \otimes \alpha$ is an isomorphism.

Therefore

$$U \otimes_R N' \xrightarrow{\mathrm{id}_U \otimes f} U \otimes_R N \xrightarrow{\mathrm{id}_U \otimes g} U \otimes_R N'' \longrightarrow 0$$

is exact. $\square$

Theorem 3.3.10. *Let ${}_RU_S$ be a left-R-right-S-bimodule. Then the functor*

$$- \otimes_R U \ : \ Mod - R \longrightarrow Mod - S$$

is right exact.

Chapter 4

Lecture 4: Projectives, Injectives

4.1 Projective, Injective and Flat Modules

Definition 4.1.1. *Let R be a ring.*

- *A left R-module P is called* projective *if the functor*

$$\mathrm{Hom}_R(P, -) : R - \mathrm{Mod} \longrightarrow \mathbf{Ab}$$

is exact.

- *A left R-module I is called* injective *if the contravariant functor*

$$\mathrm{Hom}_R(-, I) : R - \mathrm{Mod} \longrightarrow \mathbf{Ab}$$

is exact.

- *A left R-module F is called* flat *if the functor*

$$- \otimes_R F : \text{Mod-}R \longrightarrow \mathbf{Ab}$$

is exact.

Theorem 4.1.2. *Let ${}_RP$ be a left R-module. The following statements are equivalent.*

1. *Whenever we are given a commutative diagram*

$$\begin{tikzcd} & P \arrow[dl, dashed, "\widetilde{\gamma}"'] \arrow[d, "\gamma"] & \\ M \arrow[r, two heads, "g"'] & N \arrow[r] & 0 \end{tikzcd}$$

with g an epimorphism, there exists an R-homomorphism

$$\widetilde{\gamma} : P \to M$$

such that $g \circ \widetilde{\gamma} = \gamma$.

2. *For every epimorphism $f : M \to N$ of left R-modules, the induced map*

$$\operatorname{Hom}_R(P, f) : \operatorname{Hom}_R(P, M) \longrightarrow \operatorname{Hom}_R(P, N)$$

is surjective.

3. *For every ring S and every right S-module structure on P making ${}_RP_S$ an R-S-bimodule, the functor*

$$\operatorname{Hom}_R(P, -) : R - \mathrm{Mod} \longrightarrow S - \mathrm{Mod}$$

is exact.

4. *For every short exact sequence*

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

of left R-modules, the sequence

$$0 \longrightarrow \operatorname{Hom}_R(P, M') \xrightarrow{\operatorname{Hom}_R(P,u)} \operatorname{Hom}_R(P, M) \xrightarrow{\operatorname{Hom}_R(P,v)} \operatorname{Hom}_R(P, M'') \longrightarrow 0$$

is exact.

Proof. We first recall two standard facts.

1. For every left R-module P, the functor

$$\operatorname{Hom}_R(P, -) : R\text{-Mod} \to \mathbf{Ab}$$

is left exact.

2. If ${}_RP_S$ is an R-S-bimodule and M is a left R-module, then $\operatorname{Hom}_R(P, M)$ is naturally a left S-module via

$$(s \cdot \varphi)(p) := \varphi(ps) \qquad (s \in S,\ \varphi \in \operatorname{Hom}_R(P, M),\ p \in P).$$

Moreover, for every R-homomorphism $h : M \to N$, the induced map

$$\operatorname{Hom}_R(P, h) : \operatorname{Hom}_R(P, M) \to \operatorname{Hom}_R(P, N)$$

is S-linear.

(1) $\Longleftrightarrow$(2). Let $f : M \to N$ be an epimorphism. For $\gamma \in \operatorname{Hom}_R(P, N)$, saying that there exists $\widetilde{\gamma} : P \to M$ such that

$$f \circ \widetilde{\gamma} = \gamma$$

is exactly saying that γ lies in the image of

$$\operatorname{Hom}_R(P, f) : \operatorname{Hom}_R(P, M) \to \operatorname{Hom}_R(P, N).$$

Hence (1) and (2) are equivalent.

(2) $\implies$(4). Let

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

be a short exact sequence of left R-modules. Since $\mathrm{Hom}_R(P, -)$ is left exact, the sequence

$$0 \longrightarrow \mathrm{Hom}_R(P, M') \xrightarrow{\mathrm{Hom}_R(P,u)} \mathrm{Hom}_R(P, M) \xrightarrow{\mathrm{Hom}_R(P,v)} \mathrm{Hom}_R(P, M'')$$

is exact. Since v is an epimorphism, (2) implies that $\mathrm{Hom}_R(P, v)$ is surjective. Therefore

$$0 \longrightarrow \mathrm{Hom}_R(P, M') \xrightarrow{\mathrm{Hom}_R(P,u)} \mathrm{Hom}_R(P, M) \xrightarrow{\mathrm{Hom}_R(P,v)} \mathrm{Hom}_R(P, M'') \longrightarrow 0$$

is exact, proving (4).

(4) $\implies$(2). Let $f : M \to N$ be an epimorphism. Apply (4) to the short exact sequence

$$0 \longrightarrow \ker(f) \longrightarrow M \xrightarrow{f} N \longrightarrow 0.$$

Then

$$\mathrm{Hom}_R(P, f) : \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N)$$

is surjective. Thus (2) holds.

(3) $\implies$(4). Take $S = \mathbb{Z}$, with the canonical right $\mathbb{Z}$-module structure on P. Then ${}_RP_{\mathbb{Z}}$ is an R-$\mathbb{Z}$-bimodule, and $\mathbb{Z}$-Mod $=$ **Ab**. Hence (3) gives that

$$\mathrm{Hom}_R(P, -) : R\text{-Mod} \to \mathbf{Ab}$$

is exact, which is exactly statement (4).

(4) $\implies$(3). Fix a ring S and a right S-module structure on P making ${}_RP_S$ an R-S-bimodule. Let

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

be a short exact sequence of left R-modules. By (4), the induced sequence

$$0 \longrightarrow \mathrm{Hom}_R(P, M') \xrightarrow{\mathrm{Hom}_R(P,u)} \mathrm{Hom}_R(P, M) \xrightarrow{\mathrm{Hom}_R(P,v)} \mathrm{Hom}_R(P, M'') \longrightarrow 0$$

is exact as a sequence of abelian groups. By the remark above, each term carries a natural left S-module structure and each map is S-linear. Since exactness in S-Mod is equivalent to exactness of the underlying abelian groups, the above sequence is exact in S-Mod. Therefore

$$\mathrm{Hom}_R(P, -) : R\text{-Mod} \to S\text{-Mod}$$

is exact. Hence (3) holds.

Therefore (1), (2), (3), and (4) are equivalent. $\square$

Proposition 4.1.3. *Free modules are projective.*

Proof. Let $F = \bigoplus_{i\in I} Re_i$ be a free left R-module with basis $\{e_i\}_{i\in I}$. Let $g : M \to N$ be an epimorphism and let $\gamma : F \to N$ be an R-homomorphism. For each $i \in I$, choose $m_i \in M$ such that

$$g(m_i) = \gamma(e_i).$$

Since F is free, there exists a unique R-homomorphism

$$\widetilde{\gamma} : F \to M$$

such that $\widetilde{\gamma}(e_i) = m_i$ for all $i \in I$. Then

$$g\widetilde{\gamma}(e_i) = g(m_i) = \gamma(e_i)$$

for every basis element, so $g\widetilde{\gamma} = \gamma$. Hence F is projective. $\square$

Theorem 4.1.4. *Let ${}_RP$ be a left R-module. The following statements are equivalent.*

1. *P is projective.*
2. *Every epimorphism*
$$f : M \longrightarrow P \longrightarrow 0$$
splits, that is, there exists an R-homomorphism $g : P \to M$ such that
$$f \circ g = \mathrm{id}_P\,.$$
3. *P is isomorphic to a direct summand of a free module.*

Proof. **(1) $\Longrightarrow$(2).** Apply the lifting property to the diagram

$$\begin{tikzcd} & P \arrow[dl, dashed, "g"'] \arrow[d, "\mathrm{id}_P"] & \\ M \arrow[r, two heads, "f"'] & P \arrow[r] & 0 \end{tikzcd}$$

to obtain $f \circ g = \mathrm{id}_P$.

(2) $\Longrightarrow$(3). Every module is a homomorphic image of a free module, so choose an epimorphism

$$\pi : F \to P$$

with F free. By (2), π splits, so there exists $s : P \to F$ such that

$$\pi \circ s = \mathrm{id}_P\,.$$

Therefore

$$F \cong \ker\pi \oplus s(P) \cong \ker\pi \oplus P,$$

and hence P is a direct summand of the free module F.

(3) $\Longrightarrow$(1). Write

$$F \cong P \oplus Q$$

with F free. Since free modules are projective, it suffices to show that a direct summand of a projective module is projective. Let $f : M \to N$ be an epimorphism and

$$\alpha : P \to N$$

an R-homomorphism. Define

$$\widehat{\alpha} : P \oplus Q \to N, \qquad \widehat{\alpha}(p, q) = \alpha(p).$$

Because F is projective, there exists

$$\widehat{\beta} : P \oplus Q \to M$$

such that $f \circ \widehat{\beta} = \widehat{\alpha}$. Let

$$\iota : P \to P \oplus Q, \qquad p \mapsto (p, 0).$$

Then

$$f \circ (\widehat{\beta} \circ \iota) = \widehat{\alpha} \circ \iota = \alpha.$$

Hence P is projective. □

Corollary 4.1.5. *A left R-module P is finitely generated and projective if and only if for some R-module P' and some integer $n \geq 0$ there is an R-isomorphism*

$$P \oplus P' \cong R^n.$$

Proof. If $P \oplus P' \cong R^n$, then P is a direct summand of a finite free module, hence finitely generated and projective.

Conversely, assume that P is finitely generated and projective. By the theorem above, there exists a free module

$$F = \bigoplus_{i \in I} Re_i$$

and an R-module Q such that

$$F \cong P \oplus Q.$$

Let $p : F \to P$ and $s : P \to F$ be the projection and inclusion corresponding to this direct sum decomposition. Choose generators $x_1, \dots, x_m$ of P. Each $s(x_j)$ is a finite linear combination of the basis elements $\{e_i\}_{i \in I}$, so there exists a finite subset $J \subseteq I$ such that

$$s(x_1), \dots, s(x_m) \in F_J := \bigoplus_{j \in J} Re_j.$$

Since the x_j generate P, we have $s(P) \subseteq F_J$. Hence the restriction

$$p|_{F_J} : F_J \to P$$

is surjective, and the map $s : P \to F_J$ is a section of $p|_{F_J}$. Therefore

$$F_J \cong P \oplus \ker(p|_{F_J}).$$

Since $F_J \cong R^{|J|}$, the result follows. $\square$

4.2 Schanuel Lemma

Theorem 4.2.1 (Schanuel Lemma). *Let*

$$0 \longrightarrow K \longrightarrow P \xrightarrow{\lambda} M \longrightarrow 0$$

and

$$0 \longrightarrow K' \longrightarrow P' \xrightarrow{\lambda'} M \longrightarrow 0$$

be short exact sequences of left R-modules, with P and P' projective. Then

$$K' \oplus P \cong K \oplus P'.$$

Proof. Define

$$Y := P \times_M P' = \{(p, p') \in P \oplus P' \mid \lambda(p) = \lambda'(p')\}.$$

Let

$$\mu : Y \to P, \qquad \mu(p, p') = p,$$

and

$$\mu' : Y \to P', \qquad \mu'(p, p') = p'.$$

Then Y fits into the pullback diagram

$$\begin{tikzcd} Y \arrow[r, "\mu'"] \arrow[d, "\mu"'] & P' \arrow[d, "\lambda'"] \\ P \arrow[r, "\lambda"'] & M \end{tikzcd}$$

that is, the square commutes and Y is the fiber product of λ and λ'.

We claim that there are short exact sequences

$$0 \longrightarrow K' \xrightarrow{i'} Y \xrightarrow{\mu} P \longrightarrow 0, \qquad i'(k') = (0, k'),$$

and

$$0 \longrightarrow K \xrightarrow{i} Y \xrightarrow{\mu'} P' \longrightarrow 0, \qquad i(k) = (k, 0).$$

First, μ is surjective. Indeed, let $p \in P$. Since λ' is surjective, there exists $p' \in P'$ such that

$$\lambda'(p') = \lambda(p).$$

Then $(p, p') \in Y$, so $\mu(p, p') = p$.

Next,

$$\ker(\mu) = \{(0, p') \in Y \mid \lambda'(p') = 0\} = \{(0, k') \mid k' \in K'\} = \operatorname{Im}(i').$$

Hence

$$0 \longrightarrow K' \xrightarrow{i'} Y \xrightarrow{\mu} P \longrightarrow 0$$

is exact.

Similarly, μ' is surjective, and

$$\ker(\mu') = \{(p, 0) \in Y \mid \lambda(p) = 0\} = \{(k, 0) \mid k \in K\} = \operatorname{Im}(i).$$

Thus

$$0 \longrightarrow K \xrightarrow{i} Y \xrightarrow{\mu'} P' \longrightarrow 0$$

is exact.

Since P is projective and $\mu : Y \to P$ is an epimorphism, there exists an R-homomorphism $s : P \to Y$ such that

$$\mu \circ s = \mathrm{id}_P.$$

This is expressed by the commutative diagram

$$\begin{tikzcd} & Y \arrow[d, two heads, "\mu"] \\ P \arrow[ur, dashed, "s"] \arrow[r, "\mathrm{id}_P"'] & P \end{tikzcd}$$

so the sequence

$$0 \longrightarrow K' \longrightarrow Y \xrightarrow{\mu} P \longrightarrow 0$$

splits. Therefore,

$$Y \cong K' \oplus P.$$

Likewise, since P' is projective and $\mu' : Y \to P'$ is an epimorphism, there exists $s' : P' \to Y$ such that

$$\mu' \circ s' = \mathrm{id}_{P'},$$

and hence

$$0 \longrightarrow K \longrightarrow Y \xrightarrow{\mu'} P' \longrightarrow 0$$

splits. Therefore,

$$Y \cong K \oplus P'.$$

Combining the two decompositions of Y, we obtain

$$K' \oplus P \cong Y \cong K \oplus P',$$

and hence

$$K' \oplus P \cong K \oplus P'.$$

□

Lemma 4.2.2. *Let M be a left R-module. Then*

$$\mathrm{Hom}_R({}_RR, M) \cong M$$

as left R-modules.

Proof. Define

$$\Phi : \mathrm{Hom}_R({}_RR, M) \to M, \qquad f \mapsto f(1).$$

This map is R-linear. Define also

$$\Psi : M \to \mathrm{Hom}_R({}_RR, M), \qquad m \mapsto (r \mapsto rm).$$

For $f \in \mathrm{Hom}_R({}_RR, M)$ and $r \in R$,

$$\Psi(\Phi(f))(r) = rf(1) = f(r \cdot 1) = f(r),$$

so $\Psi\Phi = \mathrm{id}$. Also

$$\Phi(\Psi(m)) = \Psi(m)(1) = 1m = m,$$

hence Φ and Ψ are inverse isomorphisms. $\square$

Corollary 4.2.3. *For any ring R with 1,*

$$\mathrm{End}_R({}_RR) \cong R^{\mathrm{op}}$$

as rings.

Proof. By the lemma, every $f \in \mathrm{End}_R({}_RR)$ is determined by $f(1)$, and

$$f(r) = rf(1)$$

for all $r \in R$. Thus

$$\Theta : \mathrm{End}_R({}_RR) \to R^{\mathrm{op}}, \qquad f \mapsto f(1)$$

is bijective. Moreover, for $f, g \in \mathrm{End}_R({}_RR)$,

$$(f \circ g)(1) = f(g(1)) = g(1)f(1),$$

which is multiplication in the opposite ring. Therefore Θ is a ring isomorphism. $\square$

4.3 Projective Morphisms

Lemma 4.3.1. *Let R be a commutative ring with 1, and let X, Y, M be R-modules. Then there is an R-module homomorphism*

$$\Theta_M^{X,Y} : \mathrm{Hom}_R(X, M) \otimes_R \mathrm{Hom}_R(M, Y) \longrightarrow \mathrm{Hom}_R(X, Y), \qquad \varphi \otimes \psi \longmapsto \psi \circ \varphi.$$

> *Its image is precisely the set of all morphisms $f : X \to Y$ which factor through some finite direct power M^n of M.*

Proof. First define

$$b : \mathrm{Hom}_R(X, M) \times \mathrm{Hom}_R(M, Y) \longrightarrow \mathrm{Hom}_R(X, Y), \qquad b(\varphi, \psi) = \psi \circ \varphi.$$

Since R is commutative, composition is R-balanced in the sense that

$$b(r\varphi, \psi) = \psi \circ (r\varphi) = r(\psi \circ \varphi) = b(\varphi, r\psi)$$

for all $r \in R$, $\varphi \in \mathrm{Hom}_R(X, M)$, and $\psi \in \mathrm{Hom}_R(M, Y)$; moreover b is additive in each variable. Hence b induces a unique R-linear map

$$\Theta_M^{X,Y} : \mathrm{Hom}_R(X, M) \otimes_R \mathrm{Hom}_R(M, Y) \to \mathrm{Hom}_R(X, Y)$$

such that

$$\Theta_M^{X,Y}(\varphi \otimes \psi) = \psi \circ \varphi.$$

Now let

$$f = \Theta_M^{X,Y}\Big(\sum_{i=1}^n \varphi_i \otimes \psi_i\Big) = \sum_{i=1}^n \psi_i \circ \varphi_i.$$

Define morphisms

$$\alpha : X \to M^n, \qquad x \longmapsto (\varphi_1(x), \ldots, \varphi_n(x)),$$

and

$$\beta : M^n \to Y, \qquad (m_1, \ldots, m_n) \longmapsto \sum_{i=1}^n \psi_i(m_i).$$

Then for every $x \in X$,

$$(\beta \circ \alpha)(x) = \beta\big(\varphi_1(x), \ldots, \varphi_n(x)\big) = \sum_{i=1}^n \psi_i(\varphi_i(x)) = f(x).$$

Thus $f = \beta \circ \alpha$, so every element in the image of $\Theta_M^{X,Y}$ factors through some finite direct power M^n.

Conversely, suppose that a morphism $f : X \to Y$ factors through M^n, say

$$f : X \xrightarrow{u} M^n \xrightarrow{v} Y.$$

Let $\pi_i : M^n \to M$ be the i th projection and $\iota_i : M \to M^n$ the i th inclusion. Since

$$\mathrm{id}_{M^n} = \sum_{i=1}^n \iota_i \circ \pi_i,$$

we obtain

$$f = v \circ u = v \circ \mathrm{id}_{M^n} \circ u = \sum_{i=1}^{n} v \circ \iota_i \circ \pi_i \circ u = \sum_{i=1}^{n} (v \circ \iota_i) \circ (\pi_i \circ u).$$

Therefore

$$f = \Theta_M^{X,Y}\Big(\sum_{i=1}^{n} (\pi_i \circ u) \otimes (v \circ \iota_i)\Big),$$

so f lies in the image of $\Theta_M^{X,Y}$.

Hence the image of $\Theta_M^{X,Y}$ is exactly the set of morphisms $X \to Y$ that factor through some finite direct power M^n. □

Definition 4.3.2. *Let R be a commutative ring with 1. For an R-module X, write*

$$X^* := \mathrm{Hom}_R(X, R).$$

For R-modules X and Y, define

$$\tau_{X,Y} : X^* \otimes_R Y \longrightarrow \mathrm{Hom}_R(X, Y), \qquad \varphi \otimes y \longmapsto \big(x \mapsto \varphi(x)y\big).$$

Definition 4.3.3. *A morphism $f : X \to Y$ is called a* projective morphism *if it factors through a finitely generated projective module.*

Proposition 4.3.4. *Let R be a commutative ring with 1. A morphism $f : X \to Y$ is projective if and only if*

$$f \in \mathrm{Im}(\tau_{X,Y}).$$

Proof. By the previous lemma with $M = R$, the image of $\tau_{X,Y}$ consists exactly of those morphisms that factor through some finite free module R^n.

Since finite free modules are finitely generated projective, every morphism in $\mathrm{Im}(\tau_{X,Y})$ is projective.

Conversely, suppose $f : X \to Y$ factors through a finitely generated projective module P:

$$X \xrightarrow{u} P \xrightarrow{v} Y.$$

Choose Q and n such that

$$P \oplus Q \cong R^n.$$

Let $i : P \to R^n$ and $p : R^n \to P$ be the inclusion and projection corresponding to this decomposition, so that $p \circ i = \mathrm{id}_P$. Then

$$f = v \circ p \circ i \circ u,$$

hence f factors through R^n. Therefore $f \in \mathrm{Im}(\tau_{X,Y})$. □

Definition 4.3.5. *For R-modules X and Y, we write*

$$\operatorname{Hom}_R^{\mathrm{pr}}(X,Y)$$

for the set of projective morphisms from X to Y.

Example 4.3.6. *Let k be a field, and consider the infinite-dimensional k-vector space*

$$V = \bigoplus_{n\geq 1} k e_n.$$

Then the identity morphism

$$\mathrm{id}_V : V \longrightarrow V$$

is not a projective morphism. In other words,

$$\mathrm{id}_V \notin \operatorname{Hom}_k^{\mathrm{pr}}(V,V).$$

Indeed, suppose for contradiction that id_V is a projective morphism. Then, by definition, there exists a finitely generated projective k-module P and k-linear maps

$$V \xrightarrow{\alpha} P \xrightarrow{\beta} V$$

such that

$$\mathrm{id}_V = \beta \circ \alpha.$$

Since k is a field, every finitely generated projective k-module is a finite-dimensional k-vector space. Hence P is finite-dimensional over k.

Therefore

$$\operatorname{Im}(\beta\circ\alpha) \subseteq \operatorname{Im}\beta$$

is finite-dimensional. But

$$\operatorname{Im}(\beta\circ\alpha) = \operatorname{Im}(\mathrm{id}_V) = V.$$

Thus V would be finite-dimensional, contradicting the definition

$$V = \bigoplus_{n\geq 1} k e_n.$$

Therefore id_V is not a projective morphism.

Equivalently, over a field k, projective morphisms are precisely finite-rank linear maps. Since id_V has infinite rank, it is not projective.

Lemma 4.3.7. *For any R-module M, the set*

$$\operatorname{Hom}_R^{\mathrm{pr}}(M,M)$$

is a two-sided ideal of the ring $\mathrm{End}_R(M)$*.*

Proof. We first show that $\mathrm{Hom}_R^{\mathrm{pr}}(M, M)$ is an additive subgroup of $\mathrm{End}_R(M)$.

Let $f, g \in \mathrm{Hom}_R^{\mathrm{pr}}(M, M)$. Then there exist finitely generated projective R-modules P, Q and morphisms

$$M \xrightarrow{u} P \xrightarrow{v} M, \qquad M \xrightarrow{u'} Q \xrightarrow{v'} M$$

such that

$$f = v \circ u, \qquad g = v' \circ u'.$$

Consider the morphisms

$$\alpha : M \to P \oplus Q, \qquad \alpha(x) = (u(x), u'(x)),$$

and

$$\beta : P \oplus Q \to M, \qquad \beta(p, q) = v(p) + v'(q).$$

Then

$$(f + g)(x) = v(u(x)) + v'(u'(x)) = \beta(\alpha(x))$$

for all $x \in M$, so

$$f + g = \beta \circ \alpha.$$

Since $P \oplus Q$ is again a finitely generated projective R-module, it follows that $f + g \in \mathrm{Hom}_R^{\mathrm{pr}}(M, M)$.

Also, the zero endomorphism factors through the zero module, and if $f = v \circ u$, then

$$-f = (-v) \circ u,$$

so $\mathrm{Hom}_R^{\mathrm{pr}}(M, M)$ is an additive subgroup of $\mathrm{End}_R(M)$.

Now let $f \in \mathrm{Hom}_R^{\mathrm{pr}}(M, M)$ and let $a, b \in \mathrm{End}_R(M)$. Choose a factorization

$$M \xrightarrow{u} P \xrightarrow{v} M$$

with P finitely generated projective and $f = v \circ u$. Then

$$a \circ f \circ b = a \circ v \circ u \circ b = (a \circ v) \circ (u \circ b),$$

so $a \circ f \circ b$ still factors through the same finitely generated projective module P. Hence

$$a \circ f \circ b \in \mathrm{Hom}_R^{\mathrm{pr}}(M, M).$$

Therefore $\mathrm{Hom}_R^{\mathrm{pr}}(M, M)$ is a two-sided ideal of $\mathrm{End}_R(M)$. □

Theorem 4.3.8. *Let* R *be a commutative ring with* 1*, and let* P *be an* R*-module. The following statements are equivalent.*

1. P *is a finitely generated projective module.*

2.
$$\mathrm{Hom}_R^{\mathrm{pr}}(P,P)=\mathrm{End}_R(P).$$

3. *For every R-module X, the morphism*
$$\tau_{X,P}:X^*\otimes_R P\longrightarrow \mathrm{Hom}_R(X,P)$$
is an isomorphism.

4. *For every R-module X, the morphism*
$$\tau_{P,X}:P^*\otimes_R X\longrightarrow \mathrm{Hom}_R(P,X)$$
is an isomorphism.

5. *The morphism*
$$\tau_{P,P}:P^*\otimes_R P\longrightarrow \mathrm{End}_R(P)$$
is an isomorphism.

Proof. **(1) $\Longrightarrow$(2).** Since P itself is finitely generated projective, the identity morphism
$$\mathrm{id}_P:P\to P$$
is projective. By the previous lemma, $\mathrm{Hom}_R^{\mathrm{pr}}(P,P)$ is a two-sided ideal of $\mathrm{End}_R(P)$ containing id_P, and therefore equals the whole endomorphism ring.

(3) $\Longrightarrow$(5) and **(4) $\Longrightarrow$(5)** are immediate by taking $X=P$.

(2) $\Longrightarrow$(3). By (2), id_P is projective, so there exist $\varphi_1,\dots,\varphi_n\in P^*$ and $p_1,\dots,p_n\in P$ such that
$$\mathrm{id}_P=\tau_{P,P}\Big(\sum_{i=1}^n\varphi_i\otimes p_i\Big).$$
Equivalently,
$$p=\sum_{i=1}^n\varphi_i(p)p_i\qquad\text{for all } p\in P.$$
For any R-module X, define
$$\sigma_X:\mathrm{Hom}_R(X,P)\to X^*\otimes_R P,\qquad \alpha\mapsto\sum_{i=1}^n(\varphi_i\circ\alpha)\otimes p_i.$$
Then for $\alpha\in\mathrm{Hom}_R(X,P)$ and $x\in X$,
$$\tau_{X,P}(\sigma_X(\alpha))(x)=\sum_{i=1}^n(\varphi_i\circ\alpha)(x)p_i=\sum_{i=1}^n\varphi_i(\alpha(x))p_i=\alpha(x),$$
so $\tau_{X,P}\circ\sigma_X=\mathrm{id}$.

Conversely, let

$$\xi = \sum_{j=1}^{m} \psi_j \otimes q_j \in X^* \otimes_R P.$$

Then

$$\begin{aligned}
\sigma_X(\tau_{X,P}(\xi)) &= \sum_{i=1}^{n} \Big(\varphi_i \circ \tau_{X,P}(\xi)\Big) \otimes p_i \\
&= \sum_{i=1}^{n} \left(x \mapsto \varphi_i\Big(\sum_{j=1}^{m} \psi_j(x) q_j\Big) \right) \otimes p_i \\
&= \sum_{i=1}^{n} \sum_{j=1}^{m} \big(\psi_j \varphi_i(q_j)\big) \otimes p_i \\
&= \sum_{j=1}^{m} \psi_j \otimes \Big(\sum_{i=1}^{n} \varphi_i(q_j) p_i\Big) \\
&= \sum_{j=1}^{m} \psi_j \otimes q_j \\
&= \xi.
\end{aligned}$$

Therefore σ_X is the inverse of $\tau_{X,P}$, and $\tau_{X,P}$ is an isomorphism.

(2) $\Longrightarrow$(4). Using the same decomposition of id_P, define for any R-module X

$$\rho_X : \mathrm{Hom}_R(P, X) \to P^* \otimes_R X, \qquad \beta \mapsto \sum_{i=1}^{n} \varphi_i \otimes \beta(p_i).$$

For $\beta \in \mathrm{Hom}_R(P, X)$ and $p \in P$,

$$\tau_{P,X}(\rho_X(\beta))(p) = \sum_{i=1}^{n} \varphi_i(p)\beta(p_i) = \beta\Big(\sum_{i=1}^{n} \varphi_i(p) p_i\Big) = \beta(p),$$

so $\tau_{P,X}\rho_X = \mathrm{id}$. Conversely, for

$$\eta = \sum_{j=1}^{m} \varphi'_j \otimes x_j \in P^* \otimes_R X,$$

we have

$$\begin{aligned}
\rho_X(\tau_{P,X}(\eta)) &= \sum_{i=1}^{n} \varphi_i \otimes \tau_{P,X}(\eta)(p_i) \\
&= \sum_{i=1}^{n} \varphi_i \otimes \Big(\sum_{j=1}^{m} \varphi'_j(p_i)x_j\Big) \\
&= \sum_{j=1}^{m} \Big(\sum_{i=1}^{n} \varphi'_j(p_i)\varphi_i\Big) \otimes x_j \\
&= \sum_{j=1}^{m} \varphi'_j \otimes x_j \\
&= \eta,
\end{aligned}$$

because

$$\varphi'_j(p) = \varphi'_j\Big(\sum_{i=1}^{n} \varphi_i(p)p_i\Big) = \sum_{i=1}^{n} \varphi_i(p)\varphi'_j(p_i)$$

for all $p \in P$. Hence $\tau_{P,X}$ is an isomorphism.

(5) $\Longrightarrow$(1). Since $\tau_{P,P}$ is surjective, we can write

$$\mathrm{id}_P = \tau_{P,P}\Big(\sum_{i=1}^{n} \varphi_i \otimes p_i\Big)$$

for some $\varphi_i \in P^*$ and $p_i \in P$. Define

$$\iota : P \to R^n, \qquad \iota(p) = (\varphi_1(p), \ldots, \varphi_n(p)),$$

and

$$\pi : R^n \to P, \qquad \pi(a_1, \ldots, a_n) = \sum_{i=1}^{n} a_i p_i.$$

Then

$$(\pi \circ \iota)(p) = \sum_{i=1}^{n} \varphi_i(p)p_i = p$$

for all $p \in P$. Thus $\pi \circ \iota = \mathrm{id}_P$, so P is a direct summand of the finite free module R^n. Therefore P is finitely generated and projective. □

4.4 Projective Covers

Definition 4.4.1. *Let R be a ring with 1. An epimorphism*

$$f : U \to V$$

is called an essential epimorphism *if no proper submodule of U is mapped surjectively* *onto V by f. Equivalently, whenever*

$$g : W \to U$$

is a morphism such that $f \circ g$ is an epimorphism, then g is an epimorphism.

Proof. Suppose first that no proper submodule of U maps onto V. Let $g : W \to U$ satisfy $f \circ g$ epimorphic. Then

$$f(g(W)) = V.$$

By assumption, $g(W)$ cannot be a proper submodule of U, so $g(W) = U$ and g is surjective.

Conversely, assume that whenever $f \circ g$ is epi, then g is epi. Let $U_0 \subsetneq U$ be a proper submodule and let

$$\iota : U_0 \hookrightarrow U$$

be the inclusion. If $f(U_0) = V$, then $f \circ \iota$ is epi, hence ι is epi, which is impossible. Thus $f(U_0) \neq V$. $\square$

Definition 4.4.2. *A* projective cover *of an R-module U is an essential epimorphism*

$$P \to U$$

with P projective.

Proposition 4.4.3. *Let*

$$U \xrightarrow{f} V \xrightarrow{g} W$$

be R-homomorphisms. If any two of f, g, and $g \circ f$ are essential epimorphisms, then so is the third.

Proof. **If f and g are essential, then $g \circ f$ is essential.** Let $U_0 \subsetneq U$ be a proper submodule. Since f is essential,

$$f(U_0) \neq V.$$

As g is essential, $g(f(U_0)) \neq W$. Hence $(g \circ f)(U_0) \neq W$, so $g \circ f$ is essential.

If g and $g \circ f$ are essential, then f is essential. Since $g \circ f$ is epic, f is epic. Let $U_0 \subsetneq U$ be proper. If $f(U_0) = V$, then

$$(g \circ f)(U_0) = W,$$

contradicting that $g \circ f$ is essential. Thus $f(U_0) \neq V$, so f is essential.

If f and $g \circ f$ are essential, then g is essential. Since $g \circ f$ is epic and f is epic, g is epic. Let $V_0 \subsetneq V$ be a proper submodule, and set

$$U_0 := f^{-1}(V_0).$$

Because f is surjective, U_0 is a proper submodule of U. Since $g \circ f$ is essential,

$$(g \circ f)(U_0) \neq W.$$

But

$$(g \circ f)(U_0) = g(V_0),$$

so $g(V_0) \neq W$. Hence g is essential. $\square$

4.5 Noetherian and Artinian Modules

Definition 4.5.1. *An R-module M is called* Noetherian *if for every ascending chain of submodules*

$$L_1 \subseteq L_2 \subseteq \cdots$$

there exists n_0 such that

$$L_n = L_{n_0} \qquad \text{for all } n \geq n_0.$$

Lemma 4.5.2. *For an R-module M, the following are equivalent.*

1. *M is Noetherian.*
2. *Every submodule of M is finitely generated.*
3. *Every non-empty set of submodules of M has a maximal element.*

Proof. **(1) $\Longrightarrow$(3).** Suppose $\mathcal{S}$ is a non-empty set of submodules of M having no maximal element. Choose $L_1 \in \mathcal{S}$. Since L_1 is not maximal, there exists $L_2 \in \mathcal{S}$ with

$$L_1 \subsetneq L_2.$$

Continuing inductively, we obtain a strictly ascending chain

$$L_1 \subsetneq L_2 \subsetneq L_3 \subsetneq \cdots,$$

contradicting that M is Noetherian.

(3) $\Longrightarrow$(2). Let $N \leq M$. Consider the set

$$\mathcal{T} := \{L \leq N \mid L \text{ is finitely generated}\}.$$

This set is non-empty because $0 \in \mathcal{T}$. By (3), $\mathcal{T}$ has a maximal element, say L. If $L \neq N$, choose $x \in N \setminus L$. Then

$$L + Rx$$

is a finitely generated submodule of N strictly containing L, contradicting the maximality of L. Hence $L = N$, so N is finitely generated.

(2) $\Longrightarrow$(1). Let
$$L_1 \subseteq L_2 \subseteq \cdots$$
be an ascending chain of submodules, and set
$$L := \bigcup_{n\geq 1} L_n.$$
Then L is a submodule of M, so by (2) it is finitely generated, say
$$L = Rx_1 + \cdots + Rx_m.$$
Each generator x_j lies in some L_{n_j}. If $N = \max\{n_1, \ldots, n_m\}$, then
$$x_1, \ldots, x_m \in L_N,$$
so $L = L_N$. Hence
$$L_n = L_N \qquad \text{for all } n \geq N,$$
and the chain stabilizes. □

Proposition 4.5.3. *Let*
$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$
be a short exact sequence of R-modules. Then M is Noetherian if and only if both M' and M'' are Noetherian.

Proof. Let $\pi : M \to M''$ be the quotient map, and identify M' with its image in M.

First assume that M is Noetherian. Since M' is a submodule of M, it is Noetherian. Now let $N'' \leq M''$ be any submodule. Then $\pi^{-1}(N'')$ is a submodule of M, hence is finitely generated. Since $\pi(\pi^{-1}(N'')) = N''$, it follows that N'' is finitely generated. Therefore M'' is Noetherian.

Conversely, assume that both M' and M'' are Noetherian. Let $N \leq M$ be any submodule. Then $N \cap M'$ is a submodule of M', so it is finitely generated. Also $\pi(N)$ is a submodule of M'', so it is finitely generated. Choose elements $x_1, \ldots, x_r \in N$ such that
$$\pi(x_1), \ldots, \pi(x_r)$$
generate $\pi(N)$, and choose elements $y_1, \ldots, y_s \in N \cap M'$ which generate $N \cap M'$.

We claim that
$$N = Rx_1 + \cdots + Rx_r + Ry_1 + \cdots + Ry_s.$$
Indeed, let $x \in N$. Since $\pi(x) \in \pi(N)$, there exist $a_1, \ldots, a_r \in R$ such that
$$\pi(x) = \sum_{i=1}^{r} a_i \pi(x_i) = \pi\left(\sum_{i=1}^{r} a_i x_i\right).$$

Hence

$$x - \sum_{i=1}^{r} a_i x_i \in \ker(\pi) \cap N = M' \cap N.$$

Since $y_1, \ldots, y_s$ generate $N \cap M'$, there exist $b_1, \ldots, b_s \in R$ such that

$$x - \sum_{i=1}^{r} a_i x_i = \sum_{j=1}^{s} b_j y_j.$$

Therefore

$$x = \sum_{i=1}^{r} a_i x_i + \sum_{j=1}^{s} b_j y_j,$$

proving the claim. Thus N is finitely generated.

Since every submodule $N \leq M$ is finitely generated, M is Noetherian. □

Definition 4.5.4. *An R-module M is called* Artinian *if for every descending chain of submodules*

$$L_1 \supseteq L_2 \supseteq \cdots$$

there exists n_0 such that

$$L_n = L_{n_0} \qquad \textit{for all } n \geq n_0.$$

Lemma 4.5.5. *An R-module M is Artinian if and only if every non-empty set of submodules of M contains a minimal element.*

Proof. Recall that M is Artinian if every descending chain of submodules

$$L_1 \supseteq L_2 \supseteq L_3 \supseteq \cdots$$

stabilizes.

First assume that M is Artinian. Let $\mathcal{S}$ be a non-empty set of submodules of M. We prove that $\mathcal{S}$ contains a minimal element.

Suppose, for contradiction, that $\mathcal{S}$ has no minimal element. Choose $L_1 \in \mathcal{S}$. Since L_1 is not minimal in $\mathcal{S}$, there exists $L_2 \in \mathcal{S}$ such that

$$L_2 \subsetneq L_1.$$

Similarly, since L_2 is not minimal in $\mathcal{S}$, there exists $L_3 \in \mathcal{S}$ such that

$$L_3 \subsetneq L_2.$$

Continuing inductively, we obtain a strictly descending chain

$$L_1 \supsetneq L_2 \supsetneq L_3 \supsetneq \cdots$$

of submodules of M. This contradicts the assumption that M is Artinian. Hence $\mathcal{S}$ must contain a minimal element.

Conversely, assume that every non-empty set of submodules of M contains a minimal element. Let

$$L_1 \supseteq L_2 \supseteq L_3 \supseteq \cdots$$

be a descending chain of submodules of M. Consider the non-empty set

$$\mathcal{S} := \{L_n \mid n \geq 1\}.$$

By assumption, $\mathcal{S}$ contains a minimal element, say L_N.

Since the chain is descending, for every $n \geq N$ we have

$$L_n \subseteq L_N.$$

But L_N is minimal among the elements of $\mathcal{S}$, and each L_n also belongs to $\mathcal{S}$. Therefore

$$L_n = L_N$$

for every $n \geq N$. Hence the descending chain stabilizes.

Therefore M is Artinian. □

Remark 4.5.6. *If an R-module M is Noetherian (respectively Artinian), then we also say that M satisfies the ascending chain condition (respectively descending chain condition) on submodules, abbreviated as ACC (respectively DCC).*

Proposition 4.5.7. *Let*

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 0$$

be a short exact sequence of R-modules. Then M is Artinian if and only if both M' and M'' are Artinian.

Proof. Since the sequence is exact, we may identify M' with its image $\iota(M') \subseteq M$. Thus M' is regarded as a submodule of M, and $M'' \cong M/M'$ via the quotient map

$$\pi : M \longrightarrow M''.$$

First assume that M is Artinian.

We show that M' is Artinian. Let

$$L_1 \supseteq L_2 \supseteq L_3 \supseteq \cdots$$

be a descending chain of submodules of M'. Since each L_i is also a submodule of M, this is a descending chain of submodules of M. Because M is Artinian, the chain stabilizes. Hence M' is Artinian.

Next we show that M'' is Artinian. Let

$$N_1 \supseteq N_2 \supseteq N_3 \supseteq \cdots$$

be a descending chain of submodules of M''. Consider their inverse images under π:

$$\pi^{-1}(N_1) \supseteq \pi^{-1}(N_2) \supseteq \pi^{-1}(N_3) \supseteq \cdots.$$

This is a descending chain of submodules of M. Since M is Artinian, there exists n_0 such that

$$\pi^{-1}(N_n) = \pi^{-1}(N_{n_0}) \qquad \text{for all } n \geq n_0.$$

Applying π to both sides, and using the fact that each N_n is contained in $\mathrm{Im}(\pi) = M''$, we get

$$N_n = N_{n_0} \qquad \text{for all } n \geq n_0.$$

Therefore M'' is Artinian.

Conversely, assume that both M' and M'' are Artinian. We prove that M is Artinian.

Let

$$L_1 \supseteq L_2 \supseteq L_3 \supseteq \cdots$$

be a descending chain of submodules of M. Intersecting with M', we obtain a descending chain of submodules of M':

$$L_1 \cap M' \supseteq L_2 \cap M' \supseteq L_3 \cap M' \supseteq \cdots.$$

Since M' is Artinian, this chain stabilizes. Hence there exists a such that

$$L_n \cap M' = L_a \cap M' \qquad \text{for all } n \geq a.$$

Applying π to the original chain, we obtain a descending chain of submodules of M'':

$$\pi(L_1) \supseteq \pi(L_2) \supseteq \pi(L_3) \supseteq \cdots.$$

Since M'' is Artinian, this chain stabilizes. Hence there exists b such that

$$\pi(L_n) = \pi(L_b) \qquad \text{for all } n \geq b.$$

Let

$$N = \max\{a, b\}.$$

We claim that

$$L_n = L_N \qquad \text{for all } n \geq N.$$

Since the chain is descending, we automatically have

$$L_n \subseteq L_N \qquad \text{for all } n \geq N.$$

It remains to prove the reverse inclusion.

Fix $n \geq N$, and take any element $x \in L_N$. Since

$$\pi(L_n) = \pi(L_N),$$

there exists $y \in L_n$ such that

$$\pi(y) = \pi(x).$$

Therefore

$$\pi(x - y) = 0,$$

so

$$x - y \in \ker(\pi) = M'.$$

Also, since $x \in L_N$ and $y \in L_n \subseteq L_N$, we have

$$x - y \in L_N.$$

Hence

$$x - y \in L_N \cap M'.$$

But the chain of intersections has stabilized, so

$$L_N \cap M' = L_n \cap M'.$$

Thus

$$x - y \in L_n \cap M' \subseteq L_n.$$

Since $y \in L_n$, it follows that

$$x = y + (x - y) \in L_n.$$

Therefore $L_N \subseteq L_n$. Combining this with $L_n \subseteq L_N$, we get

$$L_n = L_N \qquad \text{for all } n \geq N.$$

Hence every descending chain of submodules of M stabilizes, so M is Artinian.

Therefore M is Artinian if and only if both M' and M'' are Artinian. □

Definition 4.5.8. *An R-module M is called* simple *(or* irreducible*) if it has exactly two submodules:* 0 *and* M*.*

Chapter 5

Lecture 5: Semisimplicity

5.1 Semisimple Modules and the Radical

Definition 5.1.1. *An R-module M is called* semisimple *if it is a direct sum of simple submodules.*

Theorem 5.1.2 (Schur's lemma)**.** *Let S_1 and S_2 be simple R-modules, and let*

$$\varphi : S_1 \to S_2$$

be an R-homomorphism. Then either $\varphi = 0$ or φ is an isomorphism. In particular, $\mathrm{End}_R(S)$ is a division ring for every simple R-module S.

Proof. Assume $\varphi \neq 0$. Since $\ker(\varphi)$ is a submodule of the simple module S_1, we have $\ker(\varphi) = 0$ or $\ker(\varphi) = S_1$. The latter is impossible because $\varphi \neq 0$, hence $\ker(\varphi) = 0$ and φ is injective.

Likewise, $\mathrm{Im}(\varphi)$ is a nonzero submodule of the simple module S_2, so $\mathrm{Im}(\varphi) = S_2$. Thus φ is surjective and therefore an isomorphism.

Taking $S_1 = S_2 = S$, every nonzero endomorphism of S is invertible, which means that $\mathrm{End}_R(S)$ is a division ring. □

Lemma 5.1.3. *Let S be a simple left R-module. Then S is a quotient of the regular left module ${}_RR$. More precisely, for every nonzero $x \in S$ there is an isomorphism*

$$S \cong R/\operatorname{Ann}_R(x).$$

In particular, $\operatorname{Ann}_R(x)$ is a maximal left ideal of R.

Proof. Fix $0 \neq x \in S$ and define

$$\pi_x : R \to S, \qquad r \mapsto rx.$$

Since S is simple and Rx is a nonzero submodule of S, we have $Rx = S$. Hence π_x is surjective. Its kernel is exactly

$$\ker(\pi_x) = \operatorname{Ann}_R(x) := \{r \in R \mid rx = 0\}.$$

By the first isomorphism theorem,

$$S \cong R/\operatorname{Ann}_R(x).$$

Because S is simple, the quotient $R/\operatorname{Ann}_R(x)$ is simple as a left R-module, so $\operatorname{Ann}_R(x)$ is a maximal left ideal. $\square$

Definition 5.1.4. *Let U be an R-module. A proper submodule $M \subsetneq U$ is called a* maximal submodule *if whenever*

$$M \subseteq N \subseteq U,$$

then either $N = M$ or $N = U$.

Lemma 5.1.5. *Let W be a proper submodule of an R-module U. Assume that U/W is finitely generated. Then there exists a maximal submodule M of U such that*

$$W \subseteq M \subsetneq U.$$

Proof. Consider the set

$$\mathcal{S} := \{N \leq U \mid W \subseteq N \subsetneq U\},$$

ordered by inclusion. Since $W \in \mathcal{S}$, the set is non-empty. Let $\{N_\lambda\}_{\lambda\in\Lambda}$ be a totally ordered subset of $\mathcal{S}$, and set

$$N := \bigcup_{\lambda\in\Lambda} N_\lambda.$$

Then N is a submodule containing W. We claim that $N \neq U$. Indeed, write

$$U/W = R\overline{u}_1 + \cdots + R\overline{u}_n.$$

If $N = U$, then each u_i lies in some N_{λ_i}. Because the family is totally ordered, there exists μ such that

$$N_{\lambda_1}, \ldots, N_{\lambda_n} \subseteq N_\mu.$$

Hence $u_1, \ldots, u_n \in N_\mu$, so $U/W \subseteq N_\mu/W$, which implies $N_\mu = U$, contradicting $N_\mu \in \mathcal{S}$. Therefore N is an upper bound for the chain in $\mathcal{S}$.

By Zorn's lemma, $\mathcal{S}$ has a maximal element M, and this is precisely a maximal submodule of U containing W. $\square$

Corollary 5.1.6. *Every proper left ideal of R is contained in a maximal left ideal.*

Proof. Apply the lemma to the left regular module ${}_RR$ and to the proper submodule $W = I$. $\square$

Definition 5.1.7. *Let U be an R-module. The* radical *of U is the intersection of all maximal submodules of U, and is denoted by* $\operatorname{Rad}(U)$. *If U has no maximal submodules, we set* $\operatorname{Rad}(U) = U$.

Theorem 5.1.8 (Nakayama's lemma). *Let U be an R-module and let $V \subseteq U$ be a submodule such that U/V is finitely generated. Then*

$$V + \operatorname{Rad}(U) = U \quad\iff\quad V = U.$$

Proof. If $V = U$, then the equality is obvious.

Conversely, assume $V \neq U$. Since U/V is finitely generated, the previous lemma yields a maximal submodule M of U with

$$V \subseteq M \subsetneq U.$$

By definition,

$$\operatorname{Rad}(U) \subseteq M.$$

Therefore

$$V + \operatorname{Rad}(U) \subseteq M \subsetneq U,$$

so $V + \operatorname{Rad}(U) \neq U$. This proves the contrapositive. $\square$

Corollary 5.1.9. *If U is a finitely generated R-module, then the quotient map*

$$\pi : U \to U/\operatorname{Rad}(U)$$

is an essential epimorphism.

Proof. Let $W \leq U$ be a submodule such that $\pi(W) = U/\operatorname{Rad}(U)$. Then

$$W + \operatorname{Rad}(U) = U.$$

Since U/W is a quotient of the finitely generated module U, it is finitely generated. Nakayama's lemma therefore gives $W = U$, so π is essential. $\square$

Lemma 5.1.10. *Let M and N be finitely generated R-modules, and let*

$$f : M \to N$$

be an R-homomorphism.

1. *We have*

$$f(\operatorname{Rad}(M)) \subseteq \operatorname{Rad}(N).$$

2. *If f is surjective and* $\ker(f) \subseteq \mathrm{Rad}(M)$, *then*

$$\mathrm{Rad}(N) = f(\mathrm{Rad}(M)).$$

Proof. **(1).** Let $x \in \mathrm{Rad}(M)$. Suppose that $f(x) \notin \mathrm{Rad}(N)$. Then there exists a maximal submodule L of N such that

$$f(x) \notin L.$$

Let

$$\rho : N \to N/L$$

be the quotient map. Since N/L is simple and $\rho(f(x)) \neq 0$, the morphism $\rho \circ f : M \to N/L$ is nonzero, hence surjective. Therefore

$$M/\ker(\rho \circ f) \cong N/L$$

is simple, so $\ker(\rho \circ f)$ is a maximal submodule of M. But

$$x \notin \ker(\rho \circ f),$$

contradicting $x \in \mathrm{Rad}(M)$. Hence $f(x) \in \mathrm{Rad}(N)$.

(2). By (1),

$$f(\mathrm{Rad}(M)) \subseteq \mathrm{Rad}(N).$$

For the reverse inclusion, let $x \in M$ with $f(x) \in \mathrm{Rad}(N)$. If $x \notin \mathrm{Rad}(M)$, choose a maximal submodule K of M with

$$x \notin K.$$

Since $\ker(f) \subseteq \mathrm{Rad}(M) \subseteq K$, the map f induces a surjection

$$\overline{f} : M/K \to N/f(K).$$

Because M/K is simple and $x \notin K$, the quotient $N/f(K)$ is nonzero and hence simple. Thus $f(K)$ is a maximal submodule of N. But $f(x) \notin f(K)$, so $f(x) \notin \mathrm{Rad}(N)$, contradiction. Therefore

$$f^{-1}(\mathrm{Rad}(N)) = \mathrm{Rad}(M),$$

and applying f gives the desired equality

$$\mathrm{Rad}(N) = f(\mathrm{Rad}(M)).$$

$\square$

Lemma 5.1.11. *Let M and N be finitely generated R-modules, and let*

$$f : M \to N$$

be an R-homomorphism. Then f is an essential epimorphism if and only if the induced morphism

$$\overline{f} : M/\operatorname{Rad}(M) \to N/\operatorname{Rad}(N)$$

is an essential epimorphism.

Proof. First assume that f is an essential epimorphism. In particular, f is surjective. By the previous lemma,

$$f(\operatorname{Rad}(M)) \subseteq \operatorname{Rad}(N),$$

so $\overline{f}$ is well defined and surjective. Let

$$X/\operatorname{Rad}(M) \subseteq M/\operatorname{Rad}(M)$$

be a submodule such that $\overline{f}(X/\operatorname{Rad}(M)) = N/\operatorname{Rad}(N)$. Then

$$f(X) + \operatorname{Rad}(N) = N.$$

Since $N/f(X)$ is a quotient of the finitely generated module N, Nakayama's lemma gives $f(X) = N$. As f is essential, we conclude that $X = M$. Hence $\overline{f}$ is essential.

Conversely, assume that $\overline{f}$ is an essential epimorphism. Since $\overline{f}$ is surjective, we have

$$f(M) + \operatorname{Rad}(N) = N.$$

By Nakayama's lemma applied to the submodule $f(M) \subseteq N$, it follows that $f(M) = N$, so f is surjective. Now let $X \leq M$ satisfy $f(X) = N$. Then

$$\overline{f}\big((X + \operatorname{Rad}(M))/\operatorname{Rad}(M)\big) = N/\operatorname{Rad}(N).$$

Since $\overline{f}$ is essential, we obtain

$$X + \operatorname{Rad}(M) = M.$$

Again Nakayama's lemma yields $X = M$. Hence f is an essential epimorphism. □

5.2 Composition Series and Jordan–Hölder Theory

Definition 5.2.1. *A finite chain of submodules*

$$U = U_0 \supsetneq U_1 \supsetneq \cdots \supsetneq U_n = 0$$

is called a composition series *of U if each factor*

$$U_i/U_{i+1}$$

is simple. The integer n is called the composition length *of U.*

Lemma 5.2.2. *A nonzero R-module U has a composition series if and only if U is both Noetherian and Artinian.*

Proof. Suppose first that

$$U = U_0 \supsetneq U_1 \supsetneq \cdots \supsetneq U_n = 0$$

is a composition series. We argue by induction on n. If $n = 1$, then U is simple, hence both Noetherian and Artinian. If $n > 1$, then U_1 has a composition series of length $n - 1$, so by induction U_1 is Noetherian and Artinian. The quotient U/U_1 is simple, hence also Noetherian and Artinian. By the permanence of both properties in short exact sequences, U is Noetherian and Artinian.

Conversely, assume that U is Noetherian and Artinian. Since $U \neq 0$, choose a maximal proper submodule U_1 of U. Then U/U_1 is simple. Because U_1 is a submodule of both a Noetherian and an Artinian module, it is again Noetherian and Artinian. Repeating the argument inductively and using the Artinian condition to rule out an infinite descending chain, we obtain a composition series. □

Lemma 5.2.3. *Suppose that*

$$U = S_1 + \cdots + S_n$$

where each S_i is a simple submodule of U. If $V \leq U$, then there exists a subset $I \subseteq \{1, \ldots, n\}$ such that

$$U = V \oplus \bigoplus_{i \in I} S_i.$$

In particular, every submodule of a finite direct sum of simple modules is a direct summand.

Proof. Choose a subset $I \subseteq \{1, \ldots, n\}$ maximal with the property that

$$V \oplus \bigoplus_{i \in I} S_i$$

is a direct sum, and set

$$W := V \oplus \bigoplus_{i \in I} S_i.$$

We claim that $W = U$. If not, then there exists $j \notin I$ such that

$$S_j \nsubseteq W.$$

Now $W \cap S_j$ is a submodule of the simple module S_j, so either

$$W \cap S_j = 0 \qquad \text{or} \qquad W \cap S_j = S_j.$$

The second case is impossible because it would imply $S_j \subseteq W$. Hence $W \cap S_j = 0$, and

therefore

$$W \oplus S_j$$

is a direct sum, contradicting the maximality of I. Thus $W = U$. $\square$

Theorem 5.2.4. *Let U be an R-module with a composition series. The following are equivalent.*

1. *U is semisimple.*
2. *U is a sum of simple submodules.*
3. *Every submodule of U is a direct summand of U.*

Proof. **(1) $\Longrightarrow$(2).** This is immediate from the definition of a semisimple module.

(2) $\Longrightarrow$(3). Since U has finite composition length, a finite number of simple summands already generate U. Thus

$$U = S_1 + \cdots + S_n$$

for simple submodules S_i. The previous lemma shows that every submodule of U is a direct summand.

(3) $\Longrightarrow$(1). Let

$$U = U_0 \supsetneq U_1 \supsetneq \cdots \supsetneq U_n = 0$$

be a composition series. Since U_1 is a direct summand of U, there exists a submodule S_1 such that

$$U = U_1 \oplus S_1.$$

Because U/U_1 is simple, the submodule S_1 is simple. Repeating the argument inside U_1, we obtain simple submodules $S_2, \ldots, S_n$ with

$$U_i = U_{i+1} \oplus S_{i+1}$$

for all i. Hence

$$U = S_1 \oplus \cdots \oplus S_n,$$

so U is semisimple. $\square$

Corollary 5.2.5. *Let U be an R-module of finite composition length.*

1. *The sum of all simple submodules of U is semisimple, hence is the unique largest semisimple submodule of U.*
2. *If S is a simple submodule of U, then the sum of all submodules of U isomorphic to S is a submodule of U isomorphic to a direct sum of copies of S. It is the unique largest submodule of U with this property.*

Proof. Since U has finite composition length, U is both Noetherian and Artinian. In particular, every submodule of U is finitely generated.

(1). Let

$$\Sigma := \sum_{\substack{T \leq U \\ T \text{ simple}}} T$$

be the sum of all simple submodules of U.

Since $\Sigma \leq U$ and U is Noetherian, the submodule Σ is finitely generated. Thus there exist elements

$$x_1, \ldots, x_m \in \Sigma$$

such that

$$\Sigma = Rx_1 + \cdots + Rx_m.$$

By the definition of a sum of submodules, each $x_j \in \Sigma$ lies in a finite sum of simple submodules of U. Hence for each j, there exist simple submodules

$$T_{j1}, \ldots, T_{jr_j} \leq U$$

such that

$$x_j \in T_{j1} + \cdots + T_{jr_j}.$$

Collecting all these finitely many simple submodules, we obtain simple submodules

$$T_1, \ldots, T_n \leq U$$

such that

$$x_1, \ldots, x_m \in T_1 + \cdots + T_n.$$

Therefore

$$\Sigma = Rx_1 + \cdots + Rx_m \subseteq T_1 + \cdots + T_n \subseteq \Sigma.$$

Hence

$$\Sigma = T_1 + \cdots + T_n.$$

Since Σ is a finite sum of simple submodules, by the previous theorem Σ is semisimple.

Now let $W \leq U$ be any semisimple submodule. Then W is a sum of simple submodules of W. But every simple submodule of W is also a simple submodule of U. Hence every simple summand appearing in W is contained in the defining sum of Σ. Therefore

$$W \subseteq \Sigma.$$

Thus Σ contains every semisimple submodule of U, and so Σ is the unique largest semisimple submodule of U.

(2). Let $S \leq U$ be a simple submodule, and define

$$\Sigma_S := \sum_{\substack{T \leq U \\ T \cong S}} T.$$

This is the sum of all submodules of U which are isomorphic to S.

Since $\Sigma_S \leq U$ and U is Noetherian, Σ_S is finitely generated. Thus there exist elements

$$y_1, \ldots, y_\ell \in \Sigma_S$$

such that

$$\Sigma_S = Ry_1 + \cdots + Ry_\ell.$$

Again, by the definition of a sum of submodules, each y_j lies in a finite sum of submodules of U isomorphic to S. Hence, after collecting all the finitely many such submodules involved, there exist submodules

$$S_1, \ldots, S_m \leq U$$

with

$$S_i \cong S \qquad \text{for all } i,$$

such that

$$\Sigma_S = S_1 + \cdots + S_m.$$

Each S_i is simple, since it is isomorphic to the simple module S. Hence Σ_S is a finite sum of simple submodules. By the previous theorem, Σ_S is semisimple. More precisely, applying the preceding lemma to the finite sum

$$\Sigma_S = S_1 + \cdots + S_m$$

with $V = 0$, we may choose a subset $I \subseteq \{1, \ldots, m\}$ such that

$$\Sigma_S = \bigoplus_{i \in I} S_i.$$

Since each $S_i \cong S$, we obtain

$$\Sigma_S \cong S^{\oplus |I|}.$$

Thus Σ_S is isomorphic to a direct sum of copies of S.

Finally, let $W \leq U$ be any submodule which is isomorphic to a direct sum of copies of S. Then W is the sum of its simple submodules, each of which is isomorphic to S. Since these simple submodules are also submodules of U, they are all contained in the defining sum of Σ_S. Hence

$$W \subseteq \Sigma_S.$$

Therefore Σ_S contains every submodule of U which is isomorphic to a direct sum of copies of S. Hence Σ_S is the unique largest submodule of U with this property. □

Theorem 5.2.6 (Zassenhaus lemma). *Let $A' \subseteq A$ and $B' \subseteq B$ be submodules of an R-module M. Then*

$$\frac{A' + (A \cap B)}{A' + (A \cap B')} \cong \frac{B' + (A \cap B)}{B' + (A' \cap B)}.$$

Proof. Set

$$C := (A \cap B') + (A' \cap B).$$

We shall show that both quotient modules are naturally isomorphic to

$$\frac{A \cap B}{C}.$$

First consider the quotient

$$\frac{A' + (A \cap B)}{A' + (A \cap B')}.$$

Apply the second isomorphism theorem to the submodules

$$X := A \cap B, \qquad Y := A' + (A \cap B')$$

of M. Since

$$X + Y = (A \cap B) + A' + (A \cap B') = A' + (A \cap B),$$

we obtain

$$\frac{A' + (A \cap B)}{A' + (A \cap B')} = \frac{X + Y}{Y} \cong \frac{X}{X \cap Y}.$$

It remains to compute $X \cap Y$. We claim that

$$(A \cap B) \cap \big(A' + (A \cap B')\big) = (A' \cap B) + (A \cap B').$$

The inclusion "$\supseteq$" is immediate, because

$$A' \cap B \subseteq A \cap B, \qquad A \cap B' \subseteq A \cap B,$$

and both summands are contained in $A' + (A \cap B')$.

Conversely, let

$$x \in (A \cap B) \cap \big(A' + (A \cap B')\big).$$

Then $x \in A \cap B$, and we may write

$$x = a' + c$$

with $a' \in A'$ and $c \in A \cap B'$. Since $x \in B$ and $c \in B' \subseteq B$, we have

$$a' = x - c \in B.$$

Hence

$$a' \in A' \cap B.$$

Therefore

$$x = a' + c \in (A' \cap B) + (A \cap B').$$

This proves the claimed equality. Consequently,

$$\frac{A' + (A \cap B)}{A' + (A \cap B')} \cong \frac{A \cap B}{(A' \cap B) + (A \cap B')}.$$

Similarly, consider the quotient

$$\frac{B' + (A \cap B)}{B' + (A' \cap B)}.$$

Apply the second isomorphism theorem to

$$X := A \cap B, \qquad Y := B' + (A' \cap B).$$

Then

$$X + Y = (A \cap B) + B' + (A' \cap B) = B' + (A \cap B),$$

so

$$\frac{B' + (A \cap B)}{B' + (A' \cap B)} \cong \frac{A \cap B}{(A \cap B) \cap \big(B' + (A' \cap B)\big)}.$$

We now compute the intersection. We claim that

$$(A \cap B) \cap \big(B' + (A' \cap B)\big) = (A \cap B') + (A' \cap B).$$

Again the inclusion "$\supseteq$" is clear. Conversely, let

$$x \in (A \cap B) \cap \big(B' + (A' \cap B)\big).$$

Then $x \in A \cap B$, and we may write

$$x = b' + d$$

with $b' \in B'$ and $d \in A' \cap B$. Since $x \in A$ and $d \in A' \subseteq A$, we get

$$b' = x - d \in A.$$

Hence

$$b' \in A \cap B'.$$

Therefore

$$x = b' + d \in (A \cap B') + (A' \cap B).$$

This proves the second intersection equality. Thus

$$\frac{B' + (A \cap B)}{B' + (A' \cap B)} \cong \frac{A \cap B}{(A \cap B') + (A' \cap B)}.$$

We have shown that both quotient modules are naturally isomorphic to the same quotient

$$\frac{A \cap B}{(A \cap B') + (A' \cap B)}.$$

Hence

$$\frac{A' + (A \cap B)}{A' + (A \cap B')} \cong \frac{B' + (A \cap B)}{B' + (A' \cap B)}.$$

□

Definition 5.2.7. *Let*

$$U = U_0 \supseteq U_1 \supseteq \cdots \supseteq U_m = 0$$

be a finite submodule series of an R-module U. A refinement of this series is another finite submodule series obtained by inserting additional submodules between the terms U_{i-1} and U_i.

Two finite submodule series are said to be equivalent if their nonzero successive quotient modules are isomorphic up to a permutation.

Theorem 5.2.8 (Schreier refinement theorem)**.** *Any two finite submodule series of an R-module admit equivalent refinements.*

Proof. Let the two finite submodule series be

$$U = U_0 \supseteq U_1 \supseteq \cdots \supseteq U_m = 0$$

and

$$U = V_0 \supseteq V_1 \supseteq \cdots \supseteq V_n = 0.$$

We shall construct refinements of both series and then compare their successive quotient modules.

For each $1 \leq i \leq m$ and $0 \leq j \leq n$, define

$$U_{i,j} := U_i + (U_{i-1} \cap V_j).$$

Since

$$V_0 = U \qquad \text{and} \qquad V_n = 0,$$

we have

$$U_{i,0} = U_i + (U_{i-1} \cap V_0) = U_i + U_{i-1} = U_{i-1},$$

and

$$U_{i,n} = U_i + (U_{i-1} \cap V_n) = U_i.$$

Moreover, since

$$V_0 \supseteq V_1 \supseteq \cdots \supseteq V_n,$$

we get

$$U_{i,0} \supseteq U_{i,1} \supseteq \cdots \supseteq U_{i,n}.$$

Thus, for each fixed i, the chain

$$U_{i-1} = U_{i,0} \supseteq U_{i,1} \supseteq \cdots \supseteq U_{i,n} = U_i$$

refines the interval

$$U_{i-1} \supseteq U_i.$$

Splicing these chains together, we obtain a refinement of the first series:

$$U = U_{1,0} \supseteq U_{1,1} \supseteq \cdots \supseteq U_{1,n} = U_1 = U_{2,0} \supseteq \cdots \supseteq U_{m,n} = 0.$$

Similarly, for each $1 \le j \le n$ and $0 \le i \le m$, define

$$V_{j,i} := V_j + (V_{j-1} \cap U_i).$$

Then

$$V_{j,0} = V_j + (V_{j-1} \cap U_0) = V_j + V_{j-1} = V_{j-1},$$

and

$$V_{j,m} = V_j + (V_{j-1} \cap U_m) = V_j.$$

Hence, for each fixed j, we have a chain

$$V_{j-1} = V_{j,0} \supseteq V_{j,1} \supseteq \cdots \supseteq V_{j,m} = V_j.$$

Splicing these chains together gives a refinement of the second series:

$$U = V_{1,0} \supseteq V_{1,1} \supseteq \cdots \supseteq V_{1,m} = V_1 = V_{2,0} \supseteq \cdots \supseteq V_{n,m} = 0.$$

We now compare the successive quotients of these two refinements. Fix indices

$$1 \le i \le m, \qquad 1 \le j \le n.$$

Apply the Zassenhaus lemma to the four submodules

$$A' = U_i, \qquad A = U_{i-1}, \qquad B' = V_j, \qquad B = V_{j-1}.$$

Since $U_i \subseteq U_{i-1}$ and $V_j \subseteq V_{j-1}$, the Zassenhaus lemma gives

$$\frac{U_i + (U_{i-1} \cap V_{j-1})}{U_i + (U_{i-1} \cap V_j)} \cong \frac{V_j + (U_{i-1} \cap V_{j-1})}{V_j + (U_i \cap V_{j-1})}.$$

In terms of the notation introduced above, this is precisely

$$\frac{U_{i,j-1}}{U_{i,j}} \cong \frac{V_{j,i-1}}{V_{j,i}}.$$

Therefore every successive quotient in the refinement of the first series is isomorphic to a corresponding successive quotient in the refinement of the second series. The correspondence is given by the pairs of indices

$$(i,j) \longleftrightarrow (j,i).$$

Hence the two constructed refinements are equivalent.

If some consecutive terms in the constructed refinements coincide, then the corresponding quotient is zero. Deleting these repeated terms on both sides does not affect the list of nonzero quotient modules. Thus the two original finite submodule series admit equivalent refinements. □

Theorem 5.2.9 (Jordan–Hölder theorem)**.** *Let U be an R-module with a composition series. Then any two composition series of U have the same length, and their composition factors are the same up to permutation and isomorphism.*

Proof. Let

$$U = U_0 \supsetneq U_1 \supsetneq \cdots \supsetneq U_m = 0$$

and

$$U = V_0 \supsetneq V_1 \supsetneq \cdots \supsetneq V_n = 0$$

be two composition series of U. Thus each quotient

$$U_{i-1}/U_i$$

and each quotient

$$V_{j-1}/V_j$$

is simple.

By the Schreier refinement theorem, these two finite submodule series admit equivalent refinements. We now observe that a composition series has no proper refinement.

Indeed, suppose that one could insert a submodule W strictly between two consecutive terms of the first composition series:

$$U_{i-1} \supsetneq W \supsetneq U_i.$$

Then W/U_i would be a nonzero proper submodule of

$$U_{i-1}/U_i.$$

This contradicts the fact that U_{i-1}/U_i is simple. Hence no such intermediate submodule W exists. Therefore any refinement of the first composition series only inserts repeated terms. After deleting repetitions, the refinement is just the original composition series. The same argument applies to the second composition series.

Since the Schreier refinements of the two composition series are equivalent, and since those refinements reduce back to the original composition series after deleting repeated terms, the original composition series themselves are equivalent.

Therefore the number of nonzero successive quotient modules in the two series is the same. Hence

$$m = n.$$

Moreover, there exists a permutation $\sigma \in S_m$ such that

$$U_{i-1}/U_i \cong V_{\sigma(i)-1}/V_{\sigma(i)}$$

for every $1 \leq i \leq m$.

Thus any two composition series of U have the same length and the same composition factors up to permutation and isomorphism. $\square$

5.3 Socles, Tops and Projective Covers

Definition 5.3.1 (Soc of a module)**.** *The unique largest semisimple submodule of an R-module U is called the* socle *of U and is denoted by*

$$\mathrm{Soc}(U).$$

Definition 5.3.2 (Top of a module)**.** *Let U be an R-module. The top, or head, of U is defined by*

$$\mathrm{Top}(U) := U/\,\mathrm{Rad}(U).$$

Example 5.3.3. *If U is semisimple, then*

$$\mathrm{Rad}(U) = 0, \qquad \mathrm{Soc}(U) = U, \qquad \mathrm{Top}(U) = U.$$

In particular, if S is simple, then

$$\mathrm{Rad}(S) = 0, \qquad \mathrm{Soc}(S) = S, \qquad \mathrm{Top}(S) = S.$$

Example 5.3.4. *Let*

$$A = k[x]/(x^n),$$

and regard A as a left A-module. Then

$$J(A) = (x),$$

so

$$\mathrm{Rad}(A) = xA, \qquad \mathrm{Top}(A) = A/xA \cong k.$$

On the other hand,

$$\mathrm{Soc}(A) = \{a \in A \mid xa = 0\} = kx^{n-1}.$$

Thus

$$\mathrm{Top}(A) \cong k, \qquad \mathrm{Soc}(A) \cong k,$$

but they occur at opposite ends of the Loewy filtration.

Remark 5.3.5. *In general,* $\mathrm{Top}(U)$ *need not be semisimple for an arbitrary module U. However, if U is finitely generated and Artinian, then* $\mathrm{Top}(U)$ *is semisimple.*

Lemma 5.3.6. *Let*

$$U = S_1^{\oplus a_1} \oplus \cdots \oplus S_r^{\oplus a_r}$$

be a semisimple R-module, where the S_i are pairwise nonisomorphic simple modules.

Then the summand $S_i^{\oplus a_i}$ is the S_i-isotypic component of U, namely the sum of all submodules of U isomorphic to S_i. In particular, it is uniquely determined by U and the isomorphism class of S_i.

Proof. This is precisely Corollary 5.2.5(2) applied to $S = S_i$. ☐

Lemma 5.3.7. *Let U be an R-module.*

1. *Suppose $M_1, \dots, M_n$ are maximal submodules of U. Then there exists a subset $I \subseteq \{1, \dots, n\}$ such that*

$$U/(M_1 \cap \cdots \cap M_n) \cong \bigoplus_{i \in I} U/M_i.$$

   In particular, $U/(M_1 \cap \cdots \cap M_n)$ is semisimple.

2. *If U is finitely generated and Artinian, then $U/\operatorname{Rad}(U)$ is semisimple, and $\operatorname{Rad}(U)$ is the unique smallest submodule of U with semisimple quotient.*

Proof. **(1).** Choose a subset $I \subseteq \{1, \dots, n\}$ maximal with the property that the diagonal map

$$\delta_I : U \longrightarrow \bigoplus_{i \in I} U/M_i, \qquad u \longmapsto (u + M_i)_{i \in I}$$

is surjective. Such a subset exists because $\{1, \dots, n\}$ is finite; for instance $I = \varnothing$ is allowed.

We have

$$\ker(\delta_I) = \bigcap_{i \in I} M_i.$$

Put

$$K := \bigcap_{i \in I} M_i.$$

We claim that

$$K \subseteq M_j \qquad \text{for every } j \notin I.$$

Suppose not. Then for some $j \notin I$ we have

$$K \nsubseteq M_j.$$

Since M_j is maximal, this implies

$$K + M_j = U.$$

Hence the natural map

$$K \longrightarrow U/M_j$$

is surjective.

We now show that

$$U \longrightarrow \left(\bigoplus_{i \in I} U/M_i \right) \oplus U/M_j$$

is surjective. Indeed, given

$$\big((u_i + M_i)_{i\in I},\, u_j + M_j\big),$$

first use the surjectivity of δ_I to choose $u \in U$ such that

$$u + M_i = u_i + M_i \qquad \text{for all } i \in I.$$

Since $K \to U/M_j$ is surjective, choose $k \in K$ such that

$$k + M_j = (u_j - u) + M_j.$$

Then $u + k$ maps to the prescribed element in every component: for $i \in I$ we have $k \in K \subseteq M_i$, so

$$u + k + M_i = u + M_i = u_i + M_i,$$

and for the j-component,

$$u + k + M_j = u_j + M_j.$$

This contradicts the maximality of I.

Therefore

$$\bigcap_{i\in I} M_i \subseteq M_j \qquad \text{for every } j \notin I.$$

Hence

$$\bigcap_{i\in I} M_i = M_1 \cap \cdots \cap M_n.$$

By the first isomorphism theorem,

$$U/(M_1 \cap \cdots \cap M_n) = U/\ker(\delta_I) \cong \bigoplus_{i\in I} U/M_i.$$

Since each M_i is maximal, each quotient U/M_i is simple. Therefore the right hand side is semisimple.

(2). If $U = 0$, the statement is trivial. Assume $U \neq 0$.

Since U is finitely generated, every proper submodule of U is contained in a maximal submodule. Indeed, if $W \subsetneq U$, then U/W is finitely generated, so by the maximal-submodule lemma there exists a maximal submodule M of U such that

$$W \subseteq M \subsetneq U.$$

We now show that the intersection of all maximal submodules is in fact a finite intersection. Consider the set

$$\mathcal{F} := \{M_1 \cap \cdots \cap M_t \mid t \geq 1,\ M_1, \ldots, M_t \text{ maximal submodules of } U\}.$$

This is a nonempty set of submodules of U. Since U is Artinian, every nonempty set of submodules has a minimal element. Choose a minimal element

$$K = M_1 \cap \cdots \cap M_t$$

in $\mathcal{F}$.

Let M be any maximal submodule of U. Then

$$K \cap M = M_1 \cap \cdots \cap M_t \cap M$$

also belongs to $\mathcal{F}$, and

$$K \cap M \subseteq K.$$

By the minimality of K, we must have

$$K \cap M = K.$$

Hence

$$K \subseteq M.$$

Since this holds for every maximal submodule M of U, we get

$$K \subseteq \operatorname{Rad}(U).$$

The opposite inclusion is immediate because K is an intersection of maximal submodules. Therefore

$$\operatorname{Rad}(U) = K = M_1 \cap \cdots \cap M_t.$$

By part (1), it follows that

$$U/\operatorname{Rad}(U)$$

is semisimple.

It remains to prove the minimality statement. Let $V \leq U$ be a submodule such that U/V is semisimple. We show that

$$\operatorname{Rad}(U) \subseteq V.$$

Since U/V is semisimple, its radical is zero:

$$\operatorname{Rad}(U/V) = 0.$$

Maximal submodules of U/V correspond exactly to maximal submodules of U containing V. Therefore

$$\operatorname{Rad}(U/V) = \bigcap_{\substack{M \text{ maximal in } U \\ V \subseteq M}} M/V.$$

Thus

$$0 = \operatorname{Rad}(U/V) = \left(\bigcap_{\substack{M \text{ maximal in } U \\ V \subseteq M}} M \right) /V.$$

Hence
$$\bigcap_{\substack{M \text{ maximal in } U \\ V\subseteq M}} M = V.$$

Since $\operatorname{Rad}(U)$ is the intersection of all maximal submodules of U, it is contained in the intersection over the smaller family of maximal submodules containing V. Therefore

$$\operatorname{Rad}(U) \subseteq V.$$

Hence $\operatorname{Rad}(U)$ is contained in every submodule V such that U/V is semisimple. Since we already proved that $U/\operatorname{Rad}(U)$ is semisimple, $\operatorname{Rad}(U)$ is the unique smallest submodule of U with semisimple quotient. □

Corollary 5.3.8 (Universal property of the top)**.** *Let U be a finitely generated Artinian R-module. Then $\operatorname{Top}(U)$ is the largest semisimple quotient of U in the following sense: if S is semisimple and*
$$f : U \to S$$
is an R-module homomorphism, then f factors uniquely through the canonical quotient map
$$\pi : U \to \operatorname{Top}(U).$$
Equivalently, there exists a unique homomorphism

$$\overline{f} : \operatorname{Top}(U) \to S$$

such that
$$f = \overline{f} \circ \pi.$$

Proof. Recall that
$$\operatorname{Top}(U) = U/\operatorname{Rad}(U),$$

and let
$$\pi : U \longrightarrow \operatorname{Top}(U)$$

be the canonical quotient map.

Let S be semisimple and let
$$f : U \longrightarrow S$$

be an R-module homomorphism. Since S is semisimple, every submodule of S is semisimple. Hence
$$\operatorname{Im}(f) \leq S$$

is semisimple.

By the first isomorphism theorem, we have

$$U/\ker(f) \cong \operatorname{Im}(f).$$

Therefore $U/\ker(f)$ is semisimple. Thus $\ker(f)$ is a submodule of U such that the quotient $U/\ker(f)$ is semisimple.

Since U is finitely generated and Artinian, $\mathrm{Rad}(U)$ is the unique smallest submodule of U with semisimple quotient. Hence

$$\mathrm{Rad}(U) \subseteq \ker(f).$$

Therefore, by the universal property of the quotient module, the map f factors through $U/\mathrm{Rad}(U)$. That is, there exists an R-module homomorphism

$$\overline{f} : \mathrm{Top}(U) \longrightarrow S$$

such that

$$f = \overline{f} \circ \pi.$$

Finally, since π is surjective, such a homomorphism $\overline{f}$ is unique. Indeed, if

$$g : \mathrm{Top}(U) \longrightarrow S$$

also satisfies

$$f = g \circ \pi,$$

then

$$\overline{f} \circ \pi = g \circ \pi.$$

Since π is surjective, it follows that

$$\overline{f} = g.$$

Hence f factors uniquely through π, and therefore $\mathrm{Top}(U)$ is the largest semisimple quotient of U. □

Example 5.3.9. *Suppose U has finite length. Since $\mathrm{Top}(U)$ is semisimple, we may write*

$$\mathrm{Top}(U) \cong S_1^{\oplus a_1} \oplus \cdots \oplus S_r^{\oplus a_r},$$

where the S_i are pairwise nonisomorphic simple modules. The integers a_i measure how many copies of S_i occur in the top layer of U.

Lemma 5.3.10. *For all R-modules U and V,*

$$\mathrm{Rad}(U \oplus V) = \mathrm{Rad}(U) \oplus \mathrm{Rad}(V),$$

$$\mathrm{Soc}(U \oplus V) = \mathrm{Soc}(U) \oplus \mathrm{Soc}(V),$$

$$\mathrm{Top}(U \oplus V) \cong \mathrm{Top}(U) \oplus \mathrm{Top}(V).$$

Proof. We first prove the statement for radicals.

Let

$$p_U : U \oplus V \longrightarrow U, \qquad p_V : U \oplus V \longrightarrow V$$

be the canonical projections. We claim that

$$p_U\big(\mathrm{Rad}(U \oplus V)\big) \subseteq \mathrm{Rad}(U), \qquad p_V\big(\mathrm{Rad}(U \oplus V)\big) \subseteq \mathrm{Rad}(V).$$

Indeed, let $x \in \mathrm{Rad}(U \oplus V)$. Suppose, for contradiction, that $p_U(x) \notin \mathrm{Rad}(U)$. Then there exists a maximal submodule $M \subsetneq U$ such that

$$p_U(x) \notin M.$$

Hence

$$M \oplus V$$

is a maximal submodule of $U \oplus V$, because

$$(U \oplus V)/(M \oplus V) \cong U/M$$

is simple. But $x \notin M \oplus V$, contradicting $x \in \mathrm{Rad}(U \oplus V)$. Therefore

$$p_U(x) \in \mathrm{Rad}(U).$$

The proof for p_V is identical. Thus

$$\mathrm{Rad}(U \oplus V) \subseteq \mathrm{Rad}(U) \oplus \mathrm{Rad}(V).$$

Conversely, let

$$u \in \mathrm{Rad}(U), \qquad v \in \mathrm{Rad}(V).$$

We show that $(u, v) \in \mathrm{Rad}(U \oplus V)$. Let

$$L \subsetneq U \oplus V$$

be a maximal submodule. Then

$$(U \oplus V)/L$$

is simple. Let

$$q : U \oplus V \longrightarrow (U \oplus V)/L$$

be the quotient map, and let

$$i_U : U \longrightarrow U \oplus V, \qquad i_V : V \longrightarrow U \oplus V$$

be the canonical inclusions.

Since $(U \oplus V)/L$ is simple, the homomorphism

$$q \circ i_U : U \longrightarrow (U \oplus V)/L$$

is either zero or surjective. In either case, its kernel contains $\mathrm{Rad}(U)$. Hence

$$q(i_U(u)) = 0.$$

Similarly,

$$q(i_V(v)) = 0.$$

Therefore

$$q(u,v) = q(i_U(u) + i_V(v)) = q(i_U(u)) + q(i_V(v)) = 0.$$

Hence

$$(u,v) \in L.$$

Since this holds for every maximal submodule $L \subsetneq U \oplus V$, we get

$$(u,v) \in \mathrm{Rad}(U \oplus V).$$

Thus

$$\mathrm{Rad}(U) \oplus \mathrm{Rad}(V) \subseteq \mathrm{Rad}(U \oplus V).$$

Combining the two inclusions gives

$$\mathrm{Rad}(U \oplus V) = \mathrm{Rad}(U) \oplus \mathrm{Rad}(V).$$

Next we prove the statement for socles.

Since $\mathrm{Soc}(U)$ and $\mathrm{Soc}(V)$ are semisimple submodules of U and V, respectively, their direct sum

$$\mathrm{Soc}(U) \oplus \mathrm{Soc}(V)$$

is a semisimple submodule of $U \oplus V$. Hence

$$\mathrm{Soc}(U) \oplus \mathrm{Soc}(V) \subseteq \mathrm{Soc}(U \oplus V).$$

Conversely, let S be a simple submodule of $U \oplus V$. Consider the restrictions of the two projections:

$$p_U|_S : S \longrightarrow U, \qquad p_V|_S : S \longrightarrow V.$$

Since S is simple, the images

$$p_U(S) \leq U, \qquad p_V(S) \leq V$$

are either zero or simple submodules. Hence

$$p_U(S) \subseteq \mathrm{Soc}(U), \qquad p_V(S) \subseteq \mathrm{Soc}(V).$$

For every $s \in S$, we have

$$s = (p_U(s), p_V(s)).$$

Therefore

$$s \in \mathrm{Soc}(U) \oplus \mathrm{Soc}(V),$$

and so

$$S \subseteq \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Since $\mathrm{Soc}(U \oplus V)$ is the sum of all simple submodules of $U \oplus V$, it follows that

$$\mathrm{Soc}(U \oplus V) \subseteq \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Hence

$$\mathrm{Soc}(U \oplus V) = \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Finally, by definition,

$$\mathrm{Top}(X) = X/\,\mathrm{Rad}(X)$$

for any R-module X. Therefore, using the radical formula proved above, we get

$$\mathrm{Top}(U \oplus V) = (U \oplus V)/\,\mathrm{Rad}(U \oplus V)$$

$$= (U \oplus V)/\big(\mathrm{Rad}(U) \oplus \mathrm{Rad}(V)\big).$$

The canonical quotient isomorphism gives

$$(U \oplus V)/\big(\mathrm{Rad}(U) \oplus \mathrm{Rad}(V)\big) \cong U/\,\mathrm{Rad}(U) \oplus V/\,\mathrm{Rad}(V).$$

Hence

$$\mathrm{Top}(U \oplus V) \cong \mathrm{Top}(U) \oplus \mathrm{Top}(V).$$

□

Proposition 5.3.11. *Let*

$$f : P \to U$$

be a projective cover, and assume that P and U are finitely generated Artinian. Then the induced morphism

$$\overline{f} : \mathrm{Top}(P) \to \mathrm{Top}(U)$$

is an isomorphism.

Proof. Let

$$\pi_P : P \longrightarrow \mathrm{Top}(P) = P/\,\mathrm{Rad}(P)$$

and

$$\pi_U : U \longrightarrow \mathrm{Top}(U) = U/\,\mathrm{Rad}(U)$$

be the canonical quotient maps.

Since P and U are finitely generated, we have

$$f(\mathrm{Rad}(P)) \subseteq \mathrm{Rad}(U).$$

Hence f induces a well-defined homomorphism

$$\overline{f} : \mathrm{Top}(P) \longrightarrow \mathrm{Top}(U)$$

such that

$$\overline{f} \circ \pi_P = \pi_U \circ f.$$

Since $f : P \to U$ is a projective cover, it is in particular an epimorphism. Therefore the induced map

$$\overline{f} : \mathrm{Top}(P) \to \mathrm{Top}(U)$$

is also an epimorphism.

It remains to prove that $\overline{f}$ is injective. Suppose, for contradiction, that

$$K := \ker(\overline{f}) \neq 0.$$

Since P is finitely generated and Artinian, $\mathrm{Top}(P)$ is semisimple. Therefore K is a direct summand of $\mathrm{Top}(P)$. Hence there exists a submodule $C \leq \mathrm{Top}(P)$ such that

$$\mathrm{Top}(P) = K \oplus C.$$

Since $K \neq 0$, we have $C \subsetneq \mathrm{Top}(P)$.

Let

$$W := \pi_P^{-1}(C).$$

Then W is a proper submodule of P, because

$$\pi_P(W) = C \subsetneq \mathrm{Top}(P).$$

We claim that $f(W) = U$. Since $\overline{f}$ is surjective and

$$\mathrm{Top}(P) = K \oplus C,$$

while $K = \ker(\overline{f})$, the restriction

$$\overline{f}|_C : C \longrightarrow \mathrm{Top}(U)$$

is still surjective. Hence

$$\overline{f}(C) = \mathrm{Top}(U).$$

Using the commutative relation

$$\overline{f} \circ \pi_P = \pi_U \circ f,$$

we get

$$\pi_U(f(W)) = \overline{f}(\pi_P(W)) = \overline{f}(C) = \mathrm{Top}(U).$$

Therefore

$$f(W) + \mathrm{Rad}(U) = U.$$

Since U is finitely generated, Nakayama's lemma gives

$$f(W) = U.$$

But $f : P \to U$ is an essential epimorphism, and $W \subseteq P$ is a submodule satisfying $f(W) = U$. Hence

$$W = P,$$

contradicting the fact that $W \subsetneq P$.

Therefore

$$\ker(\overline{f}) = 0.$$

Thus $\overline{f}$ is both injective and surjective, hence an isomorphism:

$$\mathrm{Top}(P) \cong \mathrm{Top}(U).$$

□

Proposition 5.3.12. *Suppose that*

$$f : P \to U$$

is a projective cover of an R-module U, and that

$$g : Q \to U$$

is an epimorphism with Q projective.

1. *There exists a decomposition*

$$Q = Q_1 \oplus Q_2$$

such that, with respect to this decomposition,

$$g = (g_1, 0),$$

and there is an isomorphism $\gamma : Q_1 \to P$ satisfying

$$f \circ \gamma = g_1.$$

2. *If a projective cover of U exists, then it is unique up to isomorphism.*

Proof. Since Q is projective and $f : P \to U$ is surjective, there exists

$$\alpha : Q \to P$$

such that

$$f \circ \alpha = g.$$

Since P is projective and $g : Q \to U$ is surjective, there exists

$$\beta : P \to Q$$

such that

$$g \circ \beta = f.$$

Consequently,

$$f = f \circ (\alpha\beta).$$

Because f is an essential epimorphism, the endomorphism $\alpha\beta$ of P must be surjective. Since P is projective, $\alpha\beta$ splits. Let

$$P = \ker(\alpha\beta) \oplus P'.$$

If $\ker(\alpha\beta) \neq 0$, then P' is a proper submodule of P and

$$f(P') = f(\alpha\beta(P')) = f(P) = U,$$

contradicting the essentiality of f. Hence $\ker(\alpha\beta) = 0$, so $\alpha\beta$ is an automorphism.

Replacing β by $\beta \circ (\alpha\beta)^{-1}$, we may assume that

$$\alpha\beta = \mathrm{id}_P .$$

Thus β is a split monomorphism and

$$Q = \beta(P) \oplus \ker(\alpha).$$

Set

$$Q_1 := \beta(P), \qquad Q_2 := \ker(\alpha),$$

and let $\gamma : Q_1 \to P$ be the inverse of the isomorphism $\beta : P \to Q_1$. Then

$$g|_{Q_1} = f \circ \gamma.$$

Moreover,

$$g(Q_2) = f(\alpha(Q_2)) = 0,$$

so $g = (g_1, 0)$ with $g_1 = f \circ \gamma$. This proves (1).

For (2), if

$$f : P \to U \qquad \text{and} \qquad f' : P' \to U$$

are projective covers, apply (1) twice: once with $Q = P'$ and once with $Q = P$. We deduce that P is isomorphic to a direct summand of P' and P' is isomorphic to a direct summand of P. The decompositions obtained in (1) force the complementary summands to map trivially to U, hence to vanish by essentiality. Therefore

$$P \cong P',$$

so the projective cover is unique up to isomorphism. $\square$

Chapter 6

Lecture 6

6.1 Projective covers, radicals, and finiteness conditions

Corollary 6.1.1. *For finitely generated projective modules P and Q,*

$$P \cong Q \iff \mathrm{Top}(P) \cong \mathrm{Top}(Q).$$

Proof. Let

$$T := P/\operatorname{Rad}(P).$$

By Nakayama lemma, the quotient map $P \to T$ is an essential epimorphism, so P is the projective cover of T. Similarly $Q \to Q/\operatorname{Rad}(Q)$ is a projective cover. Then apply uniqueness of projective covers from the previous lecture. □

Definition 6.1.2. *A unital ring R is called left Noetherian if ${}_RR$ is Noetherian as a left R-module.*

Remark 6.1.3. *Similar definitions are given for right Noetherian and for left/right Artinian rings.*

Remark 6.1.4. *one-sided Noetherian / Artinian conditions generally do not imply the opposite-sided versions*

Example 6.1.5. *There exists a ring which is left Noetherian but not right Noetherian.*

Let

$$R = \left\{ \begin{pmatrix} a & b \\ 0 & n \end{pmatrix} \middle| \ a, b \in \mathbb{Q}, \ n \in \mathbb{Z} \right\}.$$

Then R is left Noetherian, but not right Noetherian.

Proof. Let

$$e_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad e_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then, as left R-modules,

$${}_RR = Re_{11} \oplus Re_{22}.$$

We first show that both Re_{11} and Re_{22} are Noetherian as left R-modules.

For Re_{11}, we have

$$Re_{11} = \left\{ \begin{pmatrix} q & 0 \\ 0 & 0 \end{pmatrix} \,\middle|\, q \in \mathbb{Q} \right\}.$$

This is a simple left R-module. Indeed, if

$$0 \neq \begin{pmatrix} q & 0 \\ 0 & 0 \end{pmatrix} \in Re_{11},$$

then for any $x \in \mathbb{Q}$,

$$\begin{pmatrix} xq^{-1} & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} q & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} x & 0 \\ 0 & 0 \end{pmatrix}.$$

Hence every nonzero submodule of Re_{11} is all of Re_{11}, so Re_{11} is simple, in particular Noetherian.

Now consider

$$Re_{22} = \left\{ \begin{pmatrix} 0 & q \\ 0 & n \end{pmatrix} \,\middle|\, q \in \mathbb{Q},\ n \in \mathbb{Z} \right\}.$$

Define

$$\pi : Re_{22} \longrightarrow \mathbb{Z}, \qquad \pi\left(\begin{pmatrix} 0 & q \\ 0 & n \end{pmatrix}\right) = n.$$

We regard $\mathbb{Z}$ as a left R-module via

$$\begin{pmatrix} a & b \\ 0 & n \end{pmatrix} \cdot m = nm.$$

Then π is a surjective R-homomorphism. Its kernel is

$$\ker \pi = \left\{ \begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} \,\middle|\, q \in \mathbb{Q} \right\}.$$

This is again a simple left R-module, because

$$\begin{pmatrix} a & b \\ 0 & n \end{pmatrix} \begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & aq \\ 0 & 0 \end{pmatrix},$$

so the action is through multiplication by elements of $\mathbb{Q}$. Moreover, ${}_R\mathbb{Z}$ is Noetherian, since its submodules are exactly the ideals of $\mathbb{Z}$, and $\mathbb{Z}$ is Noetherian.

Therefore Re_{22} is an extension of two Noetherian left R-modules, hence is Noetherian. It follows that

$${}_RR = Re_{11} \oplus Re_{22}$$

is a Noetherian left R-module. Thus R is left Noetherian.

Next we show that R is not right Noetherian.

For each integer $k \geq 0$, define

$$I_k = \left\{ \begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} \,\middle|\, q \in 2^{-k}\mathbb{Z} \right\}.$$

We claim that each I_k is a right ideal of R. Indeed, if

$$\begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} \in I_k, \qquad \begin{pmatrix} a & b \\ 0 & n \end{pmatrix} \in R,$$

then

$$\begin{pmatrix} 0 & q \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & n \end{pmatrix} = \begin{pmatrix} 0 & qn \\ 0 & 0 \end{pmatrix}.$$

Since $q \in 2^{-k}\mathbb{Z}$ and $n \in \mathbb{Z}$, we have $qn \in 2^{-k}\mathbb{Z}$, so the product again lies in I_k.

Moreover,

$$I_0 \subsetneq I_1 \subsetneq I_2 \subsetneq \cdots$$

is a strictly ascending chain of right ideals, because

$$2^{-(k+1)} \in 2^{-(k+1)}\mathbb{Z} \quad \text{but} \quad 2^{-(k+1)} \notin 2^{-k}\mathbb{Z}.$$

Hence R does not satisfy the ascending chain condition on right ideals.

Therefore R is not right Noetherian. ☐

Lemma 6.1.6. *Let R be a left Artinian ring. Then its Jacobson radical $J(R)$ is nilpotent.*

Proof. Set

$$J := J(R).$$

Since R is left Artinian, the descending chain of left ideals

$$J \supseteq J^2 \supseteq J^3 \supseteq \cdots$$

stabilizes. Hence there exists some $n \geq 1$ such that

$$J^n = J^{n+1}.$$

Let

$$N := J^n.$$

We claim that $N = 0$.

Suppose, for contradiction, that $N \neq 0$. Then

$$N^2 = J^{2n} = J^n = N,$$

since the chain has stabilized.

Now consider the set

$$\Sigma = \{ I \leq {}_R R \mid NI \neq 0 \}$$

of left ideals of R. This set is nonempty because

$$N^2 = N \neq 0,$$

so $N \in \Sigma$.

Since R is left Artinian, the set of left ideals satisfies the descending chain condition, so Σ contains a minimal element. Let $I_0 \in \Sigma$ be such a minimal left ideal.

Because $NI_0 \neq 0$, there exists some $a \in I_0$ such that

$$Na \neq 0.$$

Then Na is a left ideal of R contained in I_0, and moreover

$$N(Na) = N^2 a = Na \neq 0.$$

Thus $Na \in \Sigma$. By the minimality of I_0, we must have

$$Na = I_0.$$

Since $a \in I_0 = Na$, there exists $b \in N$ such that

$$a = ba.$$

Therefore

$$(1 - b)a = 0.$$

But $b \in N \subseteq J(R)$, and for every element of the Jacobson radical, $1 - b$ has a left inverse. Hence there exists $c \in R$ such that

$$c(1 - b) = 1.$$

Multiplying the equation $(1 - b)a = 0$ on the left by c, we get

$$a = c(1 - b)a = 0,$$

contradicting the choice of a with $Na \neq 0$.

This contradiction shows that $N = 0$. Hence

$$J^n = 0,$$

so $J(R)$ is nilpotent. □

Theorem 6.1.7 (Levitzki)**.** *Every left Artinian ring is left Noetherian.*

Proof. Let R be a left Artinian ring, and let $J = J(R)$ be its Jacobson radical.

Since R is left Artinian, the quotient ring R/J is also left Artinian. Moreover, $J(R/J) = 0$, hence R/J is semisimple. Therefore every left R/J-module is semisimple.

Also, since R is left Artinian, J is nilpotent. Thus there exists $n \geq 1$ such that

$$J^n = 0.$$

Consider the finite chain of left ideals

$$0 = J^n \subseteq J^{n-1} \subseteq \cdots \subseteq J \subseteq R.$$

For each $i = 0, 1, \ldots, n-1$, the factor module J^i/J^{i+1} is naturally a left R/J-module, because

$$J \cdot J^i \subseteq J^{i+1}.$$

Hence J^i/J^{i+1} is semisimple.

On the other hand, J^i/J^{i+1} is Artinian, being a quotient of the Artinian left R-module ${}_RR$. A semisimple Artinian module must be a finite direct sum of simple modules, so it is Noetherian. Therefore each factor

$$J^i/J^{i+1}$$

is Noetherian.

Now we argue by descending induction on i. Since $J^n = 0$ is Noetherian, and for each $i = 0, 1, \ldots, n-1$ we have a short exact sequence

$$0 \longrightarrow J^{i+1} \longrightarrow J^i \longrightarrow J^i/J^{i+1} \longrightarrow 0,$$

it follows that if J^{i+1} is Noetherian, then so is J^i. Hence

$$J^{n-1},\ J^{n-2},\ \ldots,\ J,\ R$$

are all Noetherian as left R-modules.

In particular, ${}_RR$ is Noetherian. Therefore R is a left Noetherian ring. □

Theorem 6.1.8. *If R is left Noetherian/Artinian, then every finitely generated left R-module is Noetherian/Artinian.*

Proof. Prove for Noetherian case: If M is generated by m elements, then $M = Rm_1 + \ldots + Rm_n$, each Rm_i is a homomorphic image of ${}_RR$. Hence Rm_i is Noetherian. Finite sums of Noetherian modules are Noetherian, so M is Noetherian. □

6.2 Artin-Wedderburn

Theorem 6.2.1. *Let*

$$ {}_RV = V_1 \oplus \cdots \oplus V_n $$

be a direct sum of R-modules, and suppose all V_i are isomorphic to each other. Then

$$ \mathrm{End}_R({}_RV) \cong M_n\big(\mathrm{End}_R(V_i)\big). $$

Proof. For each i, let

$$ \iota_i : V_i \hookrightarrow V \qquad \text{and} \qquad \pi_i : V \to V_i $$

be the canonical inclusion and projection. Fix isomorphisms

$$ \theta_i : V_1 \xrightarrow{\sim} V_i \qquad (i = 1, \ldots, n). $$

For $\sigma \in \mathrm{End}_R(V)$, define

$$ \varphi_{ij}(\sigma) := \theta_i^{-1}\pi_i\,\sigma\,\iota_j\theta_j \in \mathrm{End}_R(V_1). $$

Thus we obtain a map

$$ \Phi : \mathrm{End}_R(V) \longrightarrow M_n(E_1), \qquad \sigma \longmapsto \big(\varphi_{ij}(\sigma)\big), $$

where

$$ E_1 := \mathrm{End}_R(V_1). $$

We first check that Φ is a ring homomorphism.

Additivity is immediate. For multiplicativity, let $\sigma, \beta \in \mathrm{End}_R(V)$. Since

$$ 1_V = \sum_{j=1}^{n} \iota_j\pi_j, $$

we have

$$ \begin{aligned} \varphi_{ik}(\beta\sigma) &= \theta_i^{-1}\pi_i\,\beta\sigma\,\iota_k\theta_k \\ &= \theta_i^{-1}\pi_i\,\beta \left(\sum_{j=1}^{n} \iota_j\pi_j\right) \sigma\,\iota_k\theta_k \\ &= \sum_{j=1}^{n} \theta_i^{-1}\pi_i\,\beta\,\iota_j\pi_j\,\sigma\,\iota_k\theta_k \\ &= \sum_{j=1}^{n} \big(\theta_i^{-1}\pi_i\,\beta\,\iota_j\theta_j\big)\big(\theta_j^{-1}\pi_j\,\sigma\,\iota_k\theta_k\big) \\ &= \sum_{j=1}^{n} \varphi_{ij}(\beta)\varphi_{jk}(\sigma). \end{aligned} $$

Hence

$$\Phi(\beta\sigma) = \Phi(\beta)\Phi(\sigma).$$

So Φ is a ring homomorphism.

Next we show that Φ is injective. Suppose $\Phi(\sigma) = 0$. Then

$$\varphi_{ij}(\sigma) = 0 \qquad \text{for all } i, j,$$

so

$$\theta_i^{-1}\pi_i\,\sigma\,\iota_j\theta_j = 0 \qquad \text{for all } i, j.$$

Since each θ_i and θ_j is an isomorphism, this implies

$$\pi_i\,\sigma\,\iota_j = 0 \qquad \text{for all } i, j.$$

Therefore $\pi_i(\sigma(v)) = 0$ for every $v \in V_j$ and every i, j. Hence

$$\sigma(V_j) = 0 \qquad \text{for all } j.$$

Because

$$V = V_1 \oplus \cdots \oplus V_n,$$

it follows that $\sigma = 0$. Thus Φ is injective.

Finally we prove surjectivity. Let

$$(a_{ij}) \in M_n(E_1).$$

Define

$$\sigma := \sum_{i,j=1}^{n} \iota_i\theta_i\,a_{ij}\,\theta_j^{-1}\pi_j \in \mathrm{End}_R(V).$$

Then for each i, j,

$$\begin{aligned}
\varphi_{ij}(\sigma) &= \theta_i^{-1}\pi_i\left(\sum_{r,s=1}^{n} \iota_r\theta_r\,a_{rs}\,\theta_s^{-1}\pi_s\right)\iota_j\theta_j \\
&= \sum_{r,s=1}^{n} \theta_i^{-1}(\pi_i\iota_r)\theta_r\,a_{rs}\,\theta_s^{-1}(\pi_s\iota_j)\theta_j.
\end{aligned}$$

Now

$$\pi_i\iota_r = \begin{cases} 1_{V_i}, & i = r, \\ 0, & i \neq r, \end{cases} \qquad \pi_s\iota_j = \begin{cases} 1_{V_j}, & s = j, \\ 0, & s \neq j, \end{cases}$$

so all terms vanish except the one with $r = i$ and $s = j$. Hence

$$\varphi_{ij}(\sigma) = \theta_i^{-1}\theta_i\,a_{ij}\,\theta_j^{-1}\theta_j = a_{ij}.$$

Therefore

$$\Phi(\sigma) = (a_{ij}),$$

and Φ is surjective.

Thus Φ is a ring isomorphism, and we conclude that

$$\mathrm{End}_R(V) \cong M_n\big(\mathrm{End}_R(V_1)\big).$$

$\square$

Definition 6.2.2. *A ring R is called semisimple if ${}_RR$ is semisimple as a left R-module.*

Lemma 6.2.3. *For a unital ring R, the following are equivalent: (1) R is semisimple. (2) Every left R-module is semisimple.*

Proof. If ${}_RR$ is semisimple, then every free module is semisimple, and every module is a quotient of a free module. Use closure properties of semisimple modules to deduce that every left R-module is semisimple. The converse is immediate by applying the statement to ${}_RR$ itself. $\square$

Lemma 6.2.4. *For an R-module ${}_RM$, there is a standard identification*

$$\mathrm{Hom}_R({}_RR_R, {}_RM) \cong {}_RM, \qquad f \longmapsto f(1).$$

Proof. Define

$$\Phi : \mathrm{Hom}_R({}_RR_R, {}_RM) \longrightarrow {}_RM, \qquad \Phi(f) = f(1).$$

We first show that Φ is bijective.

For any $f \in \mathrm{Hom}_R({}_RR_R, {}_RM)$ and any $r \in R$, since f is left R-linear, we have

$$f(r) = f(r \cdot 1) = r\, f(1).$$

Thus f is completely determined by the element $f(1) \in M$.

Conversely, for each $m \in M$, define

$$\Psi(m) : {}_RR_R \longrightarrow {}_RM, \qquad \Psi(m)(r) = rm \quad (r \in R).$$

Then $\Psi(m)$ is left R-linear, because for all $a, r \in R$,

$$\Psi(m)(ar) = (ar)m = a(rm) = a\, \Psi(m)(r).$$

Hence $\Psi(m) \in \mathrm{Hom}_R({}_RR_R, {}_RM)$.

Now for $m \in M$,

$$(\Phi \circ \Psi)(m) = \Phi(\Psi(m)) = \Psi(m)(1) = 1m = m.$$

And for $f \in \mathrm{Hom}_R({}_RR_R, {}_RM)$ and $r \in R$,

$$(\Psi \circ \Phi)(f)(r) = \Psi(f(1))(r) = r\, f(1) = f(r).$$

So $\Psi \circ \Phi = \mathrm{id}$ and $\Phi \circ \Psi = \mathrm{id}$, hence Φ is a bijection.

Finally, we check that this identification is R-linear. Recall that $\mathrm{Hom}_R({}_RR_R, {}_RM)$ carries the left R-module structure

$$(a \cdot f)(r) = f(ra) \qquad (a, r \in R).$$

Then for $a \in R$ and $f \in \mathrm{Hom}_R({}_RR_R, {}_RM)$,

$$\Phi(a \cdot f) = (a \cdot f)(1) = f(1a) = f(a) = a\, f(1) = a\, \Phi(f).$$

Therefore Φ is an isomorphism of left R-modules:

$$\mathrm{Hom}_R({}_RR_R, {}_RM) \cong {}_RM.$$

□

Corollary 6.2.5. *For any ring R, there is a ring isomorphism*

$$\mathrm{End}_R({}_RR) \cong R^{\mathrm{op}}, \qquad f \longmapsto f(1).$$

Proof. Consider the map

$$\Phi : \mathrm{End}_R({}_RR) \longrightarrow R, \qquad \Phi(f) = f(1).$$

By the preceding lemma with $M = R$, this map is a bijection of sets.

We now check the ring structure. Let $f_1, f_2 \in \mathrm{End}_R({}_RR)$. Then

$$\Phi(f_1 \circ f_2) = (f_1 \circ f_2)(1) = f_1\big(f_2(1)\big).$$

Since f_1 is left R-linear, for any $r \in R$ we have

$$f_1(r) = f_1(r \cdot 1) = r\, f_1(1).$$

Hence

$$f_1\big(f_2(1)\big) = f_2(1)\, f_1(1).$$

Therefore

$$\Phi(f_1 \circ f_2) = \Phi(f_2)\Phi(f_1),$$

where the product on the right is the multiplication in R. This means exactly that

$$\Phi(f_1 \circ f_2) = \Phi(f_1) \cdot_{\mathrm{op}} \Phi(f_2)$$

when the codomain is viewed as R^{op}.

Thus Φ is a ring isomorphism

$$\mathrm{End}_R({}_RR) \cong R^{\mathrm{op}}.$$

∎

Lemma 6.2.6. *For any ring R and any $n \geq 1$, there is a ring isomorphism*

$$M_n(R)^{\mathrm{op}} \cong M_n\big(R^{\mathrm{op}}\big), \qquad A \longmapsto A^\top.$$

Proof. Define

$$\Psi : M_n(R)^{\mathrm{op}} \longrightarrow M_n\big(R^{\mathrm{op}}\big), \qquad \Psi(A) = A^\top.$$

This map is clearly additive and bijective, since transpose is its own inverse.

It remains to check multiplicativity. Let

$$A = (a_{ij}), \qquad B = (b_{ij}) \in M_n(R).$$

Recall that multiplication in $M_n(R)^{\mathrm{op}}$ is reversed, so

$$A \cdot_{\mathrm{op}} B = BA$$

with the usual matrix product on the right-hand side.

Now fix i, j. Then

$$\big(\Psi(A \cdot_{\mathrm{op}} B)\big)_{ij} = \big((BA)^\top\big)_{ij} = (BA)_{ji} = \sum_{k=1}^{n} b_{jk} a_{ki}.$$

On the other hand, in the ring R^{op} the product is reversed, so

$$\big(\Psi(A)\Psi(B)\big)_{ij} = \sum_{k=1}^{n} (A^\top)_{ik} \cdot_{R^{\mathrm{op}}} (B^\top)_{kj} = \sum_{k=1}^{n} a_{ki} \cdot_{R^{\mathrm{op}}} b_{jk} = \sum_{k=1}^{n} b_{jk} a_{ki}.$$

Thus

$$\Psi(A \cdot_{\mathrm{op}} B) = \Psi(A)\Psi(B).$$

So Ψ is a ring homomorphism, hence a ring isomorphism.

Therefore

$$M_n(R)^{\mathrm{op}} \cong M_n\big(R^{\mathrm{op}}\big).$$

∎

Theorem 6.2.7. *(Artin-Wedderburn) Let A be a semisimple Artinian ring. Then*

$$A \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r),$$

where each D_i is a division ring. More concretely, if

$${}_A A \cong S_1^{\oplus n_1} \oplus \cdots \oplus S_r^{\oplus n_r}$$

with pairwise nonisomorphic simple A-modules S_i, then

$$D_i = \operatorname{End}_A(S_i)^{op}.$$

Proof. Decompose $\operatorname{End}_A(A)$ using the decomposition of the regular module. Use

$$\operatorname{Hom}_A(S_i, S_j) = 0 \quad (i \neq j)$$

and Schur's lemma to identify the diagonal blocks with division rings. Then use the previous matrix-ring result to rewrite

$$\operatorname{End}_A\big(S_i^{\oplus n_i}\big)$$

as

$$M_{n_i}\big(\operatorname{End}_A(S_i)\big) = M_{n_i}(D_i^{op}).$$

Therefore

$$A \cong \operatorname{End}_A(A)^{op} \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r).$$

□

Definition 6.2.8. *Let R be a commutative ring with 1, and let A be an associative ring with 1. An R-algebra is an R-module A whose multiplication is compatible with the R-action. Equivalently, there is a ring homomorphism*

$$R \to A$$

whose image lies in the center $Z(A)$.

Remark 6.2.9. *If $R = k$ is a field and A is finite-dimensional over k, then A is called a finite-dimensional k-algebra. Such an algebra is Noetherian and Artinian as a ring/module object.*

Example 6.2.10. *Let R be a commutative ring with 1. The following are R-algebras:*

1. *$R[x_1, \ldots, x_n]$.*
2. *$M_n(R)$.*
3. *Let G be a finite group. The group algebra*

$$RG = \left\{ \sum_{g \in G} a_g g \,\middle|\, a_g \in R \right\}$$

is an R-algebra, with multiplication defined by

$$\left(\sum_{g\in G} a_g g\right)\left(\sum_{h \in G} b_h h\right) = \sum_{r \in G}\left(\sum_{\substack{g,h\in G\\ gh=r}} a_g b_h\right) r.$$

Lemma 6.2.11. *Let A be a k-algebra, where k is a field, and let S be a simple A-module. Then $\operatorname{End}_A(S)$ is a division ring, and it is naturally a k-algebra.*

Proof. We first show that $\operatorname{End}_A(S)$ is a division ring.

Let
$$0 \neq f \in \operatorname{End}_A(S).$$

Since f is A-linear, both $\ker(f)$ and $\operatorname{im}(f)$ are A-submodules of S. Because S is simple and $f \neq 0$, we have $\operatorname{im}(f) \neq 0$, hence
$$\operatorname{im}(f) = S.$$

Also $\ker(f) \neq S$ since $f \neq 0$, so by simplicity,
$$\ker(f) = 0.$$

Thus f is bijective. Since the inverse of a bijective A-linear map is again A-linear, $f^{-1} \in \operatorname{End}_A(S)$. Therefore every nonzero element of $\operatorname{End}_A(S)$ is invertible, so $\operatorname{End}_A(S)$ is a division ring.

Next we show that $\operatorname{End}_A(S)$ is naturally a k-algebra.

Because A is a k-algebra, S becomes a k-vector space via
$$\lambda \cdot s := (\lambda 1_A)s \qquad (\lambda \in k,\ s \in S).$$

Now let $f \in \operatorname{End}_A(S)$. Then for $\lambda \in k$ and $s \in S$,
$$f(\lambda s) = f\big((\lambda 1_A)s\big) = (\lambda 1_A)f(s) = \lambda f(s).$$

Hence every A-linear endomorphism of S is automatically k-linear. So
$$\operatorname{End}_A(S) \subseteq \operatorname{End}_k(S)$$
as a k-subspace. Equivalently, there is a k-algebra homomorphism
$$k \longrightarrow \operatorname{End}_A(S), \qquad \lambda \longmapsto \lambda\, \mathrm{id}_S,$$
whose image lies in the center of $\operatorname{End}_A(S)$. Therefore $\operatorname{End}_A(S)$ is naturally a k-algebra. $\square$

Lemma 6.2.12. *If A is finite-dimensional over k, then $\operatorname{End}_A(S)$ is finite-dimensional over k.*

Proof. Choose $0 \neq s \in S$. Since S is simple, the cyclic submodule As is nonzero, hence
$$As = S.$$

Consider the map
$$\varphi : A \longrightarrow S, \qquad a \longmapsto as.$$

This is a surjective k-linear map. Therefore

$$\dim_k S \leq \dim_k A < \infty.$$

So S is finite-dimensional over k.

Since every A-endomorphism of S is k-linear, we have

$$\operatorname{End}_A(S) \subseteq \operatorname{End}_k(S)$$

as k-vector spaces. Hence

$$\dim_k \operatorname{End}_A(S) \leq \dim_k \operatorname{End}_k(S) = (\dim_k S)^2 < \infty.$$

Thus $\operatorname{End}_A(S)$ is finite-dimensional over k. □

Lemma 6.2.13. *If A is finite-dimensional over k and k is algebraically closed, then*

$$\operatorname{End}_A(S) = k.$$

Proof. By the previous lemma, S is finite-dimensional over k, and $\operatorname{End}_A(S)$ is a finite-dimensional k-algebra.

Let

$$f \in \operatorname{End}_A(S).$$

Since S is a finite-dimensional k-vector space and k is algebraically closed, the k-linear operator $f : S \to S$ has an eigenvalue $\lambda \in k$. Hence

$$f - \lambda \operatorname{id}_S$$

has a nonzero kernel, so it is not injective.

But $f - \lambda \operatorname{id}_S$ is still an element of $\operatorname{End}_A(S)$. Since $\operatorname{End}_A(S)$ is a division ring by the first lemma, every nonzero element is invertible, hence injective. Therefore

$$f - \lambda \operatorname{id}_S = 0.$$

So

$$f = \lambda \operatorname{id}_S .$$

This shows that every A-endomorphism of S is scalar. Under the natural embedding

$$k \hookrightarrow \operatorname{End}_A(S), \qquad \lambda \mapsto \lambda \operatorname{id}_S,$$

we therefore obtain

$$\operatorname{End}_A(S) = k.$$

□

Theorem 6.2.14. *If k is algebraically closed and A is a finite-dimensional semisimple k-algebra, then*

$$A \cong M_{n_1}(k) \oplus \cdots \oplus M_{n_r}(k).$$

Proof. Since A is finite-dimensional over k, it is Artinian. Hence A is a semisimple Artinian ring.

Decompose the left regular A-module into simple summands:

$${}_A A \cong S_1^{\oplus n_1} \oplus \cdots \oplus S_r^{\oplus n_r},$$

Here $S_1, \dots, S_r$ are pairwise non-isomorphic simple left A-modules. By the Artin–Wedderburn theorem,

$$A \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r),$$

where

$$D_i = \mathrm{End}_A(S_i)^{\mathrm{op}}.$$

It remains to identify the division rings D_i. Since A is a finite-dimensional k-algebra, each simple A-module S_i is finite-dimensional over k. Indeed, for any $0 \neq s \in S_i$, the map

$$A \longrightarrow S_i, \qquad a \longmapsto as$$

is a surjective k-linear map, because S_i is simple. Therefore

$$\dim_k S_i \leq \dim_k A < \infty.$$

Now $\mathrm{End}_A(S_i)$ is a division ring by Schur's lemma. Moreover, every A-endomorphism of S_i is k-linear. Since k is algebraically closed and S_i is finite-dimensional over k, every $f \in \mathrm{End}_A(S_i)$ has an eigenvalue $\lambda \in k$. Thus

$$f - \lambda\, \mathrm{id}_{S_i}$$

has nonzero kernel, so it is not injective. But $f - \lambda\, \mathrm{id}_{S_i} \in \mathrm{End}_A(S_i)$, and $\mathrm{End}_A(S_i)$ is a division ring. Hence a nonzero element of $\mathrm{End}_A(S_i)$ must be invertible, in particular injective. Therefore

$$f - \lambda\, \mathrm{id}_{S_i} = 0,$$

and so

$$f = \lambda\, \mathrm{id}_{S_i}.$$

This proves that

$$\mathrm{End}_A(S_i) = k.$$

Hence

$$D_i = \mathrm{End}_A(S_i)^{\mathrm{op}} \cong k^{\mathrm{op}} = k,$$

since k is commutative. Substituting this into the Artin–Wedderburn decomposition gives

$$A \cong M_{n_1}(k) \oplus \cdots \oplus M_{n_r}(k).$$

$\square$

Corollary 6.2.15. *Let A be a finite-dimensional semisimple k-algebra, where k is a field.*

In any decomposition

$${}_AA \cong S_1^{n_1} \oplus \cdots \oplus S_r^{n_r},$$

where the S_i's are pairwise non-isomorphic simple A-modules, we have that $S_1, \ldots, S_r$ form a complete set of representatives of the isomorphism classes of simple A-modules.

If moreover $k = \overline{k}$, then $n_i = \dim_k S_i$ and

$$\dim_k A = n_1^2 + \cdots + n_r^2.$$

Proof. Let S be a simple module. Then

$$S \cong A/M$$

where M is a maximal submodule.

By semisimplicity,

$${}_AA = M \oplus S.$$

Hence all simple submodules appear in $\{S_1, \ldots, S_r\}$.

Since $k = \overline{k}$, we have

$$A \cong M_{n_1}(k) \oplus \cdots \oplus M_{n_r}(k).$$

Moreover,

$$M_{n_i}(k) = \underbrace{\left\{\begin{pmatrix} * & \cdots & 0 \\ \vdots & \ddots & \vdots \\ * & \cdots & 0 \end{pmatrix}\right\} \oplus \cdots \oplus \left\{\begin{pmatrix} 0 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & * \end{pmatrix}\right\}}_{\text{each of them is a simple } M_{n_i}(k)\text{-module.}}$$

$\square$

6.3 k-algebras, indecomposables, and idempotents

Definition 6.3.1. *A nonzero R-module M is called indecomposable if every decomposition*

$$M = M_1 \oplus M_2$$

forces $M_1 = 0$ or $M_2 = 0$.

Theorem 6.3.2 (Krull-Schmidt)**.** *Let M be an R-module with a composition series. Suppose*

$$M \cong V_1 \oplus \cdots \oplus V_m \cong U_1 \oplus \cdots \oplus U_n$$

where all V_i and U_j are indecomposable. Then

$$m = n,$$

and after reordering the summands one has

$$V_i \cong U_i \quad \text{for all } i.$$

Proof. Replacing the given isomorphisms by identifications, we may assume

$$M = V_1 \oplus \cdots \oplus V_m = U_1 \oplus \cdots \oplus U_n.$$

If $M = 0$, the statement is trivial. So assume $M \neq 0$.

Since M has a composition series, it has finite length. Hence each direct summand V_i and U_j also has finite length. Moreover, for every indecomposable module X of finite length, Fitting's lemma implies that every endomorphism of X is either invertible or nilpotent; therefore $\mathrm{End}_R(X)$ is a local ring.

We prove the theorem by induction on the length $\ell(M)$ of M. If $\ell(M) = 1$, then M is simple, hence indecomposable, so necessarily $m = n = 1$ and $V_1 \cong U_1$.

Now assume $\ell(M) > 1$ and that the statement is known for all modules of smaller length. Set

$$A := V_1, \qquad B := V_2 \oplus \cdots \oplus V_m,$$

so that

$$M = A \oplus B.$$

Let

$$\iota : A \to M, \qquad \pi : M \to A$$

be the canonical inclusion and projection. For the decomposition

$$M = U_1 \oplus \cdots \oplus U_n,$$

let

$$\kappa_j : U_j \to M, \qquad p_j : M \to U_j$$

be the canonical inclusion and projection for $1 \leq j \leq n$.

Since

$$1_M = \sum_{j=1}^{n} \kappa_j p_j,$$

we get

$$1_A = \pi\iota = \pi\Big(\sum_{j=1}^{n} \kappa_j p_j\Big)\iota = \sum_{j=1}^{n} (\pi\kappa_j)(p_j\iota).$$

For each j, put
$$u_j := (\pi\kappa_j)(p_j\iota) \in \mathrm{End}_R(A).$$
If all u_j were noninvertible, then, since $\mathrm{End}_R(A)$ is local, their sum would also be noninvertible. But their sum is 1_A, a contradiction. Hence u_j is a unit for some j. After reordering the U_j, we may assume that u_1 is a unit.

Set
$$a := \pi\kappa_1 : U_1 \to A, \qquad b := p_1\iota : A \to U_1.$$
Then
$$ab = u_1$$
is an automorphism of A. Consider
$$e := b(ab)^{-1}a \in \mathrm{End}_R(U_1).$$
Then
$$e^2 = b(ab)^{-1}a\, b(ab)^{-1}a = b(ab)^{-1}(ab)(ab)^{-1}a = b(ab)^{-1}a = e,$$
so e is an idempotent. Also,
$$eb = b(ab)^{-1}ab = b \neq 0,$$
hence $e \neq 0$. Since $\mathrm{End}_R(U_1)$ is local, its only idempotents are 0 and 1, therefore $e = 1_{U_1}$.

Thus
$$b(ab)^{-1}a = 1_{U_1},$$
so a is injective. On the other hand, ab is surjective, hence a is surjective as well. Therefore a is an isomorphism, and we obtain
$$A \cong U_1.$$

Now define
$$f := \kappa_1 a^{-1} : A \to M.$$
Then
$$\pi f = \pi\kappa_1 a^{-1} = aa^{-1} = 1_A.$$
Hence
$$M = f(A) \oplus B.$$
Indeed, for any $x \in M$,
$$x = f(\pi(x)) + \big(x - f(\pi(x))\big),$$
and
$$\pi\big(x - f(\pi(x))\big) = \pi(x) - \pi f(\pi(x)) = 0,$$
so $x - f(\pi(x)) \in \ker\pi = B$. Also, if $f(a) \in B = \ker\pi$, then
$$a = \pi f(a) = 0,$$

so the sum is direct. Since $f(A) = \kappa_1(U_1) = U_1$, we have

$$M = U_1 \oplus B.$$

Therefore

$$B \cong M/U_1 \cong U_2 \oplus \cdots \oplus U_n.$$

But also

$$B = V_2 \oplus \cdots \oplus V_m.$$

Now $\ell(B) < \ell(M)$, so by the induction hypothesis,

$$m - 1 = n - 1,$$

and after reordering $U_2, \ldots, U_n$, we have

$$V_i \cong U_i \qquad (i = 2, \ldots, m).$$

Together with $V_1 \cong U_1$, this gives

$$m = n, \qquad V_i \cong U_i \quad \text{for all } i$$

after a suitable reordering.

This completes the proof. $\square$

Definition 6.3.3. *An idempotent of a ring R is a nonzero element*

$$e \in R$$

such that

$$e^2 = e.$$

Remark 6.3.4. *If e is an idempotent, then the corner ring*

$$eRe$$

is a ring with identity element e.

Definition 6.3.5. *Two idempotents $e_1, e_2 \in R$ are called orthogonal if*

$$e_1e_2 = e_2e_1 = 0.$$

More generally, finitely many idempotents are orthogonal if they are pairwise orthogonal.

Lemma 6.3.6. *If $e_1, \ldots, e_n$ are orthogonal idempotents, then*

$$e_1 + \cdots + e_n$$

is again an idempotent.

Proof. Let
$$e := e_1 + \cdots + e_n.$$
Since each e_i is an idempotent, we have $e_i^2 = e_i$ for all i. Since the *idempotents* are orthogonal, for $i \neq j$ we have
$$e_i e_j = 0.$$
Therefore
$$e^2 = (e_1 + \cdots + e_n)^2 = \sum_{i=1}^{n} e_i^2 + \sum_{i \neq j} e_i e_j = \sum_{i=1}^{n} e_i + \sum_{i \neq j} 0 = e_1 + \cdots + e_n = e.$$
Hence e is an idempotent. □

Definition 6.3.7. *An idempotent decomposition of an idempotent e is an expression*
$$e = e_1 + \cdots + e_n$$
where the e_i are orthogonal idempotents.

Remark 6.3.8. *If $e = e_1 + e_2$ is an idempotent decomposition, then e_1 and e_2 are in eRe.*

Definition 6.3.9. *An idempotent is called primitive if it cannot be written as a sum of two nonzero orthogonal idempotents.*

Definition 6.3.10. *A primitive orthogonal decomposition is an idempotent decomposition in which all summands are primitive.*

Example 6.3.11. *If e is an idempotent in a unital ring, then*
$$1 = e + (1 - e)$$
is an idempotent decomposition of 1.

Example 6.3.12. *Let e be an idempotent, $Re = \{ae | a \in R\}$ is a left module of R.*

Theorem 6.3.13. *Let e be an idempotent of R.*

If
$$e = e_1 + \cdots + e_n$$
is an idempotent decomposition, then
$$Re = Re_1 \oplus \cdots \oplus Re_n.$$

Conversely, if Re is a direct sum of left ideals

$$Re = I_1 \oplus \cdots \oplus I_n,$$

then there exists an idempotent decomposition

$$e = e_1 + \cdots + e_n$$

with $e_i \in I_i$ and $I_i = Re_i$.

In this way, the idempotent decompositions of e are in one-to-one correspondence with the direct sum decompositions of the left R-module Re.

Proof. Assume first that

$$e = e_1 + \cdots + e_n$$

is an idempotent decomposition. Since

$$e_i e = e e_i = e_i,$$

we have

$$Re_i \subseteq Re.$$

Indeed, if $x \in Re_i$, then $xe = x$.

Now let $a \in R$. Then

$$ae = ae_1 + \cdots + ae_n \in Re_1 + \cdots + Re_n,$$

so

$$Re \subseteq Re_1 + \cdots + Re_n.$$

The reverse inclusion is clear, hence

$$Re = Re_1 + \cdots + Re_n.$$

To show that the sum is direct, suppose

$$a_1 e_1 + \cdots + a_n e_n = 0.$$

Multiplying on the right by e_i, we get

$$a_i e_i = 0$$

for each i, since $e_j e_i = 0$ for $j \neq i$. Therefore the expression is unique, and

$$Re = Re_1 \oplus \cdots \oplus Re_n.$$

Conversely, suppose

$$Re = I_1 \oplus \cdots \oplus I_n.$$

Since $e \in Re$, we may write

$$e = e_1 + \cdots + e_n, \qquad e_i \in I_i.$$

For any $a_i \in I_i \subseteq Re$, we have

$$a_i = a_i e = a_i e_1 + \cdots + a_i e_n.$$

Since $e_j \in I_j$, we have

$$Re_j \subseteq I_j,$$

hence

$$a_i e_j \in I_j.$$

But $a_i \in I_i$, and the decomposition

$$Re = I_1 \oplus \cdots \oplus I_n$$

is direct. Therefore

$$a_i = a_i e_i, \qquad a_i e_j = 0 \quad (j \neq i).$$

It follows that

$$I_i \subseteq Re_i.$$

On the other hand, since $e_i \in I_i$ and I_i is a left ideal, we have

$$Re_i \subseteq I_i.$$

Thus

$$I_i = Re_i.$$

Now take $a_i = e_i$. Then

$$e_i = e_i^2, \qquad e_i e_j = 0 \quad (j \neq i).$$

Hence

$$e = e_1 + \cdots + e_n$$

is an idempotent decomposition. □

Corollary 6.3.14. *Let e be an idempotent of R. Then the following are equivalent:*

(1) e is primitive.

(2) Re is indecomposable as a left R-module.

(3) The only idempotents in eRe are 0 and e.

Proof. **(1) $\iff$ (2).** This follows immediately from the previous theorem, which gives a bijection between idempotent decompositions of e and direct sum decompositions of the left ideal Re.

(1) $\implies$ (3). Suppose that $e' \in eRe$ is an idempotent with

$$e' \neq 0, \qquad e' \neq e.$$

Since $e' \in eRe$, we have

$$ee' = e' = e'e.$$

Hence

$$e = e' + (e - e').$$

We claim that this is an idempotent decomposition of e.

First,

$$(e - e')^2 = e^2 - ee' - e'e + (e')^2 = e - e' - e' + e' = e - e'.$$

So $e - e'$ is an idempotent.

Next,

$$e'(e - e') = e'e - (e')^2 = e' - e' = 0,$$

and similarly

$$(e - e')e' = ee' - (e')^2 = e' - e' = 0.$$

Thus e' and $e - e'$ are orthogonal idempotents. Since both are nonzero, this contradicts the primitiveness of e.

Therefore no such e' exists, and the only idempotents in eRe are 0 and e.

(3) $\implies$ (1). Suppose that e is not primitive. Then there exist nonzero orthogonal idempotents e_1, e_2 such that

$$e = e_1 + e_2.$$

Since

$$ee_1e = e_1, \qquad ee_2e = e_2,$$

we have

$$e_1, e_2 \in eRe.$$

But e_1 is a nonzero idempotent different from e, contradicting (3). Hence e must be primitive. □

Chapter 7

Lecture 7

7.1 Idempotents and Module Homomorphisms

Theorem 7.1.1. *Let R be a ring, let $e, f \in R$ be idempotents, and let ${}_RV$ be a left R-module.*

1. *There is a natural isomorphism of abelian groups*
$$\operatorname{Hom}_R(Re, V) \cong eV.$$
In particular,
$$\operatorname{Hom}_R(Re, Rf) \cong eRf.$$

2. *There is an isomorphism of rings*
$$\operatorname{End}_R(Re) \cong (eRe)^{op}.$$

Proof. **(1)** Define
$$g : \operatorname{Hom}_R(Re, V) \longrightarrow eV, \qquad \varphi \longmapsto \varphi(e).$$
First we check that g is well-defined. If $\varphi \in \operatorname{Hom}_R(Re, V)$, then
$$\varphi(e) = \varphi(e^2) = e\varphi(e),$$
so indeed $\varphi(e) \in eV$.

Next, for any $a \in R$ and any $\varphi \in \operatorname{Hom}_R(Re, V)$, we have
$$\varphi(ae) = a\varphi(e).$$
Thus φ is uniquely determined by the single value $\varphi(e)$. Hence g is injective.

To show surjectivity, let $v \in eV$. Define
$$\psi_v : Re \longrightarrow V, \qquad ae \longmapsto av.$$

This is well-defined, since if $ae = be$, then

$$av = (ae)v = (be)v = bv.$$

It is also R-linear. Moreover,

$$g(\psi_v) = \psi_v(e) = ev = v,$$

because $v \in eV$. Hence g is surjective.

Finally, g is a homomorphism of abelian groups, since for $\varphi, \psi \in \mathrm{Hom}_R(Re, V)$,

$$g(\varphi + \psi) = (\varphi + \psi)(e) = \varphi(e) + \psi(e) = g(\varphi) + g(\psi).$$

Therefore g is an isomorphism of abelian groups:

$$\mathrm{Hom}_R(Re, V) \cong eV.$$

Taking $V = Rf$, we obtain

$$\mathrm{Hom}_R(Re, Rf) \cong e(Rf) = eRf.$$

(2) Apply part **(1)** with $V = Re$. Then we get an isomorphism of abelian groups

$$\Phi : \mathrm{End}_R(Re) \longrightarrow eRe, \qquad \sigma \longmapsto \sigma(e).$$

Now let $\sigma, \tau \in \mathrm{End}_R(Re)$. Then

$$\Phi(\sigma \circ \tau) = (\sigma \circ \tau)(e) = \sigma(\tau(e)).$$

Since $\tau(e) \in Re$, we have $\tau(e) = \tau(e)e$, and hence

$$\sigma(\tau(e)) = \sigma(\tau(e)e) = \tau(e)\sigma(e).$$

Therefore

$$\Phi(\sigma \circ \tau) = \Phi(\tau)\Phi(\sigma).$$

So composition in $\mathrm{End}_R(Re)$ corresponds to multiplication in eRe with the order reversed. Hence

$$\mathrm{End}_R(Re) \cong (eRe)^{op}$$

as rings. $\square$

Theorem 7.1.2. *Let e, f be idempotents of a ring R. Then the followings are equivalent:*

1. *$Re \cong Rf$ as left R-modules.*

2. *$eR \cong fR$ as right R-modules.*

3. *There exist elements*

$$a \in eRf, \qquad b \in fRe$$

such that

$$ab = e, \qquad ba = f.$$

Proof. By passing to the opposite ring R^{op}, statement (2) is the right-handed analogue of (1). Thus it suffices to prove that (1) and (3) are equivalent.

(1) $\Rightarrow$ (3). Assume that there is an isomorphism of left R-modules

$$\varphi : Re \xrightarrow{\sim} Rf.$$

Set

$$a := \varphi(e), \qquad b := \varphi^{-1}(f).$$

Then

$$a = \varphi(e) = \varphi(ee) = e\varphi(e) = ea,$$

and since $a \in Rf$, we also have $af = a$. Hence

$$a \in eRf.$$

Similarly,

$$b = \varphi^{-1}(f) = \varphi^{-1}(ff) = f\varphi^{-1}(f) = fb,$$

and since $b \in Re$, we have $be = b$. Therefore

$$b \in fRe.$$

Now compute:

$$f = \varphi\big(\varphi^{-1}(f)\big) = \varphi(b) = \varphi(be) = b\varphi(e) = ba,$$

and

$$e = \varphi^{-1}(\varphi(e)) = \varphi^{-1}(a) = \varphi^{-1}(af) = a\varphi^{-1}(f) = ab.$$

Thus

$$ab = e, \qquad ba = f.$$

(3) $\Rightarrow$ (1). Assume there exist $a \in eRf$ and $b \in fRe$ such that

$$ab = e, \qquad ba = f.$$

Since $a \in eRf$ and $b \in fRe$, we have

$$ea = a, \qquad af = a, \qquad fb = b, \qquad be = b.$$

Define maps

$$\varphi_a : Re \longrightarrow Rf, \qquad xe \longmapsto xa,$$

and

$$\varphi_b : Rf \longrightarrow Re, \qquad xf \longmapsto xb.$$

These are well-defined left R-module homomorphisms.

Now for any $xe \in Re$,

$$(\varphi_b \circ \varphi_a)(xe) = \varphi_b(xa) = xab = xe,$$

since $ab = e$. Likewise, for any $xf \in Rf$,

$$(\varphi_a \circ \varphi_b)(xf) = \varphi_a(xb) = xba = xf,$$

since $ba = f$.

Hence φ_a and φ_b are inverse isomorphisms, so

$$Re \cong Rf$$

as left R-modules.

Therefore (1) and (3) are equivalent, and by symmetry so are all three statements. □

7.2 Jacobson Radical of a Ring

Definition 7.2.1 (Jacobson radical)**.** *Let R be a ring with identity. The* Jacobson radical *of R is defined by*

$$J(R) := \bigcap \{\, M \mid M \text{ is a maximal left ideal of } R \,\}.$$

Remark 7.2.2 (Annihilators)**.** *Let ${}_R M$ be a left R-module.*

1. *For $x \in M$, define*

   $$\operatorname{Ann}_R(x) := \{\, r \in R \mid rx = 0 \,\}.$$

   This is a left ideal of R.

2. *For a subset $X \subseteq M$, define*

   $$\operatorname{Ann}_R(X) := \bigcap_{x \in X} \operatorname{Ann}_R(x).$$

3. *If $N \leq M$ is a submodule, then*

   $$\operatorname{Ann}_R(N) := \bigcap_{x \in N} \operatorname{Ann}_R(x)$$

   is a two-sided ideal of R.

Theorem 7.2.3 (Properties of the Jacobson radical)**.** *Let R be a ring with identity. Then the following hold.*

1. $$J(R) = \bigcap\{\, \mathrm{Ann}_R(S) \mid S \text{ is a simple left } R\text{-module}\,\}.$$

   In particular, $J(R)$ is a two-sided ideal of R.

2. *For $a \in R$,*

   $$a \in J(R) \quad \Longleftrightarrow \quad 1 - xa \text{ has a left inverse for every } x \in R.$$

   Equivalently, for every $x \in R$ there exists $b \in R$ such that

   $$b(1 - xa) = 1.$$

3. *$J(R)$ is the largest two-sided ideal I of R satisfying the condition*

   $$a \in I \implies 1 - a \text{ is a unit in } R.$$

4. *$J(R)$ coincides with the intersection of all maximal right ideals of R.*

Proof. **(1)** Let S be a simple left R-module, and choose $0 \neq v \in S$. Since S is simple,

$$S = Rv.$$

Consider the surjective R-homomorphism

$$R \longrightarrow S, \qquad r \longmapsto rv.$$

Its kernel is $\mathrm{Ann}_R(v)$, so

$$S \cong R/\mathrm{Ann}_R(v).$$

Because S is simple, $\mathrm{Ann}_R(v)$ is a maximal left ideal of R. Hence

$$J(R) \subseteq \mathrm{Ann}_R(v)$$

for every nonzero $v \in S$, and therefore

$$J(R) \subseteq \mathrm{Ann}_R(S)$$

for every simple left R-module S.

Conversely, suppose that

$$x \in \bigcap\{\, \mathrm{Ann}_R(S) \mid S \text{ simple left } R\text{-module}\,\}.$$

Let M be any maximal left ideal of R. Then R/M is a simple left R-module, so

$$x \in \mathrm{Ann}_R(R/M).$$

In particular,

$$x(1+M)=0,$$

hence

$$x+M=0,$$

so $x \in M$. Since this holds for every maximal left ideal M, we obtain

$$x \in J(R).$$

Thus

$$J(R)=\bigcap\{\operatorname{Ann}_R(S) \mid S \text{ simple left } R\text{-module}\}.$$

Finally, each $\operatorname{Ann}_R(S)$ is a two-sided ideal, so their intersection $J(R)$ is also a two-sided ideal.

(2) Assume first that $a \in J(R)$. Suppose, for contradiction, that there exists $x \in R$ such that $1-xa$ has no left inverse. Then the principal left ideal

$$R(1-xa)$$

is a proper left ideal of R, so it is contained in some maximal left ideal M.

Since $a \in J(R)$, we have $a \in M$, and because M is a left ideal,

$$xa \in M.$$

Also,

$$1-xa \in M.$$

Therefore

$$1=(1-xa)+xa \in M,$$

a contradiction. Hence $1-xa$ has a left inverse for every $x \in R$.

Conversely, assume that for every $x \in R$, the element $1-xa$ has a left inverse. Suppose that $a \notin J(R)$. Then there exists a maximal left ideal M such that

$$a \notin M.$$

Hence

$$M+Ra=R,$$

so there exist $b \in M$ and $x \in R$ such that

$$b+xa=1.$$

Thus

$$1-xa=b \in M.$$

But an element of a proper left ideal cannot have a left inverse, for if $c(1 - xa) = 1$, then

$$1 \in M,$$

contradicting the propriety of M. This contradiction shows that $a \in J(R)$.

(3) First let $a \in J(R)$. By part **(2)** with $x = 1$, there exists $b \in R$ such that

$$b(1 - a) = 1.$$

Then

$$b = 1 + ba = 1 - (-b)a.$$

Since $a \in J(R)$, part **(2)** again implies that b has a left inverse. So there exists $c \in R$ such that

$$cb = 1.$$

Now

$$c = cb(1 - a) = 1 - a.$$

Hence

$$(1 - a)b = 1,$$

and together with $b(1 - a) = 1$, this shows that $1 - a$ is a unit in R.

Thus $J(R)$ satisfies the stated property.

Now let I be a two-sided ideal of R such that

$$a \in I \implies 1 - a \text{ is a unit in } R.$$

Suppose $I \not\subseteq J(R)$. Then there exists a maximal left ideal M such that

$$I \not\subseteq M.$$

Choose $a \in I \setminus M$. Then

$$M + Ra = R,$$

so there exist $m \in M$ and $r \in R$ such that

$$m + ra = 1.$$

Since I is a two-sided ideal and $a \in I$, we have

$$ra \in I.$$

Therefore

$$1 - ra$$

is a unit in R. But

$$1 - ra = m \in M,$$

and a proper left ideal cannot contain a unit. This is a contradiction. Hence

$$I \subseteq J(R).$$

So $J(R)$ is the largest such two-sided ideal.

(4) Let

$$J_r(R) := \bigcap \{ N \mid N \text{ is a maximal right ideal of } R \}.$$

Applying the preceding arguments to the opposite ring R^{op}, we see that $J_r(R)$ is also the largest two-sided ideal I of R such that

$$a \in I \implies 1 - a \text{ is a unit in } R.$$

By part **(3)**, this property characterizes $J(R)$ uniquely. Therefore

$$J_r(R) = J(R).$$

Hence $J(R)$ coincides with the intersection of all maximal right ideals of R. $\square$

Definition 7.2.4. *Let R be a ring with 1.*

1. *An element $a \in R$ is called* nilpotent *if*

$$a^n = 0$$

for some integer $n \geq 1$.

2. *A left ideal $I \subseteq R$ is called* nilpotent *if*

$$I^n = 0$$

for some integer $n \geq 1$.

3. *A left ideal $I \subseteq R$ is called a* nil left ideal *if every element of I is nilpotent.*

Lemma 7.2.5. *If $a \in R$ is nilpotent, say $a^n = 0$, then $1 - a$ is a unit in R, with inverse*

$$(1-a)^{-1} = 1 + a + a^2 + \cdots + a^{n-1}.$$

Proof. Since $a^n = 0$, we have

$$(1-a)(1 + a + a^2 + \cdots + a^{n-1}) = 1 - a^n = 1,$$

and similarly

$$(1 + a + a^2 + \cdots + a^{n-1})(1-a) = 1 - a^n = 1.$$

Hence $1 - a$ is a unit. $\square$

Lemma 7.2.6. *Every nilpotent left ideal is a nil left ideal.*

Proof. Suppose $I^n = 0$ and let $a \in I$. Then

$$a^n \in I^n = 0,$$

so a is nilpotent. Hence every element of I is nilpotent. □

Remark 7.2.7. *We shall use the following standard characterization of the Jacobson radical:*

$$a \in J(R) \quad \Longleftrightarrow \quad \text{for every } x \in R,\ 1 - xa \text{ has a left inverse in } R.$$

In particular, if $a \in J(R)$, then $1 - a$ is a unit.

Theorem 7.2.8. *Let I be a nil left ideal of a ring R. Then*

$$I \subseteq J(R).$$

Proof. Let $a \in I$. For every $x \in R$, since I is a left ideal we have $xa \in I$. Because I is nil, the element xa is nilpotent. By the previous lemma, $1 - xa$ is a unit in R, hence in particular it has a left inverse.

Therefore, by the characterization of $J(R)$,

$$a \in J(R).$$

Since $a \in I$ was arbitrary, it follows that

$$I \subseteq J(R).$$

□

7.3 Artinian Rings and the Jacobson Radical

Theorem 7.3.1. *Let A be an Artinian ring. Then $J(A)$ is nilpotent.*

Proof. Consider the descending chain of left ideals

$$J(A) \supseteq J(A)^2 \supseteq J(A)^3 \supseteq \cdots.$$

Since A is Artinian, this chain stabilizes. Hence there exists $n \geq 1$ such that

$$J(A)^n = J(A)^{n+1} = J(A)^{n+2} = \cdots.$$

Set

$$N := J(A)^n.$$

Then

$$N^2 = J(A)^{2n} = J(A)^n = N,$$

because all powers from the nth onward are equal.

We claim that $N = 0$. Suppose not. Consider the set

$$\Sigma := \{ L \leq_A A \mid NL \neq 0 \}$$

of left ideals L of A such that $NL \neq 0$. Since

$$NN = N \neq 0,$$

the set Σ is nonempty. As A is Artinian, Σ has a minimal element; call it L_0.

Since $NL_0 \neq 0$, there exists $a \in L_0$ such that

$$Na \neq 0.$$

Now Na is a left ideal of A contained in L_0, and

$$N(Na) = N^2 a = Na \neq 0.$$

Thus $Na \in \Sigma$. By minimality of L_0, we must have

$$L_0 = Na.$$

In particular, $a \in Na$, so there exists $b \in N$ such that

$$a = ba.$$

Hence

$$(1 - b)a = 0.$$

But $b \in N \subseteq J(A)$, so $1 - b$ is a unit. Multiplying by its inverse gives $a = 0$, a contradiction to $Na \neq 0$.

Therefore $N = 0$, that is,

$$J(A)^n = 0.$$

So $J(A)$ is nilpotent. □

Corollary 7.3.2. *Let A be an Artinian ring. Then $J(A)$ is the largest nilpotent left ideal of A.*

Proof. By the theorem above, $J(A)$ is itself nilpotent.

Now let I be any nilpotent left ideal of A. By the earlier lemma, I is a nil left ideal. Hence by the theorem on nil left ideals,

$$I \subseteq J(A).$$

Thus $J(A)$ contains every nilpotent left ideal, so it is the largest one. $\square$

Theorem 7.3.3. *Let R be a ring and let I be a two-sided ideal of R such that*

$$I \subseteq J(R).$$

Then

$$J(R/I) = J(R)/I.$$

Proof. We prove the two inclusions separately.

First inclusion: $J(R)/I \subseteq J(R/I)$.

Let $a \in J(R)$. For any $x \in R$, the element $1 - xa$ has a left inverse in R. So there exists $b \in R$ such that

$$b(1 - xa) = 1.$$

Passing to the quotient modulo I, we get

$$(b + I)\big((1 - xa) + I\big) = 1 + I.$$

Hence $(1 - xa) + I$ has a left inverse in R/I. Since this holds for every $x \in R$, the characterization of the Jacobson radical gives

$$a + I \in J(R/I).$$

Thus

$$J(R)/I \subseteq J(R/I).$$

Second inclusion: $J(R/I) \subseteq J(R)/I$.

Let $a + I \in J(R/I)$. We show that $a \in J(R)$. Let $x \in R$ be arbitrary. Since $a + I \in J(R/I)$, the element

$$(1 - xa) + I$$

has a left inverse in R/I. Therefore there exists $b \in R$ such that

$$(b + I)\big((1 - xa) + I\big) = 1 + I,$$

that is,

$$b(1 - xa) = 1 - c$$

for some $c \in I$.

Since $I \subseteq J(R)$, we have $c \in J(R)$, so $1 - c$ has a left inverse in R. Thus there exists $d \in R$ such that

$$d(1 - c) = 1.$$

Multiplying the equation $b(1 - xa) = 1 - c$ on the left by d, we obtain

$$(db)(1 - xa) = 1.$$

Hence $1 - xa$ has a left inverse in R. Since $x \in R$ was arbitrary, the characterization of $J(R)$ implies

$$a \in J(R).$$

Therefore

$$a + I \in J(R)/I.$$

Combining the two inclusions, we conclude that

$$J(R/I) = J(R)/I.$$

□

Definition 7.3.4. *A ring A is called* semisimple *if the left regular module ${}_AA$ is semisimple.*

Theorem 7.3.5. *Let A be an Artinian ring. Then A is semisimple if and only if*

$$J(A) = 0.$$

Proof. *(⇒)* Suppose A is semisimple. Then ${}_AA$ is a direct sum of simple left A-modules:

$${}_AA \cong S_1 \oplus \cdots \oplus S_r.$$

By the standard description of $J(A)$ as the intersection of the annihilators of all simple left A-modules, every element of $J(A)$ annihilates each S_i. Hence

$$J(A)A = 0.$$

But $1 \in A$, so for any $x \in J(A)$ we have

$$x = x \cdot 1 = 0.$$

Therefore

$$J(A) = 0.$$

(⇐) Suppose now that $J(A) = 0$. Let

$$\mathcal{S} := \{\, M_1 \cap \cdots \cap M_r \mid r \geq 1,\ M_1, \ldots, M_r \text{ maximal left ideals of } A\}\,.$$

Since A is Artinian, the set $\mathcal{S}$ has a minimal element with respect to inclusion. Choose such a minimal element and write it as

$$L = M_1 \cap \cdots \cap M_r.$$

We claim that $L = 0$. Suppose $L \neq 0$. Since

$$J(A) = \bigcap \{ M \mid M \text{ maximal left ideal of } A \} = 0,$$

there exists a maximal left ideal M such that

$$L \not\subseteq M.$$

Hence

$$L \cap M \subsetneq L.$$

But $L \cap M$ is again a finite intersection of maximal left ideals, contradicting the minimality of L. Therefore $L = 0$.

Now consider the natural A-module homomorphism

$$\varphi : A \longrightarrow A/M_1 \oplus \cdots \oplus A/M_r, \qquad a \longmapsto (a + M_1, \ldots, a + M_r).$$

Its kernel is

$$\ker \varphi = M_1 \cap \cdots \cap M_r = L = 0,$$

so φ is injective.

Each A/M_i is a simple left A-module, hence the direct sum

$$A/M_1 \oplus \cdots \oplus A/M_r$$

is semisimple. Since every submodule of a semisimple module is semisimple, it follows that ${}_AA$ is semisimple. Therefore A is semisimple. $\square$

Corollary 7.3.6. *If A is an Artinian ring, then*

$$A/J(A)$$

is semisimple.

Proof. Since A is Artinian, so is the quotient ring $A/J(A)$. By the previous theorem, it suffices to show that

$$J\big(A/J(A)\big) = 0.$$

But by the quotient formula,

$$J\big(A/J(A)\big) = J(A)/J(A) = 0.$$

Hence $A/J(A)$ is semisimple. $\square$

Corollary 7.3.7. *Let A be an Artinian ring. Then $J(A)$ is the unique two-sided ideal U of A such that*

1. *U is nilpotent;*

2. A/U is semisimple.

Proof. First, $J(A)$ has these two properties: it is nilpotent by the nilpotency theorem above, and $A/J(A)$ is semisimple by the previous corollary.

Now let U be a two-sided ideal satisfying (1) and (2). Since U is nilpotent, it is a nilpotent left ideal, hence

$$U \subseteq J(A).$$

Because A/U is semisimple, we have

$$J(A/U) = 0.$$

On the other hand, since $U \subseteq J(A)$, the quotient formula gives

$$J(A/U) = J(A)/U.$$

Therefore

$$J(A)/U = 0,$$

so

$$J(A) = U.$$

Thus $J(A)$ is unique. $\square$

7.4 Radical and Socle Series

Proposition 7.4.1. *Let A be an Artinian ring, and let U be a finitely generated left A-module.*

1. *For a submodule $V \leq U$, the following are equivalent:*
   - *(a)* $V = \mathrm{Rad}(U)$*;*
   - *(b) V is the smallest submodule of U such that U/V is semisimple;*
   - *(c)* $V = J(A)U$*.*
2. *For a submodule $V \leq U$, the following are equivalent:*
   - *(a)* $V = \mathrm{Soc}(U)$*;*
   - *(b) V is the largest semisimple submodule of U;*
   - *(c)*
   $$V = \{\, u \in U \mid J(A)u = 0 \,\}.$$

Proof. We first record two standard facts.

Fact 1. Since A is Artinian, the quotient ring $A/J(A)$ is semisimple. Hence every $A/J(A)$-module is semisimple.

Fact 2. If S is a simple left A-module, then

$$J(A)S = 0.$$

Indeed, by the characterization

$$J(A) = \bigcap \{ \operatorname{Ann}_A(T) \mid T \text{ simple left } A\text{-module} \},$$

we have $J(A) \subseteq \operatorname{Ann}_A(S)$.

(1) We prove the equivalence of (a), (b), (c).

(c) $\Rightarrow$ (b). Set

$$W := J(A)U.$$

Then

$$J(A)(U/W) = 0,$$

so U/W is naturally an $A/J(A)$-module. By Fact 1, U/W is semisimple.

Now let $V' \leq U$ be any submodule such that U/V' is semisimple. Since every simple module is annihilated by $J(A)$, every semisimple module is annihilated by $J(A)$ as well. Hence

$$J(A)(U/V') = 0.$$

Equivalently,

$$J(A)U \subseteq V'.$$

Thus $W = J(A)U$ is the smallest submodule of U with semisimple quotient.

(b) $\Rightarrow$ (c). Assume that V is the smallest submodule of U such that U/V is semisimple.

Since $U/J(A)U$ is semisimple by the previous paragraph, the minimality of V gives

$$V \subseteq J(A)U.$$

On the other hand, U/V is semisimple, so it is annihilated by $J(A)$. Hence

$$J(A)(U/V) = 0,$$

which means

$$J(A)U \subseteq V.$$

Therefore

$$V = J(A)U.$$

(c) $\Leftrightarrow$ (a). Let $W = J(A)U$. We show that $W = \operatorname{Rad}(U)$.

First, let M be a maximal submodule of U. Then U/M is simple, hence semisimple. By the minimality proved above for $J(A)U$, we must have

$$J(A)U \subseteq M.$$

Since this is true for every maximal submodule M of U, it follows that

$$J(A)U \subseteq \operatorname{Rad}(U).$$

For the reverse inclusion, suppose there exists

$$u \in \operatorname{Rad}(U) \setminus J(A)U.$$

Let

$$\overline{U} := U/J(A)U, \qquad \overline{u} := u + J(A)U \neq 0.$$

We already know that $\overline{U}$ is semisimple, so we may write

$$\overline{U} = \bigoplus_{i \in I} S_i$$

with each S_i simple. Since $\overline{u} \neq 0$, some component of $\overline{u}$ in this direct sum is nonzero; choose $i_0 \in I$ such that the S_{i_0}-component of $\overline{u}$ is nonzero.

Set

$$\overline{M} := \bigoplus_{i \neq i_0} S_i.$$

Then $\overline{M}$ is a maximal submodule of $\overline{U}$ and

$$\overline{u} \notin \overline{M}.$$

Let M be the inverse image of $\overline{M}$ under the quotient map

$$\pi : U \twoheadrightarrow \overline{U}.$$

Then M is a maximal submodule of U, because

$$U/M \cong \overline{U}/\overline{M} \cong S_{i_0}$$

is simple. But $u \notin M$, contradicting $u \in \operatorname{Rad}(U)$, since $\operatorname{Rad}(U)$ is contained in every maximal submodule of U.

Hence

$$\operatorname{Rad}(U) \subseteq J(A)U.$$

Together with the opposite inclusion, we get

$$\operatorname{Rad}(U) = J(A)U.$$

Combining the implications above, we conclude that (a), (b), and (c) are equivalent.

(2) Recall that

$$\operatorname{Soc}(U) = \sum \{\, S \leq U \mid S \text{ is a simple submodule} \,\}.$$

Let

$$T := \{\, u \in U \mid J(A)u = 0 \,\}.$$

We first show that T is a submodule of U. If $u, v \in T$ and $a \in A$, then

$$J(A)(u + v) = J(A)u + J(A)v = 0,$$

and for every $j \in J(A)$,

$$j(au) = (ja)u = 0,$$

because $J(A)$ is a two-sided ideal and $ja \in J(A)$. Thus $au \in T$, so $T \leq U$.

(c) $\Rightarrow$ (b). Assume $V = T$.

Since $J(A)T = 0$, the module T is naturally an $A/J(A)$-module. By Fact 1, T is semisimple.

Now let $W \leq U$ be any semisimple submodule. Then W is a sum of simple submodules, and by Fact 2 each simple submodule is annihilated by $J(A)$. Hence

$$J(A)W = 0,$$

so

$$W \subseteq T.$$

Therefore T is the largest semisimple submodule of U.

(b) $\Rightarrow$ (c). Assume that V is the largest semisimple submodule of U.

The submodule

$$T = \{\, u \in U \mid J(A)u = 0 \,\}$$

is semisimple by the previous paragraph, so by maximality of V we get

$$T \subseteq V.$$

On the other hand, V is semisimple, hence $J(A)V = 0$, so

$$V \subseteq T.$$

Thus

$$V = T.$$

(a) $\Leftrightarrow$ (c). Let $T = \{u \in U \mid J(A)u = 0\}$ as above.

Every simple submodule $S \leq U$ satisfies $J(A)S = 0$ by Fact 2, so

$$S \subseteq T.$$

Taking the sum over all simple submodules of U, we obtain

$$\mathrm{Soc}(U) \subseteq T.$$

Conversely, T is semisimple, so it is a sum of simple submodules of U. Hence

$$T \subseteq \operatorname{Soc}(U).$$

Therefore

$$\operatorname{Soc}(U) = T.$$

So (a), (b), and (c) are equivalent. $\square$

Let A be an Artinian ring, and let U be a finitely generated left A-module. Recall the basic identities

$$\operatorname{Rad}(U) = J(A)U, \qquad \operatorname{Soc}(U) = \{u \in U \mid J(A)u = 0\}.$$

Definition 7.4.2. *Define inductively*

$$\operatorname{Rad}^0(U) := U, \qquad \operatorname{Rad}^n(U) := \operatorname{Rad}(\operatorname{Rad}^{n-1}(U)) \quad (n \geq 1),$$

and

$$\operatorname{Soc}^0(U) := 0, \qquad \operatorname{Soc}^n(U)/\operatorname{Soc}^{n-1}(U) := \operatorname{Soc}\big(U/\operatorname{Soc}^{n-1}(U)\big) \quad (n \geq 1).$$

The descending chain

$$U = \operatorname{Rad}^0(U) \supseteq \operatorname{Rad}(U) \supseteq \operatorname{Rad}^2(U) \supseteq \cdots$$

is called the radical series *of U, and the ascending chain*

$$0 = \operatorname{Soc}^0(U) \subseteq \operatorname{Soc}(U) \subseteq \operatorname{Soc}^2(U) \subseteq \cdots$$

is called the socle series *of U.*

Proposition 7.4.3. *For every integer $n \geq 1$,*

$$\operatorname{Rad}^n(U) = J(A)^n U,$$

and

$$\operatorname{Soc}^n(U) = \{u \in U \mid J(A)^n u = 0\}.$$

Proof. We prove both formulas by induction on n.

Radical formula. For $n = 1$, this is exactly the identity

$$\operatorname{Rad}(U) = J(A)U.$$

Assume now that $\operatorname{Rad}^{n-1}(U) = J(A)^{n-1}U$. Since $\operatorname{Rad}^{n-1}(U)$ is a finitely generated A-

module, we may apply the same identity to it:

$$\operatorname{Rad}^n(U) = \operatorname{Rad}(\operatorname{Rad}^{n-1}(U)) = J(A) \operatorname{Rad}^{n-1}(U) = J(A)\bigl(J(A)^{n-1}U\bigr) = J(A)^n U.$$

Socle formula. For $n = 1$, this is exactly the identity

$$\operatorname{Soc}(U) = \{u \in U \mid J(A)u = 0\}.$$

Assume now that

$$\operatorname{Soc}^{n-1}(U) = \{u \in U \mid J(A)^{n-1}u = 0\}.$$

By definition,

$$\operatorname{Soc}^n(U)/\operatorname{Soc}^{n-1}(U) = \operatorname{Soc}\bigl(U/\operatorname{Soc}^{n-1}(U)\bigr).$$

Applying the case $n = 1$ to the module $U/\operatorname{Soc}^{n-1}(U)$, we get

$$\operatorname{Soc}\bigl(U/\operatorname{Soc}^{n-1}(U)\bigr) = \left\{u + \operatorname{Soc}^{n-1}(U) \in U/\operatorname{Soc}^{n-1}(U) \;\middle|\; J(A)u \subseteq \operatorname{Soc}^{n-1}(U)\right\}.$$

By the induction hypothesis, the condition

$$J(A)u \subseteq \operatorname{Soc}^{n-1}(U)$$

is equivalent to

$$J(A)^{n-1}(J(A)u) = 0,$$

that is,

$$J(A)^n u = 0.$$

Hence the preimage of $\operatorname{Soc}(U/\operatorname{Soc}^{n-1}(U))$ in U is precisely

$$\{u \in U \mid J(A)^n u = 0\}.$$

Therefore

$$\operatorname{Soc}^n(U) = \{u \in U \mid J(A)^n u = 0\}.$$

This completes the induction. □

Remark 7.4.4. *For $n \geq 1$, the quotient*

$$\operatorname{Rad}^{n-1}(U)/\operatorname{Rad}^n(U)$$

is called the n-th radical layer *of U, and*

$$\operatorname{Soc}^n(U)/\operatorname{Soc}^{n-1}(U)$$

is called the n-th socle layer *of U.*

Remark 7.4.5. *The n-th radical layer is the top of the previous radical term:*

$$\operatorname{Rad}^{n-1}(U)/\operatorname{Rad}^n(U) = \operatorname{Top}\big(\operatorname{Rad}^{n-1}(U)\big).$$

Lemma 7.4.6. *For every $n \geq 1$ and any finitely generated A-modules U, V,*

$$\operatorname{Rad}^n(U \oplus V) = \operatorname{Rad}^n(U) \oplus \operatorname{Rad}^n(V),$$

and

$$\operatorname{Soc}^n(U \oplus V) = \operatorname{Soc}^n(U) \oplus \operatorname{Soc}^n(V).$$

Proof. Using Proposition 7.4.3,

$$\operatorname{Rad}^n(U \oplus V) = J(A)^n(U \oplus V) = J(A)^n U \oplus J(A)^n V = \operatorname{Rad}^n(U) \oplus \operatorname{Rad}^n(V).$$

Similarly,

$$\operatorname{Soc}^n(U \oplus V) = \{(u, v) \in U \oplus V \mid J(A)^n(u, v) = 0\}.$$

But

$$J(A)^n(u, v) = 0 \iff J(A)^n u = 0 \text{ and } J(A)^n v = 0.$$

Hence

$$\operatorname{Soc}^n(U \oplus V) = \{u \in U \mid J(A)^n u = 0\} \oplus \{v \in V \mid J(A)^n v = 0\} = \operatorname{Soc}^n(U) \oplus \operatorname{Soc}^n(V).$$

□

Theorem 7.4.7. *Let A be an Artinian ring, and let U be a finitely generated left A-module.*

1. *The radical series of U is the fastest descending chain*

$$U = U_0 \supseteq U_1 \supseteq U_2 \supseteq \cdots$$

such that every factor U_{r-1}/U_r is semisimple. More precisely, if

$$U = U_0 \supseteq U_1 \supseteq U_2 \supseteq \cdots$$

is any descending chain with each U_{r-1}/U_r semisimple, then for every $r \geq 0$,

$$\operatorname{Rad}^r(U) \subseteq U_r.$$

2. *The socle series of U is the fastest ascending chain*

$$0 = V_0 \subseteq V_1 \subseteq V_2 \subseteq \cdots$$

such that every factor V_r/V_{r-1} is semisimple. More precisely, if

$$0 = V_0 \subseteq V_1 \subseteq V_2 \subseteq \cdots$$

is any ascending chain with each V_r/V_{r-1} semisimple, then for every $r \geq 0$,

$$V_r \subseteq \mathrm{Soc}^r(U).$$

3. *Both series terminate. If m and n are the minimal integers such that*

$$\mathrm{Rad}^m(U) = 0 \qquad \textit{and} \qquad \mathrm{Soc}^n(U) = U,$$

then

$$m = n.$$

Proof. **(1)** Let

$$U = U_0 \supseteq U_1 \supseteq U_2 \supseteq \cdots$$

be a descending chain such that each quotient U_{r-1}/U_r is semisimple. We prove by induction on r that

$$\mathrm{Rad}^r(U) \subseteq U_r.$$

For $r = 0$, this is clear, since $\mathrm{Rad}^0(U) = U = U_0$.

Assume $r \geq 1$ and that

$$\mathrm{Rad}^{r-1}(U) \subseteq U_{r-1}.$$

Consider the quotient

$$\mathrm{Rad}^{r-1}(U) / \big(\mathrm{Rad}^{r-1}(U) \cap U_r\big).$$

By the second isomorphism theorem,

$$\mathrm{Rad}^{r-1}(U) / \big(\mathrm{Rad}^{r-1}(U) \cap U_r\big) \cong \big(\mathrm{Rad}^{r-1}(U) + U_r\big)/U_r.$$

Since $\mathrm{Rad}^{r-1}(U) \subseteq U_{r-1}$, we have

$$\big(\mathrm{Rad}^{r-1}(U) + U_r\big)/U_r \subseteq U_{r-1}/U_r.$$

But U_{r-1}/U_r is semisimple, and any submodule of a semisimple module is semisimple. Hence

$$\mathrm{Rad}^{r-1}(U) / \big(\mathrm{Rad}^{r-1}(U) \cap U_r\big)$$

is semisimple.

By definition of the radical as the smallest submodule with semisimple quotient, it follows that

$$\mathrm{Rad}\big(\mathrm{Rad}^{r-1}(U)\big) \subseteq \mathrm{Rad}^{r-1}(U) \cap U_r.$$

That is,

$$\mathrm{Rad}^r(U) \subseteq U_r.$$

This proves (1).

(2) Let
$$0 = V_0 \subseteq V_1 \subseteq V_2 \subseteq \cdots$$
be an ascending chain such that each quotient V_r/V_{r-1} is semisimple. We prove by induction on r that
$$V_r \subseteq \operatorname{Soc}^r(U).$$

For $r = 0$, this is clear.

Assume $r \geq 1$ and that
$$V_{r-1} \subseteq \operatorname{Soc}^{r-1}(U).$$
Consider the image of V_r in the quotient $U/\operatorname{Soc}^{r-1}(U)$:
$$\big(V_r + \operatorname{Soc}^{r-1}(U)\big)/\operatorname{Soc}^{r-1}(U).$$
By the second isomorphism theorem,
$$\big(V_r + \operatorname{Soc}^{r-1}(U)\big)/\operatorname{Soc}^{r-1}(U) \cong V_r/\big(V_r \cap \operatorname{Soc}^{r-1}(U)\big).$$
Since $V_{r-1} \subseteq \operatorname{Soc}^{r-1}(U)$, we have
$$V_{r-1} \subseteq V_r \cap \operatorname{Soc}^{r-1}(U),$$
so
$$V_r/\big(V_r \cap \operatorname{Soc}^{r-1}(U)\big)$$
is a quotient of V_r/V_{r-1}. Because V_r/V_{r-1} is semisimple, so is every quotient of it. Hence
$$\big(V_r + \operatorname{Soc}^{r-1}(U)\big)/\operatorname{Soc}^{r-1}(U)$$
is a semisimple submodule of
$$U/\operatorname{Soc}^{r-1}(U).$$
By the maximality of the socle,
$$\big(V_r + \operatorname{Soc}^{r-1}(U)\big)/\operatorname{Soc}^{r-1}(U) \subseteq \operatorname{Soc}\big(U/\operatorname{Soc}^{r-1}(U)\big) = \operatorname{Soc}^r(U)/\operatorname{Soc}^{r-1}(U).$$
Therefore
$$V_r \subseteq \operatorname{Soc}^r(U).$$
This proves (2).

(3) Since A is Artinian, $J(A)$ is nilpotent. Choose $N \geq 1$ such that
$$J(A)^N = 0.$$

Then by Proposition 7.4.3,

$$\operatorname{Rad}^N(U) = J(A)^N U = 0,$$

and

$$\operatorname{Soc}^N(U) = \{u \in U \mid J(A)^N u = 0\} = U.$$

So both series terminate.

Now let m be minimal such that $\operatorname{Rad}^m(U) = 0$. By Proposition 7.4.3,

$$J(A)^m U = 0.$$

Hence every $u \in U$ satisfies $J(A)^m u = 0$, so again by Proposition 7.4.3,

$$\operatorname{Soc}^m(U) = U.$$

Thus $n \le m$.

Conversely, let n be minimal such that $\operatorname{Soc}^n(U) = U$. Then for every $u \in U$,

$$J(A)^n u = 0.$$

So

$$J(A)^n U = 0,$$

hence

$$\operatorname{Rad}^n(U) = J(A)^n U = 0.$$

Therefore $m \le n$.

Combining the two inequalities, we obtain

$$m = n.$$

□

Definition 7.4.8. *The common integer in Theorem 7.4.7(3) is called the* Loewy length *of U.*

7.5 Lifting Idempotents

Definition 7.5.1. *Let R be a ring and let I be a two-sided ideal of R.*

If $f \in R$ is an idempotent and $f \notin I$, then $e = f + I$ is an idempotent of R/I. We say that an idempotent $f \in R$ lifts *e.*

Remark 7.5.2. *For a general ideal I, an idempotent of R/I need not lift to an idempotent of R. The next theorem shows that lifting is always possible when I is nilpotent.*

Theorem 7.5.3. *Let R be a ring, let I be a nilpotent two-sided ideal of R, and let*

$$e \in R/I$$

be an idempotent. Then there exists an idempotent $f \in R$ such that

$$f + I = e.$$

Moreover, if e is primitive, then every lift f of e is primitive.

Proof. Since I is nilpotent, there exists $r \geq 1$ such that

$$I^r = 0.$$

We shall construct idempotents

$$e_i \in R/I^i \qquad (1 \leq i \leq r)$$

such that

$$e_1 = e$$

and for each $i \geq 2$, the image of e_i in R/I^{i-1} is e_{i-1}.

Assume inductively that $e_{i-1} \in R/I^{i-1}$ is an idempotent, with $2 \leq i \leq r$. Choose any element

$$a \in R/I^i$$

whose image in R/I^{i-1} is e_{i-1}. Since e_{i-1} is idempotent, we have

$$a^2 - a \in I^{i-1}/I^i.$$

Now

$$\left(I^{i-1}/I^i\right)^2 = 0,$$

because

$$I^{2i-2} \subseteq I^i \qquad (i \geq 2).$$

Hence

$$(a^2 - a)^2 = 0 \qquad \text{in } R/I^i.$$

Define

$$e_i := 3a^2 - 2a^3 \in R/I^i.$$

First, e_i maps to e_{i-1} in R/I^{i-1}, because

$$e_i - a = 3a^2 - 2a^3 - a = -(2a - 1)(a^2 - a) \in I^{i-1}/I^i.$$

Next, e_i is idempotent. Indeed, the polynomial identity

$$(3t^2 - 2t^3)^2 - (3t^2 - 2t^3) = -(3 - 2t)(1 + 2t)(t^2 - t)^2$$

gives

$$e_i^2 - e_i = -(3 - 2a)(1 + 2a)(a^2 - a)^2 = 0$$

in R/I^i.

Thus, by induction, we obtain an idempotent

$$e_r \in R/I^r = R.$$

Let

$$f := e_r.$$

Then f is an idempotent of R whose image in R/I is e.

Now suppose that e is primitive and that f is a lift of e. Assume that

$$f = f_1 + f_2$$

is a decomposition of f into orthogonal idempotents:

$$f_1^2 = f_1, \qquad f_2^2 = f_2, \qquad f_1 f_2 = f_2 f_1 = 0.$$

Reducing modulo I, we get

$$e = (f_1 + I) + (f_2 + I),$$

and this is again a decomposition into orthogonal idempotents in R/I. Since e is primitive, one of $f_1 + I$ or $f_2 + I$ must be zero. Without loss of generality, assume

$$f_1 + I = 0.$$

Then $f_1 \in I$.

But f_1 is idempotent, so for every $m \geq 1$,

$$f_1 = f_1^m \in I^m.$$

Taking $m = r$, we get

$$f_1 \in I^r = 0,$$

hence $f_1 = 0$. Therefore f cannot be written as a sum of two nonzero orthogonal idempotents, so f is primitive. □

Chapter 8

Lecture 8

8.1 Lifting Idempotent Decompositions

Corollary 8.1.1. *Let I be a nilpotent two-sided ideal of R, and let*

$$1 = e_1 + \cdots + e_n$$

be an idempotent decomposition in R/I. Then there exists an idempotent decomposition

$$1 = f_1 + \cdots + f_n$$

in R such that

$$f_i + I = e_i \qquad (1 \leq i \leq n).$$

If moreover the e_i are primitive, then so are the f_i.

Proof. We argue by induction on n.

If $n = 1$, this is exactly Theorem 7.5.3.

Assume the result holds for decompositions with fewer than n summands. Write

$$1 = e_1 + E, \qquad \text{where} \qquad E = e_2 + \cdots + e_n$$

in R/I. Then E is an idempotent orthogonal to e_1.

By Theorem 7.5.3, e_1 lifts to an idempotent $f \in R$ such that

$$e_1 = f + I.$$

Set

$$F := 1 - f.$$

Then F is an idempotent and

$$F + I = 1 - (f + I) = 1 - e_1 = E,$$

so F lifts E.

Now F is the identity element of the corner ring FRF, and FIF is a nilpotent two-sided ideal of FRF. Consider the composite homomorphism

$$FRF \hookrightarrow R \twoheadrightarrow R/I.$$

Its kernel is $FRF \cap I$. We claim that

$$FRF \cap I = FIF.$$

Indeed, if $x \in FIF$, then clearly $x \in FRF \cap I$. Conversely, if $x \in FRF \cap I$, then since $x \in FRF$ we have

$$x = FxF,$$

and because $x \in I$ and I is a two-sided ideal, it follows that

$$x = FxF \in FIF.$$

Thus the claim holds.

Hence we obtain an injective homomorphism

$$FRF/FIF \hookrightarrow R/I.$$

Its image is

$$E(R/I)E.$$

Indeed, the image of $FrF \in FRF$ is

$$(F + I)(r + I)(F + I) = E(r + I)E,$$

and every element of $E(R/I)E$ has this form. Therefore

$$FRF/FIF \cong E(R/I)E.$$

In the ring $E(R/I)E$, whose identity element is E, we have the idempotent decomposition

$$E = e_2 + \cdots + e_n.$$

So, by the induction hypothesis applied to the ring FRF and its nilpotent ideal FIF, there exists an idempotent decomposition

$$F = f_2 + \cdots + f_n$$

in FRF such that

$$f_i + I = e_i \qquad (2 \le i \le n).$$

Finally, letting

$$f_1 := f,$$

we get

$$1 = f_1 + F = f_1 + f_2 + \cdots + f_n,$$

which is an idempotent decomposition in R, and each f_i lifts e_i.

Moreover, since each $f_i \in FRF$ for $i \geq 2$, we have

$$Ff_i = f_i = f_iF, \qquad f_1f_i = f_if_1 = 0 \qquad (i \geq 2),$$

so the sum is orthogonal.

If the e_i are primitive, then f_1 is primitive by Theorem 7.5.3, and $f_2, \ldots, f_n$ are primitive by the induction hypothesis. Hence all f_i are primitive. □

Corollary 8.1.2. *Let I be a nilpotent two-sided ideal of R, and let $f \in R$ be an idempotent. Then f is primitive in R if and only if $f + I$ is primitive in R/I.*

Proof. First suppose that $f + I$ is primitive in R/I. Since f is an idempotent lift of $f + I$, Theorem 7.5.3 implies that f is primitive in R.

Conversely, suppose that f is primitive in R. We show that $f + I$ is primitive in R/I. Assume that

$$f + I = \overline{e}_1 + \overline{e}_2$$

is an idempotent decomposition in R/I, where

$$\overline{e}_1^2 = \overline{e}_1, \qquad \overline{e}_2^2 = \overline{e}_2, \qquad \overline{e}_1\overline{e}_2 = \overline{e}_2\overline{e}_1 = 0.$$

Since $\overline{e}_1 + \overline{e}_2 = f + I$, we have

$$\overline{e}_j(f + I) = \overline{e}_j = (f + I)\overline{e}_j \qquad (j = 1, 2).$$

Hence $\overline{e}_1, \overline{e}_2$ lie in the corner ring

$$(f + I)(R/I)(f + I).$$

There is a natural isomorphism

$$fRf/fIf \cong (f + I)(R/I)(f + I).$$

Moreover, fIf is a nilpotent two-sided ideal of the ring fRf.

Thus, by the lifting theorem for idempotent decompositions, the decomposition

$$f + I = \overline{e}_1 + \overline{e}_2$$

lifts to an idempotent decomposition

$$f = f_1 + f_2$$

in $fRf \subseteq R$, with

$$f_j + I = \bar{e}_j \qquad (j = 1, 2).$$

Since f is primitive in R, one of f_1, f_2 must be zero. Therefore one of $\bar{e}_1, \bar{e}_2$ is zero.

Hence $f+I$ admits no nontrivial idempotent decomposition in R/I, so $f+I$ is primitive.

□

8.2 Projective Covers over Artinian Rings

Lemma 8.2.1. *Let A be a semisimple Artinian ring. Then the following statements hold.*

1. *There exists a primitive idempotent decomposition*

$$1 = e_1 + \cdots + e_n.$$

This decomposition corresponds to a decomposition of the left regular module

$$_AA = Ae_1 \oplus \cdots \oplus Ae_n,$$

where each Ae_i is a simple left A-module.

2. *For every simple left A-module S, there exists a primitive idempotent $e \in A$ such that*

$$S \cong Ae.$$

3. *For every simple left A-module S, there exists some $1 \leq i \leq n$ such that*

$$e_iS \neq 0.$$

Moreover, if T is a simple left A-module with $T \not\cong S$, then

$$e_iT = 0.$$

Proof. **(1)** Since A is semisimple Artinian, the left regular module $_AA$ is semisimple and Artinian. Hence it is a finite direct sum of simple left ideals:

$$_AA = I_1 \oplus \cdots \oplus I_n.$$

By the correspondence between direct sum decompositions of $_AA$ and idempotent decompositions of 1, there exist pairwise orthogonal idempotents

$$e_1, \ldots, e_n \in A$$

such that

$$1 = e_1 + \cdots + e_n$$

and

$$I_i = Ae_i \qquad (1 \leq i \leq n).$$

Since each I_i is simple, each Ae_i is simple and hence indecomposable. Therefore each e_i is primitive.

(2) Let S be a simple left A-module. Choose $0 \neq s \in S$. Then

$$S = As,$$

so the map

$$A \longrightarrow S, \qquad a \longmapsto as$$

is a surjective A-module homomorphism.

Using the decomposition from part (1), we have

$$A = Ae_1 \oplus \cdots \oplus Ae_n.$$

Since the map $A \to S$ is nonzero and surjective, its restriction to at least one summand Ae_i is nonzero. Thus there exists i such that we have a nonzero homomorphism

$$Ae_i \longrightarrow S.$$

Because both Ae_i and S are simple, this nonzero homomorphism is an isomorphism. Therefore

$$S \cong Ae_i.$$

Taking $e = e_i$, we obtain the desired primitive idempotent.

(3) Let S be a simple left A-module. Since

$$1 = e_1 + \cdots + e_n,$$

we have

$$S = 1S = (e_1 + \cdots + e_n)S = e_1S + \cdots + e_nS.$$

Since $S \neq 0$, there exists some i such that

$$e_iS \neq 0.$$

For this i, we claim that if T is a simple left A-module with $T \not\cong S$, then

$$e_iT = 0.$$

Indeed, by the standard isomorphism

$$\mathrm{Hom}_A(Ae_i, T) \cong e_iT, \qquad \varphi \longmapsto \varphi(e_i),$$

it is enough to show that

$$\mathrm{Hom}_A(Ae_i, T) = 0.$$

Since $e_i S \neq 0$, the same isomorphism gives

$$\mathrm{Hom}_A(Ae_i, S) \cong e_i S \neq 0.$$

Thus there is a nonzero homomorphism

$$Ae_i \longrightarrow S.$$

Because Ae_i and S are simple, this homomorphism is an isomorphism. Hence

$$Ae_i \cong S.$$

If $T \ncong S$, then $T \ncong Ae_i$. Therefore, by Schur's lemma,

$$\mathrm{Hom}_A(Ae_i, T) = 0.$$

Consequently,

$$e_i T = 0.$$

□

Theorem 8.2.2. *Let A be an Artinian ring, and let S be a simple left A-module. Then the followings hold.*

1. *There exists an indecomposable projective left A-module P_S such that*

$$P_S/\,\mathrm{Rad}(P_S) \cong S.$$

Moreover, P_S is of the form

$$P_S = Af,$$

where f is a primitive idempotent of A.

2. *The idempotent f has the following property:*

$$fS \neq 0,$$

and if T is any simple left A-module with $T \ncong S$, then

$$fT = 0.$$

3. *The module P_S is the projective cover of S.*

Proof. Let

$$\overline{A} := A/J(A).$$

Since A is Artinian, the ideal $J(A)$ is nilpotent, and $\overline{A}$ is a semisimple Artinian ring.

By the structure theory of semisimple Artinian rings, there exists a primitive idempotent

$$e \in \overline{A}$$

such that

$$eS \neq 0.$$

Equivalently, the simple $\overline{A}$-module $\overline{A}e$ is isomorphic to S.

Since $J(A)$ is nilpotent, idempotents lift modulo $J(A)$. Hence there exists an idempotent $f \in A$ such that

$$f + J(A) = e.$$

Moreover, because e is primitive, the lift f is also primitive.

Now define

$$P_S := Af.$$

Since f is an idempotent, Af is a direct summand of the left regular module ${}_AA$. Indeed,

$$A = Af \oplus A(1 - f).$$

Hence Af is projective. Since f is primitive, Af is indecomposable. Therefore $P_S = Af$ is an indecomposable projective left A-module.

Consider the natural map

$$Af \longrightarrow \overline{A}e$$

defined by

$$af \longmapsto (a + J(A))(f + J(A)).$$

Since $f + J(A) = e$, this map is explicitly

$$af \longmapsto (a + J(A))e.$$

It is a surjective A-module homomorphism. Its kernel is

$$J(A)f.$$

Indeed, af maps to zero if and only if

$$(a + J(A))(f + J(A)) = 0,$$

which is equivalent to

$$af \in J(A).$$

Since $af \in Af$, the kernel is

$$Af \cap J(A) = J(A)f.$$

Thus
$$Af/J(A)f \cong \overline{A}e.$$
Because $\overline{A}e \cong S$, we get
$$P_S/J(A)P_S \cong S.$$
Since A is Artinian and P_S is finitely generated, we have
$$\mathrm{Rad}(P_S) = J(A)P_S.$$
Therefore
$$P_S/\,\mathrm{Rad}(P_S) \cong S.$$

Next we verify the stated property of f. Since every simple A-module is annihilated by $J(A)$, the action of f on any simple A-module is the same as the action of
$$e = f + J(A)$$
through the quotient algebra $\overline{A} = A/J(A)$. Hence
$$fS = eS \neq 0.$$
If T is a simple A-module with $T \ncong S$, then, by the choice of e in the semisimple algebra $\overline{A}$, we have
$$eT = 0.$$
Therefore
$$fT = 0.$$

Finally, the natural epimorphism
$$P_S = Af \longrightarrow P_S/\,\mathrm{Rad}(P_S) \cong S$$
is essential by Nakayama's lemma. Since P_S is projective, this epimorphism is a projective cover of S.

Thus P_S is the projective cover of S. □

Theorem 8.2.3. *Let A be an Artinian ring. Up to isomorphism, the indecomposable projective left A-modules are exactly the modules*
$$P_S,$$
where S runs through the isomorphism classes of simple left A-modules and P_S denotes the projective cover of S.

Moreover,
$$P_S \cong P_T \qquad \Longleftrightarrow \qquad S \cong T.$$
Each P_S occurs as a direct summand of the left regular module ${}_AA$, with multi-

plicity equal to the multiplicity of S as a direct summand of

$$A/J(A).$$

Equivalently, one has a decomposition

$$_AA \cong \bigoplus_S P_S^{n_S},$$

where S runs through the isomorphism classes of simple left A-modules and

$$A/J(A) \cong \bigoplus_S S^{n_S}.$$

Furthermore,

$$n_S = \dim_{D_S} S, \qquad D_S := \operatorname{End}_A(S),$$

where S is regarded as a right D_S-vector space by

$$s \cdot \varphi := \varphi(s).$$

In particular, if A is a finite-dimensional k-algebra over an algebraically closed field k, then

$$n_S = \dim_k S.$$

Proof. We first classify the indecomposable projective modules.

Let P be a finitely generated indecomposable projective left A-module. Since A is Artinian, P has finite length, and therefore

$$P/\operatorname{Rad}(P)$$

is semisimple. Hence we may write

$$P/\operatorname{Rad}(P) \cong S_1 \oplus \cdots \oplus S_t,$$

where each S_i is simple.

The canonical epimorphism

$$P \longrightarrow P/\operatorname{Rad}(P)$$

is essential by Nakayama's lemma. Since P is projective, this epimorphism is a projective cover of

$$S_1 \oplus \cdots \oplus S_t.$$

For each simple left A-module T, let

$$\pi_T : P_T \longrightarrow T$$

denote the projective cover of T, whose existence was proved in Theorem 8.2.2. Thus P_T is an indecomposable projective left A-module and

$$P_T/\operatorname{Rad}(P_T) \cong T.$$

In particular, for each $i = 1, \ldots, t$, we have a projective cover

$$\pi_{S_i} : P_{S_i} \longrightarrow S_i.$$

Therefore the direct sum map

$$\pi_{S_1} \oplus \cdots \oplus \pi_{S_t} : P_{S_1} \oplus \cdots \oplus P_{S_t} \longrightarrow S_1 \oplus \cdots \oplus S_t$$

is also a projective cover. By uniqueness of projective covers, we obtain

$$P \cong P_{S_1} \oplus \cdots \oplus P_{S_t}.$$

Since P is indecomposable, we must have $t = 1$. Hence

$$P \cong P_{S_1}.$$

Thus every indecomposable projective module is isomorphic to P_S for some simple module S.

Conversely, for every simple module S, the projective cover P_S is indecomposable. Indeed, by the previous theorem, P_S has the form

$$P_S \cong Af$$

for some primitive idempotent $f \in A$. Since f is primitive, the module Af is indecomposable.

Next we show that

$$P_S \cong P_T \qquad \Longleftrightarrow \qquad S \cong T.$$

If $P_S \cong P_T$, then passing to radical quotients gives

$$P_S/\operatorname{Rad}(P_S) \cong P_T/\operatorname{Rad}(P_T).$$

But

$$P_S/\operatorname{Rad}(P_S) \cong S, \qquad P_T/\operatorname{Rad}(P_T) \cong T,$$

so $S \cong T$. The converse is immediate from the uniqueness of projective covers.

Now consider the left regular module ${}_AA$. Since A is Artinian, the natural epimorphism

$$A \longrightarrow A/J(A)$$

is essential by Nakayama's lemma. Since ${}_AA$ is projective, this epimorphism is a projective cover of $A/J(A)$.

Write the semisimple A-module $A/J(A)$ as

$$A/J(A) \cong \bigoplus_S S^{n_S},$$

where S runs through the isomorphism classes of simple left A-modules. Then

$$\bigoplus_S P_S^{n_S} \longrightarrow \bigoplus_S S^{n_S}$$

is a projective cover. By uniqueness of projective covers, we get

$${}_A A \cong \bigoplus_S P_S^{n_S}.$$

Thus the multiplicity of P_S in ${}_A A$ is exactly the multiplicity of S in $A/J(A)$.

It remains to identify the integer n_S. Put

$$\overline{A} := A/J(A).$$

Since A is Artinian, the quotient $\overline{A}$ is semisimple Artinian. Hence the Wedderburn–Artin theorem applies to $\overline{A}$, not necessarily to A itself.

Every simple left A-module S is annihilated by $J(A)$, and therefore may be regarded naturally as a simple left $\overline{A}$-module. Conversely, every simple left $\overline{A}$-module is naturally a simple left A-module via the quotient map $A \to \overline{A}$. Moreover,

$$\mathrm{End}_{\overline{A}}(S) = \mathrm{End}_A(S).$$

By the Wedderburn–Artin theorem applied to $\overline{A}$, the multiplicity of S in the left regular $\overline{A}$-module ${}_{\overline{A}}\overline{A}$ is

$$n_S = \dim_{D_S} S,$$

where

$$D_S := \mathrm{End}_{\overline{A}}(S)^{\mathrm{op}} = \mathrm{End}_A(S)^{\mathrm{op}}.$$

Here S is regarded as a right D_S-vector space by

$$s \cdot \varphi = \varphi(s), \qquad s \in S,\ \varphi \in \mathrm{End}_A(S).$$

Equivalently, this is the usual right action of $\mathrm{End}_A(S)^{\mathrm{op}}$ on S.

Finally, if A is a finite-dimensional k-algebra over an algebraically closed field k, then Schur's lemma gives

$$\mathrm{End}_A(S) \cong k.$$

Thus also

$$D_S \cong k,$$

and hence

$$n_S = \dim_{D_S} S = \dim_k S.$$

$\square$

Corollary 8.2.4. *Let A be an Artinian ring, and let U be a finitely generated left A-module. Then U has a projective cover.*

Proof. Since A is Artinian and U is finitely generated, the quotient

$$U/\operatorname{Rad}(U)$$

is semisimple. Hence we may write

$$U/\operatorname{Rad}(U) \cong S_1 \oplus \cdots \oplus S_n,$$

where each S_i is a simple left A-module.

For each i, let

$$\pi_i : P_{S_i} \longrightarrow S_i$$

be the projective cover of S_i. Then the direct sum

$$\pi := \pi_1 \oplus \cdots \oplus \pi_n : P_{S_1} \oplus \cdots \oplus P_{S_n} \longrightarrow S_1 \oplus \cdots \oplus S_n$$

is a projective cover.

Let

$$q : U \longrightarrow U/\operatorname{Rad}(U)$$

be the canonical epimorphism. After identifying

$$U/\operatorname{Rad}(U) \cong S_1 \oplus \cdots \oplus S_n,$$

we may regard π as an epimorphism

$$\pi : P_{S_1} \oplus \cdots \oplus P_{S_n} \longrightarrow U/\operatorname{Rad}(U).$$

Since

$$P := P_{S_1} \oplus \cdots \oplus P_{S_n}$$

is projective and q is surjective, there exists an A-module homomorphism

$$\widetilde{\pi} : P \longrightarrow U$$

such that the following diagram commutes:

$$\begin{tikzcd} & P \arrow[dl, dashed, "\widetilde{\pi}"'] \arrow[d, "\pi"] & \\ U \arrow[r, "q"'] & U/\operatorname{Rad}(U) \arrow[r] & 0. \end{tikzcd}$$

That is,

$$q \circ \widetilde{\pi} = \pi.$$

We claim that $\widetilde{\pi}$ is an essential epimorphism. Since

$$q \circ \widetilde{\pi} = \pi,$$

and since both q and π are essential epimorphisms, it follows from the two-out-of-three property for essential epimorphisms that $\widetilde{\pi}$ is also an essential epimorphism.

Therefore

$$\widetilde{\pi} : P_{S_1} \oplus \cdots \oplus P_{S_n} \longrightarrow U$$

is an essential epimorphism from a projective module onto U. Hence it is a projective cover of U. $\square$

Lemma 8.2.5. *Let A be a ring with identity.*

1. *If A is semisimple, then A is left Artinian if and only if A is left Noetherian.*

2. *If A is left Artinian, then A is left Noetherian.*

Proof. **(1)** Assume that A is semisimple. Then the left regular module ${}_AA$ is semisimple, so it is a direct sum of simple left A-modules:

$${}_AA = \bigoplus_{\lambda \in \Lambda} S_\lambda.$$

For a semisimple module, being Noetherian is equivalent to being a finite direct sum of simple modules, and the same is true for being Artinian. Indeed, if the index set Λ is infinite, then we can form an infinite strictly increasing chain

$$S_{\lambda_1} \subset S_{\lambda_1} \oplus S_{\lambda_2} \subset S_{\lambda_1} \oplus S_{\lambda_2} \oplus S_{\lambda_3} \subset \cdots,$$

so the module is not Noetherian. Similarly, we can form an infinite strictly decreasing chain by removing summands one by one, so the module is not Artinian.

Thus a semisimple module is Noetherian if and only if it is Artinian, namely if and only if it is a finite direct sum of simple modules. Applying this to the left regular module ${}_AA$, we obtain that A is left Artinian if and only if A is left Noetherian.

(2) Assume that A is left Artinian. Let

$$J = J(A)$$

be the Jacobson radical of A. Since A is left Artinian, the radical J is nilpotent. Hence there exists $n \geq 1$ such that

$$J^n = 0.$$

Therefore we have a finite filtration of left A-modules

$$A \supseteq J \supseteq J^2 \supseteq \cdots \supseteq J^n = 0.$$

It is enough to show that each quotient

$$J^i/J^{i+1}$$

is Noetherian as a left A-module. Since J annihilates J^i/J^{i+1}, each quotient is naturally a module over

$$A/J.$$

Moreover, A/J is semisimple Artinian. Hence every A/J-module is semisimple.

Because J^i/J^{i+1} is a quotient of the left Artinian module J^i, it is Artinian. Thus J^i/J^{i+1} is an Artinian semisimple module. By part (1), it is Noetherian.

Now using the finite filtration

$$A \supseteq J \supseteq J^2 \supseteq \cdots \supseteq J^n = 0$$

and the fact that a module is Noetherian whenever it has a finite filtration whose successive quotients are Noetherian, we conclude that ${}_A A$ is Noetherian.

Therefore A is left Noetherian. □

8.3 Multiplicities and the Cartan Matrix

Definition 8.3.1 (Multiplicity of a simple module)**.** *Let A be an Artinian ring, and let V be a finitely generated left A-module. Then V has a composition series*

$$0 = V_n \subset V_{n-1} \subset \cdots \subset V_1 \subset V_0 = V,$$

where each quotient

$$V_{i-1}/V_i$$

is a simple A-module.

For a simple A-module S, the number of composition factors

$$V_{i-1}/V_i$$

which are isomorphic to S is called the multiplicity of S in V*. We denote it by*

$$[V : S].$$

Theorem 8.3.2. *Let A be an Artinian ring, and let $f \in A$ be a primitive idempotent. Put*

$$J = J(A), \qquad \overline{A} = A/J, \qquad e = f + J \in \overline{A}.$$

Assume that e is a primitive idempotent in $\overline{A}$. Then, for every finitely generated left A-module V, the multiplicity of the simple A-module $\overline{A}e$ in V is equal to the *composition length of fV as a left fAf-module. In other words,*

$$[V : \overline{A}e] = \ell_{fAf}(fV).$$

Proof. Let

$$V = V_0 \supset V_1 \supset \cdots \supset V_n = 0$$

be a composition series of V. Applying f gives a descending chain of left fAf-modules

$$fV = fV_0 \supseteq fV_1 \supseteq \cdots \supseteq fV_n = 0.$$

For each i, the quotient V_i/V_{i+1} is a simple A-module, hence it is also a simple $\overline{A}$-module because $J(A)$ annihilates every simple A-module. We claim that

$$V_i/V_{i+1} \cong \overline{A}e \quad \iff \quad f(V_i/V_{i+1}) \neq 0.$$

Indeed, the action of f on a simple A-module is the same as the action of $e = f + J$. Hence, for a simple A-module S,

$$fS \neq 0 \quad \iff \quad eS \neq 0.$$

Moreover,

$$eS \cong \mathrm{Hom}_{\overline{A}}(\overline{A}e, S),$$

and since $\overline{A}e$ and S are simple $\overline{A}$-modules, this space is nonzero if and only if

$$S \cong \overline{A}e.$$

Thus the claim follows.

Now for each i, we have

$$f(V_i/V_{i+1}) \cong \frac{fV_i + V_{i+1}}{V_{i+1}} \cong \frac{fV_i}{fV_i \cap V_{i+1}}.$$

Since f is idempotent, one has

$$fV_i \cap V_{i+1} = fV_{i+1}.$$

Indeed, if $x \in fV_i \cap V_{i+1}$, then $x = fv$ for some $v \in V_i$, and since $x \in V_{i+1}$, we get

$$x = fx \in fV_{i+1}.$$

The reverse inclusion is clear. Therefore

$$f(V_i/V_{i+1}) \cong \frac{fV_i}{fV_{i+1}}.$$

Hence every quotient

$$fV_i/fV_{i+1}$$

is either 0 or a simple fAf-module. Moreover, it is nonzero precisely when

$$V_i/V_{i+1} \cong \overline{A}e.$$

Therefore, after deleting the zero quotients from the chain

$$fV = fV_0 \supseteq fV_1 \supseteq \cdots \supseteq fV_n = 0,$$

we obtain a composition series of fV as an fAf-module.

Consequently, the composition length of fV as an fAf-module is exactly the number of composition factors of V isomorphic to $\overline{A}e$. Hence

$$\ell_{fAf}(fV) = [V : \overline{A}e].$$

□

Definition 8.3.3 (Cartan invariant and Cartan matrix)**.** *Let A be an Artinian ring, and let*

$$J = J(A), \qquad \overline{A} = A/J.$$

Let

$$1 = e_1 + \cdots + e_n$$

be a primitive idempotent decomposition of 1 *in $\overline{A}$. Choose a lifting*

$$1 = f_1 + \cdots + f_n$$

to a primitive idempotent decomposition of 1 *in A, so that*

$$e_i = f_i + J(A) \qquad \text{for all } 1 \leq i \leq n.$$

For $1 \leq i, j \leq n$, define

$$c_{ij} := \left[Af_j : \overline{A}e_i\right],$$

that is, c_{ij} is the multiplicity of the simple A-module $\overline{A}e_i$ as a composition factor of the A-module Af_j.

The integer c_{ij} is called the Cartan invariant *of A. The matrix*

$$C_A = (c_{ij})_{1 \leq i,j \leq n}$$

is called the Cartan matrix *of A.*

Theorem 8.3.4. *Let A be an Artinian ring, and let*

$$J = J(A), \qquad \overline{A} = A/J.$$

Let

$$1 = e_1 + \cdots + e_n$$

be a primitive idempotent decomposition in $\overline{A}$, *and let*

$$1 = f_1 + \cdots + f_n$$

be a primitive idempotent decomposition in A *such that*

$$e_i = f_i + J(A) \qquad \textit{for all } 1 \leq i \leq n.$$

Let

$$c_{ij} = [Af_j : \overline{A}e_i]$$

be the Cartan invariant of A. *Then*

$$c_{ij} \neq 0 \quad \Longleftrightarrow \quad f_i A f_j \neq 0.$$

Proof. By the multiplicity theorem for primitive idempotents, for any finitely generated A-module V, the multiplicity of $\overline{A}e_i$ in V is equal to the composition length of f_iV as an f_iAf_i-module. That is,

$$[V : \overline{A}e_i] = \ell_{f_iAf_i}(f_iV).$$

Now take

$$V = Af_j.$$

Then

$$f_iV = f_i(Af_j) = f_iAf_j.$$

Therefore

$$c_{ij} = [Af_j : \overline{A}e_i] = \ell_{f_iAf_i}(f_iAf_j).$$

Since f_iAf_j is a finite length f_iAf_i-module, its composition length is nonzero if and only if the module itself is nonzero. Hence

$$c_{ij} \neq 0 \quad \Longleftrightarrow \quad \ell_{f_iAf_i}(f_iAf_j) \neq 0 \quad \Longleftrightarrow \quad f_iAf_j \neq 0.$$

This proves the theorem. $\square$

Proposition 8.3.5. *Let* A *be a finite-dimensional* k*-algebra over a field* k. *Let* S *be a simple left* A*-module with projective cover*

$$P_S \twoheadrightarrow S,$$

and let U *be a finite-dimensional left* A*-module.*

1. *If T is a simple left A-module, then*

$$\dim_k \operatorname{Hom}_A(P_S, T) = \begin{cases} \dim_k \operatorname{End}_A(S), & S \cong T, \\ 0, & S \not\cong T. \end{cases}$$

2. *The multiplicity of S in U is given by*

$$[U : S] = \frac{\dim_k \operatorname{Hom}_A(P_S, U)}{\dim_k \operatorname{End}_A(S)}.$$

3. *If $e \in A$ is an idempotent, then*

$$\dim_k \operatorname{Hom}_A(Ae, U) = \dim_k eU.$$

4. *Let $e_S, e_T \in A$ be idempotents such that*

$$P_S = Ae_S, \qquad P_T = Ae_T$$

are projective covers of S and T, respectively. If

$$c_{ST} := [P_T : S]$$

denotes the Cartan invariant, then

$$c_{ST} = \frac{\dim_k \operatorname{Hom}_A(P_S, P_T)}{\dim_k \operatorname{End}_A(S)} = \frac{\dim_k e_S A e_T}{\dim_k \operatorname{End}_A(S)}.$$

In particular, if k is algebraically closed, then

$$c_{ST} = \dim_k \operatorname{Hom}_A(P_S, P_T) = \dim_k e_S A e_T.$$

Proof. Since $P_S \twoheadrightarrow S$ is a projective cover, we have

$$P_S / \operatorname{Rad}(P_S) \cong S.$$

(1) Let T be a simple left A-module. Every homomorphism

$$\varphi : P_S \to T$$

vanishes on $\operatorname{Rad}(P_S)$. Indeed, if $\varphi = 0$, this is clear. If $\varphi \neq 0$, then φ is surjective because T is simple, hence $\ker \varphi$ is a maximal submodule of P_S. Therefore

$$\operatorname{Rad}(P_S) \subseteq \ker \varphi.$$

Thus every map $P_S \to T$ factors uniquely through $P_S / \operatorname{Rad}(P_S)$. Hence

$$\operatorname{Hom}_A(P_S, T) \cong \operatorname{Hom}_A(P_S / \operatorname{Rad}(P_S), T) \cong \operatorname{Hom}_A(S, T).$$

By Schur's lemma,

$$\operatorname{Hom}_A(S,T)=0$$

if $S\not\cong T$, while

$$\operatorname{Hom}_A(S,S)=\operatorname{End}_A(S)$$

if $S\cong T$. Therefore

$$\dim_k \operatorname{Hom}_A(P_S,T)=\begin{cases}\dim_k \operatorname{End}_A(S), & S\cong T,\\ 0, & S\not\cong T.\end{cases}$$

(2) Let

$$U=U_0\supset U_1\supset\cdots\supset U_m=0$$

be a composition series of U. Since P_S is projective, the functor

$$\operatorname{Hom}_A(P_S,-)$$

is exact. Therefore, for each i, applying $\operatorname{Hom}_A(P_S,-)$ to

$$0\longrightarrow U_{i+1}\longrightarrow U_i\longrightarrow U_i/U_{i+1}\longrightarrow 0$$

gives an exact sequence

$$0\longrightarrow \operatorname{Hom}_A(P_S,U_{i+1})\longrightarrow \operatorname{Hom}_A(P_S,U_i)\longrightarrow \operatorname{Hom}_A(P_S,U_i/U_{i+1})\longrightarrow 0.$$

Hence

$$\dim_k \operatorname{Hom}_A(P_S,U)=\sum_{i=0}^{m-1}\dim_k \operatorname{Hom}_A(P_S,U_i/U_{i+1}).$$

By part (1), each composition factor U_i/U_{i+1} contributes $\dim_k \operatorname{End}_A(S)$ precisely when

$$U_i/U_{i+1}\cong S,$$

and contributes 0 otherwise. Therefore

$$\dim_k \operatorname{Hom}_A(P_S,U)=[U:S]\cdot\dim_k \operatorname{End}_A(S).$$

Thus

$$[U:S]=\frac{\dim_k \operatorname{Hom}_A(P_S,U)}{\dim_k \operatorname{End}_A(S)}.$$

(3) Define a map

$$\Phi:\operatorname{Hom}_A(Ae,U)\longrightarrow eU$$

by

$$\Phi(\varphi)=\varphi(e).$$

Since

$$\varphi(e) = \varphi(e^2) = e\varphi(e),$$

we indeed have $\varphi(e) \in eU$.

Conversely, for $u \in eU$, define

$$\psi_u : Ae \longrightarrow U$$

by

$$\psi_u(ae) = au.$$

This is well-defined and A-linear. Moreover,

$$\Phi(\psi_u) = \psi_u(e) = u.$$

Also, for $\varphi \in \mathrm{Hom}_A(Ae, U)$,

$$\psi_{\varphi(e)}(ae) = a\varphi(e) = \varphi(ae).$$

Hence Φ is an isomorphism of k-vector spaces:

$$\mathrm{Hom}_A(Ae, U) \cong eU.$$

Therefore

$$\dim_k \mathrm{Hom}_A(Ae, U) = \dim_k eU.$$

(4) By definition,

$$c_{ST} = [P_T : S].$$

Applying part (2) to $U = P_T$, we obtain

$$c_{ST} = \frac{\dim_k \mathrm{Hom}_A(P_S, P_T)}{\dim_k \mathrm{End}_A(S)}.$$

Since $P_S = Ae_S$ and $P_T = Ae_T$, part (3) gives

$$\mathrm{Hom}_A(P_S, P_T) = \mathrm{Hom}_A(Ae_S, Ae_T) \cong e_S A e_T.$$

Thus

$$\dim_k \mathrm{Hom}_A(P_S, P_T) = \dim_k e_S A e_T.$$

Therefore

$$c_{ST} = \frac{\dim_k \mathrm{Hom}_A(P_S, P_T)}{\dim_k \mathrm{End}_A(S)} = \frac{\dim_k e_S A e_T}{\dim_k \mathrm{End}_A(S)}.$$

If k is algebraically closed, then by Schur's lemma

$$\mathrm{End}_A(S) \cong k,$$

so

$$\dim_k \mathrm{End}_A(S) = 1.$$

Hence

$$c_{ST} = \dim_k \operatorname{Hom}_A(P_S, P_T) = \dim_k e_S A e_T.$$

□

8.4 Integral Extensions

Definition 8.4.1 (Integral element)**.** *Let S be an integral domain, and let $R \subseteq S$ be a subring containing 1_S. Let $\alpha \in S$. We say that α is* integral over R *if there exists a monic polynomial*

$$f(X) \in R[X]$$

such that

$$f(\alpha) = 0.$$

Equivalently, there exist $a_0, a_1, \ldots, a_{n-1} \in R$ such that

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0.$$

Proposition 8.4.2 (Equivalent characterizations of integrality)**.** *Let S be an integral domain, and let $R \subseteq S$ be a subring containing 1_S. For $\alpha \in S$, the following conditions are equivalent:*

1. *α is integral over R.*

2. *$R[\alpha]$ is a finitely generated R-module.*

3. *There exists a subring T of S such that*

$$R \subseteq T, \qquad \alpha \in T,$$

and T is finitely generated as an R-module.

Proof. We prove the implications separately.

(2) ⇒ (3). Take

$$T = R[\alpha].$$

Then T is a subring of S, contains R, contains α, and is finitely generated as an R-module by assumption.

(1) ⇒ (2). Assume that α is integral over R. Then there exists a monic polynomial

$$f(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_1X + a_0 \in R[X]$$

such that $f(\alpha) = 0$. Hence

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0,$$

so

$$\alpha^n = -a_{n-1}\alpha^{n-1} - \cdots - a_1\alpha - a_0.$$

Thus α^n is an R-linear combination of

$$1, \alpha, \alpha^2, \ldots, \alpha^{n-1}.$$

Multiplying this relation by powers of α, we see inductively that every power α^m with $m \geq n$ is also an R-linear combination of

$$1, \alpha, \alpha^2, \ldots, \alpha^{n-1}.$$

Therefore every element of $R[\alpha]$ is an R-linear combination of

$$1, \alpha, \alpha^2, \ldots, \alpha^{n-1}.$$

Hence

$$R[\alpha] = R \cdot 1 + R\alpha + \cdots + R\alpha^{n-1},$$

so $R[\alpha]$ is a finitely generated R-module.

(3) $\Rightarrow$ (1). Assume that there exists a subring T of S such that

$$R \subseteq T, \qquad \alpha \in T,$$

and T is finitely generated as an R-module.

Let

$$\varepsilon_1, \ldots, \varepsilon_n$$

be a set of generators of T as an R-module. Since $\alpha \in T$ and T is a subring, for each i we have

$$\alpha\varepsilon_i \in T.$$

Hence there exist elements $a_{ji} \in R$ such that

$$\alpha\varepsilon_i = \sum_{j=1}^{n} a_{ji}\varepsilon_j, \qquad 1 \leq i \leq n.$$

Let

$$A = (a_{ji}) \in M_n(R).$$

Writing

$$\varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix},$$

the above relations can be written as

$$(\alpha I_n - A)\varepsilon = 0.$$

Multiplying both sides by the adjugate matrix

$$\operatorname{adj}(\alpha I_n - A),$$

we obtain

$$\det(\alpha I_n - A)\varepsilon = 0.$$

Thus

$$\det(\alpha I_n - A)\varepsilon_i = 0 \qquad \text{for all } i = 1, \dots, n.$$

Since $\varepsilon_1, \dots, \varepsilon_n$ generate T as an R-module, it follows that

$$\det(\alpha I_n - A)\, T = 0.$$

But $1 \in T$, so

$$\det(\alpha I_n - A) = 0.$$

Now define

$$p(X) = \det(XI_n - A) \in R[X].$$

This is a monic polynomial of degree n. Since

$$p(\alpha) = \det(\alpha I_n - A) = 0,$$

we conclude that α is integral over R. ☐

Lemma 8.4.3 (Finite generation of a generated subring)**.** *Let S be an integral domain, and let $R \subseteq S$ be a subring containing 1_S. Let M_1, M_2 be subrings of S such that*

$$R \subseteq M_i \subseteq S, \qquad i = 1, 2.$$

If M_1 and M_2 are finitely generated as R-modules, then the subring

$$R[M_1, M_2] \subseteq S$$

generated by R, M_1, M_2 is finitely generated as an R-module.

Proof. Suppose

$$M_1 = R\alpha_1 + \cdots + R\alpha_m$$

as an R-module, and

$$M_2 = R\beta_1 + \cdots + R\beta_n$$

as an R-module.

We claim that $R[M_1, M_2]$ is generated as an R-module by the finite set

$$\{\alpha_i\beta_j \mid 1 \leq i \leq m,\ 1 \leq j \leq n\}.$$

Set

$$N = \sum_{i=1}^{m} \sum_{j=1}^{n} R\alpha_i\beta_j.$$

Clearly $N \subseteq R[M_1, M_2]$. We show that N is a subring of S containing M_1 and M_2.

Since $1 \in M_2$, every $\alpha_i \in M_1$ can be written as

$$\alpha_i = \alpha_i \cdot 1 \in N.$$

Similarly, since $1 \in M_1$, every $\beta_j \in M_2$ belongs to N. Hence

$$M_1 \subseteq N, \qquad M_2 \subseteq N.$$

It remains to show that N is closed under multiplication. Take two generators $\alpha_i\beta_j$ and $\alpha_{i'}\beta_{j'}$. Then

$$(\alpha_i\beta_j)(\alpha_{i'}\beta_{j'}) = (\alpha_i\alpha_{i'})(\beta_j\beta_{j'}).$$

Since M_1 and M_2 are subrings, we have

$$\alpha_i\alpha_{i'} \in M_1, \qquad \beta_j\beta_{j'} \in M_2.$$

Therefore there exist $r_k, s_\ell \in R$ such that

$$\alpha_i\alpha_{i'} = \sum_{k=1}^{m} r_k\alpha_k, \qquad \beta_j\beta_{j'} = \sum_{\ell=1}^{n} s_\ell\beta_\ell.$$

Thus

$$(\alpha_i\alpha_{i'})(\beta_j\beta_{j'}) = \sum_{k=1}^{m} \sum_{\ell=1}^{n} r_k s_\ell \, \alpha_k\beta_\ell \in N.$$

Hence N is closed under multiplication.

Therefore N is a subring of S containing M_1 and M_2. By minimality of $R[M_1, M_2]$, we get

$$R[M_1, M_2] \subseteq N.$$

Since the reverse inclusion is clear, we have

$$R[M_1, M_2] = N.$$

Hence $R[M_1, M_2]$ is finitely generated as an R-module. □

> **Theorem 8.4.4** (Integral elements form a subring)**.** *Let S be an integral domain, and let $R \subseteq S$ be a subring containing 1_S. Then the set of integral elements of S over R forms a subring of S containing R.*

Proof. Let $\alpha, \beta \in S$ be integral over R. By the characterization of integral elements, $R[\alpha]$ and $R[\beta]$ are finitely generated R-modules. Hence, by the finite-generation lemma for generated subrings,

$$R[\alpha, \beta]$$

is also a finitely generated R-module.

Since

$$\alpha + \beta, \quad \alpha - \beta, \quad \alpha\beta \in R[\alpha, \beta],$$

again by the characterization of integrality, the elements

$$\alpha + \beta, \qquad \alpha - \beta, \qquad \alpha\beta$$

are integral over R.

Moreover, every element $r \in R$ is integral over R, since it satisfies the monic polynomial

$$X - r \in R[X].$$

Therefore the integral elements of S over R form a subring of S containing R. □

Definition 8.4.5 (Algebraic integer)**.** *Let $\mathbb{Z} \subseteq \mathbb{C}$. If $z \in \mathbb{C}$ is integral over $\mathbb{Z}$, then z is called an* algebraic integer*.*

Definition 8.4.6 (Integrally closed subring)**.** *Let S be an integral domain, and let $R \subseteq S$ be a subring containing 1_S. We say that R is* integrally closed in S *if every element of S which is integral over R already lies in R.*

Definition 8.4.7 (Integrally closed domain)**.** *Let R be an integral domain with quotient field $K = \mathrm{Frac}(R)$. We say that R is* integrally closed *if R is integrally closed in its quotient field K. Equivalently, every element of K integral over R belongs to R.*

Theorem 8.4.8. *Every unique factorization domain is integrally closed.*

Proof. Let R be a unique factorization domain, and let

$$K = \mathrm{Frac}(R)$$

be its quotient field. We show that R is integrally closed in K.

Let $\alpha \in K$ be integral over R. Write

$$\alpha = \frac{b}{a},$$

where $a, b \in R$, $a \neq 0$, and $\gcd(a, b) = 1$.

Since α is integral over R, there exist $n \geq 1$ and $c_0, c_1, \ldots, c_{n-1} \in R$ such that

$$\alpha^n + c_{n-1}\alpha^{n-1} + \cdots + c_1\alpha + c_0 = 0.$$

Substituting $\alpha = b/a$, we get

$$\left(\frac{b}{a}\right)^n + c_{n-1}\left(\frac{b}{a}\right)^{n-1} + \cdots + c_1\left(\frac{b}{a}\right) + c_0 = 0.$$

Multiplying by a^n, we obtain

$$b^n + c_{n-1}b^{n-1}a + \cdots + c_1ba^{n-1} + c_0a^n = 0.$$

Thus

$$b^n = -a\left(c_{n-1}b^{n-1} + c_{n-2}b^{n-2}a + \cdots + c_1ba^{n-2} + c_0a^{n-1}\right).$$

Hence

$$a \mid b^n.$$

If a is not a unit, then a has a prime divisor p. Since $a \mid b^n$, we have

$$p \mid b^n.$$

Because p is prime in the UFD R, it follows that

$$p \mid b.$$

But $p \mid a$ as well, contradicting the assumption that $\gcd(a, b) = 1$.

Therefore a must be a unit in R. Hence

$$\alpha = \frac{b}{a} \in R.$$

So every element of K integral over R belongs to R. Therefore R is integrally closed. □

8.5 Valuations

Definition 8.5.1 (Valuation)**.** *Let K be a field. A* valuation *on K is a function*

$$v : K^\times \longrightarrow \mathbb{R},$$

where $K^\times = K \setminus \{0\}$, satisfying the following conditions:

1. *For all $a, b \in K^\times$,*
$$v(ab) = v(a) + v(b).$$
2. *For all $a, b \in K^\times$ with $a + b \neq 0$,*
$$v(a + b) \geq \min\{v(a), v(b)\}.$$

We extend v to K by setting

$$v(0) = +\infty.$$

With this convention, the second condition can be written as

$$v(a+b) \geq \min\{v(a), v(b)\}$$

for all $a, b \in K$.

Proposition 8.5.2. *Let v be a valuation on a field K. Then*

$$v : (K^\times, \cdot) \longrightarrow (\mathbb{R}, +)$$

is a group homomorphism.

Proof. By definition, for all $a, b \in K^\times$,

$$v(ab) = v(a) + v(b).$$

Also,

$$0 = v(1) = v(aa^{-1}) = v(a) + v(a^{-1}),$$

so

$$v(a^{-1}) = -v(a).$$

Therefore v is a group homomorphism from $K^\times$ to the additive group $(\mathbb{R}, +)$. $\square$

Definition 8.5.3 (Value group)**.** *Let (K, v) be a valued field. The subgroup*

$$v(K^\times) \subseteq (\mathbb{R}, +)$$

is called the value group *of v.*

A field K together with a valuation v is called a valued field*, and is denoted by*

$$(K, v).$$

Proposition 8.5.4 (Basic properties of valuations)**.** *Let v be a valuation on a field K. Then for all $a, b \in K^\times$, the following hold:*

1. $$v(1) = 0, \qquad v(-1) = 0.$$

2. $$v(-a) = v(a).$$

3. $$v(a^{-1}) = -v(a).$$

4. *If*

$$v(b) < v(a),$$

then

$$v(a+b) = v(b).$$

Proof. First,

$$v(1) = v(1 \cdot 1) = v(1) + v(1),$$

so

$$v(1) = 0.$$

Since $(-1)^2 = 1$, we have

$$2v(-1) = v((-1)^2) = v(1) = 0,$$

and hence

$$v(-1) = 0.$$

For $a \in K^\times$,

$$v(-a) = v((-1)a) = v(-1) + v(a) = v(a).$$

Also,

$$0 = v(1) = v(aa^{-1}) = v(a) + v(a^{-1}),$$

so

$$v(a^{-1}) = -v(a).$$

Finally, suppose that $v(b) < v(a)$. By the valuation inequality,

$$v(a+b) \geq \min\{v(a), v(b)\} = v(b).$$

We claim that equality holds. If instead

$$v(a+b) > v(b),$$

then using

$$b = (a+b) - a,$$

we get

$$v(b) \geq \min\{v(a+b), v(a)\}.$$

But both $v(a+b)$ and $v(a)$ are strictly larger than $v(b)$, so the right-hand side is strictly larger than $v(b)$, a contradiction. Therefore

$$v(a+b) = v(b).$$

□

Chapter 9

Lecture 9

9.1 Local rings

Definition 9.1.1 (Local ring). *A ring R is called* local *if the set of non-units of R forms an ideal of R.*

Theorem 9.1.2. *Let R be a ring with identity. The following conditions are equivalent:*

1. *R is local.*
2. *The set of non-units of R coincides with $J(R)$. Equivalently, $J(R)$ is the unique maximal left ideal of R and also the unique maximal right ideal of R.*
3. *$R/J(R)$ is a division ring.*

Proof. Let M denote the set of non-units of R.

(1) ⇒ (2). Assume that R is local. By definition, M is an ideal of R. Since $1 \notin M$, the ideal M is proper.

We claim that M contains every proper left ideal of R. Indeed, let L be a proper left ideal of R. If L contained a unit u, then $1 = u^{-1}u \in L$, and hence $L = R$, a contradiction. Therefore every element of L is a non-unit, so

$$L \subseteq M.$$

Thus M is the unique maximal left ideal of R.

Similarly, M contains every proper right ideal of R, and hence M is also the unique maximal right ideal of R.

Since the Jacobson radical is the intersection of all maximal left ideals, we get

$$J(R) = M.$$

Hence the set of non-units of R coincides with $J(R)$.

(2) ⇒ (3). Assume that the set of non-units of R is exactly $J(R)$. Let

$$\overline{a} = a + J(R) \in R/J(R)$$

be nonzero. Then $a \notin J(R)$, so by assumption a is a unit in R. Hence there exists $a^{-1} \in R$ such that

$$aa^{-1} = a^{-1}a = 1.$$

Passing to the quotient gives

$$\overline{a}\,\overline{a^{-1}} = \overline{a^{-1}}\,\overline{a} = \overline{1}.$$

Thus every nonzero element of $R/J(R)$ is invertible, so $R/J(R)$ is a division ring.

(3) ⇒ (1). Assume that $R/J(R)$ is a division ring. We show that the set of non-units of R is precisely $J(R)$.

First, every element of $J(R)$ is a non-unit, because $J(R)$ is contained in every maximal left ideal, hence is a proper ideal.

Conversely, suppose $a \notin J(R)$. Then

$$\overline{a} = a + J(R) \neq 0$$

in $R/J(R)$. Since $R/J(R)$ is a division ring, $\overline{a}$ is invertible. Hence there exist $b, d \in R$ such that

$$\overline{b}\,\overline{a} = \overline{1}, \qquad \overline{a}\,\overline{d} = \overline{1}.$$

Equivalently, there exist $c, c' \in J(R)$ such that

$$ba = 1 - c, \qquad ad = 1 - c'.$$

Since $c, c' \in J(R)$, the elements $1 - c$ and $1 - c'$ are units in R. Therefore ba is a unit and ad is a unit.

Since ba is a unit, a has a left inverse. Indeed,

$$(ba)^{-1}ba = 1,$$

so

$$\big((ba)^{-1}b\big)a = 1.$$

Similarly, since ad is a unit, a has a right inverse:

$$a\big(d(ad)^{-1}\big) = 1.$$

Thus a has both a left inverse and a right inverse, hence a is a unit.

Therefore every element outside $J(R)$ is a unit, so the set of non-units is exactly $J(R)$. Since $J(R)$ is an ideal, the set of non-units forms an ideal. Hence R is local. □

Theorem 9.1.3 (Idempotents in a local ring)**.** *Let R be a local ring. Then R has no non-trivial idempotents. In other words, if $e \in R$ satisfies*

$$e^2 = e,$$

then

$$e = 0 \qquad \text{or} \qquad e = 1.$$

Proof. Let $e \in R$ be an idempotent, so

$$e^2 = e.$$

Then

$$e(1-e) = e - e^2 = 0.$$

We first observe that if e is a unit, then $e = 1$. Indeed, multiplying $e^2 = e$ by e^{-1} gives

$$e = 1.$$

Similarly, if $1 - e$ is a unit, then from

$$e(1-e) = 0$$

we get

$$e = 0.$$

Therefore, if $e \neq 0$ and $e \neq 1$, then neither e nor $1 - e$ is a unit. Since R is local, the set of non-units of R is precisely the Jacobson radical $J(R)$. Hence

$$e \in J(R), \qquad 1 - e \in J(R).$$

Because $J(R)$ is an ideal, it follows that

$$1 = e + (1-e) \in J(R).$$

This is impossible, since $J(R)$ is a proper ideal of R.

Hence our assumption $e \neq 0, 1$ is false. Therefore every idempotent of R is either 0 or 1. □

9.2 Additive valuations, discrete valuations, and DVRs

Definition 9.2.1 (Valuation ring and valuation ideal)**.** *Let (K, v) be a valued field. Define*

$$R_v := \{a \in K \mid v(a) \geq 0\}$$

and

$$P_v := \{a \in K \mid v(a) > 0\}.$$

Then R_v is called the valuation ring *of v, and P_v is called the* valuation ideal *of v.*

Proposition 9.2.2. *Let (K, v) be a valued field. Then:*

1. *R_v is a subring of K.*

2. *P_v is an ideal of R_v.*

3. *The group of units of R_v is*

$$R_v^\times = \{a \in K^\times \mid v(a) = 0\}.$$

4. *One has*

$$K = R_v \cup R_v^{-1},$$

where

$$R_v^{-1} := \{a^{-1} \mid a \in R_v,\ a \neq 0\}.$$

In particular, K is the field of fractions of R_v.

5. *R_v is a local ring and*

$$J(R_v) = P_v.$$

Equivalently, P_v is precisely the set of non-units of R_v.

6. *The quotient field*

$$\kappa(v) := R_v/P_v$$

is called the residue field *of v.*

Proof. First, $0, 1 \in R_v$. If $a, b \in R_v$, then

$$v(ab) = v(a) + v(b) \geq 0,$$

so $ab \in R_v$. Also,

$$v(a + b) \geq \min\{v(a), v(b)\} \geq 0,$$

so $a + b \in R_v$. Finally, since $v(-a) = v(a)$, we have $-a \in R_v$. Therefore R_v is a subring of K.

Now let $a, b \in P_v$. Then

$$v(a + b) \geq \min\{v(a), v(b)\} > 0,$$

so $a + b \in P_v$. If $r \in R_v$ and $a \in P_v$, then

$$v(ra) = v(r) + v(a) > 0.$$

Thus $ra \in P_v$, and hence P_v is an ideal of R_v.

Next, let $a \in R_v$. If a is a unit of R_v, then $a^{-1} \in R_v$, so

$$0 = v(1) = v(aa^{-1}) = v(a) + v(a^{-1}).$$

Since $v(a) \geq 0$ and $v(a^{-1}) \geq 0$, we get $v(a) = 0$. Conversely, if $v(a) = 0$, then

$$v(a^{-1}) = -v(a) = 0,$$

so $a^{-1} \in R_v$. Therefore

$$R_v^\times = \{a \in K^\times \mid v(a) = 0\}.$$

For every $a \in K^\times$, either $v(a) \geq 0$, in which case $a \in R_v$, or $v(a) < 0$, in which case

$$v(a^{-1}) = -v(a) > 0,$$

so $a^{-1} \in R_v$. Hence

$$K = R_v \cup R_v^{-1}.$$

In particular, every element of K is a fraction of elements of R_v, so

$$K = \mathrm{Frac}(R_v).$$

Finally, the non-units of R_v are exactly the elements $a \in R_v$ satisfying $v(a) > 0$, namely the elements of P_v. Hence R_v has a unique maximal ideal P_v, so R_v is local and

$$J(R_v) = P_v.$$

Consequently,

$$\kappa(v) := R_v/P_v$$

is a field. □

Definition 9.2.3 (Equivalent valuations)**.** *Let v be a valuation on K, and let $\lambda \in \mathbb{R}_{>0}$. Define*

$$v_\lambda(a) := \lambda v(a), \qquad a \in K^\times.$$

Then v_λ is again a valuation on K. We say that v is equivalent *to v_λ, and write*

$$v \sim v_\lambda.$$

Proposition 9.2.4. *Equivalent valuations have the same valuation ring, valuation ideal, group of units, and residue field.*

Proof. Let $v_\lambda = \lambda v$, where $\lambda > 0$. Then for $a \in K^\times$,

$$v_\lambda(a) \geq 0 \iff \lambda v(a) \geq 0 \iff v(a) \geq 0.$$

Hence

$$R_{v_\lambda} = R_v.$$

Similarly,

$$v_\lambda(a) > 0 \iff v(a) > 0,$$

so

$$P_{v_\lambda} = P_v.$$

Also,

$$v_\lambda(a) = 0 \iff v(a) = 0,$$

and therefore

$$R_{v_\lambda}^\times = R_v^\times.$$

Since the valuation rings and valuation ideals are the same, the residue fields are also the same:

$$R_{v_\lambda}/P_{v_\lambda} = R_v/P_v.$$

□

Definition 9.2.5 (Discrete valuation). *A valuation v of K is called* discrete *if its value group is isomorphic to $\mathbb{Z}$, that is,*

$$v(K^\times) \cong \mathbb{Z}.$$

If moreover

$$v(K^\times) = \mathbb{Z},$$

then v is called a normalized discrete valuation.

Remark 9.2.6. *If v is a discrete valuation, then after replacing v by an equivalent valuation λv for a suitable $\lambda > 0$, one may assume that*

$$v(K^\times) = \mathbb{Z}.$$

Thus every discrete valuation is equivalent to a normalized discrete valuation.

Definition 9.2.7. *An element $\pi \in R$ satisfying*

$$v(\pi) = 1$$

is called a uniformizer.

Theorem 9.2.8. *Let (K, v) be a field with a normalized discrete valuation. Let*

$$R = \{x \in K \mid v(x) \geq 0\}$$

be its valuation ring, and let

$$P = \{x \in R \mid v(x) > 0\}$$

be its maximal ideal. Suppose $\pi \in R$ satisfies $v(\pi) = 1$. Then:

1. *For every $\alpha \in K^\times$, one can write*

$$\alpha = \pi^{v(\alpha)} u$$

   for some unit $u \in R^\times$.

2. *Every nonzero ideal I of R is of the form*

$$I = (\pi^i)$$

   for some integer $i \geq 0$. In particular,

$$P = (\pi), \qquad I = P^i,$$

   In particular, R is a principal ideal domain.

3. *The ring R is integrally closed.*

Proof. **(1).** Let $\alpha \in K^\times$, and set

$$i = v(\alpha).$$

Since $v(\pi) = 1$, we have

$$v(\pi^{-i}\alpha) = v(\alpha) - iv(\pi) = i - i = 0.$$

Therefore

$$u := \pi^{-i}\alpha \in R^\times.$$

Hence

$$\alpha = \pi^i u = \pi^{v(\alpha)} u,$$

as desired.

(2). Let $I \neq 0$ be an ideal of R. Since $I \subseteq R$, for every nonzero $a \in I$ we have

$$v(a) \geq 0.$$

Thus the set

$$\{v(a) \mid 0 \neq a \in I\} \subseteq \mathbb{Z}_{\geq 0}$$

is nonempty, and therefore has a minimal element. Let

$$i = \min\{v(a) \mid 0 \neq a \in I\}.$$

Choose $a_0 \in I$ such that

$$v(a_0) = i.$$

By part (1), we may write

$$a_0 = \pi^i u$$

for some $u \in R^\times$. Hence

$$\pi^i = a_0 u^{-1} \in I.$$

Therefore

$$(\pi^i) \subseteq I.$$

Conversely, let $a \in I$ be nonzero. Write

$$v(a) = j.$$

By the minimality of i, we have $j \geq i$. Again by part (1), there exists $u \in R^\times$ such that

$$a = \pi^j u.$$

Then

$$a = \pi^i \big(\pi^{j-i} u\big).$$

Since $j - i \geq 0$, we have $\pi^{j-i} u \in R$. Therefore

$$a \in (\pi^i).$$

Hence

$$I \subseteq (\pi^i).$$

Combining the two inclusions, we obtain

$$I = (\pi^i).$$

In particular, applying this to the maximal ideal P, we get

$$P = (\pi).$$

Therefore every nonzero ideal of R is of the form

$$I = (\pi^i) = P^i.$$

Since the zero ideal is also principal, R is a principal ideal domain.

(3). By part (2), the ring R is a principal ideal domain. Hence R is a unique factorization domain. Since every unique factorization domain is integrally closed, it follows that R is integrally closed. □

Corollary 9.2.9. *Let v and v' be two discrete valuations on a field K. Then v and v' are equivalent if and only if they have the same valuation ideal, that is,*

$$P_v = P_{v'},$$

where

$$P_v = \{x \in K \mid v(x) > 0\}, \qquad P_{v'} = \{x \in K \mid v'(x) > 0\}.$$

Proof. If v and v' are equivalent, then there exists $\lambda \in \mathbb{R}_{>0}$ such that

$$v' = \lambda v.$$

Hence, for every $x \in K$,

$$v'(x) > 0 \iff \lambda v(x) > 0 \iff v(x) > 0.$$

Therefore

$$P_v = P_{v'}.$$

Conversely, suppose that

$$P_v = P_{v'} =: P.$$

We first show that the valuation ring can be recovered from the valuation ideal. Namely,

$$R_v = K \setminus P^{-1},$$

where

$$P^{-1} := \{a^{-1} \mid a \in P,\ a \neq 0\}.$$

Indeed, if $x \in R_v$, then $v(x) \geq 0$. Hence x cannot be of the form a^{-1} with $a \in P$, because then

$$v(x) = v(a^{-1}) = -v(a) < 0.$$

Thus $x \notin P^{-1}$. Conversely, if $x \notin R_v$, then $x \neq 0$ and $v(x) < 0$. Hence

$$v(x^{-1}) = -v(x) > 0,$$

so $x^{-1} \in P$, and therefore

$$x = (x^{-1})^{-1} \in P^{-1}.$$

This proves

$$R_v = K \setminus P^{-1}.$$

Since $P_v = P_{v'}$, we get

$$R_v = R_{v'} =: R.$$

Now replace v and v' by equivalent normalized discrete valuations. This does not change the valuation ring or the valuation ideal. Hence we may assume that

$$v(K^\times) = \mathbb{Z}, \qquad v'(K^\times) = \mathbb{Z}.$$

Choose $\pi \in K^\times$ such that

$$v(\pi) = 1.$$

Then π is a uniformizer for v, so by Theorem 9.0.11,

$$P = (\pi)$$

as an ideal of R.

Let π' be a uniformizer for v'. Then similarly

$$P = (\pi').$$

Since

$$(\pi) = (\pi'),$$

there exist $r, s \in R$ such that

$$\pi = r\pi', \qquad \pi' = s\pi.$$

Applying v', we obtain

$$v'(\pi) = v'(r) + v'(\pi') \geq 1,$$

and

$$1 = v'(\pi') = v'(s) + v'(\pi) \geq v'(\pi).$$

Therefore

$$v'(\pi) = 1.$$

Now let $a \in K^\times$. By Theorem 9.0.11, there exists $u \in R^\times$ such that

$$a = \pi^{v(a)}u.$$

Since $R = R_{v'}$, the units of R are exactly the elements of valuation 0 with respect to v'. Hence

$$v'(u) = 0.$$

Therefore

$$v'(a) = v(a)\, v'(\pi) + v'(u) = v(a).$$

Thus $v' = v$ after normalization.

Hence the original discrete valuations differ by multiplication by a positive real constant. Therefore v and v' are equivalent. □

Theorem 9.2.10 (Discrete valuation rings and local PIDs)**.** *Let R be an integral domain which is not a field, and let $K = \mathrm{Frac}(R)$. Then R is a discrete valuation* *ring if and only if R is a local principal ideal domain.*

Proof. Suppose first that R is a discrete valuation ring. Then there exists a normalized discrete valuation

$$v : K^{\times} \longrightarrow \mathbb{Z}$$

such that

$$R = \{x \in K \mid v(x) \geq 0\}.$$

By the structure theorem for valuation rings of normalized discrete valuations, R is a local principal ideal domain. More precisely, if $\pi \in R$ satisfies $v(\pi) = 1$, then the maximal ideal of R is

$$P = \{x \in R \mid v(x) > 0\} = (\pi),$$

and every nonzero ideal of R is of the form

$$P^n = (\pi^n)$$

for some integer $n \geq 0$. Hence R is a local PID.

Conversely, suppose that R is a local principal ideal domain. Since R is not a field, its unique maximal ideal is nonzero. Denote it by

$$P = J(R).$$

Since R is a PID, there exists a nonzero nonunit $\pi \in R$ such that

$$P = (\pi).$$

We first claim that

$$\bigcap_{n \geq 1} P^n = 0.$$

Suppose, for contradiction, that

$$0 \neq a \in \bigcap_{n \geq 1} P^n.$$

Since $P^n = (\pi^n)$, for each $n \geq 1$ there exists $a_n \in R$ such that

$$a = \pi^n a_n.$$

Then

$$\pi^n a_n = \pi^{n+1} a_{n+1}.$$

Since R is an integral domain and $\pi \neq 0$, we may cancel π^n, obtaining

$$a_n = \pi a_{n+1}.$$

Hence

$$(a_n) \subseteq (a_{n+1}).$$

This inclusion is strict. Indeed, if $(a_n) = (a_{n+1})$, then $a_{n+1} = ra_n$ for some $r \in R$. Hence

$$a_{n+1} = r\pi a_{n+1}.$$

Since $a_{n+1} \neq 0$, we get $1 = r\pi$, which means that π is a unit, a contradiction. Thus we obtain a strictly ascending chain of ideals

$$(a_1) \subsetneq (a_2) \subsetneq (a_3) \subsetneq \cdots .$$

This contradicts the fact that R is Noetherian, because every PID is Noetherian. Therefore

$$\bigcap_{n \geq 1} P^n = 0.$$

Now let $0 \neq a \in R$. Since $P^0 = R$, the set

$$\{n \geq 0 \mid a \in P^n\}$$

is nonempty. By the claim above, a cannot belong to all powers P^n. Hence there exists a unique integer $n \geq 0$ such that

$$a \in P^n \qquad \text{and} \qquad a \notin P^{n+1}.$$

Since $P^n = (\pi^n)$, we may write

$$a = \pi^n u$$

for some $u \in R$. If $u \in P$, then

$$a = \pi^n u \in \pi^n P = P^{n+1},$$

contradicting the choice of n. Thus $u \notin P$. Since R is local and P is the set of nonunits, it follows that

$$u \in R^\times.$$

Therefore every nonzero element $a \in R$ can be written uniquely in the form

$$a = \pi^n u, \qquad n \geq 0, \quad u \in R^\times.$$

We now extend this to $K^\times$. Every $x \in K^\times$ can be written as

$$x = \frac{a}{b}, \qquad a, b \in R \setminus \{0\}.$$

Write

$$a = \pi^m u, \qquad b = \pi^n w,$$

where $m, n \geq 0$ and $u, w \in R^\times$. Then

$$x = \pi^{m-n} u w^{-1}.$$

Hence every $x \in K^\times$ has a representation

$$x = \pi^r u, \qquad r \in \mathbb{Z}, \quad u \in R^\times.$$

This representation is unique. Indeed, if

$$\pi^r u = \pi^s w, \qquad u, w \in R^\times,$$

and say $r < s$, then

$$u = \pi^{s-r} w \in P,$$

contradicting that u is a unit. Hence $r = s$.

Define

$$v : K^\times \longrightarrow \mathbb{Z}$$

by

$$v(\pi^r u) = r, \qquad r \in \mathbb{Z}, \quad u \in R^\times.$$

This is well-defined by the uniqueness just proved.

We verify that v is a valuation. If

$$x = \pi^r u, \qquad y = \pi^s w,$$

with $u, w \in R^\times$, then

$$xy = \pi^{r+s} uw,$$

and $uw \in R^\times$. Therefore

$$v(xy) = r + s = v(x) + v(y).$$

Next, suppose $x + y \neq 0$. Without loss of generality, assume $r \leq s$. Then

$$x + y = \pi^r \left(u + \pi^{s-r} w \right).$$

Since

$$u + \pi^{s-r} w \in R,$$

its valuation is nonnegative whenever it is nonzero. Thus

$$v(x + y) \geq r = \min\{r, s\} = \min\{v(x), v(y)\}.$$

Hence v is a valuation.

Moreover,

$$v(\pi) = 1,$$

so

$$v(K^\times) = \mathbb{Z}.$$

Therefore v is a normalized discrete valuation.

Finally, we show that R is exactly the valuation ring of v. Let

$$R_v := \{x \in K \mid v(x) \geq 0\}.$$

If $x \in R \setminus \{0\}$, then by the decomposition above,

$$x = \pi^n u$$

with $n \geq 0$, so $v(x) = n \geq 0$. Hence $R \subseteq R_v$.

Conversely, if $x \in R_v \setminus \{0\}$, write

$$x = \pi^r u, \qquad r \in \mathbb{Z}, \quad u \in R^\times.$$

Since $v(x) = r \geq 0$, we have

$$x = \pi^r u \in R.$$

Hence $R_v \subseteq R$.

Therefore

$$R = R_v.$$

Thus R is the valuation ring of a normalized discrete valuation, and hence R is a discrete valuation ring. □

9.3 Multiplicative valuations

Definition 9.3.1 (Multiplicative valuation associated to a discrete valuation)**.** *Let K be a field, and let*

$$v : K^\times \longrightarrow \mathbb{Z}$$

be a normalized discrete valuation. We extend v to K by setting

$$v(0) = +\infty.$$

Fix a real number

$$0 < \gamma < 1.$$

Define a function

$$\varphi : K \longrightarrow \mathbb{R}_{\geq 0}$$

by

$$\varphi(a) = \begin{cases} \gamma^{v(a)}, & a \neq 0, \\ 0, & a = 0. \end{cases}$$

Then φ is called the multiplicative valuation associated to v with parameter γ.

Proposition 9.3.2. *The function*

$$\varphi : K \longrightarrow \mathbb{R}_{\geq 0}$$

defined above satisfies the following properties:

1. $\varphi(a) = 0$ *if and only if* $a = 0$.

2. *For all* $a, b \in K$,
$$\varphi(ab) = \varphi(a)\varphi(b).$$

3. *For all* $a, b \in K$,
$$\varphi(a + b) \leq \max\{\varphi(a), \varphi(b)\}.$$

Proof. The first assertion follows immediately from the definition, because

$$\gamma^{v(a)} > 0$$

whenever $a \neq 0$, while $\varphi(0) = 0$.

For the multiplicativity, if $a, b \neq 0$, then

$$v(ab) = v(a) + v(b).$$

Hence

$$\varphi(ab) = \gamma^{v(ab)} = \gamma^{v(a)+v(b)} = \gamma^{v(a)}\gamma^{v(b)} = \varphi(a)\varphi(b).$$

If $a = 0$ or $b = 0$, both sides are zero, so the formula still holds.

Finally, assume first that $a + b \neq 0$. Since v is a valuation,

$$v(a + b) \geq \min\{v(a), v(b)\}.$$

Because $0 < \gamma < 1$, the function $x \mapsto \gamma^x$ is decreasing. Therefore

$$\gamma^{v(a+b)} \leq \gamma^{\min\{v(a),v(b)\}} = \max\{\gamma^{v(a)}, \gamma^{v(b)}\}.$$

Thus

$$\varphi(a + b) \leq \max\{\varphi(a), \varphi(b)\}.$$

If $a + b = 0$, then $\varphi(a + b) = 0$, and the same inequality is obvious. $\square$

Definition 9.3.3 (Multiplicative valuation)**.** *Let* K *be a field. A function*

$$\varphi : K \longrightarrow \mathbb{R}_{\geq 0}$$

is called a multiplicative valuation, or a non-archimedean absolute value, if it satisfies:

1. $\varphi(a) = 0$ *if and only if* $a = 0$;

2.

$$\varphi(ab) = \varphi(a)\varphi(b)$$

for all $a, b \in K$;

3.

$$\varphi(a+b) \leq \max\{\varphi(a), \varphi(b)\}$$

for all $a, b \in K$.

Proposition 9.3.4 (Basic properties of multiplicative valuations)**.** *Let*

$$\varphi : K \longrightarrow \mathbb{R}_{\geq 0}$$

be a multiplicative valuation. Then for all $a, b \in K^\times$, the following hold:

1.

$$\varphi(1) = 1, \qquad \varphi(-1) = 1.$$

2.

$$\varphi(-a) = \varphi(a).$$

3.

$$\varphi(a^{-1}) = \varphi(a)^{-1}.$$

4. *If*

$$\varphi(a) < \varphi(b),$$

then

$$\varphi(a+b) = \varphi(b).$$

Proof. Since $1 \cdot 1 = 1$, we have

$$\varphi(1) = \varphi(1)^2.$$

Since $\varphi(1) \neq 0$, it follows that

$$\varphi(1) = 1.$$

Also, since $(-1)^2 = 1$, we get

$$\varphi(-1)^2 = \varphi(1) = 1.$$

Since $\varphi(-1) > 0$, we have

$$\varphi(-1) = 1.$$

Hence, for $a \in K^\times$,

$$\varphi(-a) = \varphi((-1)a) = \varphi(-1)\varphi(a) = \varphi(a).$$

Also,

$$1 = \varphi(1) = \varphi(aa^{-1}) = \varphi(a)\varphi(a^{-1}),$$

so

$$\varphi(a^{-1}) = \varphi(a)^{-1}.$$

Finally, suppose that

$$\varphi(a) < \varphi(b).$$

By the non-archimedean triangle inequality,

$$\varphi(a+b) \leq \max\{\varphi(a), \varphi(b)\} = \varphi(b).$$

On the other hand,

$$b = (a+b) - a.$$

Hence

$$\varphi(b) \leq \max\{\varphi(a+b), \varphi(a)\}.$$

If $\varphi(a+b) < \varphi(b)$, then both $\varphi(a+b)$ and $\varphi(a)$ are strictly smaller than $\varphi(b)$, contradicting the above inequality. Therefore

$$\varphi(a+b) = \varphi(b).$$

□

Definition 9.3.5 (Valuation ring and valuation ideal)**.** *Let φ be a multiplicative valuation on K. Define*

$$R_\varphi := \{a \in K \mid \varphi(a) \leq 1\},$$

and

$$P_\varphi := \{a \in K \mid \varphi(a) < 1\}.$$

Then R_φ is called the valuation ring of φ, and P_φ is called the valuation ideal of φ.

Proposition 9.3.6. *Let φ be a multiplicative valuation on K. Then R_φ is a subring of K, and P_φ is an ideal of R_φ.*

Proof. We first show that R_φ is a subring. Clearly,

$$\varphi(0) = 0 \leq 1, \qquad \varphi(1) = 1,$$

so $0, 1 \in R_\varphi$. If $a, b \in R_\varphi$, then

$$\varphi(ab) = \varphi(a)\varphi(b) \leq 1,$$

hence $ab \in R_\varphi$. Also,

$$\varphi(-a) = \varphi(a) \leq 1,$$

so $-a \in R_\varphi$. Finally,

$$\varphi(a+b) \leq \max\{\varphi(a), \varphi(b)\} \leq 1,$$

so $a + b \in R_\varphi$. Therefore R_φ is a subring of K.

Next we show that P_φ is an ideal of R_φ. If $a, b \in P_\varphi$, then

$$\varphi(a+b) \leq \max\{\varphi(a), \varphi(b)\} < 1,$$

so $a+b \in P_\varphi$. Also, if $r \in R_\varphi$ and $a \in P_\varphi$, then

$$\varphi(ra) = \varphi(r)\varphi(a) < 1.$$

Hence $ra \in P_\varphi$. Thus P_φ is an ideal of R_φ. $\square$

Remark 9.3.7. *If $\varphi(a) = \gamma^{v(a)}$ for a normalized discrete valuation v, where $0 < \gamma < 1$, then*

$$R_\varphi = \{a \in K \mid \varphi(a) \leq 1\} = \{a \in K \mid v(a) \geq 0\} = R_v,$$

and

$$P_\varphi = \{a \in K \mid \varphi(a) < 1\} = \{a \in K \mid v(a) > 0\} = P_v.$$

Proposition 9.3.8 (From multiplicative valuations to additive valuations)**.** *Let φ be a multiplicative valuation on K, and fix*

$$0 < \gamma < 1.$$

Define

$$v_\varphi : K^\times \longrightarrow \mathbb{R}$$

by

$$v_\varphi(a) := \log_\gamma \varphi(a).$$

Then v_φ is an additive valuation on K, that is,

$$v_\varphi(ab) = v_\varphi(a) + v_\varphi(b),$$

and

$$v_\varphi(a+b) \geq \min\{v_\varphi(a), v_\varphi(b)\}$$

whenever $a + b \neq 0$.

Proof. For $a, b \in K^\times$, we have

$$\varphi(ab) = \varphi(a)\varphi(b).$$

Taking logarithms with base γ, we get

$$v_\varphi(ab) = \log_\gamma \varphi(ab) = \log_\gamma \varphi(a) + \log_\gamma \varphi(b) = v_\varphi(a) + v_\varphi(b).$$

Now suppose $a + b \neq 0$. Since φ is non-archimedean,

$$\varphi(a+b) \leq \max\{\varphi(a), \varphi(b)\}.$$

Because $0 < \gamma < 1$, the function $\log_\gamma(x)$ is decreasing. Therefore

$$\log_\gamma \varphi(a+b) \geq \log_\gamma \max\{\varphi(a), \varphi(b)\}.$$

But

$$\log_\gamma \max\{\varphi(a), \varphi(b)\} = \min\{\log_\gamma \varphi(a), \log_\gamma \varphi(b)\}.$$

Hence

$$v_\varphi(a+b) \geq \min\{v_\varphi(a), v_\varphi(b)\}.$$

Thus v_φ is an additive valuation. □

Definition 9.3.9 (Value group)**.** *Let φ be a multiplicative valuation on K. Then*

$$\varphi(K^\times) \subseteq \mathbb{R}_{>0}$$

is a multiplicative subgroup of $\mathbb{R}_{>0}$. It is called the value group of φ.

Definition 9.3.10 (Equivalent multiplicative valuations)**.** *Let K be a field, and let*

$$\varphi, \psi : K \longrightarrow \mathbb{R}_{\geq 0}$$

be two multiplicative valuations on K. We say that φ and ψ are equivalent if there exists a positive real number

$$\gamma \in \mathbb{R}_{>0}$$

such that

$$\psi(a) = \varphi(a)^\gamma \qquad \text{for all } a \in K.$$

Definition 9.3.11 (Metric induced by a multiplicative valuation)**.** *Let φ be a multiplicative valuation on a field K. Define*

$$d : K \times K \longrightarrow \mathbb{R}_{\geq 0}$$

by

$$d(a,b) = \varphi(a-b).$$

Then d is a metric on K.

Proposition 9.3.12. *Let K be a field equipped with a multiplicative valuation φ. Then the function*

$$d(a,b) = \varphi(a-b)$$

defines a metric on K. In particular, K is a Hausdorff topological space with respect to the topology induced by d.

Proof. We verify the axioms of a metric.

First, for any $a, b \in K$, we have

$$d(a, b) = \varphi(a - b) \geq 0.$$

Moreover,

$$d(a, b) = 0 \iff \varphi(a - b) = 0 \iff a - b = 0 \iff a = b.$$

Next, since $\varphi(-1) = 1$, we have

$$d(a, b) = \varphi(a - b) = \varphi\big(-(b - a)\big) = \varphi(b - a) = d(b, a).$$

Finally, for any $a, b, c \in K$, the valuation inequality gives

$$d(a, c) = \varphi(a - c) = \varphi\big((a - b) + (b - c)\big) \leq \varphi(a - b) + \varphi(b - c).$$

Hence

$$d(a, c) \leq d(a, b) + d(b, c).$$

Therefore d is a metric on K. Since every metric space is Hausdorff, the field K is Hausdorff with respect to the topology induced by d. $\square$

9.4 Completion of valued fields

Definition 9.4.1 (Cauchy sequence). *Let K be a field equipped with a metric*

$$d : K \times K \longrightarrow \mathbb{R}_{\geq 0}.$$

A sequence

$$(a_n)_{n \geq 1} = (a_1, a_2, \ldots, a_n, \ldots)$$

of elements of K is called a Cauchy sequence if for every

$$\varepsilon > 0,$$

there exists an integer $N \geq 1$ such that

$$d(a_m, a_n) < \varepsilon \qquad \text{for all } m, n \geq N.$$

Definition 9.4.2 (Convergent sequence). *Let K be a field equipped with a metric d. A sequence*

$$(a_n)_{n \geq 1}$$

in K is said to converge to an element $a \in K$ if

$$\lim_{n \to \infty} d(a_n, a) = 0.$$

In this case, we write

$$\lim_{n\to\infty} a_n = a.$$

If the metric d is induced by a multiplicative valuation φ, namely

$$d(x,y) = \varphi(x-y),$$

then the above condition is equivalently

$$\lim_{n\to\infty} \varphi(a_n - a) = 0.$$

Definition 9.4.3 (Complete valued field)**.** *Let K be a field equipped with a metric d, or equivalently with a valuation inducing d. We say that K is complete if every Cauchy sequence in K converges to an element of K.*

If K is complete with respect to the metric induced by a valuation φ, then K is also called a complete valued field.

Definition 9.4.4 (Complete valuation ring)**.** *Let (K,φ) be a valued field, and let*

$$R_\varphi = \{x \in K \mid \varphi(x) \le 1\}$$

be its valuation ring in the multiplicative convention. If K is complete with respect to the metric induced by φ, then R_φ is called a complete valuation ring.

Definition 9.4.5 (Ring of Cauchy sequences)**.** *Let (K,φ) be a valued field, and let*

$$d(a,b) = \varphi(a-b)$$

be the metric induced by φ. Denote by $\mathcal{C}$ the set of all Cauchy sequences in K:

$$\mathcal{C} := \{(a_n)_{n\ge1} \mid (a_n)_{n\ge1} \text{ is a Cauchy sequence in } K\}.$$

For $(a_n),(b_n) \in \mathcal{C}$, define

$$(a_n) + (b_n) := (a_n + b_n),$$

and

$$(a_n)(b_n) := (a_n b_n).$$

With these operations, $\mathcal{C}$ is a commutative ring.

Definition 9.4.6 (Null sequences)**.** *Let*

$$\mathcal{I} := \left\{(a_n) \in \mathcal{C} \;\middle|\; \lim_{n\to\infty} a_n = 0\right\}.$$

Equivalently,

$$(a_n) \in \mathcal{I} \iff \lim_{n\to\infty} \varphi(a_n) = 0.$$

The elements of $\mathcal{I}$ are called null sequences.

Proposition 9.4.7. *The subset $\mathcal{I}$ is a maximal ideal of $\mathcal{C}$. Consequently, the quotient*

$$\widehat{K} := \mathcal{C}/\mathcal{I}$$

is a field.

Proof. It is straightforward to check that $\mathcal{I}$ is an ideal of $\mathcal{C}$.

To show that $\mathcal{I}$ is maximal, let

$$(a_n) \in \mathcal{C} \setminus \mathcal{I}.$$

Then (a_n) does not converge to 0. Since (a_n) is Cauchy, there exist $N \geq 1$ and $\varepsilon > 0$ such that

$$\varphi(a_n) \geq \varepsilon \qquad \text{for all } n \geq N.$$

Hence $a_n \neq 0$ for all sufficiently large n. Define a sequence

$$b_n = \begin{cases} 0, & n < N, \\ a_n^{-1}, & n \geq N. \end{cases}$$

Then (b_n) is again a Cauchy sequence in K, and

$$(a_n)(b_n) - 1 \in \mathcal{I}.$$

Therefore the class of (a_n) in $\mathcal{C}/\mathcal{I}$ is invertible. Thus every nonzero element of $\mathcal{C}/\mathcal{I}$ is invertible, so $\mathcal{C}/\mathcal{I}$ is a field. Hence $\mathcal{I}$ is maximal. $\square$

Definition 9.4.8 (Completion of a valued field)**.** *The field*

$$\widehat{K} := \mathcal{C}/\mathcal{I}$$

is called the completion of K with respect to the valuation φ.

Proposition 9.4.9 (Extension of the valuation)**.** *Let $(a_n) \in \mathcal{C}$. Then the sequence*

$$\big(\varphi(a_n)\big)_{n\geq 1}$$

converges in $\mathbb{R}_{\geq 0}$.

Moreover, if

$$(a_n) - (b_n) \in \mathcal{I},$$

then

$$\lim_{n\to\infty} \varphi(a_n) = \lim_{n\to\infty} \varphi(b_n).$$

Hence the rule

$$\widehat{\varphi}\big((a_n) + \mathcal{I}\big) := \lim_{n\to\infty} \varphi(a_n)$$

defines a well-defined multiplicative valuation on $\widehat{K}$*.*

Proof. Since (a_n) is Cauchy in K, for every $\varepsilon > 0$ there exists $N \geq 1$ such that

$$\varphi(a_n - a_m) < \varepsilon \qquad \text{for all } m, n \geq N.$$

By the reverse triangle inequality,

$$|\varphi(a_n) - \varphi(a_m)| \leq \varphi(a_n - a_m).$$

Therefore $\big(\varphi(a_n)\big)_{n\geq 1}$ is a Cauchy sequence in $\mathbb{R}_{\geq 0}$. Since $\mathbb{R}$ is complete, the limit

$$\lim_{n\to\infty} \varphi(a_n)$$

exists.

Now assume that

$$(a_n) - (b_n) \in \mathcal{I}.$$

Then

$$\lim_{n\to\infty} \varphi(a_n - b_n) = 0.$$

Again by the reverse triangle inequality,

$$|\varphi(a_n) - \varphi(b_n)| \leq \varphi(a_n - b_n).$$

Taking limits gives

$$\lim_{n\to\infty} \varphi(a_n) = \lim_{n\to\infty} \varphi(b_n).$$

Thus $\widehat{\varphi}$ is well-defined.

Finally, for $(a_n), (b_n) \in \mathcal{C}$, we have

$$\widehat{\varphi}\left(\big((a_n) + \mathcal{I}\big)\big((b_n) + \mathcal{I}\big)\right) = \lim_{n\to\infty} \varphi(a_n b_n) = \lim_{n\to\infty} \varphi(a_n)\varphi(b_n),$$

hence

$$\widehat{\varphi}\left(\big((a_n) + \mathcal{I}\big)\big((b_n) + \mathcal{I}\big)\right) = \widehat{\varphi}\big((a_n) + \mathcal{I}\big)\widehat{\varphi}\big((b_n) + \mathcal{I}\big).$$

Similarly, the triangle inequality for φ passes to the limit. Hence $\widehat{\varphi}$ is a multiplicative valuation on $\widehat{K}$. □

Proposition 9.4.10. *There is a natural embedding of valued fields*

$$K \hookrightarrow \widehat{K}.$$

Proof. Define
$$\iota : K \longrightarrow \widehat{K}$$
by sending $a \in K$ to the class of the constant sequence:

$$\iota(a) := (a, a, a, \dots) + \mathcal{I}.$$

This is a field homomorphism. If $\iota(a) = 0$, then the constant sequence $(a, a, a, \dots)$ belongs to $\mathcal{I}$, so
$$\lim_{n\to\infty} \varphi(a) = 0.$$
Hence $\varphi(a) = 0$, and therefore $a = 0$. Thus ι is injective.

Moreover,
$$\widehat{\varphi}(\iota(a)) = \widehat{\varphi}\big((a, a, a, \dots) + \mathcal{I}\big) = \varphi(a),$$
so the embedding preserves the valuation. $\square$

Theorem 9.4.11 (Completion of a valued field)**.** *Let (K, φ) be a valued field. Then there exists a complete valued field*

$$(\widehat{K}, \widehat{\varphi})$$

together with a valuation-preserving embedding

$$K \hookrightarrow \widehat{K}$$

such that the image of K is dense in $\widehat{K}$.

Moreover, $\widehat{K}$ is unique up to a unique K-isomorphism of valued fields.

Proof. The field $\widehat{K} = \mathcal{C}/\mathcal{I}$, together with the valuation $\widehat{\varphi}$, is complete. Indeed, every Cauchy sequence in $\widehat{K}$ can be represented by a suitable diagonal sequence of Cauchy sequences in K, and this diagonal sequence gives its limit in $\widehat{K}$.

The image of K is dense in $\widehat{K}$. If

$$x = (a_n) + \mathcal{I} \in \widehat{K},$$

then the sequence of constant classes

$$\iota(a_n) \in \widehat{K}$$

converges to x.

Finally, if L is another complete valued field containing K densely and preserving the valuation, then every Cauchy sequence in K has a unique limit in L. Thus the rule

$$(a_n) + \mathcal{I} \longmapsto \lim_{n\to\infty} a_n$$

defines a unique K-isomorphism

$$\widehat{K} \cong L$$

of valued fields. Hence the completion is unique up to K-isomorphism of valued fields. □

9.5 Residue fields and the discrete valuation topology

Theorem 9.5.1 (Residue field and completion). *Let (K, φ) be a valued field with valuation ring*

$$R = \{x \in K \mid \varphi(x) \leq 1\}$$

and valuation ideal

$$P = \{x \in K \mid \varphi(x) < 1\}.$$

Let $(\widehat{K}, \widehat{\varphi})$ be the completion of K, and let

$$\widehat{R} = \{x \in \widehat{K} \mid \widehat{\varphi}(x) \leq 1\}$$

be its valuation ring, with valuation ideal

$$\widehat{P} = \{x \in \widehat{K} \mid \widehat{\varphi}(x) < 1\}.$$

Then there is a canonical isomorphism of residue fields

$$R/P \cong \widehat{R}/\widehat{P}.$$

Proof. We identify K with its image in $\widehat{K}$. Since the embedding

$$K \hookrightarrow \widehat{K}$$

preserves the valuation, we have

$$\widehat{\varphi}|_K = \varphi.$$

Hence

$$R \subseteq \widehat{R}, \qquad P = R \cap \widehat{P}.$$

It remains to show that

$$\widehat{R} = R + \widehat{P}.$$

The inclusion

$$R + \widehat{P} \subseteq \widehat{R}$$

is clear, since $R \subseteq \widehat{R}$ and $\widehat{P} \subseteq \widehat{R}$.

Conversely, let $a \in \widehat{R}$. If $a = 0$, then $a \in R + \widehat{P}$. Suppose $a \neq 0$. Since K is dense in $\widehat{K}$, there exists a sequence

$$(a_n)_{n \geq 1}$$

in K such that

$$\lim_{n\to\infty} a_n = a.$$

Thus

$$\lim_{n\to\infty} \widehat{\varphi}(a_n - a) = 0.$$

Since $a \neq 0$, we have

$$\widehat{\varphi}(a) > 0.$$

Therefore, for some sufficiently large n,

$$\widehat{\varphi}(a_n - a) < \widehat{\varphi}(a).$$

Since

$$a_n = a + (a_n - a)$$

and

$$\widehat{\varphi}(a_n - a) < \widehat{\varphi}(a),$$

the strong triangle inequality for a non-archimedean multiplicative valuation implies

$$\widehat{\varphi}(a_n) = \widehat{\varphi}\big(a + (a_n - a)\big) = \widehat{\varphi}(a).$$

Indeed, for a multiplicative non-archimedean valuation, if

$$\widehat{\varphi}(x) < \widehat{\varphi}(y),$$

then

$$\widehat{\varphi}(x + y) = \widehat{\varphi}(y).$$

Since $a \in \widehat{R}$, we have

$$\widehat{\varphi}(a) \leq 1.$$

Hence

$$\varphi(a_n) = \widehat{\varphi}(a_n) \leq 1,$$

so

$$a_n \in R.$$

Moreover,

$$\widehat{\varphi}(a - a_n) < \widehat{\varphi}(a) \leq 1,$$

hence

$$a - a_n \in \widehat{P}.$$

Therefore

$$a = a_n + (a - a_n) \in R + \widehat{P}.$$

Thus

$$\widehat{R} \subseteq R + \widehat{P}.$$

Hence
$$\widehat{R} = R + \widehat{P}.$$

Now by the second isomorphism theorem,

$$\widehat{R}/\widehat{P} = (R + \widehat{P})/\widehat{P} \cong R/(R \cap \widehat{P}).$$

Since
$$R \cap \widehat{P} = P,$$

we obtain
$$\widehat{R}/\widehat{P} \cong R/P.$$

□

Proposition 9.5.2 (Fundamental system of neighbourhoods)**.** *Let* (K, v) *be a field with a normalized discrete valuation. Let*

$$R = \{x \in K \mid v(x) \geq 0\}$$

be its valuation ring, and let

$$P = \{x \in K \mid v(x) > 0\}$$

be its valuation ideal. Then
$$P = J(R),$$

and the decreasing sequence of ideals

$$R = P^0 \supset P \supset P^2 \supset \cdots \supset P^\ell \supset \cdots$$

forms a fundamental system of neighbourhoods of 0 *in* K*.*

Proof. Since v is a normalized discrete valuation, we have

$$v(K^\times) = \mathbb{Z}.$$

Choose a uniformizer $\pi \in R$, so that

$$v(\pi) = 1.$$

Then
$$P = (\pi),$$

and more generally

$$P^\ell = (\pi^\ell) = \{x \in R \mid v(x) \geq \ell\} \qquad \text{for every } \ell \geq 0.$$

Thus saying that an element is sufficiently close to 0 is equivalent to saying that it lies in

P^ℓ for large ℓ. Hence the ideals

$$P^\ell, \qquad \ell \geq 0,$$

form a fundamental system of neighbourhoods of 0. ☐

Corollary 9.5.3 (Cauchy criterion). *Let $(a_n)_{n\geq 1}$ be a sequence in K. Then (a_n) is a Cauchy sequence if and only if for every natural number $\ell \geq 1$, there exists a natural number $N \geq 1$ such that*

$$a_m - a_n \in P^\ell \qquad \textit{whenever } m, n \geq N.$$

Proof. By definition, (a_n) is Cauchy if and only if for every neighbourhood U of 0, there exists $N \geq 1$ such that

$$a_m - a_n \in U \qquad \text{for all } m, n \geq N.$$

Since the ideals P^ℓ form a fundamental system of neighbourhoods of 0, it is enough to test this condition on $U = P^\ell$. Hence (a_n) is Cauchy if and only if for every $\ell \geq 1$, there exists $N \geq 1$ such that

$$a_m - a_n \in P^\ell \qquad \text{for all } m, n \geq N.$$

☐

Corollary 9.5.4 (Convergence criterion). *Let $(a_n)_{n\geq 1}$ be a sequence in K, and let $a \in K$. Then*

$$\lim_{n\to\infty} a_n = a$$

if and only if for every natural number $\ell \geq 1$, there exists a natural number $N \geq 1$ such that

$$a_n - a \in P^\ell \qquad \textit{whenever } n \geq N.$$

Proof. The sequence (a_n) converges to a if and only if

$$a_n - a \longrightarrow 0.$$

Since the ideals P^ℓ form a fundamental system of neighbourhoods of 0, this is equivalent to saying that for every $\ell \geq 1$, there exists $N \geq 1$ such that

$$a_n - a \in P^\ell \qquad \text{for all } n \geq N.$$

☐

Lemma 9.5.5 (Equivalent discrete multiplicative valuations). *Let φ_1 and φ_2 be discrete multiplicative valuations of a field K. The following conditions are equivalent:*

1. *$\varphi_1 \sim \varphi_2$, that is, there exists $\lambda > 0$ such that*

$$\varphi_2(x) = \varphi_1(x)^\lambda \qquad \textit{for all } x \in K.$$

2. *The valuation ideals are the same:*

$$\{ x \in K \mid \varphi_1(x) < 1 \} = \{ x \in K \mid \varphi_2(x) < 1 \}.$$

3. *The topologies on K induced by φ_1 and φ_2 are the same.*

Proof. For $i = 1, 2$, let

$$R_i := \{ x \in K \mid \varphi_i(x) \leq 1 \}$$

be the valuation ring of φ_i, and let

$$P_i := \{ x \in K \mid \varphi_i(x) < 1 \}$$

be its valuation ideal.

We first note that condition (2) is precisely the assertion

$$P_1 = P_2.$$

(1) $\Rightarrow$ (2). Suppose that $\varphi_2 = \varphi_1^\lambda$ for some $\lambda > 0$. Then, for every $x \in K$,

$$\varphi_2(x) < 1 \Longleftrightarrow \varphi_1(x)^\lambda < 1 \Longleftrightarrow \varphi_1(x) < 1.$$

Hence $P_1 = P_2$.

(2) $\Rightarrow$ (1). Assume that

$$P_1 = P_2 =: P.$$

Since φ_1 and φ_2 are discrete valuations, their valuation rings are discrete valuation rings with maximal ideal P. Hence the common ideal P is principal. Choose a uniformizer

$$\pi \in P.$$

Then every element $x \in K^\times$ admits a unique expression

$$x = \pi^n u, \qquad n \in \mathbb{Z}, \quad u \in R_i^\times.$$

Therefore, for $i = 1, 2$,

$$\varphi_i(x) = \varphi_i(\pi)^n.$$

Put

$$\rho_i := \varphi_i(\pi).$$

Since $\pi \in P$, we have $0 < \rho_i < 1$. Choose

$$\lambda := \frac{\log \rho_2}{\log \rho_1} > 0.$$

Then

$$\rho_2 = \rho_1^\lambda.$$

Hence for every $x = \pi^n u \in K^\times$,

$$\varphi_2(x) = \rho_2^n = (\rho_1^\lambda)^n = (\rho_1^n)^\lambda = \varphi_1(x)^\lambda.$$

Thus

$$\varphi_2 = \varphi_1^\lambda,$$

so $\varphi_1 \sim \varphi_2$.

(1) $\Rightarrow$ (3). If $\varphi_2 = \varphi_1^\lambda$ with $\lambda > 0$, then for any sequence (x_n) in K,

$$\varphi_1(x_n) \to 0 \iff \varphi_1(x_n)^\lambda \to 0 \iff \varphi_2(x_n) \to 0.$$

Therefore the two valuations define the same convergent sequences to 0, and hence induce the same topology on K.

(3) $\Rightarrow$ (2). Assume that the topologies induced by φ_1 and φ_2 are the same. Let $a \in P_1$. Then

$$\varphi_1(a) < 1.$$

Hence

$$\lim_{n\to\infty} \varphi_1(a^n) = \lim_{n\to\infty} \varphi_1(a)^n = 0.$$

Thus

$$a^n \longrightarrow 0$$

with respect to the topology induced by φ_1. Since the two topologies are the same, we also have

$$a^n \longrightarrow 0$$

with respect to the topology induced by φ_2. Therefore

$$\lim_{n\to\infty} \varphi_2(a^n) = \lim_{n\to\infty} \varphi_2(a)^n = 0.$$

This implies

$$\varphi_2(a) < 1,$$

so $a \in P_2$. Hence

$$P_1 \subseteq P_2.$$

By symmetry, we also get

$$P_2 \subseteq P_1.$$

Therefore

$$P_1 = P_2.$$

We have proved

$$(1) \Rightarrow (2) \Rightarrow (1), \qquad (1) \Rightarrow (3), \qquad (3) \Rightarrow (2).$$

Hence the three conditions are equivalent. $\square$

9.6 Norms on finite-dimensional vector spaces

Definition 9.6.1 (Norm over a multiplicatively valued field)**.** *Let* (K, φ) *be a field equipped with a multiplicative valuation, and let* V *be a vector space over* K*. A function*

$$\| \cdot \| : V \longrightarrow \mathbb{R}_{\geq 0}$$

is called a non-archimedean norm on V *compatible with* φ *if the following conditions hold:*

1. *For every* $v \in V$*,*
$$\|v\| \geq 0,$$
and
$$\|v\| = 0 \iff v = 0.$$
2. *For every* $\alpha \in K$ *and every* $v \in V$*,*
$$\|\alpha v\| = \varphi(\alpha)\|v\|.$$
3. *For every* $u, v \in V$*,*
$$\|u + v\| \leq \max\{\|u\|, \|v\|\}.$$

Proposition 9.6.2. *Let* V *be a vector space over a multiplicatively valued field* (K, φ)*. Suppose that* V *is equipped with a non-archimedean norm* $\| \cdot \|$*. Then* V *becomes a metric space with metric*

$$d(u, v) := \|u - v\|.$$

Moreover, the maps

$$V \times V \longrightarrow V, \qquad (u, v) \longmapsto u + v,$$

and

$$K \times V \longrightarrow V, \qquad (\alpha, v) \longmapsto \alpha v,$$

are both continuous.

Proof. The function $d(u, v) = \|u - v\|$ is a metric. Indeed, for $u, v \in V$,

$$d(u, v) = 0 \iff \|u - v\| = 0 \iff u = v.$$

Also,

$$d(u, v) = \|u - v\| = \| - (v - u)\| = \|v - u\| = d(v, u),$$

because $\varphi(-1) = 1$. Finally, for $u, v, w \in V$,

$$d(u, w) = \|u - w\| = \|(u - v) + (v - w)\| \leq \max\{\|u - v\|, \|v - w\|\} = \max\{d(u, v), d(v, w)\}.$$

Hence d is a non-archimedean metric on V.

The addition map is continuous because

$$\|(u+v)-(u_0+v_0)\| = \|(u-u_0)+(v-v_0)\| \le \max\{\|u-u_0\|, \|v-v_0\|\}.$$

The scalar multiplication map is continuous because

$$\|\alpha v - \alpha_0 v_0\| = \|\alpha(v-v_0)+(\alpha-\alpha_0)v_0\| \le \max\{\varphi(\alpha)\|v-v_0\|, \varphi(\alpha-\alpha_0)\|v_0\|\}.$$

Since $\varphi(\alpha)$ is locally bounded near α_0, the right-hand side tends to 0 whenever $\alpha \to \alpha_0$ in K and $v \to v_0$ in V. Thus scalar multiplication is continuous. ☐

Lemma 9.6.3. *Let (K, φ) be a multiplicatively valued field, and let V be a finite-dimensional K-vector space with K-basis*

$$u_1, \dots, u_n.$$

For

$$v = a_1u_1 + \cdots + a_nu_n \in V,$$

define

$$\|v\|_0 := \max\{\, \varphi(a_i) \mid 1 \le i \le n \,\}.$$

Then $\|\cdot\|_0$ is a non-archimedean norm on V. Moreover, the following hold:

1. *Let*

   $$v_r = a_{r,1}u_1 + \cdots + a_{r,n}u_n \in V \qquad (r \ge 1).$$

   Then the sequence (v_r) is a Cauchy sequence in V if and only if each coordinate sequence $(a_{r,i})_{r\ge 1}$ is a Cauchy sequence in K.

   Also, for

   $$v = a_1u_1 + \cdots + a_nu_n \in V,$$

   one has

   $$\lim_{r\to\infty} v_r = v \Longleftrightarrow \lim_{r\to\infty} a_{r,i} = a_i \quad \text{for all } 1 \le i \le n.$$

2. *If K is complete, then V is complete with respect to the norm $\|\cdot\|_0$.*

Proof. First we verify that $\|\cdot\|_0$ is a norm. Clearly

$$\|v\|_0 \ge 0.$$

Moreover,

$$\|v\|_0 = 0 \Longleftrightarrow \varphi(a_i) = 0 \quad \text{for all } i \Longleftrightarrow a_i = 0 \quad \text{for all } i \Longleftrightarrow v = 0.$$

For $\lambda \in K$, we have

$$\|\lambda v\|_0 = \max_i \varphi(\lambda a_i) = \varphi(\lambda)\max_i \varphi(a_i) = \varphi(\lambda)\|v\|_0.$$

Finally, if

$$v = a_1u_1 + \cdots + a_nu_n, \qquad w = b_1u_1 + \cdots + b_nu_n,$$

then

$$v + w = (a_1 + b_1)u_1 + \cdots + (a_n + b_n)u_n.$$

Hence

$$\|v + w\|_0 = \max_i \varphi(a_i + b_i) \leq \max_i \max\{\varphi(a_i), \varphi(b_i)\} = \max\{\|v\|_0, \|w\|_0\}.$$

Thus $\|\cdot\|_0$ is a non-archimedean norm.

Now let

$$v_r = a_{r,1}u_1 + \cdots + a_{r,n}u_n.$$

For $r, s \geq 1$, we have

$$v_r - v_s = \sum_{i=1}^{n}(a_{r,i} - a_{s,i})u_i.$$

Therefore

$$\|v_r - v_s\|_0 = \max_{1\leq i\leq n} \varphi(a_{r,i} - a_{s,i}).$$

It follows immediately that (v_r) is Cauchy in V if and only if each $(a_{r,i})_{r\geq 1}$ is Cauchy in K.

Similarly,

$$v_r - v = \sum_{i=1}^{n}(a_{r,i} - a_i)u_i,$$

so

$$\|v_r - v\|_0 = \max_{1\leq i\leq n} \varphi(a_{r,i} - a_i).$$

Hence

$$\lim_{r\to\infty} v_r = v \iff \lim_{r\to\infty} a_{r,i} = a_i \quad \text{for all } i.$$

Finally, assume that K is complete. Let (v_r) be a Cauchy sequence in V. By the first part, each coordinate sequence $(a_{r,i})$ is Cauchy in K. Since K is complete, there exists $a_i \in K$ such that

$$\lim_{r\to\infty} a_{r,i} = a_i.$$

Put

$$v = a_1u_1 + \cdots + a_nu_n.$$

By the convergence criterion above,

$$\lim_{r\to\infty} v_r = v.$$

Therefore V is complete. $\square$

Theorem 9.6.4 (Equivalence of norms over a complete non-archimedean field)**.** *Let (K, φ) be a complete non-archimedean normed field, and let V be a finite-dimensional*

K-vector space with basis

$$u_1, \dots, u_n.$$

Define the associated supremum norm by

$$\left\| \sum_{i=1}^{n} a_i u_i \right\|_\infty := \max_{1 \le i \le n} \varphi(a_i).$$

Then, for any non-archimedean norm $\| \cdot \|$ on V, there exist constants $\lambda, \mu > 0$ such that

$$\|v\| \le \lambda \|v\|_\infty, \qquad \|v\|_\infty \le \mu \|v\|$$

for all $v \in V$. Consequently, $\| \cdot \|$ and $\| \cdot \|_\infty$ induce the same topology on V. In particular, any two norms on V induce the same topology.

Proof. Let

$$\lambda = \max_{1 \le i \le n} \|u_i\|.$$

For

$$v = \sum_{i=1}^{n} a_i u_i,$$

the non-archimedean triangle inequality gives

$$\|v\| \le \max_{1 \le i \le n} \|a_i u_i\| = \max_{1 \le i \le n} \varphi(a_i) \|u_i\| \le \lambda \|v\|_\infty.$$

Hence the first inequality holds.

It remains to prove that there exists $\mu > 0$ such that

$$\|v\|_\infty \le \mu \|v\|$$

for all $v \in V$. We prove this by induction on $n = \dim_K V$.

If $n = 1$, then every $v \in V$ can be written as $v = a_1 u_1$. Hence

$$\|v\| = \varphi(a_1) \|u_1\|, \qquad \|v\|_\infty = \varphi(a_1).$$

Therefore

$$\|v\|_\infty = \frac{1}{\|u_1\|} \|v\|,$$

so the claim holds.

Now assume $n > 1$, and suppose the result is known for all vector spaces of dimension $n - 1$. For

$$v = \sum_{i=1}^{n} a_i u_i,$$

define

$$\psi_i(v) := \varphi(a_i).$$

We claim that, for each i, there exists a constant $\mu_i > 0$ such that

$$\psi_i(v) \leq \mu_i \|v\|$$

for all $v \in V$. Once this is proved, taking

$$\mu = \max_{1 \leq i \leq n} \mu_i$$

gives

$$\|v\|_\infty = \max_{1 \leq i \leq n} \psi_i(v) \leq \mu \|v\|.$$

It suffices to prove the claim for $i = n$, since the other cases follow by reordering the basis. Put

$$V' = K u_1 \oplus \cdots \oplus K u_{n-1}.$$

By the induction hypothesis, the restrictions of $\|\cdot\|$ and $\|\cdot\|_\infty$ to V' induce the same topology.

Since K is complete, V' is complete with respect to the norm $\|\cdot\|_\infty$. Hence V' is also complete with respect to $\|\cdot\|$. Therefore V' is closed in V with respect to the topology induced by $\|\cdot\|$.

Suppose, for contradiction, that there is no constant $\mu_n > 0$ such that

$$\psi_n(v) \leq \mu_n \|v\|$$

for all $v \in V$. Then, for every integer $j \geq 1$, there exists $v_j \in V$ such that

$$\psi_n(v_j) > j \|v_j\|.$$

Since the inequality is homogeneous, we may multiply v_j by a nonzero scalar and assume that the coefficient of u_n in v_j is 1. Thus we may write

$$v_j = a_{1j} u_1 + \cdots + a_{n-1,j} u_{n-1} + u_n.$$

Then

$$\psi_n(v_j) = 1,$$

and the inequality above gives

$$\|v_j\| < \frac{1}{j}.$$

Hence

$$v_j \longrightarrow 0$$

with respect to $\|\cdot\|$.

On the other hand,

$$v_j - u_n \in V'$$

for every j, and

$$v_j - u_n \longrightarrow -u_n$$

with respect to $\|\cdot\|$. Since V' is closed in V, it follows that

$$-u_n \in V'.$$

This is impossible, because $u_1, \dots, u_n$ are linearly independent.

Therefore there exists $\mu_n > 0$ such that

$$\psi_n(v) \leq \mu_n \|v\|$$

for all $v \in V$. As explained above, the same argument applies to each coordinate ψ_i. Hence there exists $\mu > 0$ such that

$$\|v\|_\infty \leq \mu \|v\|$$

for all $v \in V$.

Combining the two inequalities, we conclude that $\|\cdot\|$ and $\|\cdot\|_\infty$ are equivalent norms. Therefore they induce the same topology on V. □

9.7 Extensions of complete discrete valuations

Definition 9.7.1 (Extension of a discretely valued field)**.** *Let $K \subseteq L$ be fields equipped with discrete multiplicative valuations*

$$\varphi : K \longrightarrow \mathbb{R}_{\geq 0}, \qquad \psi : L \longrightarrow \mathbb{R}_{\geq 0},$$

respectively.

We say that ψ is an extension of φ to L if

$$\psi|_K = \varphi.$$

In this case, we also say that the valued field (L, ψ) is an extension of the valued field (K, φ).

Theorem 9.7.2 (Extension of a complete discrete valuation)**.** *Let φ be a complete discrete multiplicative valuation on a field K, and let L/K be a finite field extension.*

Suppose that there exists an extension (L, ψ) of (K, φ). Then ψ is unique up to equivalence. Furthermore, the valued field

$$(L, \psi)$$

is complete.

Proof. Let ψ_1 and ψ_2 be two extensions of φ to L. We first prove uniqueness up to equivalence.

Since L/K is finite, L is a finite-dimensional K-vector space. Moreover, each ψ_i is a non-archimedean norm on L as a K-vector space, because for every $a \in K$ and $x \in L$,

$$\psi_i(ax) = \psi_i(a)\psi_i(x) = \varphi(a)\psi_i(x).$$

Thus ψ_i is compatible with the scalar norm φ on K.

Since (K, φ) is complete and L is finite-dimensional over K, the equivalence theorem for norms on finite-dimensional vector spaces over a complete non-archimedean field implies that the two norms ψ_1 and ψ_2 induce the same topology on L.

Now use the standard fact that two discrete valuations on a field which induce the same topology are equivalent. Indeed, if w_1, w_2 are two discrete valuations inducing the same topology, then their valuation ideals are determined topologically by

$$P_{w_i} = \{x \in L \mid x^n \to 0 \text{ as } n \to \infty\}.$$

Hence

$$P_{w_1} = P_{w_2}.$$

The valuation ring is recovered from the valuation ideal by

$$R_{w_i} = \{x \in L \mid xP_{w_i} \subseteq P_{w_i}\},$$

so

$$R_{w_1} = R_{w_2}.$$

Since the valuations are discrete, choosing a uniformizer π for w_1, every $x \in L^\times$ can be written uniquely in the form

$$x = \pi^m u, \qquad m \in \mathbb{Z}, \quad u \in R_{w_1}^\times.$$

Because the valuation rings agree, the unit groups also agree. Therefore $w_2(u) = 0$, and hence

$$w_2(x) = m\, w_2(\pi) = w_2(\pi)\, w_1(x).$$

Thus

$$w_2 = \lambda w_1$$

for some $\lambda > 0$, so w_1 and w_2 are equivalent.

Applying this to ψ_1 and ψ_2, we conclude that

$$\psi_1 \sim \psi_2.$$

Hence the extension of φ to L, if it exists, is unique up to equivalence.

It remains to prove that (L, ψ) is complete. Choose a K-basis

$$u_1, \ldots, u_n$$

of L, and define the associated supremum norm

$$\left\| \sum_{i=1}^{n} a_i u_i \right\|_\infty = \max_{1 \le i \le n} \varphi(a_i).$$

Since K is complete with respect to φ, the finite-dimensional normed K-vector space $(L, \|\cdot\|_\infty)$ is complete.

On the other hand, ψ is another non-archimedean norm on the same finite-dimensional K-vector space L. Hence, by the equivalence theorem for finite-dimensional norms, ψ and $\|\cdot\|_\infty$ induce the same topology on L. In particular, a sequence in L is ψ-Cauchy if and only if it is $\|\cdot\|_\infty$-Cauchy.

Let (x_m) be a ψ-Cauchy sequence in L. Then it is $\|\cdot\|_\infty$-Cauchy, so by completeness of $(L, \|\cdot\|_\infty)$, there exists $x \in L$ such that

$$x_m \longrightarrow x$$

with respect to $\|\cdot\|_\infty$. Since ψ and $\|\cdot\|_\infty$ induce the same topology, the same convergence holds with respect to ψ. Therefore every ψ-Cauchy sequence converges in L.

Hence (L, ψ) is complete. $\square$

Chapter 10

Lecture 10

Chapter 11

Lecture 11

Chapter 12

Homework

12.1 Homework Week 1

Exercise 12.1.1. *Let R be a ring, and let M be an R-module. Suppose that S is a ring and*

$$\theta : S \to R$$

is a ring homomorphism satisfying $\theta(1_S) = 1_R$. Let S act on M by

$$sx = \theta(s)x,$$

where $x \in M$ and $s \in S$. Prove that M is an S-module.

Proof. This action is well-defined because $\forall s \in S$, $\theta(s) \in R$ and $\theta(s)x \in M$ for all $x \in M$. We need to verify the module axioms, for all $s, t \in S$ and $x, y \in M$:

- $s(x + y) = \theta(s)(x + y) = \theta(s)x + \theta(s)y = sx + sy$
- $(s + t)x = \theta(s + t)x = \theta(s)x + \theta(t)x = sx + tx$
- $(st)x = \theta(st)x = \theta(s)\theta(t)x = \theta(s)(\theta(t)x) = s(tx)$.
- $1_S x = \theta(1_S)x = 1_R x = x$

Hence M is an S-module. □

Exercise 12.1.2. *Let R be a ring, and let M and N be two right R-modules. Let* $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ *be the set of group homomorphisms from M to N (as abelian groups). The action of R on* $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ *is given as follows: for any $a \in R$ and $\varphi \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$, define*

$$(\varphi a)(x) = \varphi(xa), \qquad \forall x \in M.$$

Prove that $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ *is a left R-module.*

Proof. This action is well defined: for every $r \in R$ and every $\varphi \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$, the map

$$\varphi r \colon M \to N, \qquad (\varphi r)(x) = \varphi(xr),$$

is again a group homomorphism.

We now verify the module axioms. Let $r_1, r_2 \in R$, let $\varphi, \psi \in \operatorname{Hom}_{\mathbb{Z}}(M, N)$, and let $x \in M$. Then

$$\begin{aligned}
((\varphi+\psi)r_1)(x) &= (\varphi+\psi)(xr_1) \\
&= \varphi(xr_1) + \psi(xr_1) \\
&= (\varphi r_1)(x) + (\psi r_1)(x), \\
(\varphi(r_1+r_2))(x) &= \varphi(x(r_1+r_2)) \\
&= \varphi(xr_1 + xr_2) \\
&= \varphi(xr_1) + \varphi(xr_2) \\
&= (\varphi r_1)(x) + (\varphi r_2)(x), \\
((\varphi r_1)r_2)(x) &= (\varphi r_1)(xr_2) \\
&= \varphi((xr_2)r_1) \\
&= \varphi(x(r_2 r_1)) \\
&= (\varphi(r_2 r_1))(x), \\
(\varphi 1_R)(x) &= \varphi(x 1_R) \\
&= \varphi(x).
\end{aligned}$$

Hence $\operatorname{Hom}_{\mathbb{Z}}(M, N)$ is a left R-module. □

Exercise 12.1.3. *Let R be a ring, and let M be an R-module. Denote*

$$\operatorname{Ann}_R M = \{r \in R \mid rm = 0,\ \forall m \in M\}.$$

We call M a faithful R-module if $\operatorname{Ann}_R M = \{0\}$. Define the action of the quotient ring $R/\operatorname{Ann}_R M$ on M by

$$\overline{r}\,x = rx,$$

where $x \in M$, $r \in R$, and $\overline{r} = r + \operatorname{Ann}_R M$. Prove that M is a faithful $R/\operatorname{Ann}_R M$-module.

Proof. First we show that the action is well-defined. Suppose

$$\bar{r} = \bar{r}' \in R/\operatorname{Ann}_R M.$$

Then $r - r' \in \operatorname{Ann}_R M$, so for every $x \in M$,

$$(r - r')x = 0.$$

Hence $rx = r'x$, so $\bar{r}\,x$ does not depend on the choice of representative.

The module axioms follow immediately from those of the R-module structure on M.

Now we prove faithfulness. Suppose $\bar{r} \in R/\operatorname{Ann}_R M$ acts trivially on M, that is,

$$\bar{r}\,x = 0 \qquad \forall x \in M.$$

By definition, this means
$$rx = 0 \qquad \forall x \in M.$$
Therefore $r \in \operatorname{Ann}_R M$, so $\bar{r} = \bar{0}$ in the quotient ring. Hence
$$\operatorname{Ann}_{R/\operatorname{Ann}_R M}(M) = \{\bar{0}\},$$
and M is faithful as an $R/\operatorname{Ann}_R M$-module. □

Exercise 12.1.4. *Let $\varphi : M \to M$ be a homomorphism of R-modules such that*
$$\varphi\varphi = \varphi.$$
Prove that
$$M = \ker\varphi \oplus \operatorname{Im}\varphi.$$

Proof. Let $x \in M$. Then
$$x = \varphi(x) + (x - \varphi(x)),$$
where $\varphi(x) \in \operatorname{Im}\varphi$ and $x - \varphi(x) \in \ker\varphi$. Hence
$$M = \operatorname{Im}\varphi + \ker\varphi.$$
It remains to show that
$$\ker\varphi \cap \operatorname{Im}\varphi = \{0\}.$$
Suppose $x \in \ker\varphi \cap \operatorname{Im}\varphi$. Then $x = \varphi(y)$ for some $y \in M$ and $\varphi(x) = 0$. Thus
$$0 = \varphi(x) = \varphi(\varphi(y)) = \varphi(y) = x,$$
so $x = 0$. Hence $\ker\varphi \cap \operatorname{Im}\varphi = \{0\}$. □

Exercise 12.1.5. *Show that any R-module is a homomorphism image of some free R-module.*

Proof. Let M be a left R-module. Consider the free R-module
$$F = \bigoplus_{x \in M} Re_x$$
with basis $\{e_x\}_{x \in M}$ indexed by the underlying set of M. Define an R-module homomorphism
$$\pi : F \to M$$
by
$$\pi(e_x) = x \qquad (x \in M),$$
and extend R-linearly. Thus
$$\pi\Big(\sum_{i=1}^{n} r_i e_{x_i}\Big) = \sum_{i=1}^{n} r_i x_i.$$

This is well-defined and R-linear.

For any $m \in M$, we have

$$\pi(e_m) = m,$$

so π is surjective. Therefore M is a homomorphic image of the free module F. Equivalently,

$$M \cong F/\ker \pi.$$

$\square$

Exercise 12.1.6. *Let $\mathbb{Q}$ be the field of rational numbers. Is the additive group $(\mathbb{Q}, +)$ a free $\mathbb{Z}$-module?*

Proof. Claim: any two non-zero element of $\mathbb{Q}$ are linearly dependent over $\mathbb{Z}$. Hence if $\mathbb{Q}$ is a free $\mathbb{Z}$-module, then it must have a basis of cardinality 1.

Proof of claim: let $0 \neq x = \frac{a}{b}$ and $0 \neq y = \frac{c}{d}$ be two elements of $\mathbb{Q}$. Then $dcx - aby = 0$. If $a = 0$ or $c = 0$, then $x = 0$ or $y = 0$, which contradicts the assumption. Hence $a, c \neq 0$ which means dc, ab are two non-zero integer coefficients. Hence x and y are linearly dependent over $\mathbb{Z}$. And if $\mathbb{Q}$ is a free $\mathbb{Z}$-module, then it must have a basis of cardinality 1.

But if $\mathbb{Q}$ is a cyclic $\mathbb{Z}$-module, then it has only one generator. Suppose it is $z = \frac{e}{f}$. Then $\mathbb{Q} = \{n\frac{e}{f} \mid n \in \mathbb{Z}\}$. But this is not possible because we can easily pick $\frac{1}{2f} \notin \mathbb{Q}$, which is absurd.

Therefore $\mathbb{Q}$ is not a free $\mathbb{Z}$-module. $\square$

12.2 Homework Week 2

Exercise 12.2.1. *Let R be a commutative ring, and suppose that φ is an endomorphism of the free R-module R^m. Prove that if φ is surjective, then φ is injective. Consider that: if φ is injective, is φ surjective?*

Proof. Let $\{e_1, \ldots, e_m\}$ be the standard basis of R^m, and let $A \in M_m(R)$ be the matrix of φ with respect to this basis. Since φ is surjective, for each j there exists $v_j \in R^m$ with $\varphi(v_j) = e_j$. Write $v_j = \sum_{i=1}^m b_{ij} e_i$ and let $B = (b_{ij}) \in M_m(R)$. Then $AB = I_m$, so

$$\det(A)\det(B) = \det(AB) = 1.$$

Hence $\det(A)$ is a unit in R. It follows that A is invertible (with $A^{-1} = \det(A)^{-1} \operatorname{adj}(A)$), so φ is an isomorphism. In particular φ is injective.

Counterexample for the converse. The converse is false in general. Let $R = \mathbb{Z}$ and $m = 1$, so $R^m = \mathbb{Z}$. Define $\varphi \colon \mathbb{Z} \to \mathbb{Z}$ by $\varphi(n) = 2n$. Then φ is injective but not surjective; for example, $1 \notin \operatorname{Im} \varphi$. $\square$

Exercise 12.2.2. *Let $\varphi : M \to N$ be a homomorphism of R-modules. Prove that:*

1. *φ is a monomorphism if and only if for any R-module T and any two R-module homomorphisms $\xi, \eta : T \to M$, $\varphi\xi = \varphi\eta$ implies $\xi = \eta$;*

2. *φ is an epimorphism if and only if for any R-module S and any two R-module homomorphisms $\rho, \theta : N \to S$, $\rho\varphi = \theta\varphi$ implies $\rho = \theta$.*

Proof. 1. ($\Rightarrow$) Suppose φ is injective and $\varphi\xi = \varphi\eta$ for $\xi, \eta : T \to M$. Then for every $t \in T$,

$$\varphi(\xi(t)) = \varphi(\eta(t)),$$

so $\xi(t) - \eta(t) \in \ker\varphi = \{0\}$. Hence $\xi(t) = \eta(t)$ for all $t \in T$, and therefore $\xi = \eta$.

($\Leftarrow$) We show $\ker\varphi = \{0\}$. Let $T = \ker\varphi$, let $\xi : T \hookrightarrow M$ be the inclusion, and let $\eta = 0 : T \to M$ be the zero map. Then $\varphi\xi(t) = \varphi(t) = 0 = \varphi\eta(t)$ for all $t \in T$, so $\varphi\xi = \varphi\eta$. By hypothesis $\xi = \eta$, which means $t = \xi(t) = \eta(t) = 0$ for all $t \in T$. Hence $\ker\varphi = \{0\}$, so φ is injective.

2. ($\Rightarrow$) Suppose φ is surjective and $\rho\varphi = \theta\varphi$. For any $n \in N$, pick $m \in M$ with $\varphi(m) = n$. Then

$$\rho(n) = \rho(\varphi(m)) = \theta(\varphi(m)) = \theta(n),$$

so $\rho = \theta$.

($\Leftarrow$) We show φ is surjective. Let $S = N/\operatorname{Im}\varphi$, let $\rho : N \to S$ be the canonical projection, and let $\theta = 0 : N \to S$ be the zero map. For any $m \in M$, we have

$$\rho(\varphi(m)) = \overline{\varphi(m)} = \bar{0} = \theta(\varphi(m)),$$

so $\rho\varphi = \theta\varphi$. By hypothesis $\rho = \theta$, i.e. the projection $N \to N/\operatorname{Im}\varphi$ is the zero map. This means $N = \operatorname{Im}\varphi$, so φ is surjective.

□

Exercise 12.2.3. *Let M and M_i $(1 \leq i \leq n)$ be R-modules. Show that $M \simeq \bigoplus_{i=1}^n M_i$ if and only if for every $i \in \{1, \dots, n\}$, there exists an $\varphi_i \in \operatorname{End}_R(M)$ such that $\operatorname{Im}\varphi_i \simeq M_i$, $\varphi_i\varphi_j = 0$ for $i \neq j$, $\varphi_i^2 = \varphi_i$, and $\varphi_1 + \varphi_2 + \cdots + \varphi_n = \mathrm{id}_M$.*

Proof. ($\rightarrow$) Suppose $f : M \xrightarrow{\sim} \bigoplus_{i=1}^n M_i$. Let $\iota_i : M_i \hookrightarrow \bigoplus_{j=1}^n M_j$ be the canonical inclusion and $\pi_i : \bigoplus_{j=1}^n M_j \twoheadrightarrow M_i$ the canonical projection. Define

$$\varphi_i = f^{-1} \circ \iota_i \circ \pi_i \circ f \in \operatorname{End}_R(M).$$

Then:

- $\varphi_i^2 = f^{-1}\iota_i\pi_i f f^{-1}\iota_i\pi_i f = f^{-1}\iota_i(\pi_i\iota_i)\pi_i f = f^{-1}\iota_i\pi_i f = \varphi_i$, since $\pi_i\iota_i = \mathrm{id}_{M_i}$.
- For $i \neq j$:$\varphi_i\varphi_j = f^{-1}\iota_i(\pi_i\iota_j)\pi_j f = 0$, since $\pi_i\iota_j = 0$ when $i \neq j$.
- $\sum_{i=1}^n \varphi_i = f^{-1}\left(\sum_{i=1}^n \iota_i\pi_i\right) f = f^{-1} \circ \mathrm{id} \circ f = \mathrm{id}_M$, since $\sum_i \iota_i\pi_i = \mathrm{id}_{\bigoplus M_j}$.
- $\operatorname{Im}\varphi_i = f^{-1} \circ \iota_i \circ \pi_i \circ f(M) \simeq M_i$.

($\Leftarrow$) Suppose such φ_i exist. Define $g : M \to \bigoplus_{i=1}^n \operatorname{Im}\varphi_i$ by $g(x) = (\varphi_1(x), \dots, \varphi_n(x))$.

Surjective: for any $(y_1, \dots, y_n) \in \bigoplus \operatorname{Im} \varphi_i$, let $x = y_1 + \cdots + y_n \in M$. Then

$$\varphi_j(x) = \sum_{i=1}^{n} \varphi_j(y_i) = \sum_{i=1}^{n} \varphi_j(\varphi_i(x_i)) = \varphi_j(\varphi_j(x_j)) = \varphi_j^2(x_j) = \varphi_j(x_j) = y_j,$$

where we used $\varphi_j\varphi_i = 0$ for $j \neq i$ and $\varphi_j^2 = \varphi_j$. Hence $g(x) = (y_1, \dots, y_n)$.

Injective: if $g(x) = 0$, then $\varphi_i(x) = 0$ for all i, so $x = \mathrm{id}_M(x) = \sum_i \varphi_i(x) = 0$.

Since $\operatorname{Im} \varphi_i \simeq M_i$, we conclude $M \simeq \bigoplus_{i=1}^n M_i$. □

Exercise 12.2.4.

1. *Prove $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/(m,n)\mathbb{Z}$, where $m, n \in \mathbb{Z}$.*

2. *Let A be an abelian group, prove $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.*

3. *Let A, B be two finitely generated abelian groups. What is $A \otimes_{\mathbb{Z}} B$?*

Proof.

1. We use the right-exact sequence $\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$. Tensoring with $\mathbb{Z}/m\mathbb{Z}$ over $\mathbb{Z}$ gives the right-exact sequence

$$\mathbb{Z}/m\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \to 0.$$

Hence $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq (\mathbb{Z}/m\mathbb{Z})/n(\mathbb{Z}/m\mathbb{Z})$. Now $n(\mathbb{Z}/m\mathbb{Z}) = (n\mathbb{Z} + m\mathbb{Z})/m\mathbb{Z}$. Since $n\mathbb{Z} + m\mathbb{Z} = (m,n)\mathbb{Z}$, we get

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \frac{\mathbb{Z}/m\mathbb{Z}}{(m,n)\mathbb{Z}/m\mathbb{Z}} \simeq \mathbb{Z}/(m,n)\mathbb{Z}.$$

2. Similarly, tensor the right-exact sequence $\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$ with A:

$$A \xrightarrow{\cdot n} A \to A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \to 0.$$

By right-exactness, $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.

3. By the fundamental theorem of finitely generated abelian groups,

$$A \simeq \mathbb{Z}^r \oplus \bigoplus_{i=1}^{s} \mathbb{Z}/d_i\mathbb{Z}, \qquad B \simeq \mathbb{Z}^t \oplus \bigoplus_{j=1}^{u} \mathbb{Z}/e_j\mathbb{Z}.$$

Since tensor product distributes over direct sums, using (1), (2), and $\mathbb{Z} \otimes_{\mathbb{Z}} M \simeq M$, we get

$$A \otimes_{\mathbb{Z}} B \simeq \mathbb{Z}^{rt} \oplus \bigoplus_{j=1}^{u} (\mathbb{Z}/e_j\mathbb{Z})^r \oplus \bigoplus_{i=1}^{s} (\mathbb{Z}/d_i\mathbb{Z})^t \oplus \bigoplus_{i=1}^{s} \bigoplus_{j=1}^{u} \mathbb{Z}/(d_i, e_j)\mathbb{Z}.$$

□

Exercise 12.2.5. *Let $\varphi : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ be the unique monomorphism. Prove that*

$$\mathrm{id} \otimes \varphi : \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$$

is a zero homomorphism.

Proof. The unique monomorphism $\varphi : \mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ is given by $\varphi(\bar{1}) = \bar{2}$. By the previous exercise, $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z}$.

It suffices to check on the generator $\bar{1} \otimes \bar{1}$ of $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$:

$$(\mathrm{id} \otimes \varphi)(\bar{1} \otimes \bar{1}) = \bar{1} \otimes \varphi(\bar{1}) = \bar{1} \otimes \bar{2} = \bar{1} \otimes 2 \cdot \bar{1} = 2 \cdot (\bar{1} \otimes \bar{1}) = (2 \cdot \bar{1}) \otimes \bar{1} = \bar{0} \otimes \bar{1} = 0,$$

where we used $2 \cdot \bar{1} = \bar{0}$ in $\mathbb{Z}/2\mathbb{Z}$. Since the generator maps to 0, the map $\mathrm{id} \otimes \varphi$ is the zero homomorphism.

This also shows that tensoring a monomorphism need not yield a monomorphism, i.e. the functor $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} -$ is not left-exact. $\square$

Exercise 12.2.6. *Let V_1 and V_2 be vector spaces over a field F, with $\dim_F V_1 = n$ and $\dim_F V_2 = m$. Let $\{\epsilon_1, \dots, \epsilon_n\}$ and $\{\eta_1, \dots, \eta_m\}$ be bases of V_1 and V_2, respectively. Suppose that $\mathscr{A} \colon V_1 \to V_1$ and $\mathscr{B} \colon V_2 \to V_2$ are F-linear maps, represented by the matrices $A = (a_{ij})_{n\times n}$ and $B = (b_{ij})_{m\times m}$ with respect to these bases. Prove that*

$$\{\epsilon_i \otimes \eta_j \mid 1 \le i \le n,\, 1 \le j \le m\}$$

is an F-basis of $V_1 \otimes_F V_2$. Determine the matrix representing the F-linear map $\mathscr{A} \otimes \mathscr{B}$ under this basis.

Proof. **Step 1.** The set $\{\epsilon_i \otimes \eta_j\}$ is a basis.

Spanning. Every element of $V_1 \otimes_F V_2$ is a finite sum of simple tensors $v \otimes w$. Writing $v = \sum_i a_i \epsilon_i$ and $w = \sum_j b_j \eta_j$, by bilinearity of the tensor product we get

$$v \otimes w = \sum_{i,j} a_i b_j \, (\epsilon_i \otimes \eta_j).$$

Hence $\{\epsilon_i \otimes \eta_j\}$ spans $V_1 \otimes_F V_2$.

Linear independence. Suppose $\sum_{i,j} c_{ij}(\epsilon_i \otimes \eta_j) = 0$. For each pair (p, q), let $f_p \in V_1^*$ and $g_q \in V_2^*$ be the dual basis functionals:$f_p(\epsilon_i) = \delta_{pi}$ and $g_q(\eta_j) = \delta_{qj}$. The bilinear map $(v, w) \mapsto f_p(v) g_q(w)$ from $V_1 \times V_2$ to F induces (by the universal property of the tensor product) an F-linear map $\Phi_{pq} : V_1 \otimes_F V_2 \to F$ with $\Phi_{pq}(\epsilon_i \otimes \eta_j) = \delta_{pi}\delta_{qj}$. Applying Φ_{pq} to the relation $\sum_{i,j} c_{ij}(\epsilon_i \otimes \eta_j) = 0$ yields $c_{pq} = 0$. Since (p, q) was arbitrary, all $c_{ij} = 0$.

Therefore $\{\epsilon_i \otimes \eta_j : 1 \le i \le n,\, 1 \le j \le m\}$ is an F-basis of $V_1 \otimes_F V_2$, and $\dim_F(V_1 \otimes_F V_2) = nm$.

Step 2: matrix of $\mathscr{A} \otimes \mathscr{B}$.

Order the basis lexicographically:$\epsilon_1 \otimes \eta_1,\ \epsilon_1 \otimes \eta_2,\ \dots,\ \epsilon_1 \otimes \eta_m,\ \epsilon_2 \otimes \eta_1,\ \dots,\ \epsilon_n \otimes \eta_m$. Since

$\mathscr{A}(\epsilon_i) = \sum_{k=1}^n a_{ki}\epsilon_k$ and $\mathscr{B}(\eta_j) = \sum_{l=1}^m b_{lj}\eta_l$, we have

$$(\mathscr{A} \otimes \mathscr{B})(\epsilon_i \otimes \eta_j) = \mathscr{A}(\epsilon_i) \otimes \mathscr{B}(\eta_j) = \sum_{k=1}^{n}\sum_{l=1}^{m} a_{ki}b_{lj}\,(\epsilon_k \otimes \eta_l).$$

The entry in row (k,l) and column (i,j) of the representing matrix is $a_{ki}b_{lj}$. Written as an $nm \times nm$ block matrix, this is the **Kronecker product**

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{pmatrix}.$$

□

12.3 Homework Week 3

Exercise 12.3.1. *Let M, N, $(M_i)_{i\in I}$, and $(N_j)_{j\in J}$ be R-modules. Prove the following isomorphisms of abelian groups:*

1.

$$\operatorname{Hom}_R\left(\bigoplus_{i\in I} M_i,\, N\right) \cong \prod_{i\in I} \operatorname{Hom}_R(M_i, N).$$

2.

$$\operatorname{Hom}_R\left(M,\, \prod_{j\in J} N_j\right) \cong \prod_{j\in J} \operatorname{Hom}_R(M, N_j).$$

Proof. We prove the two statements separately.

(1) Define

$$\Phi : \operatorname{Hom}_R\left(\bigoplus_{i\in I} M_i,\, N\right) \longrightarrow \prod_{i\in I} \operatorname{Hom}_R(M_i, N)$$

by

$$\Phi(f) = (f \circ \iota_i)_{i\in I},$$

where

$$\iota_i : M_i \to \bigoplus_{i\in I} M_i$$

is the canonical inclusion.

We show that Φ is an isomorphism.

First, Φ is a group homomorphism, since composition is compatible with addition:

$$\Phi(f+g) = \big((f+g)\circ \iota_i\big)_{i\in I} = \big(f\circ\iota_i + g\circ\iota_i\big)_{i\in I} = \Phi(f) + \Phi(g).$$

Now let $(f_i)_{i\in I} \in \prod_{i\in I} \mathrm{Hom}_R(M_i, N)$. Define

$$f : \bigoplus_{i\in I} M_i \to N$$

by

$$f\big((m_i)_{i\in I}\big) = \sum_{i\in I} f_i(m_i).$$

This is well-defined, because an element of $\bigoplus_{i\in I} M_i$ has only finitely many nonzero components, so the above sum is finite. It is immediate that f is R-linear, and

$$f \circ \iota_i = f_i \qquad \text{for all } i \in I.$$

Hence $\Phi(f) = (f_i)_{i\in I}$, so Φ is surjective.

To prove injectivity, suppose $\Phi(f) = \Phi(g)$. Then

$$f \circ \iota_i = g \circ \iota_i \qquad \text{for all } i \in I.$$

Take any $x = (m_i)_{i\in I} \in \bigoplus_{i\in I} M_i$. Since x has finite support, we may write

$$x = \sum_{i\in I} \iota_i(m_i),$$

with only finitely many nonzero terms. Therefore,

$$f(x) = \sum_{i\in I} f(\iota_i(m_i)) = \sum_{i\in I} g(\iota_i(m_i)) = g(x).$$

So $f = g$, and Φ is injective.

Thus Φ is an isomorphism:

$$\mathrm{Hom}_R\left(\bigoplus_{i\in I} M_i,\, N\right) \cong \prod_{i\in I} \mathrm{Hom}_R(M_i, N).$$

(2) Let

$$\pi_j : \prod_{j\in J} N_j \to N_j$$

be the canonical projection. Define

$$\Psi : \mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right) \longrightarrow \prod_{j\in J} \mathrm{Hom}_R(M, N_j)$$

by

$$\Psi(f) = (\pi_j \circ f)_{j\in J}.$$

Again, Ψ is a group homomorphism.

We show that Ψ is an isomorphism. Given a family

$$(g_j)_{j\in J} \in \prod_{j\in J} \mathrm{Hom}_R(M, N_j),$$

define

$$g : M \to \prod_{j\in J} N_j$$

by

$$g(m) = (g_j(m))_{j\in J}.$$

Since each g_j is R-linear, g is also R-linear. Moreover,

$$\pi_j \circ g = g_j \qquad \text{for all } j \in J,$$

so

$$\Psi(g) = (g_j)_{j\in J}.$$

Hence Ψ is surjective.

For injectivity, suppose $\Psi(f) = \Psi(g)$. Then

$$\pi_j \circ f = \pi_j \circ g \qquad \text{for all } j \in J.$$

Thus for every $m \in M$ and every $j \in J$,

$$\pi_j(f(m)) = \pi_j(g(m)).$$

Since two elements of $\prod_{j\in J} N_j$ are equal if and only if all their components are equal, it follows that

$$f(m) = g(m) \qquad \text{for all } m \in M.$$

Hence $f = g$, and Ψ is injective.

Therefore

$$\mathrm{Hom}_R\left(M, \prod_{j\in J} N_j\right) \cong \prod_{j\in J} \mathrm{Hom}_R(M, N_j).$$

This completes the proof. $\square$

Exercise 12.3.2 (Five Lemma)**.** *Suppose that the following diagram of homomorphisms of R-modules*

$$\begin{tikzcd}
M_1 \arrow[r] \arrow[d, "\varphi_1"'] & M_2 \arrow[r] \arrow[d, "\varphi_2"'] & M_3 \arrow[r] \arrow[d, "\varphi_3"'] & M_4 \arrow[r] \arrow[d, "\varphi_4"'] & M_5 \arrow[d, "\varphi_5"'] \\
N_1 \arrow[r] & N_2 \arrow[r] & N_3 \arrow[r] & N_4 \arrow[r] & N_5
\end{tikzcd}$$

is commutative and has exact rows. Prove that:

1. *If φ_1 is surjective, and φ_2 and φ_4 are injective, then φ_3 is injective.*
2. *If φ_5 is injective, and φ_2 and φ_4 are surjective, then φ_3 is surjective.*

Proof. We prove the two assertions separately.

(1) Assume that φ_1 is surjective and that φ_2, φ_4 are injective. We show that φ_3 is injective.

Let $x \in M_3$ and suppose that

$$\varphi_3(x) = 0.$$

We must prove that $x = 0$.

Let $\alpha_3 : M_3 \to M_4$ and $\beta_3 : N_3 \to N_4$ denote the third horizontal maps. By commutativity,

$$\varphi_4(\alpha_3(x)) = \beta_3(\varphi_3(x)) = \beta_3(0) = 0.$$

Since φ_4 is injective, it follows that $\alpha_3(x) = 0$. Exactness of the top row at M_3 yields

$$x = \alpha_2(y)$$

for some $y \in M_2$, where $\alpha_2 : M_2 \to M_3$ is the second horizontal map.

Again by commutativity,

$$0 = \varphi_3(x) = \varphi_3(\alpha_2(y)) = \beta_2(\varphi_2(y)),$$

where $\beta_2 : N_2 \to N_3$ is the second horizontal map. Hence $\varphi_2(y) \in \ker(\beta_2)$. By exactness of the bottom row at N_2,

$$\ker(\beta_2) = \mathrm{Im}(\beta_1),$$

so there exists $z \in N_1$ such that

$$\beta_1(z) = \varphi_2(y),$$

where $\beta_1 : N_1 \to N_2$ is the first horizontal map.

Since φ_1 is surjective, there exists $w \in M_1$ such that

$$\varphi_1(w) = z.$$

Using commutativity once more,

$$\varphi_2(\alpha_1(w)) = \beta_1(\varphi_1(w)) = \beta_1(z) = \varphi_2(y),$$

where $\alpha_1 : M_1 \to M_2$ is the first horizontal map. Because φ_2 is injective, we get

$$\alpha_1(w) = y.$$

Therefore

$$x = \alpha_2(y) = \alpha_2(\alpha_1(w)) = 0,$$

since the composition of two consecutive maps in an exact sequence is zero. Thus $\ker(\varphi_3) = 0$, so φ_3 is injective.

(2) Assume that φ_5 is injective and that φ_2, φ_4 are surjective. We show that φ_3 is surjective.

Let $n \in N_3$. We must find $m \in M_3$ such that

$$\varphi_3(m) = n.$$

Let $\beta_3 : N_3 \to N_4$ and $\alpha_3 : M_3 \to M_4$ denote the third horizontal maps. Since φ_4 is surjective, there exists $u \in M_4$ such that

$$\varphi_4(u) = \beta_3(n).$$

Let $\alpha_4 : M_4 \to M_5$ and $\beta_4 : N_4 \to N_5$ be the next horizontal maps. Then

$$\varphi_5(\alpha_4(u)) = \beta_4(\varphi_4(u)) = \beta_4(\beta_3(n)) = 0.$$

Because φ_5 is injective, it follows that $\alpha_4(u) = 0$. Exactness of the top row at M_4 shows that there exists $m \in M_3$ such that

$$\alpha_3(m) = u.$$

Consider

$$n - \varphi_3(m) \in N_3.$$

By commutativity,

$$\beta_3\big(n - \varphi_3(m)\big) = \beta_3(n) - \beta_3(\varphi_3(m)) = \beta_3(n) - \varphi_4(\alpha_3(m)) = \beta_3(n) - \varphi_4(u) = 0.$$

Hence $n - \varphi_3(m) \in \ker(\beta_3)$. By exactness of the bottom row at N_3,

$$\ker(\beta_3) = \mathrm{Im}(\beta_2),$$

so there exists $v \in N_2$ such that

$$\beta_2(v) = n - \varphi_3(m),$$

where $\beta_2 : N_2 \to N_3$ is the second horizontal map.

Since φ_2 is surjective, there exists $y \in M_2$ such that

$$\varphi_2(y) = v.$$

Then

$$\varphi_3(\alpha_2(y)) = \beta_2(\varphi_2(y)) = \beta_2(v) = n - \varphi_3(m),$$

where $\alpha_2 : M_2 \to M_3$ is the second horizontal map. Therefore

$$n = \varphi_3(m) + \varphi_3(\alpha_2(y)) = \varphi_3\big(m + \alpha_2(y)\big).$$

Thus n lies in the image of φ_3, proving that φ_3 is surjective. $\square$

Exercise 12.3.3 (Snake Lemma)**.** *Suppose that the following diagram of homomorphisms* *of R-modules*

$$\begin{tikzcd}
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d, "\varphi'"'] & M \arrow[r, "\beta"] \arrow[d, "\varphi"'] & M'' \arrow[r] \arrow[d, "\varphi''"'] & 0 \\
0 \arrow[r] & N' \arrow[r, "\mu"] & N \arrow[r, "\nu"] & N'' \arrow[r] & 0
\end{tikzcd}$$

is commutative and has exact rows. Prove that

$$0 \longrightarrow \ker\varphi' \xrightarrow{\alpha'} \ker\varphi \xrightarrow{\beta'} \ker\varphi'' \xrightarrow{\delta} \operatorname{coker}\varphi' \xrightarrow{\mu'} \operatorname{coker}\varphi \xrightarrow{\nu'} \operatorname{coker}\varphi'' \longrightarrow 0$$

is a long exact sequence, where $\alpha', \beta', \mu', \nu'$ are the homomorphisms induced by α, β, μ, ν, respectively.

Proof. Since the rows are exact, α and μ are injective, while β and ν are surjective.

We first define the connecting homomorphism

$$\delta : \ker\varphi'' \to \operatorname{coker}\varphi'.$$

Let $x \in \ker\varphi'' \subseteq M''$. Since $\beta : M \to M''$ is surjective, choose $y \in M$ such that

$$\beta(y) = x.$$

By commutativity,

$$\nu(\varphi(y)) = \varphi''(\beta(y)) = \varphi''(x) = 0.$$

Hence $\varphi(y) \in \ker\nu = \operatorname{Im}\mu$. Therefore there exists $z \in N'$ such that

$$\mu(z) = \varphi(y).$$

Define

$$\delta(x) = \overline{z} = z + \operatorname{Im}\varphi' \in \operatorname{coker}\varphi'.$$

We must check that this is well-defined. First, suppose $y_1, y_2 \in M$ both satisfy

$$\beta(y_1) = \beta(y_2) = x.$$

Then $y_1 - y_2 \in \ker\beta = \operatorname{Im}\alpha$, so there exists $m' \in M'$ such that

$$y_1 - y_2 = \alpha(m').$$

If $\mu(z_i) = \varphi(y_i)$ for $i = 1, 2$, then

$$\mu(z_1 - z_2) = \varphi(y_1) - \varphi(y_2) = \varphi(\alpha(m')) = \mu(\varphi'(m')),$$

where we used commutativity. Since μ is injective, it follows that

$$z_1 - z_2 = \varphi'(m') \in \operatorname{Im}\varphi'.$$

Thus $\overline{z_1} = \overline{z_2}$ in $\operatorname{coker} \varphi'$, so $\delta(x)$ is independent of the choice of y.

Next, if $z, z' \in N'$ both satisfy

$$\mu(z) = \mu(z') = \varphi(y),$$

then $\mu(z - z') = 0$, and since μ is injective, $z = z'$. Hence $\delta(x)$ is also independent of the choice of z.

It is straightforward to verify that δ is a homomorphism.

Now we prove exactness.

Exactness at $\ker \varphi'$. Since α induces α', we have

$$\alpha'(\ker \varphi') \subseteq \ker \varphi.$$

Because α is injective, so is α'. Hence

$$\ker \alpha' = 0.$$

Therefore the sequence is exact at $\ker \varphi'$.

Exactness at $\ker \varphi$. We show that

$$\operatorname{Im} \alpha' = \ker \beta'.$$

Take $x' \in \ker \varphi'$. Then

$$\beta'(\alpha'(x')) = \beta(\alpha(x')) = 0,$$

so $\operatorname{Im} \alpha' \subseteq \ker \beta'$.

Conversely, let $y \in \ker \varphi$ satisfy $\beta'(y) = 0$. Then $\beta(y) = 0$, hence by exactness of the top row there exists $x' \in M'$ such that

$$\alpha(x') = y.$$

Since $\varphi(y) = 0$, we get

$$0 = \varphi(\alpha(x')) = \mu(\varphi'(x')).$$

As μ is injective, $\varphi'(x') = 0$, so $x' \in \ker \varphi'$. Thus

$$y = \alpha'(x'),$$

proving $\ker \beta' \subseteq \operatorname{Im} \alpha'$.

Exactness at $\ker \varphi''$. We show that

$$\operatorname{Im} \beta' = \ker \delta.$$

First let $y \in \ker \varphi$. Then $\beta(y) \in \ker \varphi''$, since

$$\varphi''(\beta(y)) = \nu(\varphi(y)) = 0.$$

Thus $\beta'(y) \in \ker \varphi''$. To compute $\delta(\beta'(y))$, choose the lift y itself. Then $\varphi(y) = 0$, so we may take $z = 0$, and hence

$$\delta(\beta'(y)) = \overline{0} = 0.$$

Therefore $\operatorname{Im} \beta' \subseteq \ker \delta$.

Conversely, let $x \in \ker \varphi''$ and suppose $\delta(x) = 0$. Choose $y \in M$ with

$$\beta(y) = x,$$

and choose $z \in N'$ such that

$$\mu(z) = \varphi(y).$$

Since $\delta(x) = \overline{z} = 0$ in $\operatorname{coker} \varphi'$, there exists $x' \in M'$ such that

$$z = \varphi'(x').$$

Hence

$$\varphi(y - \alpha(x')) = \varphi(y) - \varphi(\alpha(x')) = \mu(z) - \mu(\varphi'(x')) = 0.$$

So $y - \alpha(x') \in \ker \varphi$, and

$$\beta'(y - \alpha(x')) = \beta(y - \alpha(x')) = \beta(y) = x,$$

because $\beta\alpha = 0$. Thus $x \in \operatorname{Im} \beta'$, proving

$$\ker \delta \subseteq \operatorname{Im} \beta'.$$

Exactness at $\operatorname{coker} \varphi'$. We show that

$$\operatorname{Im} \delta = \ker \mu'.$$

Let $x \in \ker \varphi''$, and write

$$\delta(x) = \overline{z} \in \operatorname{coker} \varphi'$$

as above, with $\mu(z) = \varphi(y)$. Then

$$\mu'(\delta(x)) = \overline{\mu(z)} = \overline{\varphi(y)} = 0$$

in $\operatorname{coker} \varphi$. Hence $\operatorname{Im} \delta \subseteq \ker \mu'$.

Conversely, let $\overline{z} \in \operatorname{coker} \varphi'$ satisfy

$$\mu'(\overline{z}) = 0.$$

This means that $\mu(z) \in \operatorname{Im} \varphi$, so there exists $y \in M$ such that

$$\varphi(y) = \mu(z).$$

Applying ν, we get

$$\varphi''(\beta(y)) = \nu(\varphi(y)) = \nu(\mu(z)) = 0.$$

Thus $\beta(y) \in \ker \varphi''$. By construction of δ,

$$\delta(\beta(y)) = \overline{z}.$$

So $\overline{z} \in \operatorname{Im} \delta$, proving

$$\ker \mu' \subseteq \operatorname{Im} \delta.$$

Exactness at $\operatorname{coker} \varphi$**.** We show that

$$\operatorname{Im} \mu' = \ker \nu'.$$

For $\overline{z} \in \operatorname{coker} \varphi'$, we have

$$\nu'(\mu'(\overline{z})) = \overline{\nu(\mu(z))} = \overline{0} = 0,$$

so $\operatorname{Im} \mu' \subseteq \ker \nu'$.

Conversely, let $\overline{n} \in \operatorname{coker} \varphi$ satisfy

$$\nu'(\overline{n}) = 0.$$

Then $\overline{\nu(n)} = 0$ in $\operatorname{coker} \varphi''$, so there exists $m'' \in M''$ such that

$$\varphi''(m'') = \nu(n).$$

Since β is surjective, choose $m \in M$ such that

$$\beta(m) = m''.$$

Then

$$\nu(n - \varphi(m)) = \nu(n) - \varphi''(\beta(m)) = \nu(n) - \varphi''(m'') = 0.$$

Hence

$$n - \varphi(m) \in \ker \nu = \operatorname{Im} \mu,$$

so there exists $z \in N'$ such that

$$\mu(z) = n - \varphi(m).$$

Therefore in $\operatorname{coker} \varphi$,

$$\overline{n} = \overline{\mu(z)} = \mu'(\overline{z}),$$

showing that $\overline{n} \in \operatorname{Im} \mu'$. Thus

$$\ker \nu' \subseteq \operatorname{Im} \mu'.$$

Exactness at $\operatorname{coker} \varphi''$ **and termination.** Since $\nu : N \to N''$ is surjective, the induced map

$$\nu' : \operatorname{coker}\varphi \to \operatorname{coker}\varphi''$$

is surjective as well: for any $\overline{n''} \in \operatorname{coker}\varphi''$, choose $n \in N$ such that

$$\nu(n) = n'',$$

then

$$\nu'(\overline{n}) = \overline{n''}.$$

Hence

$$\operatorname{Im}\nu' = \operatorname{coker}\varphi'',$$

which is equivalent to exactness at $\operatorname{coker}\varphi''$ and the final 0.

Collecting all the above, we obtain the exact sequence

$$0 \longrightarrow \ker\varphi' \xrightarrow{\alpha'} \ker\varphi \xrightarrow{\beta'} \ker\varphi'' \xrightarrow{\delta} \operatorname{coker}\varphi' \xrightarrow{\mu'} \operatorname{coker}\varphi \xrightarrow{\nu'} \operatorname{coker}\varphi'' \longrightarrow 0.$$

□

12.4 Homework Week 4

Exercise 12.4.1. *Let J be an R-module. Show that the following are equivalent.*

1. *J is an injective module.*
2. *For any R-module monomorphism $\varphi\colon J \to M$, there exists an R-module homomorphism $\psi\colon M \to J$ such that $\psi\varphi = \mathrm{id}_J$.*
3. *If J is a submodule of some R-module M, then J is a direct summand of M.*

Proof. We prove (1) ⇒ (2) ⇒ (3) ⇒ (1).

(1) ⇒ (2)**.** Assume that J is injective, and let $\varphi\colon J \to M$ be a monomorphism. Apply the defining property of injectivity to the monomorphism φ and the map $\mathrm{id}_J\colon J \to J$:

$$\begin{tikzcd} J \arrow[r, hook, "\varphi"] \arrow[d, "\mathrm{id}_J"'] & M \arrow[dl, dashed, "\exists\psi"] \\ J & \end{tikzcd}$$

Then there exists an R-module homomorphism $\psi\colon M \to J$ such that

$$\psi \circ \varphi = \mathrm{id}_J\,.$$

(2) ⇒ (3)**.** Assume (2). If $J \subseteq M$, let $\iota\colon J \hookrightarrow M$ be the inclusion map. By (2), there exists $\psi\colon M \to J$ such that

$$\psi \circ \iota = \mathrm{id}_J\,.$$

We claim that
$$M = J \oplus \ker \psi.$$
Indeed, for any $m \in M$,
$$m = \iota(\psi(m)) + \big(m - \iota(\psi(m))\big),$$
and
$$\psi\big(m - \iota(\psi(m))\big) = \psi(m) - \psi(\iota(\psi(m))) = \psi(m) - \psi(m) = 0,$$
so $m - \iota(\psi(m)) \in \ker \psi$. Hence $M = J + \ker \psi$. If $x \in J \cap \ker \psi$, then
$$x = (\psi \circ \iota)(x) = \psi(x) = 0,$$
so $J \cap \ker \psi = 0$. Therefore $M = J \oplus \ker \psi$, and J is a direct summand of M.

(3) ⇒ (1). Assume (3). To prove that J is injective, let
$$i \colon A \hookrightarrow B$$
be a monomorphism of R-modules, and let
$$f \colon A \to J$$
be an R-module homomorphism. We must show that f extends to a map $g \colon B \to J$.

Consider the quotient module
$$P = (B \oplus J)/N, \qquad N = \{(i(a), -f(a)) \mid a \in A\}.$$
Define maps
$$\eta \colon J \to P, \qquad \eta(j) = [(0, j)],$$
and
$$\beta \colon B \to P, \qquad \beta(b) = [(b, 0)].$$
We first show that η is injective. If $\eta(j) = 0$, then $(0, j) \in N$, so for some $a \in A$,
$$(0, j) = (i(a), -f(a)).$$
Hence $i(a) = 0$. Since i is injective, $a = 0$, and therefore $j = -f(0) = 0$. Thus η is a monomorphism, so we may identify J with the submodule $\eta(J) \subseteq P$.

By (3), this submodule is a direct summand of P. Therefore there exists an R-module homomorphism
$$s \colon P \to J$$
such that
$$s \circ \eta = \mathrm{id}_J .$$
Now define
$$g = s \circ \beta \colon B \to J.$$

For any $a \in A$, we have in P:

$$[(i(a), 0)] = [(0, f(a))],$$

because

$$(i(a), 0) - (0, f(a)) = (i(a), -f(a)) \in N.$$

Hence

$$g(i(a)) = s([(i(a), 0)]) = s([(0, f(a))]) = (s \circ \eta)(f(a)) = f(a).$$

So $g \circ i = f$, and f extends to B. Therefore J is injective. $\square$

Exercise 12.4.2. *Suppose that $(M_i)_{i \in I}$ are R-modules. Prove that*

$$\bigoplus_{i \in I} M_i$$

is a flat R-module if and only if each M_i is a flat R-module.

Proof. Set

$$M = \bigoplus_{i \in I} M_i.$$

First assume that each M_i is flat. Let

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be a short exact sequence of R-modules. Since each M_i is flat, for every $i \in I$ the sequence

$$0 \longrightarrow M_i \otimes_R A \longrightarrow M_i \otimes_R B \longrightarrow M_i \otimes_R C \longrightarrow 0$$

is exact. Taking direct sums over $i \in I$, we obtain an exact sequence

$$0 \longrightarrow \bigoplus_{i \in I} (M_i \otimes_R A) \longrightarrow \bigoplus_{i \in I} (M_i \otimes_R B) \longrightarrow \bigoplus_{i \in I} (M_i \otimes_R C) \longrightarrow 0.$$

Using the canonical isomorphisms

$$\left(\bigoplus_{i \in I} M_i \right) \otimes_R X \cong \bigoplus_{i \in I} (M_i \otimes_R X) \qquad (X = A, B, C),$$

we get

$$0 \longrightarrow M \otimes_R A \longrightarrow M \otimes_R B \longrightarrow M \otimes_R C \longrightarrow 0.$$

Hence M is flat.

Conversely, assume that M is flat. Fix $i \in I$, and write

$$M = M_i \oplus N_i, \qquad N_i = \bigoplus_{j \neq i} M_j.$$

Let

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be a short exact sequence of R-modules. Since M is flat, the sequence

$$0 \longrightarrow M \otimes_R A \longrightarrow M \otimes_R B \longrightarrow M \otimes_R C \longrightarrow 0$$

is exact. Using

$$M \otimes_R X \cong (M_i \otimes_R X) \oplus (N_i \otimes_R X) \qquad (X = A, B, C),$$

this becomes

$$0 \longrightarrow (M_i \otimes_R A) \oplus (N_i \otimes_R A) \longrightarrow (M_i \otimes_R B) \oplus (N_i \otimes_R B) \longrightarrow (M_i \otimes_R C) \oplus (N_i \otimes_R C) \longrightarrow 0.$$

Since kernels and images of direct sum maps decompose componentwise, it follows that

$$0 \longrightarrow M_i \otimes_R A \longrightarrow M_i \otimes_R B \longrightarrow M_i \otimes_R C \longrightarrow 0$$

is exact. Therefore M_i is flat. As i was arbitrary, each M_i is flat. $\square$

Exercise 12.4.3. *Prove that the following three conditions on an R-module V are equivalent.*

1. (Ascending chain condition.) *For every infinite chain of R-submodules of V,*

$$V_1 \subseteq V_2 \subseteq \cdots \subseteq V_n \subseteq \cdots,$$

   there is a number m such that

$$V_m = V_{m+1} = \cdots.$$

2. *Any non-empty set of submodules of V contains a maximal element with respect to inclusion.*

3. *Every R-submodule of V is finitely generated.*

Proof. We prove (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1).

(1) $\Rightarrow$ (2)**.** Assume that (2) fails. Then there exists a non-empty set $\mathcal{S}$ of submodules of V having no maximal element. Choose $V_1 \in \mathcal{S}$. Since V_1 is not maximal in $\mathcal{S}$, there exists $V_2 \in \mathcal{S}$ such that

$$V_1 \subsetneq V_2.$$

Inductively, having chosen $V_n \in \mathcal{S}$, since V_n is not maximal there exists $V_{n+1} \in \mathcal{S}$ with

$$V_n \subsetneq V_{n+1}.$$

Thus we obtain a strictly ascending chain

$$V_1 \subsetneq V_2 \subsetneq \cdots,$$

which contradicts (1). Hence every non-empty set of submodules contains a maximal element.

(2) ⇒ (3). Let U be a submodule of V. Consider the set

$$\mathcal{T} = \{N \leq U \mid N \text{ is finitely generated}\}.$$

This set is non-empty, since $0 \in \mathcal{T}$. By (2), $\mathcal{T}$ has a maximal element, say N.

We claim that $N = U$. If not, choose $u \in U \setminus N$. Then

$$N + Ru$$

is a finitely generated submodule of U properly containing N, contradicting the maximality of N in $\mathcal{T}$. Therefore $U = N$, so U is finitely generated. Hence every submodule of V is finitely generated.

(3) ⇒ (1). Let

$$V_1 \subseteq V_2 \subseteq \cdots$$

be an ascending chain of submodules of V. Set

$$U = \bigcup_{n \geq 1} V_n.$$

Since the chain is ascending, U is a submodule of V. By (3), U is finitely generated, say

$$U = Ru_1 + \cdots + Ru_r.$$

For each j, choose n_j such that $u_j \in V_{n_j}$, and let

$$m = \max\{n_1, \ldots, n_r\}.$$

Then $u_1, \ldots, u_r \in V_m$, so $U \subseteq V_m$. Since always $V_m \subseteq U$, we get $U = V_m$. Therefore for all $n \geq m$,

$$V_n = U = V_m,$$

so the chain stabilizes. Hence (1) holds. □

Exercise 12.4.4. *Let W be an R-submodule of V. Prove that V is Noetherian if and only if both W and V/W are Noetherian.*

Proof. We use the characterization that an R-module is Noetherian if and only if every submodule is finitely generated.

Assume first that V is Noetherian. Then every submodule of V is finitely generated.

Since every submodule of W is also a submodule of V, it follows that every submodule of W is finitely generated. Hence W is Noetherian.

Now let $\overline{U}$ be a submodule of V/W, and let

$$\pi \colon V \to V/W$$

be the quotient map. Then $U := \pi^{-1}(\overline{U})$ is a submodule of V, so U is finitely generated, say

$$U = Rv_1 + \cdots + Rv_n.$$

Applying π, we get

$$\overline{U} = R(v_1 + W) + \cdots + R(v_n + W).$$

Thus every submodule of V/W is finitely generated, and therefore V/W is Noetherian.

Conversely, assume that both W and V/W are Noetherian. Let U be any submodule of V. We show that U is finitely generated.

First,

$$U \cap W$$

is a submodule of W, so it is finitely generated. Choose generators

$$U \cap W = Rw_1 + \cdots + Rw_m.$$

Next, since

$$(U + W)/W$$

is a submodule of V/W, it is finitely generated as well. Choose elements $u_1, \ldots, u_n \in U$ such that

$$(U + W)/W = R(u_1 + W) + \cdots + R(u_n + W).$$

Now let $u \in U$. Since $u + W \in (U + W)/W$, there exist $a_1, \ldots, a_n \in R$ such that

$$u + W = \sum_{i=1}^{n} a_i(u_i + W).$$

Therefore

$$u - \sum_{i=1}^{n} a_i u_i \in W.$$

Because also $u, u_i \in U$, we have

$$u - \sum_{i=1}^{n} a_i u_i \in U \cap W.$$

So there exist $b_1, \ldots, b_m \in R$ such that

$$u - \sum_{i=1}^{n} a_i u_i = \sum_{j=1}^{m} b_j w_j.$$

Hence

$$u = \sum_{i=1}^{n} a_i u_i + \sum_{j=1}^{m} b_j w_j.$$

This shows that U is generated by

$$u_1, \ldots, u_n, w_1, \ldots, w_m.$$

Thus every submodule of V is finitely generated, so V is Noetherian. $\square$

Exercise 12.4.5. *Let V be a Noetherian R-module and let $f\colon V \to V$ be an epimorphism. Show that f is an R-isomorphism.*

Proof. Since f is an epimorphism, it is surjective. It therefore suffices to prove that f is injective.

For each $n \geq 1$, set

$$K_n = \ker(f^n).$$

Then

$$K_1 \subseteq K_2 \subseteq K_3 \subseteq \cdots,$$

because if $f^n(x) = 0$, then

$$f^{n+1}(x) = f(f^n(x)) = 0.$$

Since V is Noetherian, this ascending chain stabilizes. Hence there exists $m \geq 1$ such that

$$K_m = K_{m+1}.$$

We claim that $K_m = 0$. Let $x \in K_m$. Because f is surjective, f^m is also surjective, there exists $y \in V$ such that

$$f^m(y) = x.$$

Now

$$f^{2m}(y) = f^m(f^m(y)) = f^m(x) = 0,$$

so $y \in K_{2m} = K_m$. Therefore

$$x = f^m(y) = 0.$$

Thus $K_m = 0$.

Since

$$\ker(f) = K_1 \subseteq K_m = 0,$$

we obtain $\ker(f) = 0$. So f is injective. Being both surjective and injective, f is an isomorphism. $\square$

12.5 Homework Week 5

Exercise 12.5.1. *Are the following statements correct?*

(1) Any submodule of a semisimple module is also semisimple.

(2) Any semisimple Noetherian module is Artinian.

Proof. **(1)** True. Let V be a semisimple module, so

$$V = \bigoplus_{i \in I} S_i$$

where each S_i is simple. Let $W \leq V$ be a submodule. We show that W is a sum of simple submodules, hence semisimple.

Consider the set $\mathcal{S}$ of all subsets $J \subseteq I$ such that the sum

$$W + \bigoplus_{j \in J} S_j$$

is direct, i.e.

$$W \cap \bigoplus_{j \in J} S_j = 0.$$

By Zorn's lemma, there exists a maximal such subset J_0. Set

$$U := W \oplus \bigoplus_{j \in J_0} S_j.$$

For any $i \in I \setminus J_0$, the maximality of J_0 forces $U \cap S_i \neq 0$. Since S_i is simple, we have $S_i \subseteq U$. Therefore $S_i \subseteq U$ for every $i \in I$, so $V = U$. Hence

$$V = W \oplus \bigoplus_{j \in J_0} S_j,$$

and

$$W \cong V \Big/ \bigoplus_{j \in J_0} S_j \cong \bigoplus_{i \in I \setminus J_0} S_i,$$

which is semisimple.

(2) True. Let V be a semisimple Noetherian module. Since V is semisimple,

$$V = \bigoplus_{i \in I} S_i$$

Since V is Noetherian, it is finitely generated. Let $v_1, \ldots, v_n$ generate V. For each k, because $V = \bigoplus_{i \in I} S_i$ is a direct sum, the element v_k has only finitely many nonzero components, so there exists a finite subset $I_k \subseteq I$ such that

$$v_k \in \bigoplus_{i \in I_k} S_i.$$

Hence

$$V = \langle v_1, \ldots, v_n \rangle \subseteq \bigoplus_{i \in I_1 \cup \cdots \cup I_n} S_i.$$

Since the reverse inclusion is obvious, we get

$$V = \bigoplus_{i \in I_1 \cup \cdots \cup I_n} S_i,$$

and therefore V is a finite direct sum of simple modules. Each S_i is simple, hence both Noetherian and Artinian. A finite direct sum of Artinian modules is Artinian, so V is Artinian. $\square$

Exercise 12.5.2. *Let V be an R-module with a composition series. Prove that the following conditions are equivalent.*

(1) V is semisimple.

(2) Every irreducible submodule of V is a direct summand of V.

(3) Every maximal submodule of V is a direct summand of V.

Proof. We prove

$$(1) \Longrightarrow (2) \Longrightarrow (3) \Longrightarrow (1).$$

(1) $\Longrightarrow$ (2). If V is semisimple, then every submodule of V is a direct summand. In particular, every irreducible submodule of V is a direct summand.

(2) $\Longrightarrow$ (3). Let M be a maximal submodule of V. Then V/M is simple.

Since M is a maximal submodule of V, the quotient V/M is simple. Now let S be a submodule of V such that $S \not\subseteq M$. Then $S + M \neq M$, so

$$(S + M)/M \neq 0.$$

By the second isomorphism theorem,

$$S/(S \cap M) \cong (S + M)/M.$$

Since $(S + M)/M$ is a nonzero submodule of the simple module V/M, it follows that $(S + M)/M$ is simple. Hence $S/(S \cap M)$ is simple.

Choose $S \subseteq V$ minimal subject to $S \not\subseteq M$. Then $S \cap M$ is a maximal submodule of S, and by minimality of S, the quotient $S/(S \cap M)$ is simple. But $S \cap M$ is a proper submodule of S, so by minimality of S, it must be 0. Thus S is simple.

Therefore S is an irreducible submodule of V with $S \not\subseteq M$. By (2), S is a direct summand of V, say

$$V = S \oplus T.$$

Since $S \not\subseteq M$ and S is simple, we have $S \cap M = 0$. Also M is maximal and $S \not\subseteq M$, so

$$M + S = V.$$

Hence

$$V = M \oplus S,$$

and therefore M is a direct summand of V.

(3) $\Longrightarrow$ (1). Let

$$V = V_0 \supsetneq V_1 \supsetneq \cdots \supsetneq V_n = 0$$

be a composition series of V. We argue by induction on n, the composition length of V.

If $n = 1$, then V is simple, hence semisimple.

Assume $n > 1$, and suppose the statement holds for modules of composition length $< n$. Since V/V_1 is simple, V_1 is a maximal submodule of V. By (3), V_1 is a direct summand of V; thus

$$V = V_1 \oplus S$$

for some simple submodule $S \cong V/V_1$.

Now V_1 has composition length $n-1$. We claim that every maximal submodule of V_1 is a direct summand of V_1. Indeed, let N be a maximal submodule of V_1. Then

$$V/N \cong (V_1/N) \oplus S,$$

and since V_1/N is simple, $N \oplus S$ is a maximal submodule of V. By (3), $N \oplus S$ is a direct summand of V, say

$$V = (N \oplus S) \oplus T.$$

Because

$$V = V_1 \oplus S,$$

intersecting with V_1 gives

$$V_1 = N \oplus T.$$

So N is a direct summand of V_1.

Hence V_1 satisfies condition (3), and by the induction hypothesis V_1 is semisimple. Therefore

$$V = V_1 \oplus S$$

is semisimple as well.

This proves (3) $\Longrightarrow$ (1), and therefore the three conditions are equivalent. □

Exercise 12.5.3. *Prove that*

$$\mathrm{Soc}(U \oplus V) = \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Proof. We first show

$$\mathrm{Soc}(U) \oplus \mathrm{Soc}(V) \subseteq \mathrm{Soc}(U \oplus V).$$

Indeed, $\mathrm{Soc}(U)$ and $\mathrm{Soc}(V)$ are semisimple, so $\mathrm{Soc}(U) \oplus \mathrm{Soc}(V)$ is a semisimple submodule of $U \oplus V$. Hence it is contained in $\mathrm{Soc}(U \oplus V)$.

For the reverse inclusion, let S be a simple submodule of $U \oplus V$, and let

$$\pi_U : U \oplus V \to U, \qquad \pi_V : U \oplus V \to V$$

be the canonical projections. Consider the restrictions

$$\pi_U|_S : S \to U, \qquad \pi_V|_S : S \to V.$$

Since S is simple, the image of any homomorphism defined on S is either 0 or simple. Therefore $\pi_U(S)$ is either 0 or a simple submodule of U, and similarly $\pi_V(S)$ is either 0 or a simple submodule of V. Hence

$$\pi_U(S) \subseteq \mathrm{Soc}(U), \qquad \pi_V(S) \subseteq \mathrm{Soc}(V).$$

Now for any $s \in S$, we can write

$$s = (\pi_U(s), \pi_V(s)).$$

Since $\pi_U(s) \in \mathrm{Soc}(U)$ and $\pi_V(s) \in \mathrm{Soc}(V)$, it follows that

$$s \in \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Thus

$$S \subseteq \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Since $\mathrm{Soc}(U \oplus V)$ is the sum of all simple submodules of $U \oplus V$, we obtain

$$\mathrm{Soc}(U \oplus V) \subseteq \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

Combining the two inclusions, we conclude that

$$\mathrm{Soc}(U \oplus V) = \mathrm{Soc}(U) \oplus \mathrm{Soc}(V).$$

□

Exercise 12.5.4. *Let R be an arbitrary ring. Then every two-sided ideal of the matrix ring $M_n(R)$ can be expressed in the form $M_n(I)$, where I is a two-sided ideal of R. In particular, R is simple if and only if $M_n(R)$ is simple.*

Hint. For an ideal L of $M_n(R)$, let I be the set of $(1,1)$-entries of L. Show that I is an ideal of R and $L = M_n(I)$.

Proof. Let E_{ij} denote the matrix with 1 in position (i, j) and 0 elsewhere. For a two-sided ideal L of $M_n(R)$, define

$$I := \{a \in R \mid aE_{11} \in L\}.$$

We first note that for any $A = (a_{kl}) \in L$, we have

$$E_{1k} \cdot A \cdot E_{l1} = a_{kl}\, E_{11} \in L,$$

so every entry of every matrix in L belongs to I. In particular,

$$I = \{a_{kl} \mid A = (a_{kl}) \in L,\ 1 \le k, l \le n\}.$$

I **is a two-sided ideal of** R**.** If $a, b \in I$, then $aE_{11}, bE_{11} \in L$, so $(a-b)E_{11} = aE_{11} - bE_{11} \in L$, hence $a - b \in I$. For $r \in R$, we have $(rE_{11}) \cdot (aE_{11}) = raE_{11} \in L$ and $(aE_{11}) \cdot (rE_{11}) = arE_{11} \in L$, so $ra, ar \in I$. Thus I is a two-sided ideal of R.

$L = M_n(I)$**.** Since every entry of every matrix in L lies in I, we have $L \subseteq M_n(I)$. Conversely, let $aE_{ij} \in M_n(I)$ with $a \in I$. Then $aE_{11} \in L$, and

$$aE_{ij} = E_{i1} \cdot (aE_{11}) \cdot E_{1j} \in L.$$

Since every matrix in $M_n(I)$ is a sum of such elements aE_{ij}, we get $M_n(I) \subseteq L$. Therefore $L = M_n(I)$.

R **is simple** $\iff$ $M_n(R)$ **is simple.** The map $I \mapsto M_n(I)$ is a bijection from the set of two-sided ideals of R to the set of two-sided ideals of $M_n(R)$. Hence R has no nontrivial two-sided ideals if and only if $M_n(R)$ has no nontrivial two-sided ideals. $\square$

Exercise 12.5.5. *Let W be a submodule of a semisimple module V. Show that W and V/W are also semisimple.*

Proof. Since V is semisimple, write $V = \bigoplus_{i \in I} S_i$ with each S_i simple.

W **is semisimple.** By Zorn's lemma, choose a maximal subset $J \subseteq I$ such that $W \cap \bigoplus_{j \in J} S_j = 0$. Set $U = W \oplus \bigoplus_{j \in J} S_j$. For any $i \in I \setminus J$, the maximality of J gives $U \cap S_i \neq 0$, and since S_i is simple, $S_i \subseteq U$. Therefore $V \subseteq U$, so $V = U = W \oplus \bigoplus_{j \in J} S_j$. Thus

$$W \cong V \Big/ \bigoplus_{j \in J} S_j \cong \bigoplus_{i \in I \setminus J} S_i,$$

which is a direct sum of simple modules, hence semisimple.

V/W **is semisimple.** From the decomposition $V = W \oplus \bigoplus_{j \in J} S_j$, we obtain

$$V/W \cong \bigoplus_{j \in J} S_j,$$

which is a direct sum of simple modules, hence semisimple. $\square$

12.6 Homework Week 6

Exercise 12.6.1. *Let D be a division ring and n a positive integer.*

(a) Show that the natural left $M_n(D)$-module, consisting of column vectors of length n with entries in D, is a simple module.

(b) Show that $M_n(D)$ is semisimple and has up to isomorphism only one simple module.

(c) Show that every ring of the form

$$M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r)$$

is semisimple, where D_i are division rings.

(d) *Show that $M_n(D)$ is a simple ring, namely, one in which the only 2-sided ideals are the zero ideal and the whole ring.*

Proof. Let $R = M_n(D)$ and let $V = D^n$ be the natural left R-module of column vectors.

(a) We show that V is simple. Let $0 \neq W \leq V$, and choose

$$v = (v_1, \dots, v_n)^t \in W, \qquad v \neq 0.$$

Then $v_j \neq 0$ for some j. For each $i \in \{1, \dots, n\}$, let $A_i \in R$ be the matrix whose (i, j)-entry is v_j^{-1} and whose other entries are 0. Then

$$A_i v = e_i,$$

where e_i is the i-th standard basis vector in D^n. Since W is an R-submodule and $v \in W$, we get $e_i \in W$ for all i. Hence

$$W = D^n = V.$$

Therefore V is a simple left R-module.

(b) Let $E_{11}, \dots, E_{nn}$ be the standard diagonal matrix units. Then

$$1 = E_{11} + \cdots + E_{nn},$$

and hence

$$R = RE_{11} \oplus \cdots \oplus RE_{nn}.$$

Indeed, RE_{ii} is precisely the set of matrices whose only possibly nonzero column is the i-th column, so every matrix is uniquely the sum of its columns.

For each i, define

$$\phi_i : RE_{ii} \longrightarrow V$$

by sending a matrix in RE_{ii} to its i-th column. This is an isomorphism of left R-modules. By part (a), V is simple, so each RE_{ii} is simple. Thus ${}_RR$ is a direct sum of simple submodules, and therefore R is semisimple.

Now let S be a simple left R-module. Pick $0 \neq s \in S$. Then the map

$$\psi : R \longrightarrow S, \qquad A \mapsto As$$

is a nonzero R-homomorphism, hence surjective since S is simple. Because R is semisimple, every quotient of R is semisimple. Since S is simple, it must be isomorphic to one of the simple direct summands RE_{ii}. Therefore

$$S \cong RE_{ii} \cong V$$

for some i. Hence, up to isomorphism, R has only one simple left module.

(c) Let

$$A = M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r).$$

As a left A-module over itself,

$${}_AA = {}_AM_{n_1}(D_1) \oplus \cdots \oplus {}_AM_{n_r}(D_r).$$

Each summand is semisimple by part (b), and a finite direct sum of semisimple modules is semisimple. Hence A is a semisimple ring.

(d) Let I be a nonzero two-sided ideal of R. Choose $0 \neq A = (a_{pq}) \in I$, and pick indices i, j such that $a_{ij} \neq 0$. For arbitrary r, s, let E_{uv} denote the standard matrix units. Then

$$E_{ri}AE_{js} = a_{ij}E_{rs} \in I.$$

Since $a_{ij} \in D^\times$, we also have

$$(a_{ij}^{-1}E_{rr})(a_{ij}E_{rs}) = E_{rs} \in I.$$

Thus every matrix unit E_{rs} belongs to I. Therefore

$$1 = E_{11} + \cdots + E_{nn} \in I,$$

so $I = R$. Hence the only two-sided ideals of R are 0 and R, i.e. R is a simple ring.

□

Exercise 12.6.2. *Let U be an A-module for a semisimple finite-dimensional algebra A (over a field). Show that if $\mathrm{End}_A(U)$ is a division ring then U is simple.*

Proof. Since A is a semisimple finite-dimensional algebra over a field, it is a semisimple Artinian ring. Hence every left A-module is semisimple. In particular, U is a semisimple A-module.

We claim that U must be simple. Suppose not. Then U is not simple, so because U is semisimple, there exist nonzero submodules $U_1, U_2 \leq U$ such that

$$U = U_1 \oplus U_2.$$

Let

$$\pi_1 : U \to U_1$$

be the projection onto the first summand. Since U_1 is an A-submodule, π_1 is an A-module endomorphism of U, so

$$\pi_1 \in \mathrm{End}_A(U).$$

Moreover,

$$\pi_1^2 = \pi_1,$$

so π_1 is an idempotent.

Now $\pi_1 \neq 0$ because $U_1 \neq 0$, and $\pi_1 \neq 1_U$ because $U_2 \neq 0$. Thus π_1 is a nontrivial idempotent in $\mathrm{End}_A(U)$.

But a division ring has no idempotents other than 0 and 1: indeed, if $e^2 = e$ and $e \neq 0$, then multiplying by e^{-1} gives $e = 1$. This contradicts the assumption that $\mathrm{End}_A(U)$ is a division ring.

Therefore U must be simple. □

Exercise 12.6.3. *Let A be a ring. Then A is a semisimple Artinian ring if and only if every A-module is projective.*

Proof. Here all modules are left A-modules.

$$(\Rightarrow)$$

Assume that A is a semisimple Artinian ring. Then every A-module is semisimple.

Let M be any A-module. To show that M is projective, consider an arbitrary short exact sequence

$$0 \longrightarrow K \longrightarrow N \xrightarrow{\pi} M \longrightarrow 0.$$

Since N is semisimple, the submodule K is a direct summand of N. Thus there exists a submodule $N' \leq N$ such that

$$N = K \oplus N'.$$

Now $\pi|_{N'} : N' \to M$ is surjective, and

$$\ker(\pi|_{N'}) = N' \cap K = 0.$$

Hence $\pi|_{N'}$ is an isomorphism. Therefore the above short exact sequence splits. Since every short exact sequence ending in M splits, M is projective.

Since M was arbitrary, every A-module is projective.

$$(\Leftarrow)$$

Assume that every A-module is projective. Let M be any A-module and let $N \leq M$ be a submodule. Then the quotient M/N is projective, so the short exact sequence

$$0 \longrightarrow N \longrightarrow M \longrightarrow M/N \longrightarrow 0$$

splits. Hence N is a direct summand of M.

Therefore every submodule of every A-module is a direct summand. It follows that every A-module is semisimple. In particular, the left regular module ${}_AA$ is semisimple.

Now A is cyclic as a left A-module, generated by 1. Since ${}_AA$ is semisimple, it is a sum of simple submodules:

$$A = \sum_{\lambda \in \Lambda} S_\lambda.$$

Because $1 \in A$, we may write

$$1 = s_1 + \cdots + s_n \qquad (s_i \in S_{\lambda_i}).$$

Then for any $a \in A$,

$$a = a \cdot 1 = as_1 + \cdots + as_n \in S_{\lambda_1} + \cdots + S_{\lambda_n}.$$

Thus

$$A = S_{\lambda_1} + \cdots + S_{\lambda_n}.$$

Since A is semisimple, this finite sum is in fact a direct sum (after deleting repetitions if necessary):

$$A \cong S_1 \oplus \cdots \oplus S_m$$

with each S_i simple.

Hence ${}_AA$ is a finite direct sum of simple modules, so it is Artinian. Therefore A is a semisimple Artinian ring.

This proves that A is a semisimple Artinian ring if and only if every A-module is projective. □

Exercise 12.6.4. *Let k be a field. Suppose the group algebra kG is semisimple, and the simple kG-modules are $S_1, \ldots, S_r$ with degrees $d_i = \dim_k S_i$. Show that*

$$\sum_{i=1}^{r} d_i^2 \geq |G|$$

with equality if and only if $\mathrm{End}_{kG}(S_i) = k$ for all i.

Proof. Let

$$A = kG, \qquad E_i = \mathrm{End}_A(S_i) \quad (1 \leq i \leq r).$$

Since A is a semisimple finite-dimensional k-algebra, by the Wedderburn–Artin theorem there exist positive integers $n_1, \ldots, n_r$ such that

$$A \cong \bigoplus_{i=1}^{r} M_{n_i}(E_i^{\mathrm{op}}),$$

where $S_1, \ldots, S_r$ form a complete set of representatives of the isomorphism classes of simple A-modules.

For each i, the simple module S_i is naturally a right E_i-vector space of dimension n_i. Hence

$$d_i = \dim_k S_i = n_i \dim_k E_i.$$

Taking k-dimensions in the Wedderburn decomposition gives

$$|G| = \dim_k(kG) = \dim_k A = \sum_{i=1}^{r} \dim_k M_{n_i}(E_i^{\mathrm{op}}) = \sum_{i=1}^{r} n_i^2 \dim_k E_i.$$

Using $d_i = n_i \dim_k E_i$, we obtain

$$|G| = \sum_{i=1}^{r} n_i^2 \dim_k E_i = \sum_{i=1}^{r} \frac{d_i^2}{\dim_k E_i}.$$

Now each E_i is a division k-algebra, so $\dim_k E_i \geq 1$. Therefore

$$|G| = \sum_{i=1}^{r} \frac{d_i^2}{\dim_k E_i} \leq \sum_{i=1}^{r} d_i^2.$$

Thus

$$\sum_{i=1}^{r} d_i^2 \geq |G|.$$

It remains to determine when equality holds. Since each $d_i > 0$ and each $\dim_k E_i \geq 1$, we have

$$\sum_{i=1}^{r} \frac{d_i^2}{\dim_k E_i} = \sum_{i=1}^{r} d_i^2$$

if and only if

$$\dim_k E_i = 1 \qquad \text{for all } i.$$

But $E_i = \mathrm{End}_A(S_i)$ is a k-algebra, so $\dim_k E_i = 1$ is equivalent to

$$E_i = k.$$

Hence equality holds if and only if

$$\mathrm{End}_{kG}(S_i) = k \qquad \text{for all } i.$$

This proves the claim. □

Exercise 12.6.5. *Suppose that we have R-module homomorphisms*

$$U \xrightarrow{f} V \xrightarrow{g} W.$$

Show that: if gf is an essential epimorphism and f is an epimorphism, then both f and g are essential epimorphisms.

Proof. Assume that

$$U \xrightarrow{f} V \xrightarrow{g} W$$

are R-module homomorphisms such that gf is an essential epimorphism and f is an epimorphism.

We must show that both f and g are essential epimorphisms.

Step 1: f is an essential epimorphism.

Since f is already assumed to be an epimorphism, it remains to show that it is essential.

Let $U' \leq U$ be a submodule such that

$$f(U') = V.$$

Then applying g, we get

$$(gf)(U') = g(f(U')) = g(V) = W,$$

because gf is an epimorphism. Thus U' maps onto W under gf.

But gf is an *essential* epimorphism, so no proper submodule of U can map surjectively onto W. Hence we must have

$$U' = U.$$

Therefore f is an essential epimorphism.

Step 2: g is an essential epimorphism.

First, g is an epimorphism: indeed, since gf is surjective and f is surjective, for any $w \in W$ there exists $u \in U$ such that

$$g(f(u)) = w.$$

Thus $w \in g(V)$, so $g(V) = W$.

Now let $V' \leq V$ be a submodule such that

$$g(V') = W.$$

We claim that $V' = V$.

Consider the inverse image

$$f^{-1}(V') = \{u \in U \mid f(u) \in V'\}.$$

This is a submodule of U, and

$$(gf)\big(f^{-1}(V')\big) = g(V') = W.$$

Since gf is an essential epimorphism, it follows that

$$f^{-1}(V') = U.$$

Applying f to both sides gives

$$V = f(U) = f\big(f^{-1}(V')\big) \subseteq V'.$$

Hence $V' = V$.

Therefore no proper submodule of V maps surjectively onto W under g, so g is an essential epimorphism.

We conclude that both f and g are essential epimorphisms. $\square$

12.7 Homework Week 7

Exercise 12.7.1. *Let V be an A-module.*

(a) Show that an endomorphism

$$e\colon V \to V$$

is a projection onto an A-submodule W if and only if e is idempotent as an element of $\mathrm{End}_A(V)$.

(b) Show that direct sum decompositions

$$V = W_1 \oplus W_2$$

as A-modules are in bijection with idempotent decompositions

$$1 = e + f$$

in $\mathrm{End}_A(V)$.

(c) Show that $e \in \mathrm{End}_A(V)$ is primitive if and only if $e(V)$ is indecomposable.

(d) Suppose that V is semisimple with finitely many simple summands and let

$$e_1, e_2 \in \mathrm{End}_A(V)$$

be idempotents. Show that

$$e_1(V) \cong e_2(V)$$

as A-modules if and only if e_1 and e_2 are conjugate by an invertible element of $\mathrm{End}_A(V)$.

(e) Let k be a field. Show that all primitive idempotent elements in $M_n(k)$ are conjugate under the action of the unit group $GL_n(k)$, which consists of all the invertible matrices in $M_n(k)$.

Proof. (a) Suppose first that $e\colon V \to V$ is a projection onto an A-submodule W. Then $e(V) = W$ and $e(w) = w$ for every $w \in W$. Hence for any $v \in V$,

$$e^2(v) = e(e(v)) = e(v),$$

because $e(v) \in W$. Thus $e^2 = e$, so e is idempotent.

Conversely, suppose that $e^2 = e$. Set

$$W := e(V).$$

Since e is A-linear, W is an A-submodule of V. Moreover, if $w = e(v) \in W$, then

$$e(w) = e^2(v) = e(v) = w.$$

Thus e fixes W pointwise and maps V onto W, so e is a projection onto W.

In fact, one also has

$$V = e(V) \oplus \ker(e),$$

since for any $v \in V$,

$$v = e(v) + (v - e(v)), \qquad e(v - e(v)) = e(v) - e^2(v) = 0,$$

and if $x \in e(V) \cap \ker(e)$, then $x = e(v)$ for some v, hence

$$x = e(x) = 0.$$

(b) Suppose first that

$$V = W_1 \oplus W_2.$$

Define

$$e(w_1 + w_2) = w_1, \qquad f(w_1 + w_2) = w_2 \qquad (w_i \in W_i).$$

Then $e, f \in \mathrm{End}_A(V)$, and e and f are the projections onto W_1 and W_2, respectively. By part (a), both are idempotent, and clearly

$$e + f = 1.$$

Also,

$$ef = fe = 0.$$

Conversely, suppose

$$1 = e + f$$

with $e^2 = e$ and $f^2 = f$. Then

$$e = e(e + f) = e + ef,$$

so $ef = 0$. Similarly,

$$f = (e + f)f = ef + f,$$

so again $ef = 0$, and likewise

$$fe = 0.$$

By part (a), e and f are projections onto $e(V)$ and $f(V)$, respectively. For every $v \in V$,

$$v = (e + f)(v) = e(v) + f(v),$$

so

$$V = e(V) + f(V).$$

If $x \in e(V) \cap f(V)$, say $x = e(v) = f(w)$, then

$$x = e(x) = ef(w) = 0.$$

Hence

$$V = e(V) \oplus f(V).$$

These two constructions are inverse to each other, so direct sum decompositions

$$V = W_1 \oplus W_2$$

are in bijection with idempotent decompositions

$$1 = e + f$$

in $\mathrm{End}_A(V)$.

(c) We show that e is primitive if and only if $e(V)$ is indecomposable.

First suppose that e is not primitive. Then there exist nonzero idempotents $e_1, e_2 \in \mathrm{End}_A(V)$ such that

$$e = e_1 + e_2, \qquad e_1 e_2 = e_2 e_1 = 0.$$

Since

$$e e_i = (e_1 + e_2) e_i = e_i,$$

we have $e_i(V) \subseteq e(V)$ for $i = 1, 2$. Moreover, for any $v \in V$,

$$e(v) = e_1(v) + e_2(v),$$

so

$$e(V) = e_1(V) + e_2(V).$$

If $x \in e_1(V) \cap e_2(V)$, say $x = e_1(v) = e_2(w)$, then

$$x = e_1(x) = e_1 e_2(w) = 0.$$

Hence

$$e(V) = e_1(V) \oplus e_2(V).$$

Since e_1 and e_2 are nonzero, both $e_1(V)$ and $e_2(V)$ are nonzero. Thus $e(V)$ is decomposable.

Conversely, suppose that $e(V)$ is decomposable:

$$e(V) = U_1 \oplus U_2$$

with $U_1, U_2 \neq 0$. Let

$$p_i \colon e(V) \to e(V)$$

be the projection onto U_i $(i = 1, 2)$. Define

$$\widetilde{e}_i := p_i \circ e \in \mathrm{End}_A(V).$$

Since e acts as the identity on $e(V)$, we have $e \circ p_i = p_i$, and therefore

$$\widetilde{e}_i^2 = p_i e p_i e = p_i p_i e = \widetilde{e}_i.$$

Also,

$$\widetilde{e}_1 \widetilde{e}_2 = p_1 e p_2 e = p_1 p_2 e = 0, \qquad \widetilde{e}_2 \widetilde{e}_1 = 0,$$

and

$$\widetilde{e}_1 + \widetilde{e}_2 = (p_1 + p_2)e = e.$$

Since $U_i \neq 0$, each $\widetilde{e}_i \neq 0$. Thus e is not primitive.

Hence e is primitive if and only if $e(V)$ is indecomposable.

(d) Assume first that $e_2 = u e_1 u^{-1}$ for some invertible $u \in \operatorname{End}_A(V)$. Then for any $x = e_1(v) \in e_1(V)$,

$$u(x) = u e_1(v) = e_2 u(v) \in e_2(V),$$

so $u(e_1(V)) \subseteq e_2(V)$. Conversely, if $y = e_2(w) \in e_2(V)$, then

$$y = u e_1 u^{-1}(w) \in u(e_1(V)).$$

Hence

$$u(e_1(V)) = e_2(V),$$

and therefore the restriction

$$u|_{e_1(V)} \colon e_1(V) \to e_2(V)$$

is an isomorphism of A-modules. So

$$e_1(V) \cong e_2(V).$$

Conversely, suppose

$$e_1(V) \cong e_2(V).$$

Since V is semisimple with finitely many simple summands, we may write

$$V \cong \bigoplus_{j=1}^{r} S_j^{n_j},$$

where $S_1, \ldots, S_r$ are pairwise nonisomorphic simple A-modules. Because $e_i(V)$ is a submodule of a semisimple module, it is semisimple, so

$$e_i(V) \cong \bigoplus_{j=1}^{r} S_j^{m_j}$$

for some integers m_j, and the m_j are the same for $i = 1, 2$ since $e_1(V) \cong e_2(V)$.

By part (a),

$$V = e_i(V) \oplus \ker(e_i).$$

Therefore

$$\ker(e_i) \cong \bigoplus_{j=1}^{r} S_j^{n_j - m_j},$$

and hence

$$\ker(e_1) \cong \ker(e_2).$$

Choose isomorphisms

$$\alpha\colon e_1(V) \xrightarrow{\sim} e_2(V), \qquad \beta\colon \ker(e_1) \xrightarrow{\sim} \ker(e_2).$$

Using the decompositions

$$V = e_1(V) \oplus \ker(e_1) \qquad \text{and} \qquad V = e_2(V) \oplus \ker(e_2),$$

define

$$u\colon V \to V, \qquad u(x+y) = \alpha(x) + \beta(y) \quad \big(x \in e_1(V),\ y \in \ker(e_1)\big).$$

Then u is an invertible A-endomorphism of V. For $x \in e_1(V)$ and $y \in \ker(e_1)$,

$$ue_1(x+y) = u(x) = \alpha(x),$$

while

$$e_2u(x+y) = e_2\big(\alpha(x) + \beta(y)\big) = \alpha(x),$$

because e_2 is the identity on $e_2(V)$ and zero on $\ker(e_2)$. Hence

$$ue_1 = e_2u,$$

so

$$e_2 = ue_1u^{-1}.$$

Thus e_1 and e_2 are conjugate by an invertible element of $\mathrm{End}_A(V)$.

(e) Let $V = k^n$. Then

$$M_n(k) \cong \mathrm{End}_k(V).$$

Let $e \in M_n(k)$ be a primitive idempotent. By part (c), $e(V)$ is indecomposable as a k-vector space. But a nonzero k-vector space is indecomposable if and only if it has dimension 1: indeed, if $\dim_k W \geq 2$, then any nonzero proper subspace has a complement, so W is decomposable.

Therefore

$$\dim_k e(V) = 1.$$

Since e is idempotent, part (a) gives

$$V = e(V) \oplus \ker(e).$$

Choose a basis $v_1, \ldots, v_n$ of V such that v_1 spans $e(V)$ and $v_2, \ldots, v_n$ is a basis of $\ker(e)$. Then

$$e(v_1) = v_1, \qquad e(v_i) = 0 \quad (i \geq 2).$$

Hence the matrix of e with respect to this basis is

$$E_{11} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Therefore e is conjugate to E_{11} by some element of $GL_n(k)$. It follows that any two primitive idempotents in $M_n(k)$ are conjugate to each other.

□

Exercise 12.7.2. *Prove: No idempotent e of A is contained in $J(A)$. [Hint: $e(1-e) = 0$.]*

Proof. Assume that $e \in J(A)$ and that e is idempotent, so

$$e^2 = e.$$

Since $e \in J(A)$, a standard characterization of the Jacobson radical implies that

$$1 - e$$

is a unit in A.

On the other hand,

$$e(1-e) = e - e^2 = 0.$$

Multiplying this equality on the right by $(1-e)^{-1}$, we obtain

$$e = 0.$$

Therefore any idempotent contained in $J(A)$ must be zero. In particular, no nonzero idempotent of A lies in $J(A)$. □

Exercise 12.7.3. *Prove: Let A be an Artinian ring with Artinian center $Z(A)$. Then*

$$J(Z(A)) = J(A) \cap Z(A).$$

Proof. Let

$$B := Z(A).$$

We prove the two inclusions separately.

First, we show that

$$J(B) \subseteq J(A) \cap B.$$

Take $z \in J(B)$. Since B is Artinian, its Jacobson radical is nilpotent, so there exists $m \geq 1$ such that

$$J(B)^m = 0.$$

In particular,

$$z^m = 0.$$

Now let $x \in A$. Because $z \in Z(A)$, we have $xz = zx$, hence

$$(xz)^m = x^m z^m = 0.$$

So xz is nilpotent, and therefore

$$1 - xz$$

is invertible in A, with inverse

$$1 + xz + \cdots + (xz)^{m-1}.$$

By the standard characterization of the Jacobson radical, this implies

$$z \in J(A).$$

Since also $z \in B$, we get

$$z \in J(A) \cap B.$$

Thus

$$J(B) \subseteq J(A) \cap B.$$

Next, we show that

$$J(A) \cap B \subseteq J(B).$$

Take $z \in J(A) \cap B$. Since A is Artinian, $J(A)$ is nilpotent, so there exists $n \geq 1$ such that

$$J(A)^n = 0.$$

Hence

$$z^n = 0.$$

Now let $c \in B$. Since B is commutative, we have

$$(cz)^n = c^n z^n = 0.$$

Therefore cz is nilpotent in B, and so

$$1 - cz$$

is invertible in B, with inverse

$$1 + cz + \cdots + (cz)^{n-1}.$$

Again by the standard characterization of the Jacobson radical, this shows that

$$z \in J(B).$$

Hence

$$J(A) \cap B \subseteq J(B).$$

Combining the two inclusions, we conclude that

$$J(Z(A)) = J(B) = J(A) \cap B.$$

□

Exercise 12.7.4. *Prove: Let*

$$A = A_1 \oplus \cdots \oplus A_n$$

be a direct sum of 2*-sided ideals. Then*

(i)

$$Z(A) = Z(A_1) \oplus Z(A_2) \oplus \cdots \oplus Z(A_n).$$

(ii)

$$J(A) = J(A_1) \oplus J(A_2) \oplus \cdots \oplus J(A_n).$$

Proof. For $i \neq j$, since A_i and A_j are two-sided ideals and the sum is direct,

$$A_i A_j \subseteq A_i \cap A_j = 0.$$

Write

$$1_A = e_1 + \cdots + e_n, \qquad e_i \in A_i.$$

Then e_i is the identity element of A_i. Indeed, if $a_i \in A_i$, then all mixed products vanish, so

$$a_i = 1_A a_i = e_i a_i, \qquad a_i = a_i 1_A = a_i e_i.$$

Thus each A_i is a unital ring, with identity e_i. Moreover multiplication in A is coordinatewise with respect to the decomposition

$$A = A_1 \oplus \cdots \oplus A_n.$$

Hence

$$\Phi \colon A \longrightarrow A_1 \times \cdots \times A_n, \qquad a_1 + \cdots + a_n \longmapsto (a_1, \ldots, a_n)$$

is an isomorphism of rings. Since n is finite, we identify

$$A \cong A_1 \times \cdots \times A_n.$$

(i) First we prove

$$Z(A) \subseteq Z(A_1) \oplus \cdots \oplus Z(A_n).$$

Let $z \in Z(A)$, and write

$$z = z_1 + \cdots + z_n, \qquad z_i \in A_i.$$

Fix i, and let $x_i \in A_i$. Since z is central in A,

$$zx_i = x_i z.$$

But all mixed products vanish, so

$$zx_i = z_i x_i, \qquad x_i z = x_i z_i.$$

Hence $z_i x_i = x_i z_i$ for every $x_i \in A_i$, and therefore

$$z_i \in Z(A_i).$$

Since this holds for every i, we get

$$z \in Z(A_1) \oplus \cdots \oplus Z(A_n).$$

Conversely, suppose

$$z = z_1 + \cdots + z_n, \qquad z_i \in Z(A_i).$$

Let

$$a = a_1 + \cdots + a_n, \qquad a_i \in A_i.$$

Again using $A_i A_j = 0$ for $i \neq j$, we have

$$za = \sum_{i=1}^{n} z_i a_i, \qquad az = \sum_{i=1}^{n} a_i z_i.$$

Since $z_i \in Z(A_i)$, each $z_i a_i = a_i z_i$. Hence $za = az$, so $z \in Z(A)$. Therefore

$$Z(A) = Z(A_1) \oplus \cdots \oplus Z(A_n).$$

(ii) First we prove

$$J(A_1) \oplus \cdots \oplus J(A_n) \subseteq J(A).$$

Let

$$x = x_1 + \cdots + x_n, \qquad x_i \in J(A_i).$$

Take any

$$a = a_1 + \cdots + a_n \in A, \qquad a_i \in A_i.$$

Since $J(A_i)$ is an ideal of A_i, we have $a_i x_i \in J(A_i)$. Hence $e_i - a_i x_i$ is invertible in A_i. Choose $u_i \in A_i$ such that

$$u_i(e_i - a_i x_i) = e_i.$$

Put $u = u_1 + \cdots + u_n$. Since multiplication is coordinatewise,

$$u(1_A - ax) = \sum_{i=1}^{n} u_i(e_i - a_i x_i) = \sum_{i=1}^{n} e_i = 1_A.$$

Therefore $1_A - ax$ has a left inverse for every $a \in A$. By the Jacobson radical criterion,

$$x \in J(A).$$

Thus

$$J(A_1) \oplus \cdots \oplus J(A_n) \subseteq J(A).$$

Conversely, we prove

$$J(A) \subseteq J(A_1) \oplus \cdots \oplus J(A_n).$$

Let

$$x = x_1 + \cdots + x_n \in J(A), \qquad x_i \in A_i.$$

Fix i, and let $a_i \in A_i$. Since $x \in J(A)$, the element

$$1_A - a_i x = 1_A - a_i x_i$$

has a left inverse in A. Thus there exists $v = v_1 + \cdots + v_n \in A$, with $v_j \in A_j$, such that

$$v(1_A - a_i x_i) = 1_A.$$

Taking the A_i-component of this equality gives

$$v_i(e_i - a_i x_i) = e_i.$$

Hence $e_i - a_i x_i$ has a left inverse in A_i for every $a_i \in A_i$. By the Jacobson radical criterion applied inside A_i,

$$x_i \in J(A_i).$$

Since i was arbitrary,

$$x \in J(A_1) \oplus \cdots \oplus J(A_n).$$

Combining the two inclusions, we conclude that

$$J(A) = J(A_1) \oplus \cdots \oplus J(A_n).$$

□

Exercise 12.7.5. *Let*

$$\phi\colon U \to V$$

be a homomorphism of A-modules. Show that

$$\phi(\operatorname{Soc} U) \subseteq \operatorname{Soc} V,$$

and that if ϕ is an isomorphism then ϕ restricts to an isomorphism

$$\operatorname{Soc} U \to \operatorname{Soc} V.$$

Proof. Recall that

$$\operatorname{Soc}(U) = \sum \{ S \leq U \mid S \text{ is a simple } A\text{-submodule} \}.$$

We first show that

$$\phi(\operatorname{Soc} U) \subseteq \operatorname{Soc} V.$$

Let $S \leq U$ be a simple submodule. Then $\phi(S)$ is a submodule of V, and since S is simple, the restriction

$$\phi|_S\colon S \to \phi(S)$$

is either the zero map or an isomorphism. Hence $\phi(S)$ is either 0 or a simple submodule of V. In either case,

$$\phi(S) \subseteq \operatorname{Soc} V.$$

Since $\operatorname{Soc}(U)$ is the sum of all simple submodules $S \leq U$, we get

$$\phi(\operatorname{Soc} U) = \sum_{S \text{ simple } \leq U} \phi(S) \subseteq \operatorname{Soc} V.$$

Now assume that ϕ is an isomorphism. Then $\phi^{-1}\colon V \to U$ is also an A-module homomorphism. Applying the first part to ϕ^{-1}, we obtain

$$\phi^{-1}(\operatorname{Soc} V) \subseteq \operatorname{Soc} U.$$

Applying ϕ to both sides gives

$$\operatorname{Soc} V \subseteq \phi(\operatorname{Soc} U).$$

Together with the already proved inclusion

$$\phi(\operatorname{Soc} U) \subseteq \operatorname{Soc} V,$$

we conclude that

$$\phi(\operatorname{Soc} U) = \operatorname{Soc} V.$$

Therefore the restriction
$$\phi|_{\operatorname{Soc} U}\colon \operatorname{Soc} U \to \operatorname{Soc} V$$
is surjective. It is also injective because ϕ itself is injective. Hence
$$\phi|_{\operatorname{Soc} U}\colon \operatorname{Soc} U \xrightarrow{\sim} \operatorname{Soc} V$$
is an isomorphism of A-modules. □

Exercise 12.7.6. *If*
$$\varphi\colon A \to A'$$
is an epimorphism of rings, then
$$\varphi(J(A)) \subseteq J(A').$$

Proof. Let $a \in J(A)$. We show that $\varphi(a) \in J(A')$.

We use the standard characterization of the Jacobson radical:
$$x \in J(R) \quad\iff\quad \text{for every } r \in R,\ \ 1 - rx \text{ has a left inverse in } R.$$

So let $y' \in A'$. Since $\varphi\colon A \to A'$ is surjective, there exists $y \in A$ such that
$$\varphi(y) = y'.$$
Because $a \in J(A)$, the element
$$1 - ya$$
has a left inverse in A. Thus there exists $b \in A$ such that
$$b(1 - ya) = 1.$$
Apply φ to this equality. Since φ is a ring homomorphism, we get
$$\varphi(b)\,\varphi(1 - ya) = \varphi(1) = 1,$$
that is,
$$\varphi(b)\bigl(1 - \varphi(y)\varphi(a)\bigr) = 1.$$
Using $\varphi(y) = y'$, this becomes
$$\varphi(b)(1 - y'\varphi(a)) = 1.$$
Hence $1 - y'\varphi(a)$ has a left inverse in A'.

Since $y' \in A'$ was arbitrary, the above characterization implies that
$$\varphi(a) \in J(A').$$
Therefore
$$\varphi(J(A)) \subseteq J(A').$$

$\square$

12.8 Homework Week 8

Exercise 12.8.1. *Let A be a finite-dimensional algebra over a field k. Suppose that V is a finitely generated A-module, and that a certain simple A-module S occurs as a composition factor of V with multiplicity 1. Suppose that there exist nonzero homomorphisms*

$$S \longrightarrow V \qquad \text{and} \qquad V \longrightarrow S.$$

Prove that S is a direct summand of V.

Proof. Since A is finite-dimensional over k and V is finitely generated over A, the module V is finite-dimensional over k. Hence V has finite length.

Let

$$\iota : S \longrightarrow V, \qquad \rho : V \longrightarrow S$$

be nonzero A-homomorphisms. Since S is simple, ι is injective and ρ is surjective.

We claim that

$$\rho \circ \iota \neq 0.$$

Suppose otherwise that $\rho \circ \iota = 0$. Then

$$\iota(S) \subseteq \ker \rho.$$

Thus S occurs as a composition factor of $\ker \rho$. Since ρ is surjective, we have a short exact sequence

$$0 \longrightarrow \ker \rho \longrightarrow V \longrightarrow S \longrightarrow 0.$$

By additivity of composition multiplicities in short exact sequences,

$$[V : S] = [\ker \rho : S] + [S : S] \geq 1 + 1 = 2,$$

contradicting the assumption that S occurs in V with multiplicity 1. Therefore

$$\rho \circ \iota \neq 0.$$

By Schur's lemma, the nonzero endomorphism

$$\rho \circ \iota : S \longrightarrow S$$

is an isomorphism. Define

$$\sigma = \iota \circ (\rho \circ \iota)^{-1} : S \longrightarrow V.$$

Then

$$\rho \circ \sigma = \rho \circ \iota \circ (\rho \circ \iota)^{-1} = \mathrm{id}_S\,.$$

Hence $\rho : V \to S$ splits. Therefore

$$V = \ker \rho \oplus \sigma(S),$$

and since $\sigma(S) \cong S$, the simple module S is a direct summand of V. $\square$

Exercise 12.8.2. *Suppose that U, V, W are modules over a finite-dimensional algebra A over a field, and let*

$$p_W : P_W \longrightarrow W$$

be a projective cover of W. Assume that all modules involved are finite-dimensional.

(a) *Show that a homomorphism*

$$\alpha : U \longrightarrow W$$

is an essential epimorphism if and only if there is a surjective homomorphism

$$h : P_W \longrightarrow U$$

such that the composite

$$P_W \xrightarrow{h} U \xrightarrow{\alpha} W$$

is a projective cover of W. In this situation, show that

$$h : P_W \longrightarrow U$$

must be a projective cover of U.

(b) *Prove the following "extension and converse" to Nakayama's lemma: let V be any submodule of U. Then the canonical epimorphism*

$$q : U \longrightarrow U/V$$

is an essential epimorphism if and only if

$$V \subseteq \operatorname{Rad} U.$$

Proof. **(a)** Recall that an epimorphism $f : X \to Y$ is essential if whenever $X_0 \leq X$ satisfies $f(X_0) = Y$, one must have $X_0 = X$.

First suppose that

$$\alpha : U \longrightarrow W$$

is an essential epimorphism. Since P_W is projective and α is surjective, the map $p_W : P_W \to W$ lifts through α. Thus there exists an A-homomorphism

$$h : P_W \longrightarrow U$$

such that

$$\alpha \circ h = p_W.$$

Because p_W is surjective, we have

$$\alpha(h(P_W)) = W.$$

Since α is essential, this forces

$$h(P_W) = U.$$

Hence h is surjective. Moreover, the composite

$$P_W \xrightarrow{h} U \xrightarrow{\alpha} W$$

is precisely p_W, hence is a projective cover of W.

Conversely, suppose that there exists a surjective homomorphism

$$h : P_W \longrightarrow U$$

such that

$$\alpha \circ h : P_W \longrightarrow W$$

is a projective cover of W. In particular, $\alpha \circ h$ is an essential epimorphism. Since h is surjective, α is also surjective.

Let $U_0 \leq U$ be a submodule such that

$$\alpha(U_0) = W.$$

Then

$$(\alpha \circ h)\big(h^{-1}(U_0)\big) = W.$$

Since $\alpha \circ h$ is essential, we must have

$$h^{-1}(U_0) = P_W.$$

Applying h, and using the surjectivity of h, we get

$$U = h(P_W) \subseteq U_0.$$

Hence $U_0 = U$. Therefore $\alpha : U \to W$ is an essential epimorphism.

Finally, we show that in this situation

$$h : P_W \longrightarrow U$$

is a projective cover of U. We already know that P_W is projective and that h is surjective. It remains to show that h is essential.

Let $X \leq P_W$ be a submodule such that

$$h(X) = U.$$

Then

$$(\alpha \circ h)(X) = \alpha(U) = W.$$

Since $\alpha \circ h$ is essential, we obtain

$$X = P_W.$$

Thus h is an essential epimorphism. Therefore $h : P_W \to U$ is a projective cover of U.

(b) Let

$$q : U \longrightarrow U/V$$

be the canonical epimorphism.

First suppose that q is essential. We prove that

$$V \subseteq \operatorname{Rad} U.$$

If not, then there exists a maximal submodule M of U such that

$$V \nsubseteq M.$$

Since M is maximal and $V \nsubseteq M$, we have

$$M + V = U.$$

Therefore

$$q(M) = U/V.$$

But $M \neq U$, contradicting the essentiality of q. Hence

$$V \subseteq \operatorname{Rad} U.$$

Conversely, suppose that

$$V \subseteq \operatorname{Rad} U.$$

Let $X \leq U$ be a submodule such that

$$q(X) = U/V.$$

This is equivalent to

$$X + V = U.$$

We show that $X = U$. Suppose, for contradiction, that $X \neq U$. Since U is finite-dimensional, X is contained in some maximal submodule M of U. Since

$$V \subseteq \operatorname{Rad} U \subseteq M,$$

we get

$$U = X + V \subseteq M,$$

which is impossible because $M \neq U$. Therefore $X = U$, and hence q is an essential epimorphism.

Thus

$$U \longrightarrow U/V$$

is an essential epimorphism if and only if

$$V \subseteq \operatorname{Rad} U.$$

□

Exercise 12.8.3. *Let v be a valuation of a field K. Prove the following properties:*

(i) $v(\pm 1) = 0$.

(ii) $v(a) = v(-a)$.

(iii) $v(a^{-1}) = -v(a)$.

(iv) If $v(b) < v(a)$, then

$$v(a+b) = v(b).$$

Proof. We use the defining properties of a valuation:

$$v(xy) = v(x) + v(y), \qquad v(x+y) \geq \min\{v(x), v(y)\}.$$

(i) Since $1 \cdot 1 = 1$, we have

$$v(1) = v(1 \cdot 1) = v(1) + v(1).$$

Hence

$$v(1) = 0.$$

Also,

$$(-1)^2 = 1,$$

so

$$2v(-1) = v((-1)^2) = v(1) = 0.$$

Therefore

$$v(-1) = 0.$$

Thus

$$v(\pm 1) = 0.$$

(ii) For $a \in K^{\times}$,

$$-a = (-1)a.$$

Therefore

$$v(-a) = v((-1)a) = v(-1) + v(a) = v(a).$$

(iii) For $a \in K^\times$, we have
$$aa^{-1} = 1.$$
Thus
$$v(a) + v(a^{-1}) = v(aa^{-1}) = v(1) = 0.$$
Hence
$$v(a^{-1}) = -v(a).$$

(iv) Assume that $a, b \in K^\times$ and
$$v(b) < v(a).$$
First, $a + b \neq 0$; otherwise $a = -b$, and by part **(ii)** we would have
$$v(a) = v(-b) = v(b),$$
contradicting $v(b) < v(a)$.

By the valuation inequality,
$$v(a + b) \geq \min\{v(a), v(b)\} = v(b).$$
We now prove the reverse inequality. Since
$$b = (a + b) - a,$$
we have
$$v(b) = v((a + b) - a) \geq \min\{v(a + b), v(-a)\} = \min\{v(a + b), v(a)\}.$$
If $v(a + b) > v(b)$, then both $v(a + b)$ and $v(a)$ are strictly larger than $v(b)$, so
$$\min\{v(a + b), v(a)\} > v(b),$$
contradicting the previous inequality. Therefore
$$v(a + b) \leq v(b).$$
Combining this with
$$v(a + b) \geq v(b),$$
we obtain
$$v(a + b) = v(b).$$

□

12.9 Homework Midterm

Exercise 12.9.1. *Let R be a ring with 1.*

1. *(i) Let*

$$h : U \longrightarrow V$$

be a monomorphism of R-modules. Then the following two conditions are equivalent.

(a) A homomorphism

$$k : V \longrightarrow Y$$

must be a monomorphism whenever the composite map

$$U \xrightarrow{h} V \xrightarrow{k} Y$$

is a monomorphism.

(b) If $0 \neq W \subset V$, then

$$h(U) \cap W \neq 0.$$

The monomorphism h satisfying the above conditions is said to be essential. *If there is an injective R-module Q with an essential monomorphism*

$$h : V \longrightarrow Q,$$

then the diagram

$$V \xrightarrow{h} Q$$

is called an injective hull *of V.*

(ii) Prove that any R-module V has an injective hull.

2. *(i) Prove that the following holds for an idempotent $e \in R$:*

$$J(eRe) = eJ(R)e = eRe \cap J(R).$$

(ii) Prove that

$$J(M_n(R)) = M_n(J(R)).$$

Here $M_n(R)$ means the matrix ring.

3. *Prove that a submodule of a finitely generated free module over a PID is free.*

Proof of 1(i). We identify U with its image $h(U) \subseteq V$.

First assume (a). Suppose that there exists a nonzero submodule $W \subseteq V$ such that

$$h(U) \cap W = 0.$$

Let

$$q : V \longrightarrow V/W$$

be the canonical quotient map. Since $h(U) \cap W = 0$, the composite

$$U \xrightarrow{h} V \xrightarrow{q} V/W$$

is a monomorphism. Hence by (a), q must be a monomorphism. But

$$\ker q = W \neq 0,$$

a contradiction. Therefore for every nonzero submodule $W \subseteq V$, one has

$$h(U) \cap W \neq 0.$$

Thus (b) holds.

Conversely, assume (b). Let

$$k : V \longrightarrow Y$$

be an R-homomorphism such that kh is a monomorphism. We prove that k is a monomorphism.

Let

$$W = \ker k.$$

If $W \neq 0$, then by (b),

$$h(U) \cap W \neq 0.$$

Choose $0 \neq h(u) \in h(U) \cap W$. Then

$$k(h(u)) = 0.$$

Since kh is a monomorphism, this implies $u = 0$, hence $h(u) = 0$, a contradiction. Therefore $W = 0$, so k is a monomorphism.

Hence (a) and (b) are equivalent. □

Definition 12.9.2 (Essential extension)**.** *Let $V \subseteq E$ be an inclusion of R-modules. We say that V is* essential *in E, or that E is an* essential extension *of V, if for every nonzero submodule $W \subseteq E$, one has*

$$W \cap V \neq 0.$$

Equivalently, every nonzero submodule of E must meet V nontrivially.

Remark 12.9.3. *Intuitively, the condition $V \subseteq E$ being essential means that E contains no nonzero submodule which is completely disjoint from V. Thus E is obtained from V in a way that does not create an independent "new part" away from V.*

Lemma 12.9.4 (Existence of a maximal essential extension inside a fixed ambient module)**.** *Let V be an R-module, and suppose that V is embedded into an R-module*

I:

$$V \subseteq I.$$

Consider the set

$$\mathcal{S} = \{\, E \mid V \subseteq E \subseteq I, \ \textit{and } V \subseteq E \textit{ is an essential extension} \,\}.$$

Then $\mathcal{S}$ has a maximal element.

Proof. First, $\mathcal{S} \neq \varnothing$. Indeed, $V \in \mathcal{S}$, because $V \subseteq V$ is clearly essential: if $0 \neq W \subseteq V$, then

$$W \cap V = W \neq 0.$$

We partially order $\mathcal{S}$ by inclusion. To apply Zorn's lemma, we show that every chain in $\mathcal{S}$ has an upper bound in $\mathcal{S}$.

Let

$$\{E_\alpha\}_{\alpha \in A}$$

be a chain in $\mathcal{S}$. This means that for any $\alpha, \beta \in A$, either

$$E_\alpha \subseteq E_\beta \quad \text{or} \quad E_\beta \subseteq E_\alpha.$$

Define

$$E = \bigcup_{\alpha \in A} E_\alpha.$$

Because the family $\{E_\alpha\}_{\alpha \in A}$ is a chain, the union E is again an R-submodule of I. Clearly,

$$V \subseteq E \subseteq I.$$

It remains to prove that $V \subseteq E$ is essential. Let

$$0 \neq W \subseteq E$$

be a nonzero submodule. Choose a nonzero element

$$0 \neq w \in W.$$

Since $E = \bigcup_{\alpha \in A} E_\alpha$, there exists some $\alpha \in A$ such that

$$w \in E_\alpha.$$

Hence

$$Rw \subseteq E_\alpha.$$

Because $w \neq 0$ and R has identity, we have

$$Rw \neq 0.$$

Since $E_\alpha \in \mathcal{S}$, the inclusion $V \subseteq E_\alpha$ is essential. Therefore

$$Rw \cap V \neq 0.$$

But $Rw \subseteq W$, so

$$W \cap V \neq 0.$$

Thus $V \subseteq E$ is essential, and hence

$$E \in \mathcal{S}.$$

Therefore every chain in $\mathcal{S}$ has an upper bound in $\mathcal{S}$. By Zorn's lemma, $\mathcal{S}$ has a maximal element. Denote it by E. □

Remark 12.9.5. *The maximality of E means the following: if E' is another submodule of I such that*

$$E \subseteq E' \subseteq I$$

and $V \subseteq E'$ is still essential, then necessarily

$$E' = E.$$

Equivalently, there is no strictly larger submodule E' with

$$E \subsetneq E' \subseteq I$$

such that $V \subseteq E'$ remains an essential extension.

Remark 12.9.6. *In the proof of the existence of injective hulls, one usually first uses the standard fact that every R-module V can be embedded into some injective module I. Then the above lemma gives a maximal essential extension*

$$V \subseteq E \subseteq I.$$

The remaining step is to prove that this maximal essential extension E is itself injective. Once this is done, the inclusion

$$V \hookrightarrow E$$

is an injective hull of V.

Lemma 12.9.7 (Baer's criterion)**.** *Let R be a ring with identity, and let E be a left R-module. Then E is injective if and only if for every left ideal $\mathfrak{a} \subseteq R$, every R-homomorphism*

$$f : \mathfrak{a} \longrightarrow E$$

can be extended to an R-homomorphism

$$\widetilde{f} : R \longrightarrow E.$$

Equivalently, for every left ideal $\mathfrak{a} \subseteq R$, every diagram

$$\begin{tikzcd} \mathfrak{a} \arrow[r, "f"] \arrow[d, hook] & E \\ R \arrow[ur, dashed, "\widetilde{f}"'] & \end{tikzcd}$$

can be completed by some $\widetilde{f} : R \to E$.

Proof. First suppose that E is injective. Let $\mathfrak{a} \subseteq R$ be a left ideal, and let

$$f : \mathfrak{a} \longrightarrow E$$

be an R-homomorphism. Since $\mathfrak{a} \hookrightarrow R$ is a monomorphism of left R-modules, the injectivity of E implies that f extends to an R-homomorphism

$$\widetilde{f} : R \longrightarrow E.$$

Thus the stated condition holds.

Conversely, assume that every homomorphism from a left ideal of R to E can be extended to R. We prove that E is injective.

Let $A \subseteq B$ be an inclusion of left R-modules, and let

$$f : A \longrightarrow E$$

be an R-homomorphism. We will prove that f extends to B.

Consider the set $\mathcal{S}$ of pairs (C, g), where

$$A \subseteq C \subseteq B$$

is a submodule and

$$g : C \longrightarrow E$$

is an R-homomorphism satisfying

$$g|_A = f.$$

We partially order $\mathcal{S}$ by declaring

$$(C, g) \leq (C', g')$$

if

$$C \subseteq C' \quad \text{and} \quad g'|_C = g.$$

The set $\mathcal{S}$ is nonempty, since $(A, f) \in \mathcal{S}$. If

$$\{(C_i, g_i)\}_{i \in I}$$

is a chain in $\mathcal{S}$, then

$$C = \bigcup_{i \in I} C_i$$

is a submodule of B, and we may define

$$g : C \longrightarrow E$$

by

$$g(c) = g_i(c)$$

whenever $c \in C_i$. This is well-defined because the pairs form a chain. Hence (C, g) is an upper bound of the chain. By Zorn's lemma, $\mathcal{S}$ has a maximal element, say

$$(C, g).$$

We claim that $C = B$. Suppose not. Choose

$$b \in B \setminus C.$$

Define

$$\mathfrak{a} = \{ r \in R \mid rb \in C \}.$$

Then $\mathfrak{a}$ is a left ideal of R. Indeed, if $r, s \in \mathfrak{a}$, then

$$(r + s)b = rb + sb \in C,$$

and if $t \in R$, then

$$(tr)b = t(rb) \in C.$$

Define an R-homomorphism

$$h : \mathfrak{a} \longrightarrow E$$

by

$$h(r) = g(rb).$$

This is well-defined and R-linear because g is R-linear.

By the assumed condition, h extends to an R-homomorphism

$$\widetilde{h} : R \longrightarrow E.$$

Let

$$e = \widetilde{h}(1).$$

Then for every $r \in R$,

$$\widetilde{h}(r) = re,$$

and in particular, for every $r \in \mathfrak{a}$,

$$g(rb) = h(r) = \widetilde{h}(r) = re.$$

Now define

$$g' : C + Rb \longrightarrow E$$

by

$$g'(c + rb) = g(c) + re, \qquad c \in C,\ r \in R.$$

We check that g' is well-defined. Suppose

$$c + rb = c' + r'b.$$

Then

$$(r - r')b = c' - c \in C,$$

so $r - r' \in \mathfrak{a}$. Hence

$$g((r - r')b) = (r - r')e.$$

But

$$g((r - r')b) = g(c' - c) = g(c') - g(c).$$

Therefore

$$g(c') - g(c) = (r - r')e,$$

which implies

$$g(c) + re = g(c') + r'e.$$

Thus g' is well-defined. It is straightforward to check that g' is an R-homomorphism.

Moreover, g' extends g, since

$$g'(c) = g'(c + 0b) = g(c).$$

Thus

$$(C, g) < (C + Rb, g')$$

unless $C + Rb = C$. But $b \notin C$, so $C + Rb \neq C$. This contradicts the maximality of (C, g).

Therefore $C = B$. Hence $f : A \to E$ extends to a homomorphism $B \to E$. This proves that E is injective. □

Proof of 1(ii). It remains to prove that E is injective. We use Baer's criterion. Suppose, for contradiction, that E is not injective. Then there exists a left ideal

$$I \subseteq R$$

and an R-homomorphism

$$f : I \longrightarrow E$$

which cannot be extended to a homomorphism $R \to E$.

Construct an R-module

$$P = (E \oplus R)/N,$$

where

$$N = \{(f(a), -a) \mid a \in I\}.$$

The natural map

$$E \longrightarrow P, \qquad e \longmapsto (e, 0) + N$$

is injective. Indeed, if $(e, 0) \in N$, then

$$(e, 0) = (f(a), -a)$$

for some $a \in I$, hence $a = 0$ and $e = f(0) = 0$.

Choose, by Zorn's lemma, a submodule $C \subseteq P$ maximal subject to

$$C \cap E = 0.$$

Then the image of E in P/C is essential. Indeed, if D/C is a nonzero submodule of P/C, then $D \supsetneq C$. By maximality of C, we must have

$$D \cap E \neq 0.$$

Hence

$$(D/C) \cap E \neq 0.$$

Thus $E \subseteq P/C$ is an essential extension.

Since $V \subseteq E$ is essential and $E \subseteq P/C$ is essential, by transitivity $V \subseteq P/C$ is essential. But E was chosen to be a maximal essential extension of V. Therefore

$$P/C = E.$$

Let

$$x = (0, 1) + N + C \in P/C.$$

Since $P/C = E$, there exists $e \in E$ such that

$$x = e.$$

For every $a \in I$, we have in P/C

$$ax = (0, a) + N + C = (f(a), 0) + N + C = f(a).$$

On the other hand, since $x = e$, we also have

$$ax = ae.$$

Therefore

$$f(a) = ae \qquad \text{for all } a \in I.$$

Hence f extends to an R-homomorphism

$$R \longrightarrow E, \qquad r \longmapsto re,$$

contradicting the choice of f.

Thus E is injective. Since $V \subseteq E$ is an essential monomorphism and E is injective, the inclusion

$$V \hookrightarrow E$$

is an injective hull of V. $\square$

Proof of 2(i). We prove

$$J(eRe) = eJ(R)e = eRe \cap J(R).$$

First observe that

$$eJ(R)e \subseteq eRe \cap J(R),$$

because $J(R)$ is a two-sided ideal of R. Conversely, if

$$x \in eRe \cap J(R),$$

then $x = exe$, so

$$x \in eJ(R)e.$$

Hence

$$eJ(R)e = eRe \cap J(R).$$

It remains to prove

$$J(eRe) = eJ(R)e.$$

First let $a \in eJ(R)e$. Then $a \in eRe$. For every $y \in eRe$, we have

$$ya \in J(R).$$

By the usual characterization of the Jacobson radical, $1 - ya$ has a left inverse in R. Thus there exists $u \in R$ such that

$$u(1 - ya) = 1.$$

Multiplying by e on both sides gives

$$eue(e - ya) = e.$$

Hence $e - ya$ has a left inverse in the ring eRe, whose identity is e. Therefore

$$a \in J(eRe).$$

Thus

$$eJ(R)e \subseteq J(eRe).$$

Conversely, let

$$a \in J(eRe).$$

We show that $a \in J(R)$. It is enough to show that a annihilates every simple left R-module.

Let S be a simple left R-module. Then eS is either 0 or a simple left eRe-module. Indeed, if $0 \neq T \subseteq eS$ is an eRe-submodule and $0 \neq t \in T$, then $Rt = S$, and hence

$$eS = eRt = eRe\,t \subseteq T.$$

Therefore $T = eS$.

Since $a \in J(eRe)$, it annihilates every simple eRe-module. Hence

$$a(eS) = 0.$$

But $a \in eRe$, so $a = ae$. Therefore, for every $s \in S$,

$$as = aes = a(es) = 0.$$

Thus a annihilates every simple left R-module, and hence

$$a \in J(R).$$

Since also $a \in eRe$, we get

$$a \in eRe \cap J(R) = eJ(R)e.$$

Therefore

$$J(eRe) \subseteq eJ(R)e.$$

Combining the two inclusions gives

$$J(eRe) = eJ(R)e = eRe \cap J(R).$$

□

Proof of 2(ii). Let

$$S = M_n(R).$$

We prove

$$J(S) = M_n(J(R)).$$

Let E_{ij} denote the standard matrix units. Applying part $2(i)$ to the idempotent $E_{11} \in S$, we obtain

$$J(E_{11}SE_{11}) = E_{11}J(S)E_{11}.$$

But

$$E_{11}SE_{11} \cong R,$$

so

$$E_{11}J(S)E_{11} \cong J(R).$$

Now take $A = (a_{ij}) \in J(S)$. Then

$$E_{1i}AE_{j1} \in E_{11}J(S)E_{11}.$$

But

$$E_{1i}AE_{j1} = a_{ij}E_{11}.$$

Therefore $a_{ij} \in J(R)$ for all i, j. Hence

$$J(S) \subseteq M_n(J(R)).$$

Conversely, we show that

$$M_n(J(R)) \subseteq J(S).$$

It is enough to prove that for every $a \in J(R)$ and every i, j,

$$aE_{ij} \in J(S).$$

Let $X = (x_{pq}) \in S$. We must show that

$$I_n - X(aE_{ij})$$

has a left inverse in S.

Set

$$u = \begin{pmatrix} x_{1i}a \\ x_{2i}a \\ \vdots \\ x_{ni}a \end{pmatrix}, \qquad v^T = \begin{pmatrix} 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \end{pmatrix},$$

where the 1 occurs in the j-th position. Then

$$X(aE_{ij}) = uv^T,$$

and

$$v^T u = x_{ji}a.$$

Since $a \in J(R)$, the element

$$1 - x_{ji}a$$

has a left inverse in R. Hence there exists $c \in R$ such that

$$c(1 - x_{ji}a) = 1.$$

Equivalently,

$$c - c(v^T u) = 1.$$

Now compute:

$$(I_n + ucv^T)(I_n - uv^T) = I_n + u\big(c - 1 - c(v^T u)\big)v^T = I_n.$$

Thus $I_n - X(aE_{ij})$ has a left inverse for every $X \in S$. Therefore

$$aE_{ij} \in J(S).$$

Since $M_n(J(R))$ is generated additively by such elements aE_{ij}, we get

$$M_n(J(R)) \subseteq J(S).$$

Therefore

$$J(M_n(R)) = M_n(J(R)).$$

□

Proof of 3. Let R be a PID, and let F be a finitely generated free R-module. Then

$$F \cong R^n$$

for some $n \geq 0$. We prove by induction on n that every submodule of R^n is free.

For $n = 0$, the claim is trivial. For $n = 1$, every submodule of R is an ideal of R. Since R is a PID, every ideal is principal. Hence every submodule of R is free.

Assume now that $n \geq 2$, and suppose the result is true for R^{n-1}. Let

$$N \subseteq R^n$$

be a submodule. Consider the projection onto the first coordinate:

$$\pi : R^n \longrightarrow R.$$

Then $\pi(N)$ is an ideal of R. Since R is a PID, there exists $d \in R$ such that

$$\pi(N) = (d).$$

Let

$$K = N \cap \ker \pi.$$

Since

$$\ker \pi \cong R^{n-1},$$

the induction hypothesis implies that K is free.

We have an exact sequence

$$0 \longrightarrow K \longrightarrow N \xrightarrow{\pi} \pi(N) \longrightarrow 0.$$

Since

$$\pi(N) = (d)$$

is a principal ideal, it is a free R-module of rank 1 if $d \neq 0$, and it is the zero module if

$d = 0$. In either case, $\pi(N)$ is projective. Hence the short exact sequence splits. Therefore

$$N \cong K \oplus \pi(N).$$

Since both K and $\pi(N)$ are free, their direct sum is free. Hence N is free.

By induction, every submodule of a finitely generated free module over a PID is free. □

12.10 Homework Week 9

Exercise 12.10.1. *Let φ be a multiplicative valuation of a field K. Then the following hold:*

1. *The sequence (a_n) is a Cauchy sequence if and only if*

$$\lim_{n\to\infty} \varphi(a_{n+1} - a_n) = 0.$$

2. *If*

$$\lim_{n\to\infty} a_n = a \neq 0,$$

then

$$\varphi(a_n) = \varphi(a)$$

for all sufficiently large n.

Proof. We use the metric on K induced by φ, namely

$$d(x, y) = \varphi(x - y).$$

1. Suppose first that (a_n) is a Cauchy sequence. Then, for every $\varepsilon > 0$, there exists N such that for all $m, n \geq N$,

$$\varphi(a_m - a_n) < \varepsilon.$$

Taking $m = n + 1$, we get

$$\varphi(a_{n+1} - a_n) < \varepsilon$$

for all $n \geq N$. Hence

$$\lim_{n\to\infty} \varphi(a_{n+1} - a_n) = 0.$$

Conversely, suppose that

$$\lim_{n\to\infty} \varphi(a_{n+1} - a_n) = 0.$$

We show that (a_n) is Cauchy. Let $\varepsilon > 0$ be given. Then there exists N such that for all $k \geq N$,

$$\varphi(a_{k+1} - a_k) < \varepsilon.$$

If $m > n \geq N$, then

$$a_m - a_n = (a_m - a_{m-1}) + (a_{m-1} - a_{m-2}) + \cdots + (a_{n+1} - a_n).$$

Since φ is non-archimedean, we have

$$\varphi(a_m - a_n) \leq \max_{n \leq k \leq m-1} \varphi(a_{k+1} - a_k) < \varepsilon.$$

Therefore (a_n) is a Cauchy sequence.

2. Suppose that

$$\lim_{n\to\infty} a_n = a \neq 0.$$

Then

$$\lim_{n\to\infty} \varphi(a_n - a) = 0.$$

Since $a \neq 0$, we have $\varphi(a) > 0$. Hence there exists N such that for all $n \geq N$,

$$\varphi(a_n - a) < \varphi(a).$$

For such n, we write

$$a_n = a + (a_n - a).$$

By the strong triangle inequality,

$$\varphi(a_n) \leq \max\{\varphi(a), \varphi(a_n - a)\} = \varphi(a).$$

On the other hand,

$$a = a_n - (a_n - a),$$

so

$$\varphi(a) \leq \max\{\varphi(a_n), \varphi(a_n - a)\}.$$

Since $\varphi(a_n - a) < \varphi(a)$, the maximum on the right-hand side must be $\varphi(a_n)$. Thus

$$\varphi(a) \leq \varphi(a_n).$$

Combining the two inequalities gives

$$\varphi(a_n) = \varphi(a)$$

for all $n \geq N$.

$\square$

Exercise 12.10.2. *Let φ be a multiplicative valuation of a field K. Let V be a finite dimensional K-space with basis $u_1, \ldots, u_n$. For*

$$v = a_1 u_1 + \cdots + a_n u_n \in V,$$

define $\|v\|_0$ by

$$\|v\|_0 = \max\{\varphi(a_i) \mid 1 \leq i \leq n\}.$$

Then this is a norm on V. Moreover, the following hold concerning the metric induced by this norm:

1. *Let*

$$v_r = a_{r1}u_1 + \cdots + a_{rn}u_n \in V$$

be given for $r = 1, 2, \ldots$. Then the sequence (v_r) is a Cauchy sequence if and only if the sequence $(a_{ri})_r$ in K is a Cauchy sequence for every i.

Also,

$$\lim_{r\to\infty} v_r = \sum_{i=1}^{n} a_i u_i$$

if and only if

$$\lim_{r\to\infty} a_{ri} = a_i$$

for every i.

2. *If K is complete, then so is V.*

Proof. First we prove that $\|\cdot\|_0$ is a norm on V.

Let

$$v = a_1u_1 + \cdots + a_nu_n.$$

Then

$$\|v\|_0 = \max\{\varphi(a_i) \mid 1 \leq i \leq n\}.$$

Since $\varphi(a_i) \geq 0$ for every i, we have

$$\|v\|_0 \geq 0.$$

Moreover,

$$\|v\|_0 = 0$$

if and only if

$$\varphi(a_i) = 0$$

for every i, which is equivalent to

$$a_i = 0$$

for every i. Since $u_1, \ldots, u_n$ is a basis, this is equivalent to

$$v = 0.$$

Now let $\lambda \in K$. Then

$$\lambda v = \lambda a_1u_1 + \cdots + \lambda a_nu_n,$$

and therefore

$$\|\lambda v\|_0 = \max_{1\leq i\leq n} \varphi(\lambda a_i) = \max_{1\leq i\leq n} \varphi(\lambda)\varphi(a_i).$$

Hence

$$\|\lambda v\|_0 = \varphi(\lambda) \max_{1\leq i\leq n} \varphi(a_i) = \varphi(\lambda)\|v\|_0.$$

Finally, let

$$v = a_1 u_1 + \cdots + a_n u_n, \qquad w = b_1 u_1 + \cdots + b_n u_n.$$

Then

$$v + w = (a_1 + b_1)u_1 + \cdots + (a_n + b_n)u_n.$$

Thus

$$\|v + w\|_0 = \max_{1 \le i \le n} \varphi(a_i + b_i).$$

Since φ is non-archimedean,

$$\varphi(a_i + b_i) \le \max\{\varphi(a_i), \varphi(b_i)\}$$

for every i. Hence

$$\|v + w\|_0 \le \max_{1 \le i \le n} \max\{\varphi(a_i), \varphi(b_i)\}.$$

Therefore

$$\|v + w\|_0 \le \max\left\{\max_{1 \le i \le n} \varphi(a_i), \max_{1 \le i \le n} \varphi(b_i)\right\} = \max\{\|v\|_0, \|w\|_0\}.$$

So $\|\cdot\|_0$ is a non-archimedean norm on V.

1. Let

$$v_r = a_{r1} u_1 + \cdots + a_{rn} u_n.$$

For $r, s \ge 1$, we have

$$v_r - v_s = (a_{r1} - a_{s1})u_1 + \cdots + (a_{rn} - a_{sn})u_n.$$

Hence

$$\|v_r - v_s\|_0 = \max_{1 \le i \le n} \varphi(a_{ri} - a_{si}).$$

Suppose first that (v_r) is Cauchy. Let i be fixed. Since

$$\varphi(a_{ri} - a_{si}) \le \max_{1 \le j \le n} \varphi(a_{rj} - a_{sj}) = \|v_r - v_s\|_0,$$

it follows that $(a_{ri})_r$ is a Cauchy sequence in K.

Conversely, suppose that for every i, the sequence $(a_{ri})_r$ is Cauchy in K. Let $\varepsilon > 0$ be given. For each i, there exists N_i such that for all $r, s \ge N_i$,

$$\varphi(a_{ri} - a_{si}) < \varepsilon.$$

Let

$$N = \max\{N_1, \ldots, N_n\}.$$

Then for all $r, s \ge N$ and all i, we have

$$\varphi(a_{ri} - a_{si}) < \varepsilon.$$

Therefore

$$\|v_r - v_s\|_0 = \max_{1 \le i \le n} \varphi(a_{ri} - a_{si}) < \varepsilon.$$

Hence (v_r) is Cauchy in V.

Next we prove the assertion about convergence. Let

$$v = \sum_{i=1}^{n} a_i u_i.$$

Then

$$v_r - v = (a_{r1} - a_1)u_1 + \cdots + (a_{rn} - a_n)u_n.$$

Hence

$$\|v_r - v\|_0 = \max_{1 \le i \le n} \varphi(a_{ri} - a_i).$$

Therefore

$$\lim_{r \to \infty} v_r = v$$

if and only if

$$\lim_{r \to \infty} \max_{1 \le i \le n} \varphi(a_{ri} - a_i) = 0.$$

Since there are only finitely many indices i, this is equivalent to

$$\lim_{r \to \infty} \varphi(a_{ri} - a_i) = 0$$

for every i, which is precisely

$$\lim_{r \to \infty} a_{ri} = a_i$$

for every i.

2. Suppose that K is complete. We show that V is complete.

   Let (v_r) be a Cauchy sequence in V, where

   $$v_r = a_{r1}u_1 + \cdots + a_{rn}u_n.$$

   By part (1), for each i, the sequence $(a_{ri})_r$ is Cauchy in K. Since K is complete, there exists $a_i \in K$ such that

   $$\lim_{r \to \infty} a_{ri} = a_i.$$

   Define

   $$v = \sum_{i=1}^{n} a_i u_i \in V.$$

   Again by part (1), we get

   $$\lim_{r \to \infty} v_r = v.$$

   Thus every Cauchy sequence in V converges in V. Hence V is complete.

□

12.11 Homework Week 10

Exercise 12.11.1. *Let R be a ring and let V be an R-module. Let I be an ideal of R. Suppose that d_I is defined on V and on R. Then:*

1. *Addition is continuous from $V \times V$ to V.*

2. *Multiplication is continuous from $R \times V$ to V.*

3. *If d_I is defined on the R-module W and $f \in \operatorname{Hom}_R(V, W)$, then f is continuous.*

Proof. Recall that, for an R-module M, the I-adic topology induced by d_I has a basis of neighbourhoods of $x \in M$ given by

$$x + I^n M, \qquad n \geq 1.$$

1. Let $(v_0, w_0) \in V \times V$. We show that addition is continuous at (v_0, w_0). Let

$$v_0 + w_0 + I^n V$$

be a neighbourhood of $v_0 + w_0$. If

$$v - v_0 \in I^n V, \qquad w - w_0 \in I^n V,$$

then

$$(v + w) - (v_0 + w_0) = (v - v_0) + (w - w_0) \in I^n V.$$

Hence

$$v + w \in v_0 + w_0 + I^n V.$$

Therefore addition $V \times V \to V$ is continuous.

2. Let $(r_0, v_0) \in R \times V$. We show that scalar multiplication is continuous at (r_0, v_0). Let

$$r_0 v_0 + I^n V$$

be a neighbourhood of $r_0 v_0$. Suppose

$$r - r_0 \in I^n, \qquad v - v_0 \in I^n V.$$

Then

$$rv - r_0 v_0 = (r - r_0)v + r_0(v - v_0).$$

Since $r - r_0 \in I^n$, we have

$$(r - r_0)v \in I^n V.$$

Also, because $I^n V$ is an R-submodule of V,

$$r_0(v - v_0) \in I^n V.$$

Therefore

$$rv - r_0 v_0 \in I^n V.$$

Hence

$$rv \in r_0 v_0 + I^n V.$$

Thus multiplication $R \times V \to V$ is continuous.

3. Let $f \in \mathrm{Hom}_R(V, W)$. We first observe that

$$f(I^n V) \subseteq I^n W$$

for every $n \geq 1$. Indeed, if

$$x = \sum_{j=1}^{m} a_j v_j \in I^n V, \qquad a_j \in I^n,\ v_j \in V,$$

then

$$f(x) = f\left(\sum_{j=1}^{m} a_j v_j\right) = \sum_{j=1}^{m} a_j f(v_j) \in I^n W.$$

Now let $v_0 \in V$, and let

$$f(v_0) + I^n W$$

be a neighbourhood of $f(v_0)$. If

$$v - v_0 \in I^n V,$$

then

$$f(v) - f(v_0) = f(v - v_0) \in I^n W.$$

Hence

$$f(v) \in f(v_0) + I^n W.$$

Therefore f is continuous.

$\square$

Exercise 12.11.2. *Let R be a commutative ring, and let A and B be R-algebras. Prove the following algebra isomorphisms:*

1.

$$M_n(A) \otimes_R B \cong M_n(A \otimes_R B).$$

2.

$$M_n(A) \otimes_R M_m(B) \cong M_{nm}(A \otimes_R B).$$

Proof. 1. Define a map

$$\Phi : M_n(A) \otimes_R B \longrightarrow M_n(A \otimes_R B)$$

on elementary tensors by

$$\Phi\big((a_{ij}) \otimes b\big) = (a_{ij} \otimes b).$$

This is well-defined and R-linear by the universal property of the tensor product.

We construct its inverse. Every element of $M_n(A \otimes_R B)$ can be written as a matrix whose (i,j)-entry is a finite sum of elementary tensors. Define

$$\Psi : M_n(A \otimes_R B) \longrightarrow M_n(A) \otimes_R B$$

by

$$\Psi\big((x_{ij})\big) = \sum_{i,j} \sum_{\ell} a_{ij\ell} E_{ij} \otimes b_{ij\ell},$$

whenever

$$x_{ij} = \sum_{\ell} a_{ij\ell} \otimes b_{ij\ell}.$$

This is well-defined because tensor product distributes over finite direct sums. It is immediate that

$$\Phi \circ \Psi = \mathrm{id}, \qquad \Psi \circ \Phi = \mathrm{id}\,.$$

Hence Φ is an R-module isomorphism.

It remains to check that Φ is an algebra homomorphism. Let $X = (x_{ij})$, $Y = (y_{ij}) \in M_n(A)$, and let $b, b' \in B$. Then

$$\Phi\big((X \otimes b)(Y \otimes b')\big) = \Phi(XY \otimes bb') = ((XY)_{ik} \otimes bb')_{i,k}.$$

On the other hand,

$$\Phi(X \otimes b)\Phi(Y \otimes b') = (x_{ij} \otimes b)_{i,j}(y_{jk} \otimes b')_{j,k},$$

whose (i,k)-entry is

$$\sum_j (x_{ij} \otimes b)(y_{jk} \otimes b') = \sum_j x_{ij}y_{jk} \otimes bb' = (XY)_{ik} \otimes bb'.$$

Thus

$$\Phi\big((X \otimes b)(Y \otimes b')\big) = \Phi(X \otimes b)\Phi(Y \otimes b').$$

Also,

$$\Phi(I_n \otimes 1_B) = I_n.$$

Therefore

$$M_n(A) \otimes_R B \cong M_n(A \otimes_R B)$$

as R-algebras.

2. Let E_{ij} denote the standard matrix units in $M_n(A)$, and let F_{kl} denote the standard matrix units in $M_m(B)$. We identify the rows and columns of $M_{nm}(A \otimes_R B)$ with

pairs

$$(i,k), \qquad 1 \le i \le n, \quad 1 \le k \le m.$$

Let $G_{(i,k),(j,l)}$ be the corresponding standard matrix unit in $M_{nm}(A \otimes_R B)$.

Define

$$\Theta : M_n(A) \otimes_R M_m(B) \longrightarrow M_{nm}(A \otimes_R B)$$

by

$$\Theta\big((aE_{ij}) \otimes (bF_{kl})\big) = (a \otimes b)G_{(i,k),(j,l)}.$$

Since

$$M_n(A) = \bigoplus_{i,j} AE_{ij}, \qquad M_m(B) = \bigoplus_{k,l} BF_{kl},$$

and tensor product distributes over finite direct sums, Θ is an R-module isomorphism.

We check multiplicativity on elementary generators. We have

$$(aE_{ij})(a'E_{i'j'}) = \delta_{j,i'}aa'E_{ij'},$$

and

$$(bF_{kl})(b'F_{k'l'}) = \delta_{l,k'}bb'F_{kl'}.$$

Therefore

$$\begin{aligned}&\Theta\left(\big((aE_{ij}) \otimes (bF_{kl})\big)\big((a'E_{i'j'}) \otimes (b'F_{k'l'})\big)\right)\\ &= \delta_{j,i'}\delta_{l,k'}(aa' \otimes bb')G_{(i,k),(j',l')}.\end{aligned}$$

On the other hand,

$$\begin{aligned}&\Theta\big((aE_{ij}) \otimes (bF_{kl})\big)\Theta\big((a'E_{i'j'}) \otimes (b'F_{k'l'})\big)\\ &= (a \otimes b)G_{(i,k),(j,l)}(a' \otimes b')G_{(i',k'),(j',l')}\\ &= \delta_{j,i'}\delta_{l,k'}(aa' \otimes bb')G_{(i,k),(j',l')}.\end{aligned}$$

Hence Θ preserves multiplication. It also sends the identity to the identity. Therefore

$$M_n(A) \otimes_R M_m(B) \cong M_{nm}(A \otimes_R B)$$

as R-algebras.

$\square$